

Smoothed exact global optimization beyond convexity

Abstract

We study exact global optimization of nonconvex quadratic and fixed-degree polynomial problems with continuous and integer variables after one linear perturbation from a finite rational law computed before sampling. The algorithms return an exact global optimizer on every draw and never resample. Their expected bit work is polynomial in the input length and explicit curvature, width, and noise ratios for fixed structural parameters. For quadratic programs with few negative directions, aligned noise for continuous problems and Gaussian-like noise for mixed problems give FPT bounds jointly in negative inertia, integer dimension where applicable, and numerical ratios; uniform ambient noise gives a dimension-dependent exponent. Sparse polynomial interactions admit polynomial bounds for each fixed treewidth, including graph, simplex, and order constraints. A small core with tractable recourse gives FPT bounds jointly in core size and numerical ratios. Core-only noise suffices for uniformly strongly convex continuous recourse and structured convex integer recourse, with distinct sampling-precision bounds. Strong noise gives a separate component algorithm without a width parameter. The proofs combine curvature-corrected refinement, noise-dependent counting, and exact certificates; most routes use a rare same-draw fallback, while lattice and component methods finish directly. We prove limits of the sparse retention rule, a hardness implication for arbitrary sparse affine constraints, and arithmetic implications for convex quartic residual point evaluation under core-only noise. Sampled solutions give certified original-objective approximations with the numerical factors retained. Practical performance is unestablished.

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1 Introduction

Global optimization asks for a provably optimal solution of a nonconvex problem, not a locally optimal or approximately optimal one. For quadratic and polynomial objectives with continuous and integer variables this is hard in the worst case, even under strong structural restrictions. Minimizing a quadratic function with a single negative eigenvalue is NP-hard [35], and box-constrained quadratic programming remains strongly NP-hard when its interaction graph has treewidth two [19]. An exact method must also handle degenerate configurations: ties between distant solutions, flat optimal faces, and optimal values very close to a decision threshold. The analysis below treats each of them explicitly. Spatial branch-and-bound solvers handle many instances well, but they have no polynomial guarantee.

Smoothed analysis [42] measures the cost of an algorithm on a worst-case instance after a small random perturbation of its data. For discrete optimization with a linear objective, polynomial smoothed complexity is closely tied to pseudo-polynomial solvability [11, 39]. For continuous problems, a linear tilt removes degeneracy qualitatively: for semialgebraic problems under a constraint qualification, generic tilts give a unique minimizer with quadratic growth [32]. Such statements give no probability bounds, and an exact algorithm in the bit model needs more. It needs quantitative probability bounds, a finite rational law that a Turing machine can sample, and an exact way to finish the draws that remain degenerate, which under a finite law have positive probability.

This paper studies exact global optimization under one random linear perturbation. A rational base instance with objective F_0 and feasible set X is given. A vector γ is drawn once from a finite rational law that is computed from the base instance before sampling, and the task is an exact global minimizer of the sampled objective

$$F_\gamma(x) = F_0(x) + \gamma^\top x \quad \text{on } X.$$

We ask for algorithms that are correct on every draw, never resample, and have expected bit complexity polynomial in the input length for fixed structural parameters, with explicit dependence on numerical ratios between curvature, widths, and the noise scale σ . [Section 2](#) makes this model precise.

1.1 Main results

The results follow four structural routes. They use different structure, and partly different noise models. [Table 1](#) lists the main theorems.

Few negative curvature directions ([section 4](#)). For a rational quadratic program on a bounded polytope whose Hessian has k negative eigenvalues, the nonconvexity lies in a k -dimensional factor. The algorithm searches k auxiliary coordinates that carry the negative curvature and solves a convex quadratic program at each query. Under an independent Gaussian-like finite law on every original coefficient, the expected bit work is $f(n_z, k, 1 + \nu \text{diam}(\mathcal{P})/\sigma) \text{poly}(I)$ with an absolute exponent, also for bounded mixed-integer programs with n_z integer variables; here ν is the largest negative curvature and $\mathcal{P}$ the polyhedral relaxation ([theorem 4.19](#)). Noise aligned with the negative directions gives a similar fixed-parameter bound for continuous problems ([theorem 4.21](#)). Under independent uniform noise on every coefficient the bound is polynomial for each fixed k but contains a power of the dimension that grows with k ([theorem 4.20](#)), and [theorem 9.4](#) shows that the counted quantities really depend on the dimension. For $k \leq 2$ a different argument gives a bound linear in a numerical ratio ([theorem 4.9](#)). Convex piecewise-quadratic unaries with rational breakpoints plus a concave term with k supplied factor rows allow arbitrarily many integer variables ([theorems 4.29 and 4.31](#));

on integer boxes with separable convex polynomial terms, a lattice of attainable objective values gives exactness without any fallback ([theorem 8.17](#)).

Sparse interactions ([sections 5 and 6](#)). For a fixed-degree polynomial on a mixed box whose monomials fit the bags of a supplied tree decomposition with at most p coordinates per bag, the expected bit work is

$$C_0^p \left[4 + \frac{(1 + n/2)Lw_{\max}}{2\sigma} \right]^p \text{poly}_d(I)$$

under independent uniform noise on all coordinates, where L bounds the coordinate second derivatives and $w_{\max}$ the coordinate widths ([theorem 5.2](#)). The integer dimension and the negative inertia are unrestricted, and no growth or nondegeneracy assumption is made. The bound is polynomial for each fixed p , but it is not fixed-parameter in p , and [theorem 9.1](#) shows that its factor $n^{\Omega(p)}$ is necessary for the retention rule used. The method extends to equality constraints given by explicit polynomial maps or monotone implicit equations, to affine state recurrences with polynomial actuators, to products of simplices, and to order constraints $x_i \leq x_j$ with binary variables ([theorems 6.4, 6.11, 6.15 and 6.17](#)). For arbitrary sparse affine constraints no such result holds unless $\text{NP} \subseteq \text{ZPP}$ ([theorem 9.5](#)).

A small core with tractable recourse ([sections 7 and 8](#)). Suppose that fixing k continuous core coordinates leaves a residual problem that an oracle solves exactly, or with certified error, on every residual subbox: a convex quadratic or polynomial problem, a quadratic program whose residual interaction graph is a forest, or a native-integer problem with an exact oracle under bound tightening. Then the expected bit work of the core search is at most $8^k[3 + (1 + k)L/(2\sigma)]^k$ times a polynomial in the input length that includes the oracle cost, where L bounds the curvature of the core coordinates only; integer recourse adds an exact solve of a small polynomial problem of cost c_d^k times a polynomial ([theorems 7.9, 7.13 and 8.4](#)). The number of residual variables, their coupling, and their curvature enter only through the input length. Under noise on all coordinates, residual convexity suffices. Under noise on the core alone, our continuous point theorem uses a certified uniform strong-convexity modulus of continuous recourse ([theorem 7.15](#)), or, for separable convex integer flows and totally unimodular recourse, a certificate that describes all tied optimal flows at once ([theorems 8.11, 8.12 and 8.14](#)). These are sufficient conditions. The companion results described below also provide point oracles for residual-convex cubics and supplied convexifiers; general arithmetic obstructions start at degree four ([section 9.4](#)).

Strong noise ([theorem 8.16](#)). Without any width or core parameter, if the noise scale dominates the variation of every coordinate derivative, weighted by the size of the coordinate's domain and the degree of the interaction graph, then most coordinates are pinned at a bound by the sign of their derivative, and the others form small random components of the interaction graph. Each component is solved exactly by a fixed-degree algebraic procedure ([theorem 8.1](#)). Here the noise is large, not small, and the result describes strongly random linear parts rather than slight perturbations. The sufficient scale (41) is conservative: for nonquadratic continuous components it contains the large effective solver base c_d , whereas the quadratic case uses the smaller weight $a_i = 3$.

In every route the output is an exact optimizer of the sampled instance. For quadratic objectives on rational polyhedral and box domains it is rational; an implicit graph lift can be irrational. For polynomial objectives the usual output is a compact implicit descriptor: fixed integer labels and active bounds, and a small box on which the objective is certified strongly convex, so that its unique minimizer there is the global optimizer. The descriptor supports evaluation of the optimizer and the optimal value to any requested precision in time polynomial in the precision. Results that finish with a small exact algebraic solve return all coordinates of the optimizer as functions of one real algebraic number. The strong-noise route returns one such representation per random component, with the optimal value as a sum of component values. Bilinear flow coupling and interior optimal

Table 1: Main bounds, exact on every draw of one finite law. Here k is negative inertia, supplied factor row count, or core size; p is bag size; b is grid bits or Gaussian accuracy; $Q_{\text{ex}} = [3 + (1 + k/2)L/(2\sigma)]^k$ and $Q_{\text{ap}} = [3 + (1 + k)L/(2\sigma)]^k$. FPT parameters include numerical ratios. Usual outputs are listed; rare fallback outputs can be exponentially long.

Result	Structure	Noise	b	Expected bit work	Output
<i>Few negative directions</i>					
Theorem 4.19	QP on a mixed polytope	Gaussian-like	$\text{poly}(I)$	$f(n_z, k, 1 + \nu \text{diam}(\mathcal{P})/\sigma) \text{poly}(I)$	rational
Theorem 4.21	QP on a polytope	aligned	$\text{poly}(I)$	$C_0^k(1 + \mathcal{H}_{\text{al}}) \text{poly}(I)$, FPT in k and factor ratios	rational
Theorem 4.20	QP on a mixed polytope	uniform	$\text{poly}(I)$	polynomial for fixed k, n_z and numerical scales; power of n grows with k	rational
Theorem 4.9	QP on a polytope, $k \leq 2$	uniform or Gaussian-like	$\text{poly}(I)$	$\text{poly}(I)(1 + \nu S/\sigma)$	rational
Theorems 4.29 and 4.31	convex piecewise-quadratic unaries, rational knots, plus a concave factor, any n_z	aligned, uniform, Gaussian-like	$\text{poly}(I)$	law-dependent factors of the cited theorems; no factor for n_z	rational
Theorem 8.17	integer box, convex piecewise-polynomial unaries plus a concave factor	aligned	$\text{poly}(I)$	$8^k H_{\text{rat}} \text{poly}(I)$, no fallback	rational
<i>Sparse interactions</i>					
Theorem 5.2	degree- d polynomial on a mixed box, bag size p	uniform	$\text{poly}_d(I)$	$C_0^p \Theta^p \text{poly}_d(I)$ with $\Theta = 4 + (1 + n/2)Lw_{\text{max}}/(2\sigma)$	implicit; rational for QP
Theorems 6.4, 6.11, 6.15 and 6.17	graphs, actuator recurrences, simplices, orders with binaries	uniform	$\text{poly}_d(I)$	polynomial for fixed p and displayed numerical scales	charted or implicit
<i>Small core with tractable recourse</i>					
Theorem 7.9	quadratic, exact recourse on every subbox	uniform	$\text{poly}(I)$	$8^k Q_{\text{ex}} \text{poly}(I)$	rational
Theorem 7.13	certified recourse, e.g. convex residual	uniform	$\text{poly}_d(I)$	$8^k Q_{\text{ap}} \text{poly}_d(I)$	implicit
Theorem 7.15	residual Hessian $\succeq \mu I$	core only	$\text{poly}_d(I)$	$8^k Q_{\text{ap}} \text{poly}_d(I)$	implicit
Theorem 8.4	native-integer recourse, exact oracle	uniform	$\text{poly}_d(I)$	$[8^k Q_{\text{ex}} + c_d^k] \text{poly}_d(I)$	label + algebraic, size $c_d^k \text{poly}_d(I)$
Theorem 8.9, corollary 8.13	convex integer flows, interior cores or bilinear coupling	core only	$\text{poly}_d(I)$	$[8^k Q_{\text{ex}} + c_d^k] \text{poly}_d(I)$	label + algebraic
Theorems 8.12 and 8.14	convex integer flows, totally unimodular systems	core only	$f_d(k) \text{poly}_d(I)$	$f_d(k) Q_{\text{ex}} \text{poly}_d(I)$	label + algebraic, size $f_d(k) \text{poly}_d(I)$
<i>Strong noise</i>					
Theorem 8.16	degree- d polynomial on a mixed box	strong uniform	any	$\text{poly}_d(I + b)$ times a subcritical branching factor	components; rational for QP

cores need only polynomially many sampling bits ([corollary 8.13](#) and [theorem 8.9](#)).

1.2 One mechanism

The first three routes use one mechanism with four steps. Stating it once explains what each theorem needs from its structure.

Curvature-corrected refinement preserves global optimizers. Each route has a search space, namely auxiliary factor coordinates, bag tuples, or core coordinates, and a value function on it obtained by minimizing over the other variables. Minimizing over a fixed feasible set preserves an upper bound on coordinate curvature. Hence the minimum of the value function over a grid cell is at least its least corner value minus a correction proportional to the curvature bound and the squared mesh. Refining only the cells whose corrected bound does not exceed the incumbent never discards a global optimizer. This holds on every draw, including draws with ties.

Independent linear noise bounds the expected number of retained states. A retained cell has a corner whose value is within a fixed multiple of the sum of the correction and certified evaluation error of the optimum. Comparing such a corner with its two grid neighbors in one coordinate confines that coordinate's noise coefficient to an interval whose length is proportional to the curvature bound times the mesh. The interval does not depend on the other coefficients of the search space, so when these coefficients are independent the probabilities multiply. Summing over a deterministic grid gives an expected count that does not depend on the mesh. The count is taken over a grid fixed before sampling, so no independence between the noise and the adaptively retained cells is assumed. The direct product argument applies to sparse box bags, cores, and aligned noise; constrained-domain variants use their proved directional and transport bounds. Under uniform ambient noise in the low-rank route, the factor coefficients and residual noise are generally dependent; a conditional volume estimate ([lemma 4.23](#)) replaces independence there and costs a power of the dimension. For Gaussian-like noise, independence holds in the continuous Gaussian proxy.

Exact closure finishes the sampled problem. Refinement alone gives approximations. An exact answer comes from a certificate proving that a small exactly solvable problem contains a global optimizer: a region on which a parametric convex quadratic program has one formula, a strongly convex patch after fixing integer labels and active bounds, a gap that excludes every competing integer label, or a certificate that fixes the face of the core and the set of tied optimal flows. Each certificate is checked exactly and is sound on every draw.

A base-selected finite law pays for an exact fallback on the same draw. The certificate passes by a refinement depth computed from base data unless the draw lies in a rare exceptional set: small growth, a small gradient at an active bound, a near-tie between integer labels, or noise near one of finitely many fixed hyperplanes. The finite law is chosen fine enough that this set has probability at most the reciprocal of the cost factor of an exact fallback algorithm, which then runs on the same draw. The cost factor is fixed first, then the thresholds and the depth, and the law last, so that no precision depends on the sampled coefficients it is meant to control.

The common invariant is a valid curvature-corrected bound together with a noise comparison whose deterministic part excludes the tested coefficient. Direct recourse obtains it by minimizing over a feasible set independent of the searched coordinates. Graph charts and order transport replace that literal premise with their proved curvature and feasible-comparison arguments. The boundaries in [section 9](#) delimit these mechanisms. Two results finish differently: the integer lattice theorem ([theorem 8.17](#)) stops exactly once a certified gap falls below the spacing of the attainable objective values, and the strong-noise theorem ([theorem 8.16](#)) solves its random components exactly on every draw. Neither needs a rare fallback.

1.3 What the guarantees mean

Several distinctions matter for reading the theorems.

Parameters. A fixed-parameter bound has the form $f(\kappa) \text{poly}(I)$ with an exponent independent of κ ; the low-rank results under aligned or Gaussian-like noise and the core results are of this form, with the numerical ratio included in κ . The sparse results are polynomial for each fixed width but are not fixed-parameter in the width. A bound that is polynomial in a ratio such as L/σ is polynomial in its magnitude. For fixed structure, polynomially bounded ratios give polynomial work, but $(1 + L/\sigma)^{O(k)} \text{poly}(I)$ with $L/\sigma = \text{poly}(I)$ gives $I^{O(k)}$; it does not give FPT in k alone.

Every draw versus expectation. Correctness holds on every draw, including ties and draws with a continuum of optimizers. Only the work bound is an expectation, over all draws; on rare draws the work can be exponential.

The law. The finite law is constructed from the base instance; its grid size or accuracy is chosen after a fallback cost factor and a stopping depth are known. The theorems do not cover an arbitrary finite law, a coarse grid chosen independently of the instance, or an arbitrary approximation of a Gaussian. Most results use polynomially many random bits per coefficient, and for them one resolution depending only on the input length suffices ([proposition 2.3](#)); the nonlinear integer-flow results with arbitrary core faces use a number of bits that depends on the core size.

Outputs. Rational outputs are explicit. Implicit outputs are compact descriptors whose global optimality is established by a separate proof record; the descriptor has polynomial length, the expected length of the record obeys the corresponding structural and numerical work bound, and evaluation to q bits costs time polynomial in $I + q$. Algebraic outputs from a fallback can be exponentially long on rare draws. No result promises expanded minimal polynomials of small degree or exact comparisons of sums of unrelated algebraic numbers. No route promises a canonical choice among tied optimizers on every draw.

The original objective. All theorems optimize the sampled objective. [Proposition 2.6](#) turns a sampled solution into a certified interval for the original optimum, of length at most the width of the realized perturbation over X , $\max_X \gamma^\top x - \min_X \gamma^\top x$, plus the evaluation error. Choosing the noise scale so that the largest possible such width is at most $\varepsilon/2$, which for Gaussian-like laws must account for their support radius ([section 2.6](#)), for routes valid at every positive scale, gives on every draw an exactly feasible point whose original objective value is certified to be within ε of optimal. The expected work is polynomial in $1/\varepsilon$ when the structure and the other numerical parameters are fixed, and polynomial in I and $1/\varepsilon$ only when those parameters are polynomially bounded. On implicit graphs the feasible point has rational free coordinates and an exact algebraic lift. We do not claim that this improves on dedicated approximation algorithms, and the strong-noise results give no such consequence. No result solves the unperturbed problem exactly, which is consistent with the worst-case hardness cited above.

1.4 Boundaries

[Section 9](#) delimits the method by explicit families. For the sparse retention rule, connected quartic instances of treewidth $p - 1$, with fixed curvature, widths, and noise scale, retain more than $(5n/(6p))^{p/2}$ cells on every draw, so the rule admits no bound $f(p) \text{poly}(I)$ ([theorem 9.1](#)); replacing the global rounding allowance by a bag-local one deletes all optimizers ([proposition 9.2](#)). Certified conditional values give one sound way to remove the ambient dimension from this count ([proposition 5.10](#) and [corollary 9.3](#)). Under uniform ambient noise, perfectly conditioned low-rank instances have expected counts growing like $(N/k)^{k/4}$ ([theorem 9.4](#)). Objective noise does not make arbitrary sparse affine constraints of treewidth three tractable unless $\text{NP} \subseteq \text{ZPP}$, because every

exact sampled optimizer of a suitable family decides SUBSET SUM (theorem 9.5). Under core-only noise with convex quartic residual problems, a constant-accuracy residual point evaluator with expected polynomial work would give Las Vegas algorithms for Square Root Sum and PosSLP. The Square Root Sum implication already holds at treewidth two with bounded coefficients; the PosSLP construction has neither restriction (theorem 9.10). Value accuracy with polynomially many bits does not localize residual points (proposition 9.9). Single-flow certificates fail at boundary core optima on an event of probability $1/16$ (proposition 9.11). These are limitations of specified mechanisms or conditional implications, not unconditional lower bounds, and several of the instances are easy for other methods.

1.5 Relation to prior work

The ingredients of the mechanism are classical, and we use them as such: semiconcavity and corner rounding on grids, nonserial dynamic programming on tree decompositions [14], exact polynomial-time convex quadratic programming [31], convex optimization in the bit model [24], exact convex mixed-integer quadratic programming in fixed integer dimension [17], separable convex optimization over totally unimodular systems [25], quantifier elimination with block-sensitive bounds [37] and other effective real-algebraic methods [9, 10], and isolation-type anticoncentration for perturbed linear objectives [11, 39]. Generic uniqueness and growth under linear tilts of semialgebraic problems are known qualitatively [32]; section 3 quantifies them for the finite laws used here.

Among smoothed analyses of continuous global optimization, Kelner and Nikolova [28] give expected polynomial time for minimizing constant-rank quasi-concave functions over integral polytopes under a random rotation of the objective subspace, by bounding the vertices of a projected polytope. Our low-rank route perturbs the linear coefficients instead, allows indefinite quadratic objectives that are not quasi-concave, and returns exact rational optimizers. For problems with few negative eigenvalues, deterministic algorithms give approximations whose work grows polynomially in the inverse accuracy for fixed negative inertia [45]. Rational Jacobi rotations also yield approximation algorithms for mixed-integer quadratic programs with fixed integer dimension and negative inertia [18]. For bounded treewidth, linear programming hierarchies give approximations of polynomial optimization problems in time polynomial for fixed width [15]. The original-objective consequence of our results (section 1.3) is of the same type and is not claimed to be better.

Recent deterministic exact results address other structured classes. Khajavirad [29] solves sparse box polynomial problems under sign, hidden-binary, width, and component-curvature conditions; the quadratic result is in the rational bit model, whereas its higher-degree result uses a unit-cost algebraic RAM. Exact integer and mixed-integer quadratic algorithms of Ari and Hildebrand [7] have dimension- and magnitude-dependent bounds for integer data on polytopes. Ari and Hildebrand [6] treats bounded pure integer QP, and Wei [46] gives polynomial time for pure integer QP in each fixed integer dimension. These premises and parameters differ from the arbitrary-dimensional continuous residual problems and finite-noise expected bit work studied here. Noise alone gives no general tractability principle: Soltanian [41] rules out uniform expected polynomial bit time for exact ReLU verification under rounded, clipped Gaussian parameter noise, assuming $\text{NP} \not\subseteq \text{BPP}$. That result concerns network verification rather than the structured optimization classes of our theorems.

Critical-region descriptions for parametric convex QP are established machinery [44]; their existence does not bound the number of regions polynomially. Likewise, classical finite-set isolation and Pareto-count analyses concern discrete feasible alternatives [12, 34]. Our continuous growth estimates and transfers to a fixed atomic law are separate steps. Algebraic tube bounds control continuous volume; the finite law requires the additional grid-transfer argument.

The results of this paper are specific certificates, bounds, and compositions built from these ingredients. For few negative directions they are closure by regions of a parametric convex program, with a fixed-hyperplane exceptional set, and the resulting fixed-parameter expected bounds under aligned and Gaussian-like noise and the dimension-dependent bound under uniform noise (section 4). For sparse interactions they are exact closure on certified strongly convex patches after a global-error bag search, with fixed-width expected bounds, and the extensions to graph, recurrence, simplex, and order constraints (sections 5 and 6). For small cores they are the excluded-region certificate for continuous recourse, the competing-label certificate for integer recourse, and the optimal-face certificate for core-only integer flows and totally unimodular systems (sections 7 and 8). Further contributions are the strong-noise component composition (section 8) and the boundary results of section 9. Exactness on every draw of a base-chosen finite law is carried through all of these results, but we do not regard the finite-law model itself, isolation-type anticoncentration, representations of algebraic points by one root, or connected-set counting as new.

Three unpublished companion manuscripts have specific overlaps. *Exact Arithmetic in Polynomial Optimization: Values, Optimizers, and Certificates* [2] supplies all-precision value and selected-core oracles for core-noise problems with convex residuals, including coupled-polytope versions and exact rational QP reconstruction. It also treats the jointly convex specialization of those value and selected-core oracles; with no residual block it returns a canonical lexicographically least optimizer. It supplies full selected-point oracles for residual-convex cubics without a residual modulus or convexifier, and under a supplied convexifier when the transformed objective is cubic on the feasible polytope or is a fixed-degree polynomial convex on all of $\mathbb{R}^n$. Its cubic selector is the lexicographically least optimal core followed by the least-norm residual optimizer in that fiber. One base-selected law and one per-draw work factor $\Omega(\gamma)$ apply to all precision queries q for this fixed point. These law and query properties already appear in the companion. Its outputs are Cauchy names for the selected point and value; our ordinary outputs are finite patch descriptors with a separate global proof record, and need not select the same canonical point. This is a representation distinction. For uniformly strongly convex residuals, the direct recourse count also preserves the original L/σ , whereas a derived Schur convexifier can require scale $M + M^2/\mu$, where M bounds the full Hessian blocks and μ is the residual modulus (section 9.5). The finite-law transfer, section bounds, projected-growth argument, and lexicographic fallback overlap with our foundations. Its quartic point reductions, including their core-only-noise consequence, and regularization example are reproduced with proofs in theorem 9.10 and proposition 9.9. Thus quartic point obstructions do not exclude its positive cubic result.

The same companion separates several output requests. Exact recognition of an active coordinate of a uniformly strongly convex cubic decides Square Root Sum, although arbitrary 2^{-q} point approximation remains polynomial in the input length and q . An expanded minimal polynomial can be exponentially large on its path family, while a linear-size recurrence describes the optimizer. Its specified rational rectangle enclosures can also require large endpoint denominators, including when an enclosure must certify an active-bound sign. These are limits of the stated output formats; they do not exclude compact implicit descriptors.

Decomposition-aware global optimization: certified coordinate grids, conditional recourse, and structural limits [1] develops deterministic curvature-corrected coordinate grids, dynamic programming of min-marginals on a tree decomposition, conditional-recourse filtering, and boundary output. It contains deterministic antecedents of several results re-proved here. Its value-function lemma shows that a recourse problem with a fixed feasible set inherits the upper coordinate curvature of the objective (lemma 7.2(b)). Its core search with an exact conditional oracle gives the retention, value-interval and witness guarantees of proposition 7.5(a)–(c). Its certified bag filter gives parts (a) and (b) of proposition 5.10 and corollary 9.3. Its two star min-marginal families contain example 7.4

and the positive definite variant discussed after it. It also gives a coupling algorithm for totally unimodular constraints, with certified approximation for fixed-degree polynomial objectives and exact output for rational quadratic ones; its accuracy-independent operation bounds need set growth and a bound on the size of each optimal coordinate projection (section 6.5). For its box quadratic class, let $L_i = \max\{0, \sup_{\hat{X}} \partial_{ii} F\}$ and $\|d\|_L^2 = \sum_i L_i d_i^2$. Weighted growth $F(x) - \min F \geq \gamma_* \|x - x^*\|_L^2$ gives $\bar{\kappa} = \max\{1, 1/\gamma_*\}$; point growth with modulus g gives $\kappa = \max\{1, \max_i L_i/g\}$ and $\bar{\kappa} \leq \kappa$. The companion certifies 2^{-q} accuracy in $f(p, \bar{\kappa})(I + q + 1)^5$ bit work, with

$$f(p, t) = c_0(c_1 p \sqrt{t})^p t(1 + \log_2 t)^2,$$

and gives exact output in $f_1(p, \kappa)(I + 1)^{C_1}$ under point growth at a unique optimizer. Graded grids use $O(\sqrt{\bar{\kappa}} \log(n + 2))$ nodes per coordinate; its sharp example for the filtered common uniform-grid method needs $\sqrt{(n - 2)\bar{\kappa}} + 1$ in the stated stages. Under rETH its lower bound excludes a sublinear width exponent in the restricted product-time model $f(p)(\kappa I)^{o(p)}$ for unique-minimizer integer box QP with a supplied decomposition. It is a deterministic approximation boundary, not a smoothed lower bound or a matching $p/2$ exponent result. The limits of integrating the growth-dependent work under noise are discussed in section 5.4. We re-prove the deterministic parts we use. What this paper adds is their composition under noise: expected counts under the finite law without a supplied growth modulus, finite-noise bounds on closure failure, and an exact same-draw fallback and output. For its fixed-degree polynomial factors, the companion also derives a Hessian lower bound on the free coordinates at a point-growth optimizer and returns a rational face box whose restricted objective is strongly convex. Its runtime guarantee additionally assumes strict complementarity at the active continuous bounds and depends polynomially on the bit length of a reciprocal margin. The returned descriptor is implicit; its optimizer need not have rational coordinates. Our polynomial closure reuses this type of enclosure and supplies finite-noise margin tails and an exact same-draw fallback.

Sparse indicator quadratics: exact complexity and smoothed separator messages [3] treats positive-definite indicator quadratics and perturbs only binary indicator penalties by independent uniform-grid draws, with at least $2n$ labels per marginal. It returns exact answers and message dictionaries on a stated bounded boundary box on every draw, with fixed-width expected work polynomial in input length and the displayed numerical magnitudes. Removing those numerical factors would give $\text{NP} \subseteq \text{ZPP}$, even at its fixed matrix family and noise scale. These finite-noise, width, and dictionary ideas overlap. Our sparse route perturbs every original linear coefficient and treats continuous and native integer coordinates; its affine-feasibility boundary does not contradict that indicator model. The proofs needed here are included in this paper. The bounds are mathematical guarantees; practical performance is unestablished.

1.6 Organization

Section 2 fixes the model and output contracts. Section 3 develops the shared counting, tail, and fallback tools. Section 4 treats few negative directions, sections 5 and 6 sparse interactions, sections 7 and 8 small cores with continuous and integer recourse and the strong-noise route. Section 9 proves the boundary results, and section 10 discusses uses and open questions. Appendices contain the longer proofs.

2 Model, noise laws, and output contracts

This section fixes the input model, the perturbation laws, the meaning of an exact smoothed algorithm, and the output formats. Later sections state their theorems in these terms. The section

ends with two elementary facts that apply to all results: a certified bound for the original objective, and an example showing why a finite law forces some exact treatment of ties.

2.1 Instances and input length

All data are rational and encoded in binary. A *base instance* consists of an objective F_0 , a feasible set $X \subseteq \mathbb{R}^n$, a noise scale $\sigma > 0$, and the structural data required by the relevant theorem: a quadratic convexifier, a tree decomposition, a core, curvature bounds, or certificates. Its *base length* I is the total binary length of these data. It is measured before any noise is drawn; the bit lengths of sampled coefficients are accounted for separately.

Objectives. A quadratic objective is $F_0(x) = \frac{1}{2}x^\top Ax + b^\top x + c$ with a symmetric rational matrix A . A polynomial objective of total degree at most d is given by an explicit list of monomials, each a rational coefficient together with the list of its variables and their positive exponents, or as a sum of factors, each such a list on a stated set of variables (its *scope*). This sparse encoding charges a monomial for the variables it contains, not for the whole dimension. In every polynomial theorem d is a fixed constant, and $\text{poly}_d(\cdot)$ denotes a polynomial whose exponent depends on d only. Objectives are never given as arithmetic circuits. Circuits can be exponentially more compact than monomial lists, and they change the complexity of evaluation and of exact output; [section 9.4](#) uses this difference.

Domains. A *mixed box* is $X = \prod_{i=1}^n X_i$, where each X_i is either a continuous interval $[\ell_i, u_i]$ or a *native integer* interval $[\ell_i, u_i] \cap \mathbb{Z}$. Native integer bounds are encoded in binary, so the number of labels $u_i - \ell_i + 1$ can be exponential in I . Integer bounds are rounded inward, empty domains are rejected, and fixed coordinates are substituted before any algorithm starts. After this preprocessing every remaining coordinate has width $w_i = u_i - \ell_i > 0$, and the *continuous hull* of X is $\prod_i [\ell_i, u_i]$. We write n_c and n_z for the numbers of continuous and native integer coordinates. A *mixed polytope* is $P \cap (\mathbb{Z}^{n_z} \times \mathbb{R}^{n_c})$ for a bounded rational polyhedron P given by linear inequalities. Products of simplices, order polytopes, explicit and implicit graph constraints, integer flows, and totally unimodular systems are defined where they are used. Optimization statements assume $X \neq \emptyset$, or explicitly decide feasibility and report an empty set before promising an attaining optimizer.

Structural and numerical parameters. The structural parameters are the negative inertia $k = n_-(A)$ of a quadratic objective, the number k of rows in a supplied concave factor when that factor is used, the bag size p (treewidth plus one) of a supplied tree decomposition whose bags contain all factor scopes, the size k of a supplied continuous core, and, in the integer theorems, the integer dimension. The numerical parameters are curvature and width quantities compared with σ . Examples are an upper bound L on coordinate second derivatives, $\nu = \max\{0, -\lambda_{\min}(A)\}$, a convexifier scale α , and coordinate widths w_i . Each theorem states on which set a curvature bound must hold. A numerical parameter enters a bound through its magnitude, not through its encoding length.

Status of premises. Each premise of a theorem has one of three statuses. It is *verified* if the algorithm decides it within the stated work bound; examples are exact definiteness tests of rational matrices and a curvature bound computed from monomial coefficients. It is *certified* if the input contains a proof of it: the proof's encoding length is part of I , and the work of checking it is part of the stated work bound. It is *promised* if the algorithm uses it without checking. A theorem says,

for each promise, whether correctness depends on it or only the complexity analysis does. A false promise of the first kind can produce a wrong output; a false promise of the second kind leaves every output correct and can only invalidate the stated work bound.

2.2 Linear perturbations and finite laws

Every result perturbs only the linear part of the objective. The *sampled objective* is

$$F_\gamma(x) = F_0(x) + \gamma^\top x, \quad (1)$$

for a random vector $\gamma \in \mathbb{Q}^n$ drawn once. The feasible set, the nonlinear coefficients, and all structural data are not perturbed.

Definition 2.1 (Grid law). For a rational $\sigma > 0$ and a power of two $M \geq 2$, let

$$\Lambda_{\sigma,M} = \left\{ -\sigma + \frac{2\sigma j}{M-1} : j = 0, 1, \dots, M-1 \right\}$$

be the M equally spaced points of $[-\sigma, \sigma]$, including both endpoints, and let $U_{\sigma,M}$ be the uniform distribution on $\Lambda_{\sigma,M}$.

A draw from $U_{\sigma,M}$ uses exactly $\log_2 M$ fair random bits, and every atom has binary length $O(\log M + \text{size}(\sigma))$. Because $M-1$ is odd, 0 is not an atom. The atoms are equally spaced at distance $2\sigma/(M-1)$, so a closed interval of length λ contains at most $\lambda(M-1)/(2\sigma) + 1$ of them. Hence

$$U_{\sigma,M}(J) \leq \frac{\lambda}{2\sigma} + \frac{1}{M} \quad \text{for every closed interval } J \text{ of length } \lambda. \quad (2)$$

The first term is the bound for the continuous uniform law on $[-\sigma, \sigma]$. The second is an *atomic term*; it does not decrease with λ , and every finite-law estimate in this paper carries a term of this kind.

Definition 2.2 (Perturbation models). Let $\mathcal{D}$ be a finite rational law on $\mathbb{Q}$. The following models specify γ in (1).

- (i) *Ambient*: $\gamma_1, \dots, \gamma_n$ are independent with law $\mathcal{D}$.
- (ii) *Core-only*: for a supplied core $C \subseteq [n]$, the coefficients γ_i , $i \in C$, are independent with law $\mathcal{D}$, and $\gamma_i = 0$ for $i \notin C$.
- (iii) *Aligned*: for a supplied rational matrix $T \in \mathbb{Q}^{k \times n}$, $\gamma = T^\top \xi$, where $\xi_1, \dots, \xi_k$ are independent with law $\mathcal{D}$.

Row-scaled aligned noise. The low-rank lattice theorem ([theorem 8.17](#)) uses one extension of the aligned model. There $\xi_1, \dots, \xi_k$ are independent, ξ_i has the grid law $U_{\sigma_i, M}$ with a specified rational scale $\sigma_i > 0$ for row i , and all rows share one resolution M chosen from the base instance. With equal scales this is model (iii) with $\mathcal{D} = U_{\sigma, M}$. Rescaling the rows of T does not remove unequal scales, because T also enters that theorem's objective. The common resolution is essential there: it fixes the spacing of the lattice of attainable objective values. Every theorem asserts its bounds only for the law it states.

The models coincide in special cases: core-only noise with $C = [n]$, and aligned noise with $T = I$, are ambient noise. In general they differ, and a guarantee proved for one model does not transfer automatically to another. Aligned coefficients are dependent in the original coordinates and lie in

the row space of T . Core-only noise leaves every residual cost unchanged, so ties and flat directions among residual variables can persist on every draw.

Two marginal laws are used. The first is the grid law $U_{\sigma,M}$. The second is a *Gaussian-like* finite rational law $\mathcal{G}_{b,\sigma}$, whose Kolmogorov distance to the normal law $N(0, \sigma^2)$ is at most 2^{-b} and which is produced by a bounded rejection sampler from polynomially many fair bits (section 3.3). We call σ the *noise scale*: it is the half-width of the grid law and the standard deviation of the normal proxy. The support of $\mathcal{G}_{b,\sigma}$ is contained in $[-(b+20)\sigma, (b+20)\sigma]$. The exact normal law is used only as a proof device; it is not an input model, because its samples are not finite rational data. Our samplers have a bound on their worst-case number of random bits, which forces finite support. A Turing machine without such a bound can also sample a rational law with countably infinite support.

Base-chosen laws. In every theorem except the strong-noise theorem, the law is computed deterministically from the base instance before sampling: the integer M , or the accuracy b of the Gaussian-like law, is a function of the base data. As stated, the law therefore depends on the instance. The strong-noise theorem (theorem 8.16) holds for every grid size M , and its work bound depends on M explicitly. The theorems assert their bounds for this constructed law. They do not assert the same bounds for every finite law with noise scale σ , for a coarse grid chosen independently of the instance, or for an arbitrary approximation of a Gaussian. The reason is the atomic term in (2): the law must be fine enough that atoms on exceptional sets have small total probability, and “fine enough” is measured against base-only quantities, such as the cost of an exact fallback or the spacing of a value lattice. In the results with polynomial sampling precision the logarithms of these quantities are bounded by polynomials in the input length alone, so the law can also be chosen from the input length rather than from the individual instance (proposition 2.3).

Sampling precision. Write $b = \log_2 M$ for the number of random bits per perturbed coordinate of a grid law. A theorem has *polynomial sampling precision* if $b \leq \text{poly}(I)$, so that the sampled instance has length $I + n \cdot \text{poly}(I)$. A few results for nonlinear flows and totally unimodular systems need *parameter-dependent sampling precision* $b \leq f(k) \text{poly}(I)$; this enlarges the sampled instance, and the stated work bounds account for it.

Proposition 2.3 (Resolution depending only on the input length). *Fix one of the theorems of sections 4 to 8 with polynomial sampling precision, together with its fixed format constants, such as the degree and the dependency depth of graph constraints, and its oracle implementation. There are effective nondecreasing positive integer-valued polynomials P , A , and D , depending only on these choices, such that the theorem remains valid, with the same output contract and the same structural and numerical factors in its expected work bound up to a change of its polynomial factor, when its law is replaced as follows for every valid base instance of length at most I :*

- (a) a grid law by $U_{\sigma, M(I)}$ with $M(I) = 2^{P(I)}$; in the lattice theorem (theorem 8.17) the law of row i is replaced by $U_{\sigma_i, M(I)}$, keeping each specified scale σ_i and one common $M(I)$ for all rows, and the level cap is reset to $\log_2 M(I)$;
- (b) a Gaussian-like law by $\mathcal{G}_{b(I), \sigma}$ with $b(I) = A(I) + D(I)t(I)$ and $t(I) = \lceil 4 \log_2(A(I) + D(I) + 21) \rceil$, the auxiliary box and the level cap being recomputed for this support.

For the core-only flow and totally unimodular theorems with arbitrary core faces (theorems 8.12 and 8.14) the same holds for instances whose core size is at most a supplied K , with $M(I, K) = 2^{\lceil F(K)P(I) \rceil}$ for an effective nondecreasing F , and with $F(K)$ in the sampling, output, and work bounds. The integer envelope $F(K)$ is chosen with evaluation cost polynomial in $1 + \text{size}(K) + F(K)$,

and this computation is charged before sampling. The strong-noise theorem holds for every M ; its probabilities q_i must be computed for the chosen M , and its sufficient regime is preserved by every larger power of two.

Proof. Apply [corollary 3.21](#), whose routewise verification is in [appendix A.7](#). The effective bounds control logarithms of the base-only format and degree multipliers; in particular $\log(s+1) \leq \text{poly}_d(I)$ for an atom universe of size s , even when native integer ranges contribute exponentially many atoms. Coefficient heights and fallback work retain their separate polynomial dependence on the added sampled bits. The Gaussian support schedule recomputes the auxiliary box and level cap, and gives $(b(I) + 20)\sigma \leq 2^{t(I)}\sigma$. The lattice cap is reset as stated. The arbitrary-face flow and TU cases use the parameter-dependent part of the corollary with $k \leq K$. For strong noise the actual q_i and β are recomputed; the sufficient regime of [equation \(41\)](#) holds for every larger power of two. $\square$

The resolution $M(I)$ standardizes the scalar law, which is still rescaled by the supplied σ , or by the specified row scales σ_i in the lattice theorem, and applied to the coordinates that the theorem perturbs; it is not one probability measure for all instances. Taking $K = I$ in the flow case gives a law depending only on I , but its sampling cost need not stay within $f_d(k) \text{poly}_d(I)$ for the actual core size k . The geometric and oracle qualifications remain part of the proposition. The routewise widths and derivative bounds have polynomial encoding length; fixed degree alone does not ensure this for arbitrary compact semialgebraic domains. For example, $1 \leq x_0 \leq 2$, $x_i = x_{i-1}^2$ has last-coordinate width $2^{2^N} - 1$. The proposition also fixes the oracle implementation; it does not supply one for a broader class.

Small and strong noise. In most results σ is a free parameter, and the bounds degrade as σ decreases; these are statements about small perturbations of an arbitrary instance. The random-component results of [section 8](#) instead require noise that dominates the variation of every coordinate derivative. There σ is large compared with the curvature of the instance, and the results describe instances whose linear part is strongly random rather than slightly perturbed.

2.3 Exact smoothed algorithms

Computation is measured in bit operations of a deterministic Turing machine. Oracle calls, where a theorem allows them, are charged at their stated cost.

Definition 2.4 (Exact smoothed algorithm). Fix a class of base instances, an output format, a perturbation model, and a rule assigning to each base instance Π a finite rational law $\mathcal{D}_\Pi$ for the perturbation. An *exact smoothed algorithm with expected work W* is a deterministic algorithm $\mathcal{A}$ with the following properties.

- (i) For every base instance Π and every γ in the support of $\mathcal{D}_\Pi$, $\mathcal{A}(\Pi, \gamma)$ halts and returns an exact global optimizer and the optimal value of F_γ on X , in the stated format.
- (ii) The perturbation is drawn once. The algorithm does not resample, discard, or condition on any event of the draw.
- (iii) The bit work of computing $\mathcal{D}_\Pi$, sampling, and running $\mathcal{A}(\Pi, \gamma)$ has expectation at most $W(\Pi)$ over $\gamma \sim \mathcal{D}_\Pi$.

The randomness defines the instance to be solved; the algorithm itself is deterministic given the draw. Property (i) includes draws with tied or positive-dimensional optimal sets and draws on

which a fast route fails. Property (ii) excludes the shortcut of redrawing until a draw is generic, which would change the law of the solved instance and with it the meaning of the expectation. By Markov's inequality, an expected bound W implies work at most tW with probability at least $1 - 1/t$; it is not a worst-case bound, and on rare draws the work can be exponential in I .

How ties are handled. A finite law gives ties and other degenerate draws positive probability (section 2.7), so an exact smoothed algorithm needs a way to solve them. Most results use a fast route that refines a search and closes it with an exact certificate, together with an exact *fallback*: a deterministic algorithm that solves every sampled instance with bit work at most $B \cdot \text{poly}(I + b)$, where the factor B depends only on the base instance and $\log B \leq \text{poly}(I)$. The fallback runs on the same draw when the fast route has not finished by a stopping depth computed from base data. If this happens with probability at most $1/B$, the fallback contributes only a polynomial term in $I + b$ to the expected work, including the stated parameter factor when sampling precision depends on the core size (lemma 3.18). The quantities are chosen in a fixed order: first B , then the thresholds that define exceptional events, then the stopping depth, and finally the law. Two results need no rare fallback. The lattice theorem for low-rank integer problems (theorem 8.17) stops exactly once the certified gap falls below the spacing of the lattice of attainable objective values. The strong-noise component theorem (theorem 8.16) solves every random component exactly on every draw.

Expectations of adaptive searches. The fast routes refine cells adaptively, and which cells are visited depends on the draw. Expected counts are therefore never computed as if the retained cells were independent of the noise. Every visited cell is charged to a node of a deterministic grid fixed before sampling, and the expected number of nodes that satisfy a necessary near-optimality condition is bounded node by node (theorem 3.9).

2.4 Output contracts

Exact output for F_γ takes one of the following forms. Each theorem states which form it returns on ordinary draws and on fallback draws.

Definition 2.5 (Output formats).

- (a) *Rational output*: a feasible $x^* \in \mathbb{Q}^n$ with $F_\gamma(x^*) = \min_X F_\gamma$, and this value.
- (b) *Implicit patch output*: a set S of coordinates with fixed rational values z_S (native integer labels, original continuous bounds, or continuous values fixed by a sound singleton-hull certificate), a rational box Q in the remaining coordinates such that $(z_S, y) \in X$ for every $y \in Q$, and a rational $g_0 > 0$ with a verified proof that $\nabla_{yy}^2 F_\gamma(z_S, y) \succeq g_0 I$ on Q , together with the claim that the unique minimizer $\hat{y}$ of $F_\gamma(z_S, \cdot)$ on Q gives a global optimizer $\hat{x} = (z_S, \hat{y})$ of F_γ on X . Products of simplices and order polytopes use a rational relative polytope in place of Q .
- (c) *Charted output*: for feasible sets that are graphs of maps from free to dependent coordinates, an implicit patch output for the reduced objective in the free coordinates, together with a certified chart that determines every dependent coordinate from the free ones. The chart is either an explicit polynomial map, or a list of scalar equations, each with a rational bracket on which a certified positive derivative bound guarantees a unique root. The optimizer is the lift of the unique reduced patch minimizer through the chart.
- (d) *Algebraic output*: a real algebraic number θ , given by a nonzero integer polynomial and an isolating interval with rational endpoints, and rational polynomial maps that give every

continuous coordinate of one optimizer and the optimal value as functions of θ , with integer coordinates listed explicitly.

- (e) *Component output*: a partition of the unfixed coordinates into components, an algebraic output for each component, rational values for the remaining coordinates, and the optimal value written as a rational constant plus the sum of the component values.

Several distinctions are essential.

Descriptor versus proof. Implicit patch and charted outputs are compact descriptors of polynomial length. A descriptor denotes a unique point because the restricted objective is strongly convex on the patch; that point is a global optimizer because of a separate *global proof record*, which contains the pruning trace and the certificate checks. The descriptor alone does not prove global optimality. The proof record can be large on an individual draw; its expected size obeys the same bound as the expected work.

Evaluation. Given a descriptor and an integer $q \geq 0$, a *q-bit evaluation* returns a rational point within Euclidean distance 2^{-q} of the optimizer and a rational interval of width at most 2^{-q} that contains the optimal value. For patch outputs this costs $\text{poly}_d(I + q)$ bit operations: convex optimization in the bit model gives value accuracy, and the verified modulus g_0 converts value accuracy into distance. On box, polytope, and explicit-graph domains the evaluation also returns a feasible rational point with a certified objective gap. An implicit graph need not contain any rational feasible point: the equation $y^2 = 2$ with $y \in [1, 2]$ is an example. For charted outputs on implicit graphs, an exactly feasible point consists of rational free coordinates together with the algebraic dependent coordinates that the chart assigns to them; rational approximations of all coordinates need not be feasible. Evaluation does not decide exact comparisons of the optimal value with a rational threshold, and it does not identify the exact active set. The exact-arithmetic companion shows why these are different requests: exact active-coordinate recognition for a uniformly strongly convex cubic can decide Square Root Sum while point approximation remains polynomial [2]. If the patch has no free coordinate, its unique point is evaluated directly; on an implicit graph this includes evaluation of its exact dependent lift. No full-dimensional convex optimization routine is invoked on a point.

Algebraic sizes. An algebraic output fixes one optimizer: all its coordinates are functions of one root θ , so coordinates of different optimizers cannot be mixed. Its defining polynomial need not be minimal. Algebraic outputs from fallback draws have length and q -bit refinement cost at most $B \text{poly}_d(I + b + q)$, where $B = 2^{\text{poly}_d(I)}$ is base-only; their expected contribution is polynomial in the sampled bit length and requested precision under the stated law, with any stated sampling-parameter factor retained. A component output does not include a defining polynomial of the total value, whose degree can grow exponentially with the number of components, and it does not provide exact sign or equality tests for sums of unrelated algebraic numbers. Rational enclosures of the total value are obtained by refining each component.

Selection. Unless a theorem says so, the returned optimizer is the one the algorithm happens to certify. No canonical selection rule, such as the lexicographically least or minimum-norm optimizer, is promised.

2.5 Fixed-parameter and fixed-width bounds

A bound is *fixed-parameter* in a parameter κ if it has the form $f(\kappa) \cdot \text{poly}(I)$, where the exponent of the polynomial does not depend on κ . The low-negative-inertia results under aligned or Gaussian-like noise, and the core and integer-dimension results, are of this form jointly in the structural parameters and the numerical ratios, such as $\nu \text{diam}(X)/\sigma$ or L/σ , included in κ . The low-negative-inertia result

under uniform ambient noise (theorem 4.20) is not: its bound contains a power of the dimension n that grows with k . It is polynomial for fixed structure when the ratios are polynomially bounded, but not fixed-parameter in the structure. A bound is *fixed-width polynomial* if it is polynomial for each fixed value of a parameter p but its degree grows with p , for instance $[nLw_{\max}/\sigma]^{O(p)} \text{poly}(I)$. The sparse results are of this kind, and they are not fixed-parameter bounds in the width. The limitation is genuine for the method used: theorem 9.1 exhibits instances on which its retained state count is $(\Omega(n/p))^{p/2}$ on every draw.

A bound that is polynomial in a numerical ratio is polynomial in its magnitude, which can be exponential in its encoding length. The phrase *expected polynomial time* is used only when the relevant ratios are polynomially bounded in I and the structural parameters are fixed. FPT means joint parameterization by structure and the displayed numerical ratios, or parameterization by structure for each fixed bounded ratio. If a ratio R grows polynomially in I , a bound $(1 + R)^{O(k)} \text{poly}(I)$ is $I^{O(k)}$, rather than FPT in k alone.

2.6 The original objective

The theorems optimize the sampled objective F_γ . An exact optimizer of F_γ need not optimize F_0 , and nothing is claimed about its distance from the optimal set of F_0 . The following elementary bound states what a sampled solution certifies for the original problem. For compact X and $\gamma \in \mathbb{R}^n$, let

$$\omega_X(\gamma) = \max_{x \in X} \gamma^\top x - \min_{x \in X} \gamma^\top x$$

be the width of X in the direction γ .

Proposition 2.6 (Certified original-objective regret). *Let X be compact, let F_0 be continuous, and let $y \in X$ and $a \in \mathbb{R}$ satisfy*

$$a \leq \min_X F_\gamma \leq F_\gamma(y) \leq a + \delta.$$

Then the interval with endpoints

$$a - \max_{x \in X} \gamma^\top x \quad \text{and} \quad F_0(y)$$

contains $\min_X F_0$, and its length is at most $\delta + \omega_X(\gamma)$. In particular $F_0(y) - \min_X F_0 \leq \delta + \omega_X(\gamma)$.

Proof. For every $x \in X$, $F_0(x) = F_\gamma(x) - \gamma^\top x \geq a - \max_X \gamma^\top x$, which gives the lower endpoint. The upper endpoint is the value of the feasible point y . Finally,

$$F_0(y) = F_\gamma(y) - \gamma^\top y \leq a + \delta - \min_X \gamma^\top x = \left(a - \max_X \gamma^\top x \right) + \delta + \omega_X(\gamma). \quad \square$$

For an exact sampled optimizer $\hat{x}$ one may take $y = \hat{x}$, $\delta = 0$, and $a = \min_X F_\gamma$. The endpoints are then real numbers in the format of the output; for implicit, charted, algebraic, and component outputs they are generally irrational. Rational certified endpoints come from a q -bit evaluation: a feasible point y with a rational upper value, and a rational lower bound a , whose distance from each other is the evaluation error, which is added to δ . The point y must be exactly feasible. On implicit graphs it consists of rational free coordinates and their exact algebraic lift, and $F_0(y)$ is replaced by the upper end of a certified rational enclosure; a coordinatewise rational approximation of the lift may be infeasible and cannot be used.

If every perturbed coefficient satisfies $|\gamma_i| \leq \bar{\sigma}$, then $\omega_X(\gamma) \leq \bar{\sigma} \sum_i w_i$ on a mixed box, where the sum is over the perturbed coordinates; under core-only noise on a unit core box this is $k\bar{\sigma}$. For aligned noise $\omega_X(\gamma) \leq \bar{\sigma} \sum_{i=1}^k \omega_X(T_{i\cdot})$, where $T_{i\cdot}$ is the i th row of T . Here $\bar{\sigma}$ bounds the

largest magnitude in the support of the marginal law: $\bar{\sigma} = \sigma$ for grid laws and $\bar{\sigma} = (b + 20)\sigma$ for Gaussian-like laws. On a box, $\max_X \gamma^\top x$ is an endpoint calculation; on a polytope it is a linear program. On a mixed polytope one may maximize over its continuous relaxation, which keeps the lower endpoint valid and replaces ω_X by the width of the relaxation.

For calibration, use widths in the coordinates of the chosen perturbation model. Write $W_{\text{noise}} = \sum_i \text{perturbed } w_i$ for ambient or core-only noise, and $W_{\text{noise}} = \sum_i \omega_X(T_i)$ for aligned noise; polyhedral relaxations may provide upper bounds for these widths. With row-specific aligned scales the bound is instead $\sum_i \sigma_i \omega_X(T_i)$. If $W_{\text{noise}} = 0$, the perturbation is constant on X and only the evaluation error needs to be charged.

For a route valid at every positive noise scale and $W_{\text{noise}} > 0$, let $\varepsilon > 0$, ensure $\bar{\sigma} W_{\text{noise}} \leq \varepsilon/2$, and evaluate with total certificate error at most $\varepsilon/2$. This gives an exactly feasible point and a rational certificate of original-objective gap at most ε on every draw. For grid laws take $\sigma = \varepsilon/(2W_{\text{noise}})$. For Gaussian-like laws the support bound is exactly $\bar{\sigma} = (b + 20)\sigma$, with b chosen by the effective base-data schedule. Put $R = \max\{1, W_{\text{noise}}/\varepsilon\}$ and let I_0 be the length of the supplied data and accuracy, excluding the noise scale. There is an effective nondecreasing polynomial P such that the required accuracy at $\sigma = 2^{-t}$ satisfies $b(t) \leq P(I_0 + t)$, after absorbing fixed encoding constants. Choose the least integer $t \geq 0$ satisfying

$$2R(P(I_0 + t) + 20) \leq 2^t.$$

Exponential growth dominates this fixed polynomial, so $t = O(I_0 + \log_2 R + 1)$. If $t \geq 1$, failure at $t - 1$ gives $2^t < 4R(P(I_0 + t) + 20)$; if $t = 0$, $2^t = 1 \leq R$. Consequently

$$\frac{1}{\sigma} \leq R \text{ poly}(I_0 + \log_2 R + 1), \quad (b(t) + 20)\sigma W_{\text{noise}} \leq \varepsilon/2.$$

This preserves polynomial sampling precision and includes the full support factor in every numerical work bound.

Substitute this scale into the theorem's displayed bound, retaining every other numerical factor. For [theorem 5.2](#) under grid noise, for example, the bound is $C_0^p[4 + (1 + n/2)Lw_{\max}W_{\text{noise}}/\varepsilon]^p \text{poly}_d(I)$. The work is polynomial in $1/\varepsilon$ for fixed structure and fixed other numerical parameters; it is polynomial in I and $1/\varepsilon$ only when those numerical factors are polynomially bounded. Binary-encoded curvature, widths, and coupling-derived bounds may be exponentially large. The strong-noise results retain the width bound but do not give this prescribed-accuracy consequence: their lower noise-scale condition need not permit the calibration. None of these guarantees yields exact optimization of F_0 by taking a positive noise scale to zero. This is consistent with the worst-case hardness of the unperturbed problems discussed in [sections 1](#) and [9](#).

2.7 Why a finite law forces an exact treatment of ties

Under a continuous noise law, ties among optimizers and other degenerate configurations typically have probability zero. Under a finite law they can have positive probability that no refinement of the search removes. The following example is the smallest instance of this phenomenon.

Example 2.7 (A tie with probability $1/M$). Let $X = [0, 1]^2$, $F_0(x) = 2x_1x_2 - x_1 - x_2$, and let γ_1, γ_2 be independent with law $U_{\sigma, M}$, where $0 < \sigma \leq 1/4$. The event $\gamma_1 = \gamma_2$ has probability exactly $M \cdot M^{-2} = 1/M$.

On this event, put $s = 1 - \gamma_1 \in [3/4, 5/4]$. The minimum value of F_γ is $-s$, and

$$F_\gamma(x) + s = s(1 - x_1 - x_2) + 2x_1x_2.$$

If $x_1 + x_2 \leq 1$, both terms are nonnegative, and they vanish together only at $(1, 0)$ and $(0, 1)$. If $x_1 + x_2 > 1$, then $x_1 x_2 > 0$, and $(1 - x_1)(1 - x_2) \geq 0$ gives $x_1 + x_2 - 1 \leq x_1 x_2$, so $F_\gamma(x) + s \geq (2 - s)x_1 x_2 > 0$. Hence F_γ has exactly the two global minimizers $(1, 0)$ and $(0, 1)$ on every draw of this event.

Consequently no certificate that exhibits a strongly convex patch containing every global optimizer can succeed on this event, at any refinement depth: a strictly convex function has at most one minimizer on a convex set. The point-growth modulus is zero, so any tail bound of the form $\Pr\{g_* < \varepsilon\} \leq c\varepsilon + \beta$ for the grid law must have $\beta \geq 1/M$ for all $\varepsilon > 0$. An algorithm that is correct on every draw must solve this event by other means. The closure-based results do so with their same-draw fallback, whose expected contribution on this event is at most M^{-1} times its cost; choosing M large compared with the base-only factor B keeps this contribution polynomial. Under any absolutely continuous law for (γ_1, γ_2) the tie has probability zero, which is why continuous-noise analyses can ignore it.

The example shows why the probability estimates of the later sections have the form “continuous bound plus atomic term”, and why the grid size M is chosen only after the fallback factor and the stopping depth are known. [Example 3.11](#) shows the companion effect on counts: a fixed atom can keep the number of retained cells growing with the refinement level.

3 Local counts, growth tails, and an exact fallback

This section develops the probabilistic and algorithmic tools shared by the quadratic, sparse, and recourse results. Every statement concerns a fixed base function and an independent random linear perturbation of it. No statement uses convexity, differentiability, uniqueness of an optimizer, or quadratic growth of the base problem unless that property is listed among its hypotheses.

Throughout, I denotes the bit length of the rational base data before any perturbation is sampled. It includes the noise width σ and every supplied decomposition or certificate. A perturbation is drawn once, from a law that the algorithm computes from the base data before sampling. No algorithm in this paper discards, resamples, or conditions on a draw, and every algorithm is correct on every draw. Probability enters only the bounds on work.

The section has four parts. Theorem [3.4](#) bounds the expected number of grid nodes that survive comparisons with their grid neighbors. Together with curvature-corrected refinement (Theorem [3.9](#)), it bounds the expected number of cells that a search retains at every refinement level, uniformly in the level. Section [3.3](#) transfers such bounds from continuous proxy laws to finite rational laws that a Turing machine can sample. Theorem [3.16](#) and Lemma [3.17](#) bound the probability of small global quadratic growth and of small active gradients. Theorem [3.20](#) gives an exact algorithm for every draw of a fixed-degree polynomial problem. Its cost contains an exponential factor that depends only on the base data, and Lemma [3.18](#) shows how a rare exceptional event pays for that factor. Section [3.6](#) summarizes how the later sections combine these parts. Proofs are in [Appendix A](#).

3.1 Upper curvature and local comparison events

Let $X = \prod_{i=1}^k [a_i, b_i]$ be a box with $a_i < b_i$ and widths $w_i = b_i - a_i$. The mesh statements use $k \geq 1$. Fixed coordinates are held at their values, and a singleton box is evaluated directly before a width maximum, mesh, or accuracy threshold is formed. For a function $f : X \rightarrow \mathbb{R}$ and a coefficient vector $c \in \mathbb{R}^k$ write

$$F_c(x) = f(x) + c^\top x.$$

Definition 3.1 (Upper coordinate curvature). A function $f : X \rightarrow \mathbb{R}$ has *upper coordinate curvature at most* $L = (L_1, \dots, L_k) \in [0, \infty)^k$ if, for every i and every $x \in X$, the function $t \mapsto f(x + te_i) - L_i t^2/2$ is concave on the interval $\{t : x + te_i \in X\}$.

The property holds when $f - \frac{1}{2} \sum_i L_i x_i^2$ is concave. It is preserved by pointwise infima: if every f_ω has upper coordinate curvature at most L and $\inf_\omega f_\omega$ is finite, then $\inf_\omega f_\omega$ has upper coordinate curvature at most L , because an infimum of concave functions is concave. The auxiliary value functions of Section 4 are of this kind. Downward kinks, discontinuous derivatives, and arbitrarily large negative curvature are allowed. Only an upper bound on curvature is used.

Lemma 3.2 (Neighbor comparisons confine one coefficient). *Let $f : X \rightarrow \mathbb{R}$ be arbitrary, $v \in X$, $h > 0$, $\delta \geq 0$, and suppose that $v \pm he_i \in X$. If*

$$F_c(v) \leq F_c(v + he_i) + \delta \quad \text{and} \quad F_c(v) \leq F_c(v - he_i) + \delta,$$

then c_i belongs to the interval

$$I_i(v, h, \delta) = \left[\frac{f(v) - f(v + he_i) - \delta}{h}, \frac{f(v - he_i) - f(v) + \delta}{h} \right].$$

This interval depends on f , v , h , and δ , but not on c . If f has upper coordinate curvature at most L , its length is at most $L_i h + 2\delta/h$; an interval with negative length is empty.

Definition 3.3 (Grids, local events, and interval concentration). For integers $m_i \geq 1$ put $h_i = w_i/m_i$, $G_i = \{a_i + \ell h_i : 0 \leq \ell \leq m_i\}$, and $G = \prod_i G_i$. The *interior coordinates* of a node $v \in G$ are $Q_v = \{i : a_i < v_i < b_i\}$. For $\delta \geq 0$, the δ -*local event* E_v^δ is the set of coefficient vectors c for which $F_c(v) \leq F_c(v \pm h_i e_i) + \delta$ for all $i \in Q_v$ and both signs. The *interval concentration* of a real random variable Y is

$$\pi_Y(\ell) = \sup\{\mathbb{P}(Y \in J) : J \text{ an interval of length at most } \ell\}.$$

A node with $F_c(v) \leq \inf_X F_c + \delta$, or with $F_c(v) \leq \min_G F_c + \delta$, satisfies E_v^δ , because its grid neighbors are feasible competitors. Three concentration bounds recur. If Y has a density bounded by ϕ , then $\pi_Y(\ell) \leq \phi\ell$. If Y has the grid law $U_{\sigma, M}$ of Definition 2.1, uniform on $\{-\sigma + 2\sigma j/(M-1) : 0 \leq j < M\}$, then $\pi_Y(\ell) \leq \ell/(2\sigma) + 1/M$ by (2). If the distribution function of Y differs everywhere by at most δ_K from that of a law with density bounded by ϕ , then $\pi_Y(\ell) \leq \phi\ell + 2\delta_K$.

Theorem 3.4 (Expected local counts). *Let $f : X \rightarrow \mathbb{R}$ have upper coordinate curvature at most L , let G be the grid of Definition 3.3, and let the coordinates of c be independent, with interval concentrations $\pi_1, \dots, \pi_k$. Then for every $\delta \geq 0$*

$$\mathbb{E} \#\{v \in G : c \in E_v^\delta\} \leq \prod_{i=1}^k \left[2 + (m_i - 1) \min\{1, \pi_i(L_i h_i + 2\delta/h_i)\} \right].$$

The same bound holds conditionally on any random element that is independent of c , if f is allowed to depend on that element.

The bound concerns a deterministic grid. An adaptive search visits a subset of its nodes that depends on the draw, but every node that the search keeps for a probabilistic reason will be shown to satisfy a local event. The expectation is therefore taken over a fixed family of events, and no independence between the draw and the adaptively selected cells is used. The comparisons with grid neighbors make the other coefficients cancel, which is why independence across coordinates multiplies the per-coordinate bounds.

Definition 3.5 (Balanced nested meshes). Let $\rho_1, \dots, \rho_k > 0$ be *mesh scales*. Put $S_\rho = \max_i w_i/\rho_i$ and $h_j = S_\rho 2^{-j}$ for $j \geq 0$. At level j , coordinate i is divided into m_{ij} equal parts, where m_{ij} is the least power of two with $w_i/m_{ij} \leq \rho_i h_j$, and $h_{ij} = w_i/m_{ij}$. The level- j grid G_j and the level- j cells are the corresponding tensor grid and the boxes between consecutive grid nodes. For a curvature vector $L \in (0, \infty)^k$ put

$$E_j = \frac{1}{8} \sum_{i=1}^k L_i h_{ij}^2, \quad c_{\text{bal}} = \max_{i,l} \frac{L_l \rho_l^2}{L_i \rho_i^2}, \quad C_k = 1 + 2c_{\text{bal}}k.$$

The *isotropic* choice $\rho_i = 1$, for a common curvature $L_i = L$, gives $c_{\text{bal}} = 1$ and $C_k = 1 + 2k$.

Corollary 3.6 (Level-uniform counts). *In Definition 3.5, the partitions are nested, level 0 consists of the single cell X , and $m_{ij} \leq 2^j$. Every coordinate with $m_{ij} \geq 2$ satisfies $\rho_i h_j/2 < h_{ij} \leq \rho_i h_j$ and*

$$L_i h_{ij} + 4E_j/h_{ij} \leq C_k L_i h_{ij}.$$

Consequently, if f has upper coordinate curvature at most L and the coordinates of c are independent, then at every level j

$$\mathbb{E} \#\{v \in G_j : c \in E_v^{2E_j}\} \leq \begin{cases} \prod_{i=1}^k [2 + C_k L_i \phi_i w_i] & \text{if } c_i \text{ has a density bounded by } \phi_i, \\ \prod_{i=1}^k [3 + C_k L_i w_i/(2\sigma_i)] & \text{if } c_i \sim U_{\sigma_i, M} \text{ and } M \geq m_{ij} \text{ for all } i. \end{cases}$$

A coordinate that is not yet refined has no interior node. It contributes the factor 2, with no dependence on its width, however small. The construction divides every coordinate interval into equal parts. Clipping a uniform step at the end of an interval would create a short final gap g , and Lemma 3.2 would then produce a term δ/g that the count cannot control. The balance constant c_{bal} matters only for anisotropic curvature; it is used with $c_{\text{bal}} < 4$ in Section 4.7.

3.2 Curvature-corrected refinement and the role of closure

Lemma 3.7 (Corrected corners). *Let f have upper coordinate curvature at most L , and let $C = \prod_i [l_i, l_i + \eta_i] \subseteq X$ be a cell. For every c and every $x \in C$,*

$$\min_{v \text{ corner of } C} F_c(v) \leq F_c(x) + \frac{1}{8} \sum_{i=1}^k L_i \eta_i^2.$$

Definition 3.8 (Corrected-corner search). Fix balanced nested meshes on X , a level cap J , and exact evaluation of F_c . Level 0 processes the cell X . Level $j \geq 1$ processes the children of the cells retained at level $j-1$. For every processed cell C , evaluate F_c at all corners of C and put $\underline{F}(C) = \min_{\text{corners}} F_c - E_j$. Optionally, a *closure step* computes $\min_C F_c$ exactly together with a point attaining it; such a cell is *closed*. After all cells of level j have been processed, let U_j be the least value of F_c found so far, at corners or at closure points, stored together with a point attaining it. Retain exactly the processed cells that are not closed and satisfy $\underline{F}(C) \leq U_j$. Stop when no cell is retained or when level J has been processed.

Theorem 3.9 (Corrected-corner search). *Let $f : X \rightarrow \mathbb{R}$ be lower semicontinuous with upper coordinate curvature at most L , and put $F^* = \min_X F_c$. For every coefficient vector c the search of Definition 3.8 has the following properties.*

- (i) $\underline{F}(C) \leq \min_C F_c$ for every processed cell C .

- (ii) If no cell is retained after level j , then $U_j = F^*$ and the stored point is a global minimizer.
- (iii) $U_j \leq F^* + E_j$.
- (iv) Every cell retained at level j has a corner $v \in G_j$ with $F_c(v) \leq F^* + 2E_j$. Hence $c \in E_v^{2E_j}$, and the number of retained cells is at most $2^k \#\{v \in G_j : c \in E_v^{2E_j}\}$.
- (v) The number $\underline{U}_j = \min\{U_j, \min_{C \text{ retained}} \underline{F}(C)\}$ satisfies $\underline{U}_j \leq F^* \leq U_j$ and $U_j - \underline{U}_j \leq E_j$, with the minimum over an empty retained set defined as $+\infty$.

Consequently, under the hypotheses of Corollary 3.6, the expected number of cells retained at level j is at most 2^k times the corresponding bound of that corollary. This holds with or without closure steps.

Item (iv) is the bridge between the adaptive search and the deterministic count: a cell is retained for a probabilistic reason only through one of its corners, which is a nearly optimal node of a fixed grid.

Corollary 3.10 (Approximation with logarithmic accuracy dependence). *Use isotropic meshes with common curvature $L > 0$, let $s = \max_i w_i$, fix $\varepsilon > 0$, and put $J = \max\{0, \lceil \frac{1}{2} \log_2(kLs^2/(8\varepsilon)) \rceil\}$. Without closure steps, the search, stopped after level J or earlier if no cell is retained, returns a point x and a number $\underline{U}$ with $\underline{U} \leq F^* \leq F_c(x)$ and $F_c(x) - \underline{U} \leq \varepsilon$, for every c . If the coordinates of c are independent with densities bounded by ϕ_i , the expected number of evaluations of F_c is at most $2^k + 8^k \mathcal{H}J$, where $\mathcal{H} = \prod_i [2 + (1 + 2k)L\phi_i w_i]$. If instead $c_i \sim U_{\sigma, M}$ with $M \geq 2^J$, the same holds with $\mathcal{H} = \prod_i [3 + (1 + 2k)Lw_i/(2\sigma)]$.*

For fixed k and numerical bounds on $L\phi_i w_i$, the expected number of evaluations is logarithmic in $1/\varepsilon$. The bound counts exact evaluations of the sampled objective. Their bit cost must be added, and the conclusion concerns the sampled objective, not the unperturbed one. The finite law in the corollary is chosen for the requested accuracy. A fixed finite law does not support the same bound at arbitrarily fine levels.

Example 3.11 (A fixed atom defeats level-uniform counts). Let $k = 1$ and $X = [0, 1]$, and let the law of c have an atom c_0 with $\mathbb{P}(c = c_0) = p_0 > 0$. For $f(x) = -c_0 x$, which has upper coordinate curvature at most L for every $L > 0$, the draw $c = c_0$ makes $F_c \equiv 0$. On this draw every node of every level is a global minimizer, every cell has $\underline{F}(C) = -E_j < 0 = U_j$, and all 2^j cells of level j are retained. The expected number of retained cells is therefore at least $p_0 2^j$, which grows without bound.

The example limits this particular search: under a fixed finite law, the cells retained by indefinitely continued refinement cannot be counted uniformly in the level. It does not show that every exact algorithm needs a rare fallback. A finite law supports Corollary 3.6 only up to a level J with $M \geq m_{iJ}$ (or, for transferred bounds, up to a node budget fixed in advance), so an exact algorithm built on this search stops refining at a level computed from the base data and finishes the sampled problem in another way. Most results below use *closure steps*, which certify the exact minimum of a whole cell from a structural formula, together with an exact *fallback* on the same draw for the draws that are still unresolved at level J . The lattice theorem (Theorem 8.17) instead stops at a predetermined level at which the certified gap is below the spacing of the attainable objective values, and the strong-noise theorem (Theorem 8.16) does not use this search; neither needs a rare fallback. In all cases atoms, ties, and flat optimal faces are handled on every draw; they are never excluded by a genericity assumption.

3.3 Finite rational laws and continuous proxies

A Turing machine with fair random bits can sample laws with infinitely many rational atoms if its running time is unbounded; for example, counting the failures before the first success gives a geometric law on the nonnegative integers. The samplers in this paper are not of this kind: each uses at most a prescribed number of fair bits and bit operations in the worst case, and a sampler with this property has a finite law with rational atoms. The analysis often computes a probability under a continuous proxy law and then transfers it to the finite law that the algorithm actually samples. Two families of finite laws are used.

Definition 3.12 (Finite laws and Kolmogorov distance). The grid law $U_{\sigma,M}$ is that of Definition 2.1. For an integer $b \geq 1$ and rational $\sigma > 0$, $\mathcal{G}_{b,\sigma}$ is the law of σZ_b , where Z_b is the output of the sampler of Lemma 3.14. The *Kolmogorov distance* of laws μ, ν on $\mathbb{R}$ is $d_K(\mu, \nu) = \sup_{t \in \mathbb{R}} |\mu(-\infty, t] - \nu(-\infty, t]|$.

Lemma 3.13 (Product replacement). (a) *The uniform grid law satisfies $d_K(U_{\sigma,M}, \text{Unif}[-\sigma, \sigma]) \leq 1/M$, and the Gaussian-like law satisfies $d_K(\mathcal{G}_{b,\sigma}, N(0, \sigma^2)) \leq 2^{-b}$.*

(b) *Let $\mu_1, \dots, \mu_n$ and $\nu_1, \dots, \nu_n$ be laws on $\mathbb{R}$ with $d_K(\mu_i, \nu_i) \leq \delta_i$, and let $E \subseteq \mathbb{R}^n$ be a Borel set. Suppose that for every i and every choice of real values of the coordinates other than the i th, the section of E in the i th coordinate is a union of at most C intervals (open, closed, half-open, degenerate, or unbounded). Then*

$$\left| \left(\bigotimes_i \mu_i \right)(E) - \left(\bigotimes_i \nu_i \right)(E) \right| \leq 2C \sum_{i=1}^n \delta_i.$$

The section bound must hold for *every* choice of the other coordinates, not only for almost every choice under the proxy law, because the replacement passes through mixed products of finite and continuous marginals.

Lemma 3.14 (A bounded rational Gaussian-like sampler). *For every integer $b \geq 1$ there is a finite law $\mathcal{G}_b$ on $\mathbb{Q}$, determined by b through a fixed deterministic rule, with the following properties.*

(a) *Its support lies in $[-(b+20), b+20]$, and every atom is a rational number of bit length $O(b)$.*

(b) *$d_K(\mathcal{G}_b, N(0, 1)) \leq 2^{-b}$.*

(c) *One sample is produced from independent fair bits in $\text{poly}(b)$ bit operations in the worst case.*

The law $\mathcal{G}_{b,\sigma}$ is not a Gaussian law, and no theorem in this paper takes real Gaussian coefficients as Turing input. The Gaussian law $N(0, \sigma^2)$ serves only as a proxy in proofs: it makes orthogonal projections of the noise independent and supplies density bounds. Every theorem that uses a Gaussian-like law specifies the accuracy b as a function of the base data, computed before sampling. The bounds do not apply to an arbitrary finite approximation of a Gaussian, and in particular not to a coarse one, because Example 3.11 shows how atoms can create unbounded counts.

3.4 Global growth and active gradients

Definition 3.15 (Point growth). Let $X \subset \mathbb{R}^n$ be nonempty and compact, let $f : X \rightarrow \mathbb{R}$ be lower semicontinuous, and let $F_c = f + c^\top x$. The *point-growth modulus* $g_*(c) \in [0, \infty]$ is the supremum of the numbers $g \geq 0$ for which some minimizer x^* of F_c on X satisfies

$$F_c(x) - F_c(x^*) \geq g \|x - x^*\|^2 \quad (x \in X).$$

If X is a singleton, $g_* = \infty$. If F_c has two distinct minimizers, then $g_* = 0$. Write $w_i = \max_X x_i - \min_X x_i$ and $S = \sum_i w_i$.

Point growth is measured from one minimizer. It differs from growth measured by the distance to the set of minimizers, which can remain positive in the presence of ties.

Theorem 3.16 (Sharp growth tail). *Let $X \subset \mathbb{R}^n$ be nonempty and compact, let $f : X \rightarrow \mathbb{R}$ be lower semicontinuous, and let c have independent coordinates with densities bounded by $\phi_1, \dots, \phi_n$. Then for every $\varepsilon > 0$*

$$\mathbb{P}\{g_*(c) < \varepsilon\} \leq 2\varepsilon \sum_{i=1}^n \phi_i w_i.$$

In particular, if c_i is uniform on $[-\sigma, \sigma]$ for every i , then $\mathbb{P}\{g_(c) < \varepsilon\} \leq \varepsilon S/\sigma$. If X is not a singleton, then for every $\rho \in (0, 1]$, with probability at least $1 - \rho$ the minimizer of F_c is unique and $g_*(c) \geq \rho/(2 \sum_i \phi_i w_i)$. The constant 2 cannot be decreased: for $X = [-w/2, w/2]$, $f = 0$, and c uniform on $[-\sigma, \sigma]$, $\mathbb{P}\{g_*(c) < \varepsilon\} = 2\varepsilon\phi w$ with $\phi = 1/(2\sigma)$ whenever $0 < \varepsilon \leq \sigma/w$.*

The theorem assumes neither convexity, differentiability, nor definability of f or X . Its proof works in coefficient space. The function $H(a) = \max_{x \in X} \{a^\top x - f(x) + \varepsilon \|x\|^2\}$ is convex, and wherever H is differentiable at the proximal point of $-c$, the tilted problem has growth at least ε . The proximal map is a 1-Lipschitz gradient map, so the area formula shows that its Jacobian is singular almost everywhere on the bad event. This bounds the indicator of the bad event by the trace of the Jacobian of the proximal residual, whose integral along each coordinate is at most $2\varepsilon w_i$.

Two comparisons clarify the scope. For a finite binary feasible set and independent continuous coefficients, Beier and Vöcking bound the density of the gap between the best and second-best values by the same expression $2 \sum_i \phi_i$ [11]. That statement compares values, not values relative to squared distances, and it does not apply to continuous feasible sets. Qualitative genericity, stability, and growth results are known for semialgebraic programs [32]; they do not give this quantitative tail on arbitrary compact sets with a lower semicontinuous objective. Theorem 3.16 quantifies the probability of small growth.

The theorem concerns continuous laws. It does not by itself give an exact algorithm in the bit model. For example, minimizing $x^2/2 + cx$ over $[-1, 1]$ with continuous c has the interior minimizer $-c$, which is irrational with probability one when it is interior. A finite rational law has atoms, and ties can occur with positive probability. The next lemma gives finite-law tails when the problem is semialgebraic.

For the polynomial statements, let $\mathcal{D} \subset \mathbb{R}^{n+m}$ be a nonempty compact set described by a quantifier-free formula with at most s atomic polynomial sign conditions of degree at most d and rational coefficients. Let $F \in \mathbb{Q}[x, y]$ have degree at most d . The *reduced objective* $f(x) = \min\{F(x, y) : (x, y) \in \mathcal{D}\}$ is lower semicontinuous on the compact projection X of $\mathcal{D}$. The noise acts on x only: $F_\gamma(x, y) = F(x, y) + \gamma^\top x$, and $g_*(\gamma)$ is the point-growth modulus of $f + \gamma^\top x$ on X . When $m = 0$, $f = F$. A bounded integer coordinate $x_j \in \{\ell_j, \dots, u_j\}$ is described by the disjunction $\bigvee_{t=\ell_j}^{u_j} (x_j = t)$, which adds $u_j - \ell_j + 1$ atoms but no variables.

Lemma 3.17 (Finite-law growth and active-gradient tails). *There is an absolute constant a_R with the following properties. Put*

$$H_s = ((2s + 1) \max\{d, 2\})^{a_R(n+m+1)^2}, \quad C_{\text{tail}} = 2H_s^3 + 1.$$

- (a) (Sections.) *Fix $\varepsilon > 0$, an index $i \leq n$, and real values of γ_l for $l \neq i$. The set of $t \in \mathbb{R}$ for which $g_*(\gamma) < \varepsilon$ when $\gamma_i = t$ is a union of at most C_{tail} intervals. The bound is uniform in ε and in the fixed values.*

(b) (Uniform grids.) If $\gamma_1, \dots, \gamma_n$ are independent with law $U_{\sigma, M}$, then for every $\varepsilon > 0$

$$\mathbb{P}\{g_* < \varepsilon\} \leq \frac{S\varepsilon}{\sigma} + \frac{2nC_{\text{tail}}}{M}, \quad \mathbb{P}\{g_* = 0\} \leq \frac{2nC_{\text{tail}}}{M}.$$

(c) (Gaussian-like laws.) If $\gamma_1, \dots, \gamma_n$ are independent with $d_K(\gamma_i, N(0, \sigma^2)) \leq \delta_K$, then for every $\varepsilon > 0$

$$\mathbb{P}\{g_* < \varepsilon\} \leq \sqrt{2/\pi} \frac{S\varepsilon}{\sigma} + 2nC_{\text{tail}}\delta_K.$$

(d) (Active gradients on mixed boxes.) Let $m = 0$ and let $\mathcal{D} = X$ be a product of n_c closed intervals of positive length and of bounded integer ranges with R_Z integer points in total. Put $D = \max\{1, d - 1\}$ and $K_{\text{act}} = n_c 3^{n_c} R_Z D^{n_c}$. For $\tau > 0$, the probability that $g_* > 0$ and the minimizer x^* has a continuous coordinate x_i^* at an endpoint of its interval with $|\partial_i F_\gamma(x^*)| \leq \tau$ is at most $K_{\text{act}}(\tau/\sigma + 1/M)$ under the law of (b), and at most $K_{\text{act}}(\sqrt{2/\pi} \tau/\sigma + 2\delta_K)$ under the laws of (c).

The constant a_R comes from the block-sensitive quantifier-elimination theorem of Renegar [37]. Its exponent depends on the product of the two block sizes, not on a tower over the number of variables; a bound of cylindrical-decomposition type would not give a sampling precision of polynomial length. For fixed d , if $\log(s+1)$ is polynomial in I , then $\log C_{\text{tail}}$ is polynomial in I . This holds for mixed boxes with binary-encoded integer ranges, since those contribute $\sum_j (u_j - \ell_j + 1)$ atoms. The value of a_R is not extracted here; the sampling rules below use one fixed admissible value. The section bound is used only through the logarithm of C_{tail} , which determines a number of sampled bits. For quadratic objectives with $m = 0$ an elementary argument gives the explicit bound $8(F+1)^2$ in place of C_{tail} , where F counts candidate faces (Lemma 4.8).

Lemma 3.18 (Paying for a rare fallback). (a) (Accounting.) Let an algorithm act on one draw from a law fixed before sampling, and let its work on a draw be $W_0 + \mathbf{1}_\mathcal{E} W_1$. Suppose that $W_1 \leq B P(L_{\text{max}})$ on every draw, where $B \geq 1$ and the polynomial P are fixed before sampling and L_{max} bounds the bit length of every sampled instance, and that $\mathbb{P}(\mathcal{E}) \leq 1/B$. Then $\mathbb{E}[W_0 + \mathbf{1}_\mathcal{E} W_1] \leq \mathbb{E}W_0 + P(L_{\text{max}})$.

(b) (Budget for polynomial mixed boxes.) In the setting of Lemma 3.17(d), assume $S > 0$, let $\rho \in (0, 1]$, and put

$$g_0 = \frac{\rho\sigma}{S}, \quad \tau = \frac{\rho\sigma}{\max\{1, K_{\text{act}}\}}.$$

Let $\mathcal{E}_{\text{bad}}$ be the event that $g_* < g_0$, or that $g_* \geq g_0$ and some active continuous coordinate has $|\partial_i F_\gamma(x^*)| \leq \tau$. If $\gamma_i \sim U_{\sigma, M}$ with $M \geq (2nC_{\text{tail}} + K_{\text{act}})/\rho$, or if $d_K(\gamma_i, N(0, \sigma^2)) \leq \rho/(2nC_{\text{tail}} + 2K_{\text{act}})$, then $\mathbb{P}(\mathcal{E}_{\text{bad}}) \leq 3\rho$.

If $S = 0$, the mixed box is a single point. It is evaluated directly before any threshold is chosen, and neither the tails nor a fallback are needed; part (b) divides by S and is used only when $S > 0$.

Part (a) cancels the combinatorial multiplier B of a fallback, not its whole bit cost: the remaining polynomial factor must depend only on the bit length of the sampled instance, which is bounded before sampling. The multiplier B must cover every base-only cost of the fallback, including the conversion of its output to the required format (Theorem 3.20). This ordering, first B , then the thresholds and the level cap, and only then the law, avoids a circular choice in which a precision depends on the sampled coefficients that it is supposed to control.

3.5 An exact fallback for fixed-degree polynomial problems

Definition 3.19 (Explicit mixed semialgebraic instances). Fix a degree bound $d \geq 1$. A *base instance* consists of variables $x \in \mathbb{R}^N$ with $N \geq 1$, a set $\mathcal{Z} \subseteq \{1, \dots, N\}$ of integer variables with binary-encoded bounds $\ell_j \leq u_j$ ($j \in \mathcal{Z}$), a quantifier-free formula Φ_0 whose atoms are explicitly listed polynomial sign conditions of degree at most d with rational coefficients, and an objective $F_0 \in \mathbb{Q}[x]$ of degree at most d , given by its list of monomials with rational coefficients. Its domain is

$$\mathcal{D} = \{x \in \mathbb{R}^N : \Phi_0(x) \text{ holds and } x_j \in \mathbb{Z} \cap [\ell_j, u_j] \text{ for } j \in \mathcal{Z}\},$$

which is assumed nonempty and compact. Its bit length I counts Φ_0 , F_0 , and the bounds. A *tilted instance* adds rational coefficients γ_i , each of bit length at most b , to any subset of the variables: $F_\gamma(x) = F_0(x) + \sum_i \gamma_i x_i$. A real algebraic number θ is given in *root representation* by a nonzero $p \in \mathbb{Z}[t]$ and a rational interval, possibly a point, that contains θ and no other real root of p .

The class contains mixed boxes (Φ_0 a conjunction of bounds), bounded polytopes and mixed polytopes, and polynomial graph constraints $y = g(x)$ with explicit polynomials g . It does not contain objectives given by arithmetic circuits or by unexpanded compositions; the expanded monomial list of F_0 is part of I . A problem with no remaining variables is evaluated directly before this $N \geq 1$ elimination construction is invoked.

Theorem 3.20 (Exact fallback). *Fix $d \geq 1$. There are a deterministic algorithm, a base-computable integer $B = 2^{\text{poly}_d(I)}$, and an exponent e_d depending only on d such that the following holds for every base instance of Definition 3.19 and every tilted instance whose added coefficients have bit length at most b .*

- (i) *Within $B(I+b+1)^{e_d}$ bit operations the algorithm returns an algebraic output (Definition 2.5(d)) of the lexicographically least global minimizer x^{lex} and of the optimal value $f^* = \min_{\mathcal{D}} F_\gamma$: the integer coordinates of x^{lex} ; a squarefree polynomial $P \in \mathbb{Z}[t]$ with a rational interval isolating one real root θ of P ; and rational polynomials r_i , for the continuous coordinates $i \notin \mathcal{Z}$, and r_f with $x_i^{\text{lex}} = r_i(\theta)$ and $f^* = r_f(\theta)$. As auxiliary data it also returns a root representation of every coordinate of x^{lex} and of f^* . The degree of P and of every auxiliary defining polynomial is at most B , independently of b , and every coefficient and interval endpoint has bit length at most $B(I+b+1)^{e_d}$.*
- (ii) *For every integer $q \geq 0$ it returns, within $B(I+b+q+1)^{e_d}$ bit operations, a rational point within Euclidean distance 2^{-q} of x^{lex} and a rational interval of width at most 2^{-q} that contains f^* .*
- (iii) *If $\mathcal{D}$ is a mixed box, it also returns within the bound of (ii) a rational point $y \in \mathcal{D}$ with $\|y - x^{\text{lex}}\| \leq 2^{-q}$ and a rational number $\ell \leq f^*$ with $F_\gamma(y) - \ell \leq 2^{-q}$.*

Neither B nor e_d depends on γ , b , or q . The algorithm is correct for every γ , including draws with ties, with a continuum of minimizers, or with singular stationary sets.

The output is long on some instances: its length is at most $B(I+b+1)^{e_d}$, exponential in the base length and polynomial in the added coefficient length b . No bound independent of b is possible, since for the domain $\{1\}$ and $F_0 = 0$ the optimal value is the added coefficient itself. Lemma 3.18(a) needs exactly this separation of a base-only factor from a polynomial factor in $I+b$. The proof characterizes each coordinate of x^{lex} , and the optimal value, as the unique solution of a formula with two quantified blocks and one free variable, eliminates the quantifiers by Renegar's theorem

[37], and recovers each singleton by univariate root isolation. The resulting scalar representations all refer to the same canonical minimizer, but separately isolated coordinates do not yet form an algebraic output. A final step converts them into one shared root (Lemma A.2): in the tensor product of the univariate quotient algebras, a linear form with integer coefficients separates all tuples of roots; the characteristic polynomial P of its multiplication matrix expresses every coordinate as a polynomial function of its roots; and the scalar isolators select the root θ that corresponds to x^{lex} and f^* . The other roots of P correspond to tuples of scalar roots that need not come from any optimizer. Only the selected root carries meaning: the value identity $F_\gamma(x^{\text{lex}}) = r_f(\theta)$ holds at θ , but it need not hold as an identity modulo P . The conversion costs a polynomial in the product of the scalar degrees. Elimination bounds these degrees independently of b (only coefficient lengths grow with b), so the product is $2^{\text{poly}_d(I)}$ and the conversion fits in the factor B . That factor is fixed, with the conversion included, before any threshold or law is chosen. No minimal polynomial, unique minimizer, or finite stationary set is required. For quadratic objectives on mixed polytopes, a much smaller rational fallback suffices (Lemma 4.11).

3.6 How the tools are combined

The closure-based exact results follow one template: those of Section 4 and, with growth, label, or face certificates in place of critical regions, those of Sections 5–8 other than the two results described at the end of this subsection.

- (1) *Before sampling*, the algorithm computes from the base data a fallback multiplier B that covers every base-only cost of the fallback, the bad-event thresholds, a level cap J , and a finite law (an accuracy M or b and, for Gaussian-like laws, a support radius). All have polynomial bit length in I , except in the results with parameter-dependent sampling precision.
- (2) It draws the perturbation once.
- (3) It runs curvature-corrected refinement with exact closure steps through level J . Closure certifies the exact minimum of a cell from a structural formula valid on the whole cell. Correctness of every retained bound and of every closed cell holds on every draw (Theorem 3.9).
- (4) If unresolved cells remain at level J , it runs an exact fallback on the same draw.

The expected work of step (3) is bounded through Theorem 3.9(iv) by expected local counts of fixed grids, computed under a continuous proxy and transferred by Lemma 3.13 when necessary. The probability that step (4) runs is at most $1/B$, which pays for the fallback by Lemma 3.18(a). The structural routes differ in what plays the role of the box X and its function f , in the closure certificate, and in the event whose small probability bounds step (4): fixed hyperplane tubes and integer-label gaps for low-rank quadratic problems (Section 4), and growth and active-gradient events when closure is certified by growth. Two results complete the sampled problem differently and need no rare fallback. The lattice theorem (Theorem 8.17) stops at a predetermined level at which the certified gap is below the spacing of the attainable objective values. The strong-noise theorem (Theorem 8.16) pins coordinates by derivative signs and solves the remaining components exactly on every draw. In each case the target is the exact optimum of the *sampled* objective.

Laws that depend only on the input length. Each theorem computes its law from the individual base instance. In the results with polynomial sampling precision one resolution per input length suffices, because the base-only quantities that determine the law have logarithms bounded by a fixed polynomial in the input length and satisfy the schedule conditions below. To state this,

fix a theorem together with its fixed format constants (such as the degree and the dependency depth of graph constraints) and its oracle implementation, let $\mathcal{C}$ be its class of valid base instances, and write I_Π for the base length of $\Pi \in \mathcal{C}$. A *grid schedule* assigns to every Π a power of two M_Π . It is *monotone* if, for every power of two $M \geq M_\Pi$, the conclusions of the theorem hold for Π with $U_{\sigma,M}$ in place of U_{σ,M_Π} , all other base-selected quantities of Π (thresholds, level cap, auxiliary box) unchanged, and the polynomial factor of the work bound replaced by a fixed polynomial in $I_\Pi + \log_2 M$. A *Gaussian support schedule* computes from Π , for each trial support radius $2^t \sigma$ ($t = 0, 1, \dots$), an auxiliary box, a level cap, and a required accuracy $b_\Pi(t)$, such that for every pair (t, b) with $b \geq b_\Pi(t)$ and $b + 20 \leq 2^t$ the conclusions of the theorem hold with $\mathcal{G}_{b,\sigma}$ and the box and cap of trial t , the polynomial factor being a fixed polynomial in $I_\Pi + t + b$.

Corollary 3.21 (A resolution depending only on the input length). *Fix a theorem as above and an integer $I \geq 1$.*

- (a) *Suppose its grid schedule is monotone and $\log_2 M_\Pi \leq P_0(I_\Pi)$ for all $\Pi \in \mathcal{C}$, with an effective nondecreasing polynomial $P_0 \geq 1$. Then for every $\Pi \in \mathcal{C}$ with $I_\Pi \leq I$ the theorem holds with the law $U_{\sigma,M(I)}$, $M(I) = 2^{\lceil P_0(I) \rceil}$. The perturbation model, the scale σ supplied with Π , the other base-selected quantities of Π , correctness on every draw, the output contract, and the structural and numerical factors of the expected work bound are unchanged; the polynomial factor becomes a polynomial in I whose exponent depends only on the exponents of P_0 and of the original factor.*
- (b) *Suppose it has a Gaussian support schedule with $b_\Pi(t) \leq A(I_\Pi) + D(I_\Pi)t$ for all $\Pi \in \mathcal{C}$ and $t \geq 0$, where $A, D \geq 1$ are effective nondecreasing integer polynomials, and put*

$$H(I) = A(I) + D(I) + 21, \quad t(I) = \lceil 4 \log_2 H(I) \rceil, \quad b(I) = A(I) + D(I)t(I).$$

Then for every $\Pi \in \mathcal{C}$ with $I_\Pi \leq I$ the theorem holds with $\mathcal{G}_{b(I),\sigma}$ and the box and cap of trial $t(I)$, with the same conclusions as in (a). The support radius $(b(I) + 20)\sigma$ is at most $2^{t(I)}\sigma \leq 2H(I)^4\sigma$.

- (c) *Suppose that the hypotheses of (a) hold except that $\log_2 M_\Pi \leq F(K)P_0(I_\Pi)$ is known only for the instances whose structural parameter is at most a supplied bound K , where F is an nondecreasing positive integer-valued function, computable in $(1 + \text{size}(K) + F(K))^{c_F}$ bit operations for a fixed exponent c_F . The supplied bound K is included in the base data. Then (a) holds for these instances with $M(I, K) = 2^{\lceil F(K)P_0(I) \rceil}$, except that the bounds on the sampled coefficient lengths, on the output length, and the polynomial factor of the work bound are multiplied by $(1 + F(K))^c$, with c depending only on the exponents of P_0 , of the original bounds, and c_F .*

The hypotheses hold route by route as follows; the proofs are in Appendix A.7. The closure-based grid theorems of Sections 4–8 with polynomial sampling precision, and Theorem 4.9(a), choose M_Π as the least power of two above finitely many base-only quantities, such as 2^J and inverse failure budgets, whose logarithms are bounded by a fixed polynomial in I_Π . It uses M only through the interval masses (2), the replacement errors of Lemma 3.13, and these lower bounds, so its schedule is monotone and part (a) applies; for the quadratic theorems the verification is in Appendices B.6 and B.8. The lattice theorem (Theorem 8.17) is the one exception to the literal definition, because its level cap is $J = \log_2 M$; part (a) applies to it once the cap is reset to $\log_2 M(I)$. Part (b) applies to the Gaussian-like Theorems 4.9(b), 4.19, and 4.31. Part (c) applies to the core-only flow and totally unimodular theorems with arbitrary core faces (Theorems 8.12 and 8.14), whose sampling

precision depends on the core size k , with a supplied $K \geq k$. Taking $K = I$ gives a law that depends only on I , but its sampling cost need not stay within the bound for the actual core size; nothing here gives these two theorems a law of polynomial sampling precision. The strong-noise theorem holds for every M , but its bound depends on M through probabilities q_i that are not monotone in M : the grids of sizes 2 and 4 give the interval $[-\sigma/2, \sigma/2]$ the masses 0 and $1/2$. With a common M , its q_i and β must be computed for that M ; its sufficient regime (41), a lower bound on σ together with a lower bound on M , is preserved when M grows.

Three qualifications apply. First, the hypothesis $\log_2 M_\Pi \leq P_0(I_\Pi)$ is a property of the domains of a route, not of fixed degree alone. The fallback factor and the tail constants of Lemma 3.17 have logarithms polynomial in I for every explicit semialgebraic instance, since they depend on the number s of atoms only through $\log(s+1)$, even when native integer ranges contribute exponentially many atoms. The level caps also involve widths and curvature bounds, and on general compact semialgebraic domains these need not have polynomial bit length: the domain $1 \leq x_0 \leq 2$, $x_i = x_{i-1}^2$ ($1 \leq i \leq N$) has a description of length $O(N \log N)$ and last width $2^{2^N} - 1$. The theorems above use boxes, rational polytopes, explicit graph constraints of fixed dependency depth, and implicit graphs with supplied brackets, on which these quantities have polynomial encoding length. Second, the oracle contract of a theorem must hold for every coefficient in the support of the new law, as the contracts in this paper do, and oracle costs remain explicit factors. Third, the common law standardizes only the scalar law: it is rescaled by the supplied σ and applied to the coordinates that the theorem perturbs, so it is not one probability measure for all instances, and grid and Gaussian-like laws are not interchangeable. For Gaussian-like laws the support radius $(b(I) + 20)\sigma$ is the one that enters the original-objective bound of Proposition 2.6. Arbitrary-precision evaluation of an output uses the same draw and does not change the law; only Corollary 3.10, whose cap depends on the requested accuracy, needs a resolution depending on that accuracy. The corollary is an elementary consequence of the effective bounds used in the proofs. A finite collection of routes can share an enlarged envelope within each law family when its format constants and oracle cost models are fixed.

4 Few negative directions: quadratic and mixed-integer quadratic problems

This section treats the first structural route, in which the nonconvexity of the objective is confined to a small number k of directions. For a quadratic objective on a bounded polytope or mixed polytope, k is the negative inertia of the Hessian. A rational factorization $A = P - \alpha T^\top T$ with $P \succeq 0$ moves the nonconvex part to an auxiliary space $\mathbb{R}^k$, where the search of Section 3 runs. The remaining recourse is a convex quadratic program, a convex mixed-integer quadratic program, or a separable scalar problem, and it is solved exactly at every query.

The principal result is Theorem 4.19. Under a finite Gaussian-like law on all original linear coefficients, selected from the base data, it computes an exact optimizer of the sampled quadratic or mixed-integer quadratic program on every draw. Its expected bit work is $f(n_z, k, 1 + \nu \text{diam}(\mathcal{P})/\sigma) (I + 1)^C$ with an absolute exponent C , where n_z is the number of integer variables, ν the magnitude of the most negative eigenvalue of the Hessian, and $\mathcal{P}$ the polyhedral relaxation. The other results are variants that differ in the perturbation law or the recourse structure: noise aligned with the negative directions (Theorem 4.21), uniform noise on all coefficients, whose bound contains a power of the dimension that grows with k (Theorem 4.20), at most two negative directions with a bound linear in a numerical ratio (Theorem 4.9), and separable mixed recourse with arbitrarily many integer variables (Theorems 4.29 and 4.31). Table 2 summarizes them. Every result concerns the exact

optimum of the sampled objective. Proofs are in Appendix B.

4.1 Instances, exact primitives, and factorizations

Definition 4.1 (Quadratic instances). A *base instance* consists of a symmetric matrix $A \in \mathbb{Q}^{n \times n}$, a vector $b \in \mathbb{Q}^n$, a matrix $D \in \mathbb{Q}^{r \times n}$ and a vector $e \in \mathbb{Q}^r$ such that $\mathcal{P} = \{x : Dx \leq e\}$ is nonempty and bounded, a number $n_z \in \{0, \dots, n\}$ of integer variables, and a rational noise width $\sigma > 0$. The feasible set is $X = \mathcal{P} \cap (\mathbb{Z}^{n_z} \times \mathbb{R}^{n-n_z})$, and the objective is $F(x) = \frac{1}{2}x^\top Ax + b^\top x$. We assume $X \neq \emptyset$. If this is not promised, one convex-MIQP feasibility solve detects infeasibility before any optimum, growth threshold, or search is defined. Lower-dimensional polytopes and redundant rows are allowed. Let $n_-(A)$ be the number of negative eigenvalues of A , counted with multiplicity, and put $\nu = \max\{0, -\lambda_{\min}(A)\}$. A *perturbation* adds a linear term: *ambient* noise gives $F_\gamma(x) = F(x) + \gamma^\top x$ with $\gamma \in \mathbb{R}^n$, and noise *aligned* with a factor $T \in \mathbb{Q}^{k \times n}$ gives $F(x) + \xi^\top Tx$ with $\xi \in \mathbb{R}^k$; these are the ambient and aligned models of Definition 2.2. We write f^* for the minimum of the sampled objective over X .

Aligned noise perturbs the original coefficients by $T^\top \xi$. These coefficients are correlated and lie in a k -dimensional subspace, so aligned noise is not independent noise on the original coefficients. Both models are treated because they lead to different bounds.

Proposition 4.2 (Exact convex primitives). (a) A convex quadratic program $\min\{\frac{1}{2}x^\top Px + p^\top x : Dx \leq e\}$ with $P \succeq 0$ and rational data of bit length L_{in} is solved exactly in $\text{poly}(L_{\text{in}})$ bit operations: infeasibility and unboundedness are detected, and otherwise the optimal value and a rational optimal solution are returned [31].

(b) A convex quadratic program with n_z integer variables and the other data as in (a) is solved exactly, with the same outputs, in $f(n_z)L_{\text{in}}^{c_0}$ bit operations for a function f and an absolute constant c_0 [17, arXiv v2, Theorem 3].

(c) Rational linear programs are solved exactly in polynomial time; in particular a vertex of a nonempty rational polytope can be found [24].

When the algorithm of (b) returns an optimal solution, we fix its integer part, which lies in the box of linear-programming bounds and therefore has polynomial bit length, and re-solve the continuous slice by (a). This *polishing* returns an optimal solution of polynomial bit length, so that the possibly large factor $f(n_z)$ never enters the size of later instances. The factor $f(n_z)$ is absent when $n_z = 0$.

Definition 4.3 (Convexifying factorization). A *convexifying factorization* of A is a rational number $\alpha > 0$ and a rational matrix $T \in \mathbb{Q}^{k \times n}$ with $P = A + \alpha T^\top T \succeq 0$. It has a supplied rational *frame constant* $c_{\text{fr}} \in (0, 1]$, counted in I , if $c_{\text{fr}}I_k \preceq TT^\top \preceq I_k$. The number k of rows satisfies $k \geq n_-(A)$, with equality for the factorization of Lemma 4.4; in theorems with a supplied factorization, k denotes its number of rows.

Positive semidefiniteness of P and a supplied frame constant are verified exactly by symmetric elimination, so a supplied factorization is a checked certificate rather than a promise. No relation between $\ker P$ and $\ker T$ is required.

Lemma 4.4 (Intrinsic rational normalization). *There is a deterministic algorithm, polynomial in the bit length of A , that either certifies $A \succeq 0$, or computes rational β with $\nu \leq \beta < 2\nu$ and a rational matrix $T \in \mathbb{Q}^{k \times n}$, with $k = n_-(A)$, such that*

$$A + 2\beta T^\top T \succeq 0, \quad \frac{63}{64}I_k \preceq TT^\top \preceq I_k,$$

and the rows of T lie in the range of A . Thus $\alpha = 2\beta$ gives a convexifying factorization with $2\nu \leq \alpha < 4\nu$ and frame constant $63/64$. It uses no real eigenvectors, no bound on ν , and no lower bound on nonzero eigenvalues as input.

The construction combines exact rational Jacobi rotations, which give an orthogonal near-diagonalization [18], with an exact projection onto the range of A . The projection is essential. For $A = \text{diag}(-1, 0)$ and a unit vector $t = (\cos \theta, \sin \theta)$ with $\sin \theta \neq 0$, the determinant of $A + 2tt^\top$ is $-2\sin^2 \theta < 0$, so an approximate negative eigenvector with a component in $\ker A$ does not give a convexifier, however small that component is. For the factorization of the lemma, $P = A + 2\beta T^\top T$ is positive definite on the range of A and vanishes on $\ker A$, so $\ker P = \ker A \subseteq \ker T$. This inclusion holds automatically, but none of the proofs below uses it.

4.2 The auxiliary value function

Let $X \subset \mathbb{R}^n$ be nonempty and compact, let $F : X \rightarrow \mathbb{R}$ be continuous, let $T \in \mathbb{R}^{k \times n}$, and let $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_k)$ be positive definite; in all results except Theorem 4.31, $\Lambda = \alpha I$. For a *residual* $\zeta \in \mathbb{R}^n$ and a *factor coefficient* $\xi \in \mathbb{R}^k$ define

$$W_\zeta(a) = \min_{x \in X} \left[F(x) + \zeta^\top x + \frac{1}{2}(a - Tx)^\top \Lambda (a - Tx) \right], \quad V(a) = W_\zeta(a) + \xi^\top a,$$

and put $\gamma = T^\top \xi + \zeta$. A point attaining the minimum defining $W_\zeta(a)$ is an *attaining witness* at a . For a quadratic F with a convexifying factorization and $\Lambda = \alpha I$, the inner problem has the convex objective $\frac{1}{2}x^\top Px + (b + \zeta - \alpha T^\top a)^\top x$ up to a constant, so $W_\zeta(a)$ is one convex QP or convex MIQP.

Lemma 4.5 (Auxiliary value function). *In this setting:*

(a) (Square completion.) *For every a ,*

$$V(a) = \min_{x \in X} \left[F_\gamma(x) + \frac{1}{2}(a - Tx + \Lambda^{-1}\xi)^\top \Lambda (a - Tx + \Lambda^{-1}\xi) \right] - \frac{1}{2}\xi^\top \Lambda^{-1}\xi.$$

Consequently $V^ = \inf_{\mathbb{R}^k} V = f^* - \frac{1}{2}\xi^\top \Lambda^{-1}\xi$, attained at $Tx^* - \Lambda^{-1}\xi$ for every minimizer x^* of F_γ , and every attaining witness x_a at any a satisfies $F_\gamma(x_a) - f^* \leq V(a) - V^*$.*

(b) (Upper curvature.) *$W_\zeta(a) - \frac{1}{2}a^\top \Lambda a$ is concave, so W_ζ and V have upper coordinate curvature at most $(\lambda_1, \dots, \lambda_k)$.*

(c) (Every attaining witness gives a small active vector.) *For every a and every attaining witness x_a , the vector $g = \Lambda(a - Tx_a) + \xi$ satisfies*

$$V(z) \leq V(a) + g^\top(z - a) + \frac{1}{2}(z - a)^\top \Lambda (z - a) \quad (z \in \mathbb{R}^k), \quad g^\top \Lambda^{-1} g \leq 2(V(a) - V^*).$$

(d) (A fixed box.) *If $\ell_i \leq (Tx)_i \leq u_i$ for all $x \in X$ and $|\xi_i| \leq s_i$, then $\mathcal{A} = \prod_i [\ell_i - s_i/\lambda_i, u_i + s_i/\lambda_i]$ contains a global minimizer of V , and $\min_{\mathcal{A}} V = V^*$.*

Part (c) is the inequality that replaces differentiability of W_ζ throughout this section. It holds for *every* attaining witness, including witnesses at integer ties, on flat optimal faces, and at points where W_ζ is not differentiable. No derivative of W_ζ and no selection of a favorable subgradient is needed. For ambient noise and a factor of full row rank, the decomposition is

$$\Psi = (TT^\top)^{-1}T, \quad \xi = \Psi\gamma, \quad \zeta = \Pi_T\gamma, \quad \Pi_T = I - T^\top\Psi,$$

so that $\gamma = T^\top \xi + \zeta$ and $T\zeta = 0$. The coordinates of ξ are in general dependent, and ξ and ζ are dependent under uniform noise; Section 4.6 shows why this matters.

4.3 The inverse-growth route for at most two negative directions

When the sampled problem has positive point growth, the auxiliary problem inherits growth, and a search without closure terminates exactly by rational reconstruction. This deterministic result is combined with the growth tail.

Theorem 4.6 (Growth-conditioned exact algorithm). *There is a deterministic algorithm for continuous instances ($n_z = 0$) with $k = n_-(A) \geq 1$ and rational data of bit length L_{in} with the following properties.*

- (a) *Every output is an exact rational global minimizer together with the exact optimal value; this holds on every instance.*
- (b) *If the minimizer x^* is unique and $F(x) - F(x^*) \geq g \|x - x^*\|^2$ on X for some $g > 0$, the algorithm halts. With $\kappa = 2 + 4\nu/g$, it processes at most $4^k(2\sqrt{k\kappa} + 2)^k$ cells per refinement level, stops after at most $J_g \leq \text{poly}(L_{\text{in}}) + \log_2(1 + \kappa)$ levels, and uses at most $c_k(1 + \sqrt{\kappa})^k(J_g + 1)\text{poly}(L_{\text{in}} + J_g)$ bit operations, where c_k depends only on k .*
- (c) *For $k \leq 2$ and $Z = \max\{1, \nu/g\}$ the bound in (b) is at most $\text{poly}(L_{\text{in}}) Z^{k/2}(1 + \log Z)^{d_0}$, where d_0 and the polynomial are absolute.*

The algorithm needs neither g nor ν .

The algorithm normalizes A by Lemma 4.4, runs the corrected-corner search of Definition 3.8 on W_0 over the box of ranges of Tx , and attempts two rational reconstructions at every level: of the optimal value from the certified interval of Theorem 3.9(v), and of the common auxiliary minimizer Tx^* from the best corner. It accepts only a feasible witness whose value equals the reconstructed optimal value, so a premature reconstruction is harmless. Growth enters only the termination bound: by Lemma 4.5 and the triangle inequality, $W_0(a) - f^* \geq g_W \|a - Tx^*\|^2$ with $\alpha/g_W \leq \kappa$. Every retained cell has a near-optimal corner in the ball of radius $h_j\sqrt{k\kappa}/2$ around Tx^* , and the whole cell lies in that ball enlarged by its diameter.

Corollary 4.7 (Growth-conditioned mixed-integer algorithm). *Theorem 4.6 also holds for nonempty mixed feasible sets of Definition 4.1, with an additional factor $f(n_z)$ in its bit-work bounds, where f is the function of Proposition 4.2(b). A feasible rational optimizer and its exact value are returned whenever the algorithm halts; positive point growth guarantees termination. The polynomial exponents are independent of n_z and k , and the factor c_k remains dependent only on k . No growth modulus is needed as input.*

To use Theorem 4.6 under a finite law, the growth tail must be transferred to that law with an atom term that does not depend on the threshold.

Lemma 4.8 (Explicit growth sections for quadratics). *Let F be quadratic and let $X = \mathcal{P} \cap (\mathbb{Z}^{n_z} \times \mathbb{R}^{n-n_z})$ as in Definition 4.1, and let $F = Z \cdot 2^r$, where Z is the number of integer points in the box of linear-programming bounds of the integer variables ($Z = 1$ if $n_z = 0$). For every $\varepsilon > 0$, every coordinate i , and all fixed values of the other coefficients, the set of values of γ_i for which $g_*(\gamma) < \varepsilon$ is a union of at most $8(F + 1)^2$ intervals. Consequently, with $S = \sum_i w_i$ the sum of the coordinate widths of X ,*

$$\mathbb{P}\{g_* < \varepsilon\} \leq \frac{S\varepsilon}{\sigma} + \frac{16n(F + 1)^2}{M} \quad \text{and} \quad \mathbb{P}\{g_* < \varepsilon\} \leq \sqrt{\frac{2}{\pi}} \frac{S\varepsilon}{\sigma} + 16n(F + 1)^2 2^{-b}$$

for independent coordinates with law $U_{\sigma, M}$ and $\mathcal{G}_{b, \sigma}$, respectively, for every $\varepsilon > 0$.

Theorem 4.9 (Expected exact work for at most two negative directions). *Let the base instance be continuous ($n_z = 0$) with $k = n_-(A) \leq 2$, let S be the sum of the coordinate widths of $\mathcal{P}$, and put $B = \max\{2, 2^r\}$ and $D = 8(2^r + 1)^2$.*

- (a) *Let M be the least power of two with $M \geq 2nDB$, so $\log_2 M = O(r + \log(n + 1))$, and let $\gamma_1, \dots, \gamma_n$ be independent with law $U_{\sigma, M}$. There is an algorithm that returns an exact rational minimizer of F_γ and the exact optimal value on every draw, with expected bit work at most $(I + 1)^C(1 + \nu S/\sigma)$ for an absolute constant C .*
- (b) *The same holds for independent coordinates with law $\mathcal{G}_{b, \sigma}$, where b is the least positive integer satisfying $2^b \geq 2nDB$, hence $b = O(r + \log(n + 1))$.*

Either law may use a larger resolution with $\log M \leq \text{poly}(I)$ or $b \leq \text{poly}(I)$, respectively; the bit polynomial then includes that resolution. This permits a common law with a closure theorem while changing its sampling and coefficient lengths. The algorithm needs neither ν , S , nor the growth of the sampled instance. For $k = 0$ one exact convex QP solve suffices.

The algorithm interleaves, bit operation by bit operation, the algorithm of Theorem 4.6 and the exact fallback of Lemma 4.11 on the same draw, and stops when either finishes. The fallback costs at most $B \text{poly}(L_{\text{in}})$ on every draw. With $p = k/2$ and $Z = \max\{1, \nu/g_*\}$ ($Z = \infty$ when $g_* = 0$), the work is at most $\text{poly}(L_{\text{in}}) \min\{B, Z^p(1 + \log Z)^{d_0}\} \leq \text{poly}(L_{\text{in}})(1 + p^{-1} \log B)^{d_0} \min\{B, Z^p\}$. The growth tail with a residual $\beta \leq 1/B$ gives $\mathbb{P}\{Z > t\} \leq r_0/t + \beta$ for $t \geq 1$, where $r_0 = \nu S/\sigma$, and therefore

$$\mathbb{E} \min\{B, Z^p\} \leq 1 + r_0 \int_1^B t^{-1/p} dt + \beta B \leq \begin{cases} 2 + r_0, & k = 1, \\ 2 + r_0 \log B, & k = 2. \end{cases}$$

All atoms and all draws with $g_* = 0$ are included in the term βB .

Remark 4.10 (Why the moment argument stops at two directions). For $k \geq 3$ the same integral is $r_0(B^{1-2/k} - 1)/(1 - 2/k)$, which is exponential in r . This is a limitation of the argument, not a lower bound for any algorithm. It cannot be repaired by a sharper tail. Let $X = [-w_0/2, w_0/2] \times Y$ and $F(x_0, y) = G(y)$, where G is a quadratic with k negative eigenvalues of magnitude at most ν on a polytope Y . For a continuous uniform perturbation, moving x_0 to the other endpoint shows $g_* \leq |\gamma_0|/w_0$. Hence $\mathbb{P}\{Z > t\} \geq \nu w_0/(\sigma t)$ for $t \geq \max\{1, \nu w_0/\sigma\}$, and $\mathbb{E} \min\{B, Z^{k/2}\} \geq a_0 \int_{t_0}^B s^{-2/k} ds$ when $B > t_0$, with $a_0 = \nu w_0/\sigma$ and $t_0 = \max\{1, a_0^{k/2}\}$. For fixed $a_0 > 0$ and sufficiently large B , this is of order $(\nu w_0/\sigma) B^{1-2/k}$ for $k \geq 3$ and large B . The inverse-growth power $Z^{k/2}$ is therefore not integrable to a polynomial bound beyond two negative directions, even though the tail of Theorem 3.16 is sharp. The closure method below does not integrate an inverse growth power and has no such restriction. The two results are not comparable: Theorem 4.9 is linear in $\nu S/\sigma$, while the closure bounds are of degree k in $\nu \text{diam}(\mathcal{P})/\sigma$. They can be applied to the same sampled objective: part (b) of Theorem 4.9 holds for the law of Theorem 4.19 once its accuracy also satisfies $2^b \geq 2nDB$, a requirement of polynomial bit length that does not depend on the support.

4.4 Exact closure of cells

Lemma 4.11 (Smallest-face candidates and the quadratic fallback). *Let $G(x) = \frac{1}{2}x^\top Hx + p^\top x$ be a rational quadratic, with H symmetric of any inertia, and let $\mathcal{Q} = \{x : Dx \leq e\}$ be a nonempty bounded rational polytope.*

- (a) *There are a global minimizer $\hat{x}$ of G on $\mathcal{Q}$ and a set S of linearly independent rows active at $\hat{x}$ such that $K_S = \begin{pmatrix} H & D_S^\top \\ D_S & 0 \end{pmatrix}$ is nonsingular and $\hat{x}$ is the x -part of $K_S^{-1}(-p; e_S)$.*

- (b) Enumerating all row subsets, solving the nonsingular systems, keeping the feasible solutions, and returning one of least value finds an exact rational global minimizer in $2^r \text{poly}(L_{\text{in}})$ bit operations, on every instance. Multiplier signs and definiteness tests are unnecessary.
- (c) For a mixed polytope, enumerating the Z integer points of the box of linear-programming bounds and applying (b) to every nonempty slice is exact in $B \text{poly}(L_{\text{in}})$ bit operations with $B = \max\{2, Z2^r\}$.
- (d) For a box in $\mathbb{R}^k$, the 3^k faces (each coordinate at its lower bound, at its upper bound, or free) suffice in (b).

In particular, every rational QP and MIQP on a nonempty bounded mixed polytope has a rational global minimizer, and an optimal value, of polynomial bit length.

The proof chooses a global minimizer on a face of smallest dimension. Its tangent Hessian is positive semidefinite; a null tangent direction would keep the objective constant up to the boundary of the face and produce a minimizer on a smaller face. This argument handles singular Hessians, continua of minimizers, and lower-dimensional polytopes, but it is specific to quadratics.

Let a base instance and a convexifying factorization be fixed, with $\Lambda = \alpha I$. A *label* is the integer part z of a point of X , and the *slice* of z is the set of continuous parts x_c with $(z, x_c) \in X$. Write W_ζ^z for the function W_ζ with X replaced by the points of label z , so that $W_\zeta = \min_z W_\zeta^z$.

Lemma 4.12 (Critical pieces of the convex recourse). *For every feasible label z and every rational query (a, ζ) , polynomial-time rational computations after one exact solve of the slice problem return a set S of linearly independent rows of the slice constraints, active at an optimal slice point, such that the following hold.*

- (a) The stationarity system of the slice with active rows S is nonsingular. Its solution is affine in (a', ζ') : $x_{z,S}(a', \zeta') = \Xi_S a' + Y_S \zeta' + x_{z,S}^0$, with multipliers $\lambda_{z,S}(a', \zeta')$ affine as well.
- (b) The region $\mathcal{R}_{z,S}(\zeta') = \{a' : x_{z,S}(a', \zeta') \text{ feasible, } \lambda_{z,S}(a', \zeta') \geq 0\}$ contains a when $\zeta' = \zeta$. It is defined by at most $r + n$ affine inequalities whose normals do not depend on ζ' and whose offsets are affine in ζ' . For every $a' \in \mathcal{R}_{z,S}(\zeta')$, the point $x_{z,S}(a', \zeta')$ attains $W_{\zeta'}^z(a')$.
- (c) The quadratic $q_{z,S}(a', \zeta') = F(x_{z,S}) + \zeta'^T x_{z,S} + \frac{\alpha}{2} \|a' - T x_{z,S}\|^2$ equals $W_{\zeta'}^z$ on the region and satisfies the identity $\nabla_{a'} q_{z,S}(a', \zeta') = \alpha(a' - T x_{z,S}(a', \zeta'))$ for all (a', ζ') . Its Hessian $H_S = \alpha(I - T \Xi_S)$ does not depend on ζ' , z , b , or e , and $\|H_S\| \leq \Theta$, where $\Theta = \alpha(1 + kn\tau_T U_0)$, $\tau_T = \max_{il} |T_{il}|$, and $U_0 = (2n)! C_0^{2n}$ with C_0 the largest absolute value of the integers obtained by clearing a common denominator of the entries of P , D , and αT^T .

There are at most $R = Z \cdot 2^r$ possible pairs (z, S) . For every feasible fixed label these regions cover all real parameter pairs (a, ζ) ; only extraction at rational queries is computational. The region description does not require full dimension, an irredundant description, or a relation between $\ker P$ and $\ker T$.

The gradient identity in (c) is algebraic. It holds on all of $\mathbb{R}^k$, even when the region has empty interior; it is not obtained by differentiating W_ζ along the region. At the query, $x_{z,S}(a, \zeta)$ is an attaining witness of $W_\zeta^z(a)$, so Lemma 4.5(c) applies to the vector $\nabla q_{z,S}(a, \zeta) + \xi$ whenever z is a winning label. The bound Θ uses only the matrices of the response to a' . The sampled coefficients enter only offsets, so Θ is fixed before sampling. Its numerical size is large, but it enters the algorithm only through its logarithm.

Under the usual uniqueness and regularity assumptions, affine optimizer formulas on polyhedral critical regions, with quadratic value formulas, are standard in multiparametric quadratic programming [13, 36, 44], and the reduction of structured indefinite quadratic programs to the global minimization of a parametric convex value function, with recovery of optimizers from the minimizing parameter, appears in Ding’s thesis [20]. Lemma 4.12 is the form of this machinery used here: one region per query, extracted in polynomial time without enumerating regions, with a Hessian bound that is independent of the sampled coefficients. What the expected-work theorems add is the analysis of the cells that a region does not close.

For a continuous instance a region certifies the value of W_ζ on a whole cell. For integer variables it certifies only the value of one label, and a second certificate is needed.

Lemma 4.13 (Best-other-label gaps). *Let $[\ell_i, u_i]$ be the range of $(Tx)_i$ over $\mathcal{P}$, and put $\bar{w} = \max\{1, \sum_i (u_i - \ell_i)\}$. At a query v , let z_* be a winning label and put $\Delta(v) = \min_{z \neq z_*} W_\zeta^z(v) - W_\zeta(v)$, with $\Delta(v) = +\infty$ if no other label is feasible.*

- (a) $\Delta(v)$ and a witness of the best other label are computed by $2n_z$ additional exact convex MIQP solves, each with one extra inequality $x_i \leq z_{*,i} - 1$ or $x_i \geq z_{*,i} + 1$.
- (b) Let C be a cell with corner v and side lengths at most h . If $C \subseteq \mathcal{R}_{z_*,S}(\zeta)$ and $\Delta(v) \geq \theta_{\text{gap}} h$ with $\theta_{\text{gap}} = \alpha k \bar{w}$, then $W_\zeta = q_{z_*,S}(\cdot, \zeta)$ on C , and $x_{z_*,S}(a, \zeta)$ attains $W_\zeta(a)$ for every $a \in C$.
- (c) If $V(v) - V^* \leq 2E$ with $E \leq k\alpha h^2/8$ and $h \leq 1$, and the gap condition of (b) fails, then two different labels have feasible points with $F_\gamma \leq f^* + C_{\text{gap}} h$, where $C_{\text{gap}} = k\alpha/4 + \theta_{\text{gap}}$.

Equality in the gap test is allowed: another label may tie inside the cell without changing the certified value or the feasible witness. Equal winning labels at all corners do not suffice (Example 4.15).

Lemma 4.14 (Integer-label isolation). *Let $\mathcal{Z}$ be a nonempty finite set of integer vectors in $\prod_{i=1}^{n_z} [L_i, U_i]$, let $h : \mathcal{Z} \rightarrow \mathbb{R}$ be arbitrary, and let $\gamma \in \mathbb{R}^{n_z}$ have independent coordinates with interval concentrations π_i . Let Δ be the difference between the two smallest values of $h(z) + \gamma^\top z$ at distinct labels ($\Delta = +\infty$ if $|\mathcal{Z}| = 1$). Then for every $\varepsilon \geq 0$*

$$\mathbb{P}\{\Delta \leq \varepsilon\} \leq \sum_{i=1}^{n_z} (U_i - L_i) \pi_i (2\varepsilon).$$

In the application, $h(z)$ is the optimal value of the slice of z with its continuous noise, conditioned on that noise, so the continuous recourse may be arbitrary. The widths $U_i - L_i$ can be exponentially large, but they enter the algorithms only through logarithms. Isolation of the best discrete solution under random objective coefficients is classical [11, 39]; the lemma is the version used here, with finite laws and arbitrary label costs.

Example 4.15 (Equal corner winners do not certify a label). Let $z \in [0, 1]$ be continuous and $y \in \{0, 1\}$ integer, and let $F(z, y) = \frac{1}{2}zy + \frac{7}{8}y^2 - \frac{15}{16}y$, $\alpha = 1$, $T = (1, 1/2)$. Then $P = (\frac{1}{1} \frac{1}{2}) \succ 0$. On $a \in [0, 1]$ the label $y = 0$ has slice value 0 and the label $y = 1$ has slice value $a^2/2 - a/2 + 1/16$. The label $y = 0$ wins strictly at both endpoints, and its slice formula is valid on the whole cell, but $y = 1$ wins at $a = 1/2$. The gap test of Lemma 4.13 refuses this cell; a test based on equal corner labels would accept it and return a wrong value. Rescaling to $\alpha = 4$, $T = (1/2, 1/4)$, and the cell $[0, 1/2]$ gives the same example with $\|T\| < 1$.

Definition 4.16 (Closure search). Let a base instance, a convexifying factorization, a residual ζ , a factor coefficient ξ , a box $\mathcal{A}$ as in Lemma 4.5(d), and a level cap J be given. Run the corrected-corner search of Definition 3.8 on V over $\mathcal{A}$ with isotropic meshes and curvature α . At every corner v of a processed cell, compute $W_\zeta(v)$ exactly with a polished attaining witness, a winning label z_v , a piece (z_v, S_v) of Lemma 4.12, and, if $n_z \geq 1$, the gap $\Delta(v)$ of Lemma 4.13. A processed cell C of level j is *closed* if for some corner v all corners of C lie in $\mathcal{R}_{z_v, S_v}(\zeta)$ and, when $n_z \geq 1$, $\Delta(v) \geq \theta_{\text{gap}} h_j$. A closed cell is minimized exactly as the quadratic $q_{z_v, S_v}(\cdot, \zeta) + \xi^\top a$ on C by Lemma 4.11(d), and the minimizer's affine witness is stored. If no unclosed cell is retained at some level $j \leq J$, return the witness of the incumbent. Otherwise, after level J , run the exact fallback of Lemma 4.11(c) on the same sampled objective.

By Lemmas 4.12 and 4.13(b), every closure step is exact, so Theorem 3.9 applies to V on $\mathcal{A}$. When the search stops before the fallback, its incumbent attains V^* , and by Lemma 4.5(a) the stored witness is a global minimizer of the sampled objective. The output is therefore exact on every draw, with a rational minimizer and value of polynomial bit length. The search trace, consisting of closed cells with their pieces and discarded cells with their corner bounds, is a verifiable global certificate; for $n_z \geq 1$ its verification includes rerunning the exact mixed-integer solves.

The probabilistic analysis rests on one deterministic fact: a retained unclosed cell forces the noise close to one of finitely many hyperplanes that are fixed before sampling.

Lemma 4.17 (Unresolved cells force fixed tubes). *There is a family of at most $K = R(r + n + 1)$ affine functionals $\ell_\kappa(\xi, \zeta) = u_\kappa^\top \xi + w_\kappa^\top \zeta + c_\kappa$ with $\|u_\kappa\| = 1$, determined by the base data and the factorization, with the following property. Suppose that a cell of level j is retained and not closed, that its corner v satisfies $V(v) - V^* \leq 2E_j$, that z_v wins at v , and that some corner of the cell lies outside $\mathcal{R}_{z_v, S_v}(\zeta)$. Then $|\ell_\kappa(\xi, \zeta)| \leq \sqrt{k}(\alpha + \Theta)h_j$ for some κ . If T has full row rank and $\|T\| \leq 1$, then $\ell_\kappa(\Psi\gamma, \Pi_T\gamma) = L_\kappa^\top \gamma + c_\kappa$ with a fixed vector L_κ of norm at least 1.*

If the piece matrix H_S is invertible, the region boundary crossed inside the cell is mapped by the affine map $a \mapsto -\nabla_a q_{z, S}(a, \zeta)$ to a hyperplane in the space of ξ ; by Lemma 4.5(c), ξ lies close to the image of v . If H_S is singular, the whole image of that map lies in a hyperplane. The normals are fixed, and only the offsets depend on ζ , affinely. A region row with zero normal cannot be violated in a cell that contains the query, so no exceptional case arises. The family is a proof device; the algorithm never constructs it.

Example 4.18 (An atom that requires the fallback). Let $X = [0, 1]^2$, $F(x, y) = xy - (x - y)/2$, $T = (1/2, -1/2)$, $\alpha = 4$, and $\sigma = 1$, so $P = I$. Under aligned noise the endpoint draw $\xi = 1$, which belongs to every grid $U_{1, M}$, gives the sampled objective xy . A direct computation gives $V(a) = -1/8$ for all $a \in [-3/4, 1/4]$, with two critical pieces, $\{x = 1/2 + 2a, y = 0\}$ for $a \geq -1/4$ and $\{x = 0, y = -1/2 - 2a\}$ for $a \leq -1/4$. The fixed box is $[-3/4, 3/4]$, and the switch $a = -1/4$ lies at relative position $1/3$, which is never a dyadic grid node. The cell containing $-1/4$ is therefore never closed, and its corners are globally optimal, so it is retained at every level. The draw has probability $1/M$ and is solved by the fallback. Closure alone does not make the algorithm terminate on this draw; the fallback is necessary.

4.5 Main theorems

In the next three theorems, $[\ell_i, u_i]$ is the range of $(Tx)_i$ over $\mathcal{P}$, computed by linear programming. All sampling parameters are computed from the base data before the draw, and all have bit length polynomial in I . The algorithm of each theorem is the closure search of Definition 4.16 with a fallback on the same draw; it never rejects or resamples a draw.

Theorem 4.19 (Gaussian-like noise on all coefficients). *Let a base instance with $A \not\equiv 0$ be given, and let (α, T) , with k rows, be either the factorization of Lemma 4.4 ($k = n_-(A)$, $c_{\text{fr}} = 63/64$) or a supplied convexifying factorization with frame constant c_{fr} . There is an algorithm that computes from the base data, in time polynomial in I , an integer $b \leq \text{poly}(I)$; draws $\gamma_1, \dots, \gamma_n$ independently from $\mathcal{G}_{b,\sigma}$; and returns, on every draw, a rational global minimizer of F_γ on X and the optimal value, or reports $X = \emptyset$. Its expected number of bit operations is at most $f(n_z) C_0^k (1 + \mathcal{H}_G)(I + 1)^{C_1}$, where C_0, C_1 are absolute constants, f is the function of Proposition 4.2(b) ($f \equiv 1$ if $n_z = 0$), and*

$$\mathcal{H}_G = \prod_{i=1}^k \left[2 + c_{\text{fr}}^{-1/2} (4k + 6) + \frac{(1 + 2k)\alpha(u_i - \ell_i)}{\sigma\sqrt{2\pi}} \right].$$

For the factorization of Lemma 4.4, $\mathcal{H}_G \leq [9 + 5k + 2(1 + 2k)\nu \text{diam}(\mathcal{P})/\sigma]^k$, so the expected work is at most $f_1(n_z, k, 1 + \nu \text{diam}(\mathcal{P})/\sigma)(I + 1)^{C_1}$ with $k = n_-(A)$. If $A \succeq 0$, one exact solve by Proposition 4.2 suffices.

The theorem assumes neither growth, nor a unique minimizer, nor genericity, nor strict complementarity, nor full-dimensional feasibility, nor a supplied global certificate. The ratio $\nu \text{diam}(\mathcal{P})/\sigma$ enters through its numerical value; a short binary encoding does not make a large ratio harmless. The exponent C_1 does not depend on k or n_z , so the bound is fixed-parameter in (n_z, k) and the numerical ratio. The law $\mathcal{G}_{b,\sigma}$ is a finite rational law whose accuracy b depends on the base instance; it is not an exact Gaussian, and the theorem does not cover other approximations of a Gaussian.

Theorem 4.20 (Uniform noise on all coefficients). *Let a base instance with $A \not\equiv 0$ be given, and let (α, T) be a convexifying factorization with k rows, T of full row rank, and $\|T\| \leq 1$, for example that of Lemma 4.4. Put $w_i = u_i - \ell_i + 2\sigma \|\Psi_{i,:}\|_1 / \alpha$. There is an algorithm that computes from the base data a power of two M with $\log M \leq \text{poly}(I)$, draws $\gamma_1, \dots, \gamma_n$ independently from $U_{\sigma,M}$, and returns an exact minimizer and the optimal value on every draw, with expected bit work at most $f(n_z) C_0^k (1 + \mathcal{H}_{\text{amb}})(I + 1)^{C_1}$, where*

$$\mathcal{H}_{\text{amb}} = \prod_{i=1}^k \left[2 + \frac{(1 + 2k)\alpha\sqrt{n} w_i}{2\sigma} \right].$$

For the factorization of Lemma 4.4, $\mathcal{H}_{\text{amb}} \leq [2 + (1 + 2k)(2n + 2\nu\sqrt{n} \text{diam}(\mathcal{P})/\sigma)]^k$.

This bound is polynomial for each fixed k when the displayed ratios are polynomially bounded. It is not a fixed-parameter bound: the power of n grows with k . Section 4.6 shows that this power is a real property of the counts used in the proof.

Theorem 4.21 (Noise aligned with the factor). *Let a continuous base instance ($n_z = 0$) and an arbitrary convexifying factorization (α, T) with $k \geq 1$ rows be given, and put $w_i = u_i - \ell_i + 2\sigma/\alpha$. There is an algorithm that computes from the base data a power of two M with $\log M \leq \text{poly}(I)$, draws $\xi_1, \dots, \xi_k$ independently from $U_{\sigma,M}$, and returns an exact minimizer of $F(x) + \xi^\top T x$ and the optimal value on every draw, with expected bit work at most $C_0^k (1 + \mathcal{H}_{\text{al}})(I + 1)^{C_1}$, where*

$$\mathcal{H}_{\text{al}} = \prod_{i=1}^k \left[3 + \frac{(1 + 2k)\alpha w_i}{2\sigma} \right].$$

The aligned bound is fixed-parameter in k and the ratios $\alpha(u_i - \ell_i)/\sigma$. Its law perturbs only the k factor coefficients. Mixed-integer instances are excluded because aligned noise does not perturb the integer coefficients independently, so the label-gap failures of Lemma 4.13(c) are not controlled.

Remark 4.22 (The original objective). By Proposition 2.6, an exact minimizer $\hat{x}$ of F_γ satisfies $F(\hat{x}) - \min_X F \leq \max_{x,y \in X} \gamma^\top(x-y)$, which is at most $\bar{\sigma}W_{\text{noise}}$ when every coefficient is bounded by $\bar{\sigma}$; for $\mathcal{G}_{b,\sigma}$, $\bar{\sigma} = (b+20)\sigma$. Here $W_{\text{noise}} = \sum_{j=1}^n (\max_X x_j - \min_X x_j)$ is the original-domain width of Section 2.6, not a sum of the auxiliary widths w_i ; on a mixed polytope the coordinate widths of the continuous relaxation give an upper bound. Under aligned noise the bound is $\bar{\sigma} \sum_i (u_i - \ell_i)$. Because the work bounds grow as σ decreases, these results do not recover the unperturbed optimizer, and they claim no improvement of worst-case approximation bounds for the unperturbed problem.

4.6 Aligned, uniform, and Gaussian-like noise

The three laws lead to different counts because of the conditional law of the factor coefficients given the residual. Every local event of the search,

$$E_v : \quad V(v) \leq V(v \pm h_{ij}e_i) + 2E_j \quad (i \in Q_v),$$

confines each ξ_i with $i \in Q_v$ to an interval of length at most $(1+2k)\alpha h_{ij}$ whose position depends on ζ (Lemma 3.2 applied to W_ζ). Under aligned noise $\zeta = 0$ and the factor coefficients are independent, so Corollary 3.6 applies directly. Under uniform ambient noise, $\xi = \Psi\gamma$ is dependent on $\zeta = \Pi_T\gamma$, and the residual-dependent intervals must be handled by a volume estimate.

Lemma 4.23 (Dependent fiber volume). *Let γ be uniform on $[-\sigma, \sigma]^n$, let $T \in \mathbb{R}^{k \times n}$ have full row rank and $\|T\| \leq 1$, and let $Q \subseteq \{1, \dots, k\}$ have t elements. For each $i \in Q$ let $I_i(\zeta)$ be an interval of length at most ℓ_i depending measurably on $\zeta = \Pi_T\gamma$. Then*

$$\mathbb{P}\{\xi_i \in I_i(\zeta) \ (i \in Q)\} \leq \frac{\prod_{i \in Q} \ell_i}{(2\sigma)^t} \sum_{|K|=t} |\det T_{Q,K}| \leq \prod_{i \in Q} \frac{\sqrt{n} \ell_i}{2\sigma}.$$

The factor $\sqrt{n}$ is attained even for perfectly conditioned factors.

Proposition 4.24 (Sharpness of the fiber estimate). *Let $n = m^2$, $T = m^{-1}(1, \dots, 1)$, and γ uniform on $[-\sigma, \sigma]^n$. Then $TT^\top = 1$, $\text{Var } \xi = \sigma^2/3$, and for the interval $I(\zeta)$ of length $\ell \leq 2\sigma m$ centered at the midpoint of the conditional support of ξ , $\mathbb{P}\{\xi \in I(\zeta)\} = 1 - (1 - \ell/(2\sigma m))^n$, whose ratio to ℓ tends to $\sqrt{n}/(2\sigma)$ as $\ell \rightarrow 0$. For k disjoint blocks of this form, in dimension $N = km^2$, the ratio of the joint probability to $\prod_i \ell_i/(2\sigma)$ tends to $m^k = (N/k)^{k/2}$.*

A dimension-free variance of ξ does not imply dimension-free concentration on intervals that may depend on the residual. The loss is not confined to arbitrary intervals: it occurs for the counts of actual quadratic instances.

Proposition 4.25 (The uniform-noise counts grow with the dimension). *For every $k \geq 1$ and every even $m \geq 256$ there is a rational continuous instance with $N = km^2$ variables, $O(N)$ inequalities, $n_-(A) = k$, and a convexifying factorization with $\alpha = 1$, $TT^\top = I_k$, and $\alpha < 4\nu$, whose ranges satisfy $u_i - \ell_i = 8$, such that under independent uniform noise on all coefficients, either continuous on $[-1, 1]$ or $U_{1,M}$ with $M \geq 32$, the following hold for the meshes on the box of Theorem 4.20. At some level the expected number of nodes satisfying the local event is at least $128^{-k}(N/k)^{k/2}$, and at some level the expected number of $2E_j$ -nearly optimal nodes is at least $64^{-k}(N/k)^{k/4}$.*

The proposition is the case $\alpha = 1$ of Theorem 9.4, which is proved in Appendix G.1 for a slightly more general family. Consequently no bound of the form $f(k, R)N^C$ with an absolute exponent C holds for either count, where R records the projected widths, the frame constant, and the curvature-to-noise ratios. The proposition limits the counting argument, not the closure

algorithm, which may close the few quadratic pieces of these instances early. The ambient diameter of these instances grows with m , so the proposition does not exclude a bound parameterized by the full diameter.

Gaussian noise removes both losses. For $\gamma \sim N(0, \sigma^2 I_n)$, the projections $\xi = \Psi\gamma$ and $\zeta = \Pi_T\gamma$ are jointly Gaussian and uncorrelated, hence independent, and $\text{Cov } \xi = \sigma^2(TT^\top)^{-1}$. Conditioning on ζ does not change the law of ξ , whose density obeys a product bound that depends only on the frame constant. In addition, Lemma 4.5(c) locates the allowed values of ξ_i : combined with the two neighbor comparisons it gives $|\xi_i + \alpha(v_i - (Tx_v)_i)| \leq (1 + 2k)\alpha h_{ij}/2$, so $\xi_i \in \alpha[\ell_i - v_i, u_i - v_i] + [-(1 + 2k)\alpha h_{ij}/2, (1 + 2k)\alpha h_{ij}/2]$, an interval determined by v and the projected range rather than by the enlarged search box.

Lemma 4.26 (Gaussian-weighted local counts). *Let $\gamma \sim N(0, \sigma^2 I_n)$, let T have frame constant c_{fr} , let $\Lambda = \text{diag}(\lambda_i)$, let $[\ell_i, u_i]$ contain the range of $(Tx)_i$ over X , and use balanced nested meshes on any fixed box with the constant C_k of Definition 3.5 for the curvature vector $(\lambda_1, \dots, \lambda_k)$. At every level j ,*

$$\sum_{v \in G_j} \mathbb{P}(E_v) \leq \prod_{i=1}^k \left[2 + c_{\text{fr}}^{-1/2}(2C_k + 4) + \frac{C_k \lambda_i (u_i - \ell_i)}{\sigma \sqrt{2\pi}} \right].$$

The bound does not depend on the location or the size of the box.

The finite law $\mathcal{G}_{b,\sigma}$ has none of these independence properties. Its local-event probabilities are obtained from the Gaussian proxy by Lemma 3.13, using a uniform bound on the number of intervals in every coordinate section of a local event (Lemma B.2); its tube probabilities are transferred in the same way.

4.7 Separable mixed recourse with rational breakpoints

When the convex part is separable and the domain is a product, the recourse decomposes into scalar problems. Integer winners are then certified globally by neighbor inequalities, and the number of integer variables no longer enters the complexity.

Definition 4.27 (Separable instances). The domain is $X = \prod_{j=1}^n X_j$, where $X_j = [\ell_j, u_j]$ with $\ell_j < u_j$ (continuous) or $X_j = \mathbb{Z} \cap [\ell_j, u_j]$ with at least two points (integer); integer endpoints are rounded inward, empty domains are rejected, and fixed coordinates are removed beforehand, their contributions being absorbed into the remaining data. Each φ_j is a continuous convex function on $[\ell_j, u_j]$ given by rational breakpoints $\ell_j = \kappa_{j0} < \dots < \kappa_{js_j} = u_j$ and rational quadratics $\psi_{jm}(t) = \frac{1}{2}c_{jm}t^2 + \mu_{jm}t$ (plus constants) with $c_{jm} \geq 0$ on $[\kappa_{j,m-1}, \kappa_{jm}]$. The objective is

$$F(x) = \sum_{j=1}^n \varphi_j(x_j) - \frac{\alpha}{2} \|Tx\|^2, \quad \alpha > 0, \quad T \in \mathbb{Q}^{k \times n}.$$

The bit length I counts all pieces, breakpoints, and binary endpoints. Continuity, the curvature signs, and the order of the one-sided derivatives at the breakpoints are checked exactly.

Rational breakpoints are essential. The function $\max\{0, x^2 - 2\}$ on $[0, 2]$ has rational quadratic pieces, but $\max\{0, x^2 - 2\} - x$ has its unique minimizer at the irrational breakpoint $\sqrt{2}$. The parameters k and α are those of the supplied concave term, not the negative inertia or curvature of the whole objective, and the product domain excludes coupling constraints.

Lemma 4.28 (Scalar recourse). *For $\lambda \in \mathbb{R}$ consider $\min_{t \in X_j} [\varphi_j(t) - \lambda t]$.*

- (a) If X_j is integer, put $\delta_j(t) = \varphi_j(t+1) - \varphi_j(t)$, which is nondecreasing. A point $t \in X_j$ is optimal if and only if $\delta_j(t-1) \leq \lambda \leq \delta_j(t)$, where a condition involving a point outside X_j is omitted. An optimal t is found by binary search with $O(\log(u_j - \ell_j + 2))$ evaluations of φ_j .
- (b) If X_j is continuous, the following at most $2s_j + 1$ states cover all $\lambda \in \mathbb{R}$, and each certifies global optimality on its closed interval of λ : for a piece with $c_{jm} > 0$, the free state $t = (\lambda - \mu_{jm})/c_{jm}$ for λ between the derivatives of ψ_{jm} at $\kappa_{j,m-1}$ and κ_{jm} ; and for every breakpoint κ_{jm} , the constant state $t = \kappa_{jm}$ for λ between the left and right derivatives of φ_j at κ_{jm} , with $-\infty$ and $+\infty$ replacing the missing derivatives at ℓ_j and u_j .

With $\lambda_j(a, \zeta) = \alpha t_j^\top a - \zeta_j$, where t_j is column j of T , the inner problem defining $W_\zeta(a)$ equals $\frac{\alpha}{2} \|a\|^2 + \sum_j \min_{x_j \in X_j} [\varphi_j(x_j) - \lambda_j(a, \zeta)x_j]$. A choice of one state per coordinate gives a piece in the sense of Lemma 4.12, with a region described by at most $2n$ inequalities with normals $\pm \alpha t_j^\top$ and offsets affine in ζ , on which the piece's response attains W_ζ itself. There are at most $R_{\text{sep}} = \prod_j \text{integer } N_j \prod_j \text{continuous } (2s_j + 1)$ pieces, where $N_j = |X_j|$. A piece has Hessian $\alpha I - \alpha^2 \sum_j \text{free } t_j t_j^\top / c_j$, bounded in norm by $\Theta_{\text{sep}} = \alpha + \alpha^2 \sum_j \text{continuous } r_j^+ \|t_j\|^2$, where r_j^+ is the largest reciprocal of a positive listed curvature of φ_j (zero if none). Since every region certifies the global value, no gap test is needed. The exact fallback enumerates integer assignments and continuous pieces and minimizes the resulting quadratics on the boxes of pieces by Lemma 4.11(d); its multiplier is $B_{\text{sep}} = \max\{2, \prod_j \text{integer } N_j \prod_j \text{continuous } 3s_j\}$. All these counts have logarithms polynomial in I .

Theorem 4.29 (Separable recourse: aligned and uniform noise). *Let a separable instance with $k \geq 1$ be given, and let $[\ell_i, u_i]$ be the range of $(Tx)_i$ over X .*

- (a) Under $\xi_1, \dots, \xi_k$ drawn independently from $U_{\sigma, M}$, with M computed from the base data, there is an algorithm that returns an exact rational minimizer of $F(x) + \xi^\top Tx$ and the optimal value on every draw, with expected bit work at most $C_0^k (1 + \mathcal{H}_{\text{al}})(I + 1)^{C_1}$, where $\mathcal{H}_{\text{al}}$ is as in Theorem 4.21.
- (b) If T has full row rank and $\|T\| \leq 1$, the same holds under $\gamma_1, \dots, \gamma_n$ drawn independently from $U_{\sigma, M}$, with the bound of Theorem 4.20 without the factor $f(n_z)$.

The number of integer variables is not a parameter.

A general spectral normalization of the separable objective would destroy the separability of the convex part. A normalization that only rotates and rescales the rows of the supplied factor preserves it exactly, at the price of an anisotropic auxiliary metric.

Lemma 4.30 (Anisotropic row normalization). *Let $T \in \mathbb{Q}^{k \times n}$ have full row rank and put $\beta = \alpha \|T\|^2$. In polynomial time one computes an exactly orthogonal rational Q and dyadic numbers $d_1, \dots, d_k > 0$ such that $U = \text{diag}(d_i)^{-1}QT$ and $\Lambda = \text{diag}(\alpha d_i^2)$ satisfy*

$$U^\top \Lambda U = \alpha T^\top T, \quad \frac{1}{16} I_k \preceq U U^\top \preceq I_k, \quad \max_i \lambda_i < 8\beta.$$

The identity $U^\top \Lambda U = \alpha T^\top T$ is exact. The approximation in the Jacobi rotations is used only to bound the frame; it is neither discarded nor moved into the convex part. Small singular values of T affect only the bit lengths of the transformed data.

Theorem 4.31 (Separable recourse: Gaussian-like noise). *Let a separable instance with $k \geq 1$ and T of full row rank after preprocessing be given; no norm or conditioning promise on T is needed. Let*

U and Λ be as in Lemma 4.30 and let $[\ell_i, u_i]$ be the range of $(Ux)_i$ over X . There is an algorithm that computes from the base data an integer $b \leq \text{poly}(I)$, draws $\gamma_1, \dots, \gamma_n$ independently from $\mathcal{G}_{b,\sigma}$, and returns an exact rational minimizer of F_γ and the optimal value on every draw, with expected bit work at most $C_0^k(1 + \mathcal{H}_\Lambda)(I+1)^{C_1}$, where, with $C_k = 1 + 8k$,

$$\mathcal{H}_\Lambda = \prod_{i=1}^k \left[2 + 4(2C_k + 4) + \frac{C_k \lambda_i (u_i - \ell_i)}{\sigma \sqrt{2\pi}} \right] \leq \left[8C_k + 18 + \frac{8C_k \beta \text{diam}(X)}{\sigma \sqrt{2\pi}} \right]^k.$$

Thus the expected work is $f(k, 1 + \beta \text{diam}(X)/\sigma)(I+1)^{C_1}$ with an absolute exponent and no parameter for the number of integer variables.

The search uses meshes balanced by the curvatures: dyadic scales ρ_i with $1 \leq \lambda_i \rho_i^2 < 4$ give $c_{\text{bal}} < 4$ in Definition 3.5, so that no ratio of largest to smallest curvature enters the count. The parameter β is the curvature of the supplied concave term; it can be much larger than the negative curvature of the whole objective. The theorem does not identify a minimum-rank representation or remove dependent rows of T .

4.8 Summary of the regimes

Result	Law (fixed before sampling)	Recurse and domain	Expected bit work
Thm. 4.9	$U_{\sigma,M}$ or $\mathcal{G}_{b,\sigma}$ on all n coefficients	continuous polytope, $k \leq 2$	$\text{poly}(I)(1 + \nu S/\sigma)$; linear in the ratio
Thm. 4.21	$U_{\sigma,M}$ on the k factor coefficients	continuous polytope	$C^k(1 + \mathcal{H}_{\text{al}}) \text{poly}(I)$; parameters k and $\alpha(u_i - \ell_i)/\sigma$
Thm. 4.20	$U_{\sigma,M}$ on all n coefficients	mixed polytope	$f(n_z)C^k(1 + \mathcal{H}_{\text{amb}}) \text{poly}(I)$; a power of n growing with k
Thm. 4.19	$\mathcal{G}_{b,\sigma}$ on all n coefficients	mixed polytope	$f_1(n_z, k, 1) + \nu \text{diam}(\mathcal{P})/\sigma \text{poly}(I)$ for intrinsic normalization
Thm. 4.29	aligned or uniform	separable, product domain	as Thms. 4.21, 4.20, no n_z
Thm. 4.31	$\mathcal{G}_{b,\sigma}$ on all n coefficients	separable, product domain	$f(k, 1 + \beta \text{diam}(X)/\sigma) \text{poly}(I)$, no n_z

Table 2: Exact algorithms for sampled problems with few negative directions. All are correct on every draw and output rational minimizers and values of polynomial bit length. In the uniform-noise row, the power of n is a property of the counting surrogate (Proposition 4.25). For the intrinsic factorization, $\alpha < 4\nu$.

The polynomial exponents are absolute in all rows. The Gaussian-like theorem is the strongest general statement. The aligned theorem is the simplest and needs no normalization, but its noise is low-dimensional. The two-direction theorem has a different and, for $k \leq 2$, possibly sharper numerical dependence. The separable theorems trade a restricted convex part for an unrestricted number of integer variables.

5 Sparse polynomial optimization

This section develops the sparse-bag route. The objective is a sum of explicit polynomial factors of fixed degree on a bounded mixed box, and every factor scope lies in a bag of a supplied tree decomposition with at most p coordinates per bag. Every coordinate receives an independent

linear coefficient from one finite rational law chosen before sampling, that is, the ambient model of [definition 2.2](#) with marginal law $U_{\sigma,M}$. No convexity, uniqueness, growth, nondegeneracy, or bound on the integer dimension or the negative inertia is assumed.

The algorithm follows the common plan of the paper.

- (i) Curvature-corrected refinement of nested bag cells preserves every global optimizer. The correction is one global rounding allowance ([section 5.2](#)).
- (ii) Independent linear noise bounds the expected number of retained bag cells at every mesh by a quantity that does not depend on the mesh ([section 5.3](#)).
- (iii) A certified local convexity test on a box that contains every global optimizer finishes the sampled problem exactly ([section 5.5](#)).
- (iv) A base-selected finite law makes failure of this test rare enough to pay for an exact fallback on the same draw ([section 5.6](#)).

[Theorem 5.2](#) states the result for fixed-degree polynomials on mixed boxes. Quadratic and mixed-integer quadratic objectives are the case $d = 2$; their output is rational on every draw ([corollary 5.19](#)). [Section 6](#) extends the method to feasible sets that are not boxes.

5.1 Setting and main result

Definition 5.1 (Sparse mixed polynomial instance). A sparse mixed polynomial instance consists of the following rational data.

- (i) A mixed box $X = \prod_{i=1}^n X_i$ after the preprocessing of [section 2.1](#): $X_i = [\ell_i, u_i]$ for $i \in I_C$ and $X_i = [\ell_i, u_i] \cap \mathbb{Z}$ with integers $\ell_i < u_i$ for $i \in I_Z$, where $[n] = I_C \cup I_Z$, $n \geq 1$, and every width $w_i = u_i - \ell_i$ is positive. Write $\hat{X} = \prod_i [\ell_i, u_i]$ for the continuous hull, $W = \sum_i w_i$, $w_{\max} = \max_i w_i$, $n_c = |I_C|$, $n_z = |I_Z|$, and $N_Z = \prod_{i \in I_Z} (w_i + 1)$ for the number of integer assignments.
- (ii) Factors f_a , $a \in \mathcal{A}$, each an explicit list of monomials with rational coefficients, of total degree at most d , with variable scope $S_a \subseteq [n]$. The objective is $F_0 = \sum_a f_a$.
- (iii) A tree decomposition: a tree $\mathcal{T}$ whose nodes are bags $\beta \subseteq [n]$, such that every coordinate and every scope S_a lies in some bag and, for every i , the bags containing i form a subtree. The bag size is $p = \max_{\beta} |\beta|$.
- (iv) A noise half-width $\sigma > 0$ and a curvature bound $L > 0$ with

$$\partial_{ii} F_0(x) \leq L \quad \text{for all } x \in \hat{X} \text{ and } i \in [n]. \quad (3)$$

The base length I is the binary length of these data. The degree $d \geq 1$ is fixed.

The curvature bound is needed on the whole continuous hull, including the segments between adjacent integer labels, because the rounding and comparison arguments move one coordinate along such segments. Inputs with fixed coordinates reduce to this form by substitution, which does not increase degrees, scopes or (3); the sampled coefficient of a substituted coordinate becomes an additive constant, which is carried separately and added to reported values. If no coordinate remains, the feasible set is one rational point. The algorithm then returns this point and its sampled value before computing any schedule (step (0) of [definition 5.17](#)), so every maximum, width sum and division below concerns $n \geq 1$ coordinates of positive width.

Status of the premises. The tree decomposition is verified in time polynomial in I . The bound (3) is promised: correctness of the pruning and of the patch certificate relies on it. It can always be verified instead of promised. With $R_0 = \max\{1, \max_i |\ell_i|, \max_i |u_i|\}$ and $F_0 = \sum_\alpha c_\alpha x^\alpha$,

$$L_{\text{mon}} = \max\left\{1, \max_i \sum_\alpha |c_\alpha| \alpha_i (\alpha_i - 1) R_0^{|\alpha|-2}\right\} \quad (4)$$

satisfies (3) and has encoding length $\text{poly}_d(I)$, because $|x^{\alpha-2e_i}| \leq R_0^{|\alpha|-2}$ on $\hat{X}$ when $\alpha_i \geq 2$. Its numerical value can be much larger than a sharper supplied bound, and it is the numerical value of L that enters the work bound. A sharper bound used in an independently checkable proof record must be certified, with its verification cost added to the work. No general test of polynomial inequalities is assumed. The same monomial method gives rational numbers

$$\kappa_2 \geq \max\left\{1, \max_i \sum_k \sup_{\hat{X}} |\partial_{ik} F_0|\right\}, \quad \kappa_3 \geq \max\left\{1, \max_i \sum_{k,l} \sup_{\hat{X}} |\partial_{ikl} F_0|\right\} \quad (5)$$

of encoding length $\text{poly}_d(I)$, which the algorithm computes itself. Their magnitudes affect only the stopping depth, logarithmically.

Noise and growth. The coefficients $\gamma_1, \dots, \gamma_n$ are independent with law $U_{\sigma, M}$ of definition 2.1, $F_\gamma = F_0 + \gamma^\top x$, and $f^* = \min_X F_\gamma$; the minimum exists because X is compact. The point-growth modulus $g_*(\gamma)$ of definition 3.15 is the largest $g \geq 0$ such that some minimizer a satisfies

$$F_\gamma(x) - F_\gamma(a) \geq g \|x - a\|_2^2 \quad (x \in X). \quad (6)$$

The set of admissible g is closed, so the largest one exists. If $g_*(\gamma) > 0$ the minimizer is unique; if F_γ has two distinct minimizers, $g_*(\gamma) = 0$.

Outputs. On ordinary draws the algorithm returns a *patch output*, which has the form of the implicit patch output of definition 2.5(b). On the remaining draws it returns the exact output of the fallback of theorem 3.20. A patch output is a quadruple $\Pi = (S, \bar{x}_S, P, g_0)$: a set $S \supseteq I_Z$; rational values $\bar{x}_i \in X_i$ for $i \in S$, each of which is an integer label, an original continuous bound, or a value shared by every global minimizer because the closure test found a one-point hull interval (proposition 5.12); a rational box $P = \prod_{i \notin S} [a_i, b_i]$ with $\ell_i \leq a_i < b_i \leq u_i$, which plays the role of the box Q of definition 2.5(b); and a rational $g_0 > 0$ for which the run has verified

$$\nabla_{\bar{S}\bar{S}}^2 F_\gamma(\bar{x}_S, y) \succeq g_0 I \quad (y \in P), \quad \bar{S} = [n] \setminus S. \quad (7)$$

It denotes the unique minimizer $\hat{y}$ of $F_\gamma(\bar{x}_S, \cdot)$ on P , and the run also proves that $\hat{x} = (\bar{x}_S, \hat{y})$ is the unique global minimizer of F_γ on X . If $S = [n]$, the box P is empty and $\hat{x} = \bar{x}$. Fixing a continuous coordinate at a singleton hull value is justified by containment of every global minimizer in the hull, not by a derivative sign, and it changes neither the uniqueness argument nor the evaluation contract. The patch output has length $\text{poly}_d(I)$. The *global proof record* of the run consists of the candidate lists, incumbents and cell bounds of every level, and the data of the closure tests; it proves that every global minimizer lies in $\{\bar{x}_S\} \times P$.

Theorem 5.2 (Sparse polynomial optimization on mixed boxes). *Fix $d \geq 1$. For sparse mixed polynomial instances (definition 5.1) there is an algorithm and a power of two M , computed from the base instance with $\log_2 M \leq \text{poly}_d(I)$, such that for the ambient perturbation model with marginal law $U_{\sigma, M}$ the following hold.*

- (i) On every draw the algorithm returns an exact global optimizer of F_γ on X . On ordinary draws the output is a patch output of length $\text{poly}_d(I)$ together with its global proof record. On the remaining draws, which have probability at most $1/(2B)$, it is the exact output of the fallback of [theorem 3.20](#), computed on the same draw, where B is a base-only factor with $\log B \leq \text{poly}_d(I)$.
- (ii) The expected bit work, including the computation of M and the sampling, and the expected length and verification work of the global proof record are at most

$$C_0^p \left[4 + \frac{(1 + n/2)Lw_{\max}}{2\sigma} \right]^p \text{poly}_d(I), \quad (8)$$

where C_0 is an absolute constant.

- (iii) For every integer $q \geq 0$, a q -bit evaluation of the output returns a rational point $y \in X$ with $\|y - \hat{x}\|_2 \leq 2^{-q}$, where $\hat{x}$ is the returned optimizer, and rationals a and $F_\gamma(y)$ with $a \leq f^* \leq F_\gamma(y) \leq a + 2^{-q}$. It costs $\text{poly}_d(I + q)$ bit operations for a patch output and $B \text{poly}_d(I + q)$ for a fallback output; its expected cost over the draw is $\text{poly}_d(I + q)$.

The validity of L is the only promised premise, and it is verified when $L = L_{\text{mon}}$. No growth, uniqueness or nondegeneracy premise is used; such conditions enter only the analysis of the stopping depth.

The bound (8) is polynomial for each fixed p when $Lw_{\max}/\sigma$ is polynomially bounded in I . It is a fixed-width polynomial bound in the sense of [section 2.5](#), not a fixed-parameter bound in p : the bracket contains n , so the bound grows as n^p for fixed numerical scales. [Section 5.4](#) explains where this factor comes from, and [theorem 9.1](#) shows that it is necessary for the retention rule used here. The integer widths w_i enter through their magnitudes, so huge binary-encoded integer ranges are not covered at polynomial cost; they enter the stopping depth, the sampling precision and the fallback factor only logarithmically. The theorem concerns the sampled objective. Its consequence for the base objective is the regret bound of [proposition 2.6](#) with $\omega_X(\gamma) \leq \sigma W$. Choosing σ for a target accuracy ε multiplies the numerical ratio in (8) by a factor of order W/ε ([section 2.6](#)); the resulting work is polynomial in $1/\varepsilon$ when the other numerical parameters are fixed. Finally, the finite law matters: [example 2.7](#) is an instance in this class on which a tie occurs with probability $1/M$, so no patch certificate can succeed on every draw and the fallback is necessary.

5.2 Nested bag cells and sparse dynamic programming

Grids and cells. Let s be the least power of two with $s \geq \max\{1, w_{\max}\}$ and $h_j = s2^{-j}$ for $j \geq 0$. For $i \in I_C$, and for $i \in I_Z$ while $h_j \geq 1$, the level- j nodes of coordinate i are

$$G_{i,j} = \{\ell_i + kh_j : k \in \mathbb{Z}_{\geq 0}, \ell_i + kh_j < u_i\} \cup \{u_i\},$$

and a *coordinate cell* is $[a, a'] \cap X_i$ for consecutive nodes $a < a'$. For $i \in I_Z$ and $h_j < 1$, the nodes are $G_{i,j} = X_i$ and the coordinate cells are the singletons $\{a\}$, $a \in X_i$. While $h_j \geq 1$, h_j is an integer, so integer nodes are integers. The *endpoints* of a cell are a, a' , or a for a singleton. The level- j grid is $\mathbb{G}_j = \prod_i G_{i,j} \subseteq X$.

Lemma 5.3 (Coordinate cells). *For every coordinate i and level j :*

- (a) the coordinate cells cover X_i , and a node contained in a cell is an endpoint of it;

- (b) every node is an endpoint of at most two cells, and of exactly one if the cells are singletons;
- (c) every cell has width at most h_j ;
- (d) every level- $(j+1)$ cell lies in a level- j cell, and every level- j cell is the union of at most two level- $(j+1)$ cells.

Proof. Consecutive nodes differ by at most h_j , and the first and last nodes are ℓ_i and u_i ; this gives (a) and (c) for interval cells, and (b) holds because a node is the right endpoint of at most one cell and the left endpoint of at most one cell. For (d), $h_{j+1} = h_j/2$ gives $G_{i,j} \subseteq G_{i,j+1}$, and between consecutive level- j nodes $a < a'$ there is at most one level- $(j+1)$ node, namely $a + h_{j+1}$ if $a + h_{j+1} < a'$. When $h_j = 1 > h_{j+1}$ for $i \in I_Z$, the level- j cells are $\{a, a+1\}$, the union of the singleton cells $\{a\}$ and $\{a+1\}$; singleton cells are never refined further. $\square$

A *bag cell* of a bag β at level j is a product $C = \prod_{i \in \beta} C_i$ of level- j coordinate cells; its *corners* are the tuples of endpoints, and its *children* are the level- $(j+1)$ bag cells contained in it. By [lemma 5.3](#), a cell has at most $2^{|\beta|}$ corners and at most $2^{|\beta|}$ children, its children cover it, and a tuple of nodes is a corner of at most $2^{|\beta|}$ cells. The algorithm keeps for every bag a *candidate list* $\mathcal{W}_j(\beta)$ of level- j bag cells. It determines the *physical domain*

$$\Omega_j = \{x \in X : \text{for every bag } \beta, x_\beta \in C \text{ for some } C \in \mathcal{W}_j(\beta)\},$$

the *row list* $\mathcal{R}_j(\beta)$, the set of corners of the cells in $\mathcal{W}_j(\beta)$, and the *allowed grid* $A_j = \{y \in \mathbb{G}_j : y_\beta \in \mathcal{R}_j(\beta) \text{ for every } \beta\}$. The physical domain is used only in the proofs; the algorithm stores cells and rows.

Lemma 5.4 (Allowed grid). $A_j = \mathbb{G}_j \cap \Omega_j$.

Proof. Corners of a cell lie in the cell, so $A_j \subseteq \Omega_j$. Conversely, if $y \in \mathbb{G}_j \cap \Omega_j$ and $y_\beta \in C \in \mathcal{W}_j(\beta)$, each y_i , $i \in \beta$, is a node contained in C_i , hence an endpoint of C_i by [lemma 5.3\(a\)](#), so y_β is a corner of C . $\square$

Sparse dynamic programming. We may assume that every node of $\mathcal{T}$ has degree at most three: a bag with children $\beta_1, \dots, \beta_k$, $k \geq 3$, is replaced by a path of $k-1$ copies of itself, the l th copy having child β_l and the last copy also β_k . Copies preserve the subtree property and the bag size and at most double the number of nodes [30]. Copies of one bag have identical candidate lists throughout, because their cell bounds below coincide. Assign every factor f_a and every noise term $\gamma_i x_i$ to one bag containing its scope, and let ϕ_β be the sum of the terms assigned to β , so that $F_\gamma = \sum_\beta \phi_\beta(x_\beta)$. For a row $v \in \mathcal{R}_j(\beta)$ let

$$m_\beta(v) = \min\{F_\gamma(y) : y \in A_j, y_\beta = v\}$$

be its *min-marginal*, with $\min \emptyset = +\infty$, and let $m_j = \min_{A_j} F_\gamma$.

Lemma 5.5 (Sparse two-pass dynamic programming). *Let finite sets G_i , a tree decomposition of maximum degree three, finite row sets $\mathcal{R}(\beta) \subseteq \prod_{i \in \beta} G_i$ and costs $\phi_\beta : \mathcal{R}(\beta) \rightarrow \mathbb{Q}$ be given, and let $A = \{y \in \prod_i G_i : y_\beta \in \mathcal{R}(\beta) \forall \beta\}$. Two passes of messages compute every min-marginal $m_\beta(v) = \min\{\sum_{\beta'} \phi_{\beta'}(y_{\beta'}) : y \in A, y_\beta = v\}$, the minimum over A , and a minimizer, using $O(\sum_\beta |\mathcal{R}(\beta)|)$ evaluations of the costs, additions, comparisons and dictionary operations on separator tuples.*

Proof. For adjacent bags β, β' let $\Sigma = \beta \cap \beta'$, and let $\mathcal{T}_{\beta \rightarrow \beta'}$ be the component of $\mathcal{T}$ minus the edge $\beta\beta'$ that contains β . Define the message $\mu_{\beta \rightarrow \beta'}$ on separator tuples ς by

$$\mu_{\beta \rightarrow \beta'}(\varsigma) = \min_{v \in \mathcal{R}(\beta), v_{\Sigma} = \varsigma} \left[\phi_{\beta}(v) + \sum_{\beta''} \mu_{\beta'' \rightarrow \beta}(v_{\beta'' \cap \beta}) \right],$$

the sum over the neighbors $\beta'' \neq \beta'$ of β , and store only its finite values in a dictionary keyed by ς . By induction on the size of $\mathcal{T}_{\beta \rightarrow \beta'}$, $\mu_{\beta \rightarrow \beta'}(\varsigma)$ is the minimum of $\sum_{\beta'' \in \mathcal{T}_{\beta \rightarrow \beta'}} \phi_{\beta''}$ over the assignments to the coordinates of the bags of $\mathcal{T}_{\beta \rightarrow \beta'}$ that equal ς on Σ and whose restriction to every bag of $\mathcal{T}_{\beta \rightarrow \beta'}$ is a row of that bag. Indeed, the subtrees $\mathcal{T}_{\beta'' \rightarrow \beta}$ partition $\mathcal{T}_{\beta \rightarrow \beta'}$ minus β , and by the subtree property a coordinate occurring in $\mathcal{T}_{\beta'' \rightarrow \beta}$ and outside it lies in $\beta'' \cap \beta$; so once v is fixed on β the minimizations over the different subtrees are independent and constrained only through their separators. The same argument with all neighbors gives $m_{\beta}(v) = \phi_{\beta}(v) + \sum_{\beta''} \mu_{\beta'' \rightarrow \beta}(v_{\beta'' \cap \beta})$, the sum over all neighbors; every coordinate lies in some bag, so the assignments are complete. Messages are computed toward a root and back, recomputing each outgoing sum from at most two incoming messages instead of subtracting, so that no difference of infinities occurs. Each row is processed a bounded number of times, and stored argmin pointers give a minimizer. $\square$

This is nonserial dynamic programming on a tree decomposition [14], applied to sparse row lists. Dense tables would enumerate products of coordinate grids; here each row is a corner of a retained cell, and adjacent row lists are never multiplied. Every message value is a sum of at most one assigned cost per bag, so its bit length is polynomial in the bit length of the costs and the number of bags, independently of the number of rows.

Rounding and pruning. The next lemma is the deterministic core of the method. It rounds all coordinates at once and keeps the result inside every candidate list.

Lemma 5.6 (Fixed-cell rounding). *Let $x \in \Omega_j$, and let $C_0 \in \mathcal{W}_j(\beta_0)$ be a cell with $x_{\beta_0} \in C_0$. There is a random point $Y \in A_j$ such that Y_{β_0} is a corner of C_0 , $Y_i = x_i$ whenever $x_i \in G_{i,j}$, and*

$$\mathbb{E}F_{\gamma}(Y) \leq F_{\gamma}(x) + E_j, \quad E_j = \frac{nLh_j^2}{8}. \quad (9)$$

Proof. For each i with $x_i \in G_{i,j}$ put $Y_i = x_i$. Otherwise x_i lies strictly between consecutive nodes $a_i < a'_i$; this happens only for $i \in I_C$ or for $i \in I_Z$ with $h_j \geq 1$, so $a_i, a'_i \in X_i$. Let $Y_i = a'_i$ with probability $(x_i - a_i)/(a'_i - a_i)$ and $Y_i = a_i$ otherwise, independently over i . Then $\mathbb{E}Y_i = x_i$ and $\text{Var } Y_i = (x_i - a_i)(a'_i - x_i) \leq h_j^2/4$.

Let β be a bag and $C \in \mathcal{W}_j(\beta)$ a cell with $x_{\beta} \in C$. For $i \in \beta$, if x_i is a node then $Y_i = x_i \in C_i$ is an endpoint of C_i by lemma 5.3(a). Otherwise the only coordinate cell containing x_i is the one with endpoints a_i, a'_i , so C_i is that cell and Y_i is an endpoint. Hence Y_{β} is a corner of C . Since $x \in \Omega_j$, every bag has such a cell, so $Y \in A_j$; taking $\beta = \beta_0$, $C = C_0$ shows that Y_{β_0} is a corner of C_0 .

For the objective, put $Y^{(k)} = (Y_1, \dots, Y_k, x_{k+1}, \dots, x_n)$. Given $Y_1, \dots, Y_{k-1}$, the function $\varphi(\theta) = F_{\gamma}(Y_1, \dots, Y_{k-1}, \theta, x_{k+1}, \dots, x_n)$ on $[\ell_k, u_k]$ satisfies $\varphi'' \leq L$ by (3), because these points lie in $\hat{X}$ and the linear noise does not change second derivatives. Taylor's theorem gives $\varphi(Y_k) \leq \varphi(x_k) + \varphi'(x_k)(Y_k - x_k) + \frac{L}{2}(Y_k - x_k)^2$. As Y_k is independent of $Y_1, \dots, Y_{k-1}$ with mean x_k ,

$$\mathbb{E}[F_{\gamma}(Y^{(k)}) \mid Y_1, \dots, Y_{k-1}] \leq F_{\gamma}(Y^{(k-1)}) + \frac{L}{2} \text{Var } Y_k.$$

Taking expectations and summing over k gives (9). $\square$

For a quadratic, (9) also follows from cancellation of the off-diagonal terms in expectation. The sequential argument is needed for higher degree, for instance for the monomial $x_1^2 x_2^2$, whose expected rounding error is not a sum of diagonal terms.

For a cell $C \in \mathcal{W}_j(\beta)$ define its *cell bound*

$$q_C = \min\{m_\beta(v) : v \text{ a corner of } C\}.$$

The *incumbent* U_j is the least value of F_γ at the allowed grid points produced by the dynamic programs of levels $0, \dots, j$; it is the value of a point of X , and $U_j \leq m_j$. Initialize $U_{-1} = +\infty$ and store a feasible point attaining the smallest value so far, keeping an earlier point when its value remains smaller. Update this incumbent pair before pruning. The candidate list is updated by the *retention rule*

$$\mathcal{W}'_j(\beta) = \{C \in \mathcal{W}_j(\beta) : q_C - E_j \leq U_j\}, \quad \mathcal{W}_{j+1}(\beta) = \{\text{children of cells in } \mathcal{W}'_j(\beta)\}, \quad (10)$$

starting from $\mathcal{W}_0(\beta) = \{X_\beta\}$. Ties are retained. Let Ω'_j be the physical domain of the retained lists; it equals Ω_{j+1} , because children cover their parents.

Proposition 5.7 (Sound pruning and witnesses). *On every draw and at every level $j \geq 0$:*

- (a) *every global minimizer x^* of F_γ on X lies in Ω_j , and x^*_β lies in a retained cell of every bag;*
- (b) *$f^* \leq U_j \leq m_j \leq f^* + E_j$;*
- (c) *every $x \in \Omega_j \setminus \Omega'_j$ satisfies $F_\gamma(x) > U_j$;*
- (d) *every retained cell $C \in \mathcal{W}'_j(\beta)$ has a corner v and a point $y \in A_j$ with $y_\beta = v$ and $F_\gamma(y) = q_C \leq f^* + 2E_j$.*

Proof. We first note the cell lower bound. Let $C \in \mathcal{W}_j(\beta)$ and let $x \in \Omega_j$ have $x_\beta \in C$. [Lemma 5.6](#) with (β, C) gives $Y \in A_j$ with Y_β a corner of C , so $F_\gamma(Y) \geq m_\beta(Y_\beta) \geq q_C$ on every outcome, and

$$q_C - E_j \leq F_\gamma(x). \quad (11)$$

We prove (a) by induction; $\Omega_0 = X$. Suppose every minimizer lies in Ω_j , and let x^* be one. For each bag β choose $C \in \mathcal{W}_j(\beta)$ containing x^*_β . By (11), $q_C - E_j \leq f^* \leq U_j$, so C is retained, and its children cover it; hence $x^* \in \Omega_{j+1}$. For (b), $U_j \geq f^*$ because U_j is a value at a point of X , and rounding a minimizer by [lemma 5.6](#) gives an allowed point of expected value at most $f^* + E_j$, so $m_j \leq f^* + E_j$. For (c), $x \in \Omega_j \setminus \Omega'_j$ lies in some cell $C \in \mathcal{W}_j(\beta)$ but in no retained cell of β , so C was deleted, and (11) gives $F_\gamma(x) \geq q_C - E_j > U_j$. For (d), $q_C \leq U_j + E_j < \infty$, so $q_C = m_\beta(v)$ for some corner v , and the minimum defining $m_\beta(v)$ is attained at some $y \in A_j$. By (b), $q_C \leq U_j + E_j \leq f^* + 2E_j$. $\square$

Part (a) holds for tied and positive-dimensional optimal sets and for every draw. Part (d) gives one globally consistent feasible point for each retained cell; the counting argument below needs exactly this, and a collection of separate per-bag witnesses would not suffice. Part (c) is the global certificate: a point removed at level j has value above $U_j \geq f^*$. An incumbent found at an earlier level may lie outside the current physical domain; it is still feasible and remains a valid upper bound.

5.3 Expected size of the retained lists

The count conditions on the noise outside one bag. For a bag β , write $O = [n] \setminus \beta$ and, for $v \in \hat{X}_\beta = \prod_{i \in \beta} [\ell_i, u_i]$,

$$V_\beta(v) = \min_{z \in X_O} [F_0(v, z) + \gamma_O^\top z]. \quad (12)$$

The minimum is over the original outside domain $X_O = \prod_{i \in O} X_i$, not over a set that depends on the run. Hence V_β depends on γ_O and not on γ_β , and $f^* = \min_{v \in X_\beta} [V_\beta(v) + \gamma_\beta^\top v]$.

Lemma 5.8 (Comparison intervals). *Fix γ_O , a bag tuple $v \in X_\beta$ and $\eta \geq 0$, and suppose $V_\beta(v) + \gamma_\beta^\top v \leq f^* + \eta$. If $i \in \beta$ and $a > 0$ satisfy $v \pm ae_i \in X_\beta$, then γ_i lies in a closed interval that depends only on γ_O , v , i , a and η , and whose length is at most $La + 2\eta/a$.*

Proof. Let z^* attain the minimum in (12) at v . For $\theta \in [-a, a]$ put $\psi(\theta) = F_0(v + \theta e_i, z^*)$; the segment lies in $\hat{X}$, so $\psi'' \leq L$ and $\psi(a) + \psi(-a) - 2\psi(0) \leq La^2$. Since $V_\beta(v \pm ae_i) \leq \psi(\pm a) + \gamma_O^\top z^*$ and $V_\beta(v) = \psi(0) + \gamma_O^\top z^*$,

$$V_\beta(v + ae_i) + V_\beta(v - ae_i) - 2V_\beta(v) \leq La^2.$$

The tuples $v \pm ae_i$ are feasible, so $f^* \leq V_\beta(v \pm ae_i) + \gamma_\beta^\top(v \pm ae_i)$. Subtracting the hypothesis gives

$$\frac{V_\beta(v) - V_\beta(v + ae_i) - \eta}{a} \leq \gamma_i \leq \frac{V_\beta(v - ae_i) - V_\beta(v) + \eta}{a}.$$

The endpoints do not involve γ_β , and the difference of the endpoints is at most $La + 2\eta/a$ by the previous display. $\square$

For a bag β and level j let

$$N_j(\beta) = \{v \in \mathbb{G}_{j,\beta} : V_\beta(v) + \gamma_\beta^\top v \leq f^* + 2E_j\}, \quad \mathbb{G}_{j,\beta} = \prod_{i \in \beta} G_{i,j},$$

a subset of the complete deterministic bag grid. The algorithm never enumerates it; it serves only as a dominating count.

Proposition 5.9 (Expected retained cells). *Let $J \geq 0$, let $M \geq 2^J$, and let $0 \leq j \leq J$. On every draw, $|\mathcal{W}'_j(\beta)| \leq 2^{|\beta|} |N_j(\beta)|$. Moreover*

$$\mathbb{E} |N_j(\beta)| \leq \Theta_\beta := \prod_{i \in \beta} \left[4 + \frac{(1 + n/2)Lw_i}{2\sigma} \right]. \quad (13)$$

Proof. Let $C \in \mathcal{W}'_j(\beta)$ and let v, y be as in proposition 5.7(d). Since $y = (v, y_O)$ with $y_O \in X_O$, $V_\beta(v) + \gamma_\beta^\top v \leq F_\gamma(y) \leq f^* + 2E_j$, so $v \in N_j(\beta)$. A node tuple is a corner of at most $2^{|\beta|}$ level- j cells, which gives the first claim.

For the expectation, let $a_i = h_j$ if $i \in I_C$ or $h_j \geq 1$, and $a_i = 1$ otherwise; thus $a_i \geq h_j$, and consecutive nodes of coordinate i are a_i apart, except possibly the last two. Call a node $v_i \in G_{i,j}$ *regular* if $v_i \pm a_i \in X_i$. Regular nodes are $\ell_i + ka_i$ with $k \geq 1$ and $\ell_i + (k+1)a_i \leq u_i$, so there are at most w_i/a_i of them, and at most three nodes are not regular: ℓ_i , the last node below u_i , and u_i .

Fix $v \in \mathbb{G}_{j,\beta}$ and condition on γ_O . If $v \in N_j(\beta)$, lemma 5.8 with $\eta = 2E_j$ confines every γ_i , $i \in \beta$ with v_i regular, to an interval determined by γ_O and v , of length at most $\lambda_i = La_i + 4E_j/a_i$. The coefficients γ_i , $i \in \beta$, are independent of each other and of γ_O , so by (2)

$$\mathbb{P}\{v \in N_j(\beta) \mid \gamma_O\} \leq \prod_{i \in \beta} c_i(v_i), \quad c_i(v_i) = \begin{cases} \lambda_i/(2\sigma) + 1/M & v_i \text{ regular,} \\ 1 & \text{otherwise.} \end{cases}$$

The right side is a product of functions of single coordinates, so summing over the grid factorizes:

$$\mathbb{E} |N_j(\beta)| \leq \prod_{i \in \beta} \sum_{v_i \in G_{i,j}} c_i(v_i) \leq \prod_{i \in \beta} \left[3 + \frac{w_i}{a_i} \left(\frac{\lambda_i}{2\sigma} + \frac{1}{M} \right) \right].$$

Here $\frac{w_i \lambda_i}{2\sigma a_i} = \frac{L w_i}{2\sigma} \left(1 + \frac{n h_j^2}{2 a_i^2} \right) \leq \frac{(1+n/2) L w_i}{2\sigma}$ because $a_i \geq h_j$, and $\frac{w_i}{a_i M} \leq \frac{s}{h_j M} = \frac{2^j}{M} \leq 1$ because $j \leq J$ and $M \geq 2^J$. This proves (13). $\square$

The expectation is taken over the complete deterministic bag grid, and every retained cell is charged to a near-optimal node of that grid on every draw. The argument therefore does not assume that the adaptively retained cells are independent of the noise. It is the bag-conditional analogue of [theorem 3.4](#) and [theorem 3.9\(iv\)](#). The proof above is self-contained because the grids here are clipped at the upper bounds and the count is conditioned on the noise outside the bag; the clipped nodes are treated as exceptional and are never used in comparisons. The requirement $M \geq 2^J$ is what makes the count independent of the level: for a fixed coarse grid, the atomic term $w_i/(a_i M)$ would grow without bound as $h_j \rightarrow 0$.

5.4 Why the bound is fixed-width but not fixed-parameter

The tree decomposition controls the shape of the dynamic program: tables are indexed by bag rows, messages by separator tuples, and the number of rows of a bag is charged to Θ_β , a product over the coordinates of that bag only. Treewidth therefore governs the exponent in (8). The base of the exponential, however, contains the total dimension n . It enters through the rounding allowance: the cell bound (11) must hold for every feasible point of a cell, and its proof rounds all n coordinates of a global point simultaneously, so the allowance $E_j = n L h_j^2 / 8$ is a sum of variances over all coordinates. The witnesses of [proposition 5.7\(d\)](#) are near-optimal only up to $2E_j$, and [lemma 5.8](#) turns this tolerance into noise intervals of length $(1 + n/2) L a_i$ for every coordinate of the bag. The resulting bound on the expected number of retained rows of a bag is of order $(n L w_{\max} / \sigma)^{|\beta|}$ when this ratio is large, which is polynomial in n only for fixed p .

Two facts show that the factor belongs to the method as specified, not only to its analysis. First, replacing E_j by an allowance for the bag alone, $|\beta| L h_j^2 / 8$, is unsound: [proposition 9.2](#) gives a star quadratic with bags of size two on which the local allowance deletes the cell that contains every global optimizer, on every draw of small noise. The error that the global allowance absorbs comes from minimizing over the grid outside the bag; it varies with the bag's own coordinates and does not cancel when min-marginals are compared. Second, for the retention rule (10) and closure on the whole hull, the factor $n^{\Omega(p)}$ actually occurs: [theorem 9.1](#) gives connected instances of treewidth $p - 1$, with $L = 2$, unit widths and $\sigma = 1/100$, on which a bag retains more than $(5n/(6p))^{p/2}$ cells on every draw before the stopping depth is reached. Both results concern the retention rule and the bag-local substitution analyzed here. They are not lower bounds for other algorithms.

Certified conditional values. The factor n leaves the count when the true conditional value of a bag tuple is available with a certified error proportional to the bag size rather than to the dimension. For a bag β and $v \in \hat{X}_\beta$ write

$$W_\beta(v) = V_\beta(v) + \gamma_\beta^\top v,$$

with V_β from (12), a minimum over the original outside domain. The next proposition gives the count of a modified search in which an oracle for W_β replaces the grid min-marginals. It is a

sufficient interface, not an algorithm: an oracle that meets it must account for the optimization over every coordinate outside the bag, and none is supplied here for general sparse instances. Parts (a) and (b) specialize the deterministic certified bag filter of an unpublished companion manuscript [1] and are re-proved here. Part (c), the expected count under the finite law, is the extension made here.

Proposition 5.10 (Conditional-recourse count). *Consider a sparse mixed polynomial instance with the coordinate cells of [section 5.2](#). Let $a \geq 0$, $e_j = pLh_j^2/8$, and $0 \leq \eta_j \leq ae_j$. Modify the refinement and retention steps of [definition 5.17](#) as follows. At level j an oracle receives every corner v of every candidate cell of every bag β and returns rationals $\ell_\beta(v) \leq W_\beta(v) \leq u_\beta(v)$ with $u_\beta(v) - \ell_\beta(v) \leq \eta_j$, together with a point $x^v \in X$ such that $x_\beta^v = v$ and $F_\gamma(x^v) \leq u_\beta(v)$. The incumbent U_j is the least value $u_\beta(v)$ returned at levels $0, \dots, j$, including every call of level j . Initialize $U_{-1} = +\infty$, store its associated feasible completion, and update the incumbent before pruning. A candidate cell C of β is retained if and only if $\min\{\ell_\beta(v) : v \text{ a corner of } C\} - e_j \leq U_j$. Then the following hold.*

- (a) *On every draw and at every level, for every global minimizer x^* and every bag β , the projection x_β^* lies in a retained cell of β .*
- (b) *On every draw, $f^* \leq U_j \leq f^* + e_j + \eta_j$, and every retained cell of β has a corner v and its returned feasible completion x^v with $W_\beta(v) \leq F_\gamma(x^v) \leq f^* + 2(e_j + \eta_j)$.*
- (c) *Let $J \geq 0$, $M \geq 2^J$ and $0 \leq j \leq J$. On every draw the number of retained cells of β at level j is at most $2^{|\beta|}$ times the number of tuples $v \in \mathbb{G}_{j,\beta}$ with $W_\beta(v) \leq f^* + 2(e_j + \eta_j)$, and under the law $U_{\sigma,M}$ the expected number of these tuples is at most*

$$\prod_{i \in \beta} \left[4 + \left(1 + \frac{(1+a)p}{2} \right) \frac{Lw_i}{2\sigma} \right]. \quad (14)$$

The oracle's answers and errors may depend on the draw; the count uses the fixed envelope ae_j , not independence of the queried errors.

Proof. Fix γ_O , a coordinate $i \in \beta$ and the other bag coordinates. The function $\theta \mapsto V_\beta(v + \theta e_i)$ is a minimum over $z \in X_O$ of functions whose second derivative is at most L on the continuous hull, so $\theta \mapsto W_\beta(v + \theta e_i) - L\theta^2/2$ is concave there, being a minimum of concave functions plus a linear term. Moreover $f^* = \min_{X_\beta} W_\beta$, with equality at x_β^* for every global minimizer x^* .

(a) We argue by induction over the levels; at level 0 the only cell of β is X_β . Let C be a candidate cell of β that contains x_β^* . Round every coordinate $i \in \beta$ of x_β^* independently to the endpoints of the coordinate cell C_i with mean x_i^* , as in the proof of [lemma 5.6](#), a coordinate that is a node staying fixed. The result Y is a corner of C and $\text{Var } Y_i \leq h_j^2/4$. Rounding one coordinate at a time, the concavity just shown and Jensen's inequality give $\mathbb{E}W_\beta(Y) \leq W_\beta(x_\beta^*) + |\beta| Lh_j^2/8 \leq f^* + e_j$. Some corner v of C therefore has $\ell_\beta(v) - e_j \leq W_\beta(v) - e_j \leq f^*$, and $f^* \leq U_j$ because U_j bounds the value of a feasible point. Hence C is retained, and its child that contains x_β^* is a candidate at level $j+1$.

(b) The corner v found in (a) is queried at level j , so $U_j \leq u_\beta(v) \leq \ell_\beta(v) + \eta_j \leq f^* + e_j + \eta_j$. A retained cell has a corner v' with $\ell_\beta(v') \leq U_j + e_j$, and then $W_\beta(v') \leq F_\gamma(x^{v'}) \leq u_\beta(v') \leq \ell_\beta(v') + \eta_j \leq f^* + 2(e_j + \eta_j)$.

(c) The first claim follows from (b), since a tuple of nodes is a corner of at most $2^{|\beta|}$ cells. Dominate the qualifying tuples by the full grid count with the deterministic tolerance $\delta = 2(1+a)e_j$, which bounds $2(e_j + \eta_j)$ on every draw. Repeat the proof of [proposition 5.9](#) with this tolerance in place of $2E_j$. [Lemma 5.8](#) uses only V_β and the curvature bound, so it confines the coefficient of every regular coordinate, given γ_O and v , to an interval of length at most $La_i + 2\delta/a_i \leq$

$La_i + (1+a)pLh_j^2/(2a_i) \leq La_i(1 + (1+a)p/2)$, because $a_i \geq h_j$. The rest of that proof is unchanged and gives (14). $\square$

Compared with Θ_β of (13), the dimension n in the bracket is replaced by $(1+a)p$, so the count depends only on the bag size, the error factor a and the numerical ratios Lw_i/σ . Whether the whole search becomes fixed-parameter in the width depends on the cost of the oracle, which the proposition does not address. The grid min-marginals of section 5.2 do not meet the interface (proposition 9.2). Exact conditional values can require optimizing over almost the whole instance, and dividing a global allowance among the outside coordinates would force a finer outside grid and a dimension-dependent state count elsewhere. The recourse route of section 7 meets the interface for one search block, a small core whose conditional problem is assumed tractable; its counts are the case of a single bag. Certified conditional values are one sufficient way to remove the factor n . The barriers above do not show that they are necessary for every algorithm whose bound is fixed-parameter in the width.

Deterministic bounds under a growth condition. The companion manuscript [1] develops an alternative deterministic analysis under a quadratic-growth premise. Write $L_i = \max\{0, \max_x \partial_{ii} F(x)\}$ and $L = \max_i L_i$. Point growth $F(x) - F(x^*) \geq g \|x - x^*\|^2$ defines $\kappa = \max\{1, L/g\}$. Weighted growth instead uses $F(x) - F(x^*) \geq \gamma \sum_i L_i (x_i - x_i^*)^2$ and defines $\bar{\kappa} = \max\{1, 1/\gamma_*\}$ for the largest valid γ_* . Point growth implies $\bar{\kappa} \leq \kappa$, but weighted growth places no requirement on coordinates with $L_i = 0$. Its approximation theorem certifies accuracy 2^{-q} in $f(p, \bar{\kappa})(I + q + 1)^5$ bit operations, where

$$f(p, t) = c_0(c_1 p \sqrt{t})^p t(1 + \log_2 t)^2.$$

The condition parameter is not an algorithm input: trials stop when a valid certificate is obtained. Its exact-output theorem separately assumes point growth and a unique minimizer, and gives $f_1(p, \kappa)(I + 1)^{C_1}$ bit operations. Graded grids use $O(\sqrt{\bar{\kappa}} \log(n + 2))$ nodes per coordinate; the companion's uniform-grid family retains at least $\sqrt{(n - 2)\bar{\kappa}} + 1$ nodes in its specified stages. This is a limitation of that filtering rule. Under the randomized exponential-time hypothesis, its product-time lower bound rules out $f(p)(\kappa I)^{p/\psi(p)}$ for unbounded ψ in the stated integer-quadratic model. It does not establish a matching $p/2 + O(1)$ lower exponent or rule out every smaller linear exponent in κ . These are bounds with a numerical condition parameter, not bounds in width alone. We do not analyze graded grids under noise.

These deterministic bounds do not by themselves give an expected bound that is fixed-parameter in the width under the noise models of this paper. Their work has a high power of the inverse growth ratio. Under the noise, $Z = L/g_*$ (with $Z = +\infty$ when $g_* = 0$) satisfies only the estimate $\mathbb{P}\{Z > t\} \leq A_0/t + \epsilon_M$, where $A_0 = LW/\sigma$ and $\epsilon_M = 2nC_{\text{tail}}/M$, by lemma 5.14. For $r > 1$ and $K \geq 1$, integration gives

$$\mathbb{E} \min\{K, Z^r\} \leq 1 + \frac{rA_0}{r-1}(K^{1-1/r} - 1) + \epsilon_M K.$$

Thus the linear small-growth tail alone does not bound high inverse-growth moments without a cutoff-dependent factor. The cutoff-dependent power already arises in the flat noisy-coordinate example of remark 4.10. When the cutoff is paid by an exact fallback, K can be exponential in the input length. The count of proposition 5.9 avoids inverse-growth moments: it bounds the retained cells of every draw by near-optimal nodes of a fixed grid, whose expected number is controlled by the linear noise alone, and growth enters only the stopping depth, logarithmically, and the probability of

the fallback. The price is the factor n in the bracket. This comparison concerns the two arguments; it is not a lower bound for other algorithms, and whether some exact algorithm has expected work fixed-parameter in the width under independent linear noise remains open (section 10.3).

5.5 Certified closure on a strongly convex patch

After the retention step of level j , the algorithm tries to finish. Let $g_0 > 0$ be a rational number chosen from the base data (definition 5.15) and let κ_2, κ_3 be as in (5).

Definition 5.11 (Closure test). At level j :

- (1) For every coordinate i let Q_i be the intersection, over the bags $\beta \ni i$, of the smallest closed interval containing the i th projections of the cells of $\mathcal{W}'_j(\beta)$, and let $Q = \prod_i Q_i$.
- (2) Record $\bar{x}_i$ equal to the point of Q_i for every $i \in I_Z$ with Q_i a single point.
- (3) Let c be the midpoint of Q and r the largest half-width of the intervals Q_i . For $i \in I_C$, record $\bar{x}_i = \ell_i$ if $\partial_i F_\gamma(c) - \kappa_2 r > 0$, and $\bar{x}_i = u_i$ if $\partial_i F_\gamma(c) + \kappa_2 r < 0$.
- (4) If some integer coordinate is unrecorded, the test fails. Otherwise record $\bar{x}_i$ equal to the point of Q_i for every unrecorded $i \in I_C$ with Q_i a single point. Let S be the recorded set and $\bar{S} = [n] \setminus S$. If $\bar{S} = \emptyset$, return the point $\bar{x}$.
- (5) Otherwise let $P = \prod_{i \in \bar{S}} Q_i$, with midpoint c^P and largest half-width r^P . If the rational symmetric matrix

$$\nabla_{\bar{S}\bar{S}}^2 F_\gamma(\bar{x}_S, c^P) - (\kappa_3 r^P + g_0)I \quad (15)$$

is positive definite, return the patch output $(S, \bar{x}_S, P, g_0)$; otherwise the test fails.

Positive definiteness of a rational symmetric matrix is decided exactly by Gaussian elimination: all pivots must be positive. All quantities in the test have encoding length polynomial in I, j and $\log M$.

Proposition 5.12 (Soundness of closure). *On every draw, every equality recorded by the closure test holds at every global minimizer of F_γ on X . If the test returns a point $\bar{x}$, then $\bar{x}$ is the unique global minimizer. If it returns $\Pi = (S, \bar{x}_S, P, g_0)$, then (7) holds on P , the function $F_\gamma(\bar{x}_S, \cdot)$ has a unique minimizer $\hat{y}$ on P , and $(\bar{x}_S, \hat{y})$ is the unique global minimizer of F_γ on X .*

Proof. By proposition 5.7(a), every minimizer x^* has x^*_β in a retained cell of every bag, so $x^* \in Q$. A singleton Q_i therefore fixes x^*_i ; for $i \in I_Z$ its point is an endpoint of an integer cell, hence an integer.

For the gradient test, let $x \in Q \subseteq \hat{X}$. The mean value theorem along the segment from c to x , which lies in Q , gives $|\partial_i F_\gamma(x) - \partial_i F_\gamma(c)| \leq \sum_k \sup_{\hat{X}} |\partial_{ik} F_0| |x_k - c_k| \leq \kappa_2 r$; the linear noise does not change second derivatives. If $\partial_i F_\gamma(c) - \kappa_2 r > 0$, then $\partial_i F_\gamma(x^*) > 0$. Were $x^*_i > \ell_i$, the point $x^* - \theta e_i$ would be feasible for small $\theta > 0$, because i is continuous, and would have a smaller value. Hence $x^*_i = \ell_i$. The upper bound is symmetric.

All recorded values lie in X_i , and $P \subseteq \prod_{i \in \bar{S}} [\ell_i, u_i]$ with $\bar{S} \subseteq I_C$, so $T = \{\bar{x}_S\} \times P \subseteq X$, and every minimizer lies in T . If $\bar{S} = \emptyset$, a minimizer exists by compactness and equals $\bar{x}$. Otherwise, for $y \in P$ every entry of $\nabla_{\bar{S}\bar{S}}^2 F_\gamma(\bar{x}_S, y) - \nabla_{\bar{S}\bar{S}}^2 F_\gamma(\bar{x}_S, c^P)$ in row k is at most $r^P \sum_m \sup_{\hat{X}} |\partial_{klm} F_0|$ in absolute value, by the mean value theorem; the spectral norm of a symmetric matrix is at most its largest absolute row sum, so the difference has norm at most $\kappa_3 r^P$. Positive definiteness of (15) gives (7). Then $F_\gamma(\bar{x}_S, \cdot)$ is strictly convex on P and has a unique minimizer $\hat{y}$. Since $T \subseteq X$ contains every global minimizer and at least one, $\min_T F_\gamma = f^*$ and the global minimizers are exactly the minimizers on T , that is, $(\bar{x}_S, \hat{y})$. $\square$

The test fixes coordinates by two different arguments. The gradient rule of step (3) fixes a continuous coordinate only at an original bound, where first-order optimality in the original box gives the sign condition; an endpoint of a hull interval that is not an original bound is never used by this rule. The singleton rule of steps (2) and (4) fixes a coordinate, integer or continuous, at the value that every global minimizer shares because its hull interval is a single point; this is justified by containment ([proposition 5.7\(a\)](#)), not by a derivative. The gradient rule is applied only to continuous coordinates: an interior integer optimum can have a nonzero derivative. A positive definite continuous block does not justify closure while integer choices remain, which is why step (4) fails when an integer coordinate is unrecorded.

The test succeeds early when the sampled instance is well conditioned. The conditions in the next proposition are not tested by the algorithm; they are used only to bound the stopping depth.

Proposition 5.13 (When closure succeeds). *Suppose $g_*(\gamma) \geq g_0 > 0$, let a be the unique minimizer, and suppose that $|\partial_i F_\gamma(a)| > \tau$ for every $i \in I_C$ with $a_i \in \{\ell_i, u_i\}$. Let $A = 2 + nL/g_0$. If*

$$h_j \leq \min \left\{ \frac{1}{4A}, \frac{\tau}{4\kappa_2 A}, \frac{g_0}{4\kappa_3 A} \right\}, \quad (16)$$

then the closure test of level j returns a point or a patch output.

Proof. Let $C \in \mathcal{W}'_j(\beta)$, and let y be its witness from [proposition 5.7\(d\)](#). Growth gives $g_0 \|y - a\|^2 \leq F_\gamma(y) - f^* \leq 2E_j = nLh_j^2/4$, so $\|y - a\|_2 \leq \frac{h_j}{2} \sqrt{nL/g_0}$. Every point of C_i is within h_j of y_i , so within $h_j(1 + \frac{1}{2}\sqrt{nL/g_0}) \leq Ah_j$ of a_i . Since $a_i \in Q_i$, every Q_i lies in $[a_i - Ah_j, a_i + Ah_j]$, and $r \leq Ah_j$ and $\|c - a\|_\infty \leq r$.

By (16), $h_j \leq 1/(4A) < 1$, so integer cells are singletons. For $i \in I_Z$, the integer cell of C is $\{y_i\}$ and $|y_i - a_i| \leq Ah_j \leq 1/4$, so $y_i = a_i$. Every retained cell of every bag containing i therefore has i th projection $\{a_i\}$, $Q_i = \{a_i\}$, and step (2) records every integer coordinate.

Let $i \in I_C$ with $a_i = \ell_i$; then $\partial_i F_\gamma(a) > \tau$ by hypothesis and first-order optimality. As in the proof of [proposition 5.12](#), $\partial_i F_\gamma(c) - \kappa_2 r \geq \partial_i F_\gamma(a) - 2\kappa_2 Ah_j > \tau - \tau/2 > 0$, so step (3) records $\bar{x}_i = \ell_i$; the case $a_i = u_i$ is symmetric. By soundness no coordinate with $\ell_i < a_i < u_i$ is recorded at a bound. Hence $\bar{S}$ consists of continuous coordinates that are interior at a . If $\bar{S} = \emptyset$ the test returns a point. Otherwise, for every unit vector e supported on $\bar{S}$, the points $a \pm \theta e$ are feasible for small $\theta > 0$, and adding the two growth inequalities and letting $\theta \rightarrow 0$ gives $e^\top \nabla^2 F_\gamma(a) e \geq 2g_0$; thus $\nabla_{\bar{S}\bar{S}}^2 F_\gamma(a) \succeq 2g_0 I$. The point $(\bar{x}_S, c^P)$ differs from a only in $\bar{S}$, by at most $r^P \leq Ah_j$, so the matrix (15) is at least $(2g_0 - 2\kappa_3 Ah_j - g_0)I \succeq \frac{g_0}{2}I$. $\square$

The two-sided Taylor argument uses only the coordinates that are interior at the optimizer. A boundary optimizer whose full Hessian is indefinite is therefore covered: the active coordinates are removed before the matrix test. No lower bound on the distance of the optimizer from inactive bounds is needed.

5.6 One finite law and the same-draw fallback

Two events make the closure test late: small point growth, and an active continuous coordinate with a small derivative. Both have small probability under the grid law.

Lemma 5.14 (Finite-law tails on mixed boxes). *There is a base-computable integer $C_{\text{tail}} \geq 1$ with $\log C_{\text{tail}} \leq \text{poly}_d(I)$ such that, for every M and every $\varepsilon > 0$,*

$$\mathbb{P}\{g_* < \varepsilon\} \leq \frac{\varepsilon W}{\sigma} + \frac{2nC_{\text{tail}}}{M}. \quad (17)$$

Let $D = \max\{1, d - 1\}$ and $K = \max\{1, n_c 3^{n_c} N_Z D^{n_c}\}$. For every $\tau > 0$, the probability that $g_* > 0$ and the unique minimizer a has a coordinate $i \in I_C$ with $a_i \in \{\ell_i, u_i\}$ and $|\partial_i F_\gamma(a)| \leq \tau$ is at most $K(\tau/\sigma + 1/M)$.

Proof. This is lemma 3.17(b) and (d) with $m = 0$, $S = W$ and $R_Z = N_Z$; the first part combines theorem 3.16 for the continuous uniform law with the transfer of lemma 3.17(a). The mixed box is described by $2n_c$ linear inequalities and by disjunctions over the N_Z integer labels, so the number s of atoms satisfies $\log(s + 1) \leq \text{poly}(I)$, and the constant C_{tail} of lemma 3.17 has logarithm $\text{poly}_d(I)$ even though the number of labels can be exponential. $\square$

The second bound counts nonsingular stationary points of faces and does not assume that stationary sets are finite. It is an unconditional estimate for the intersection with $\{g_* > 0\}$, not a conditional probability given good growth.

Definition 5.15 (Base schedule). Before sampling, compute from the base instance:

- (1) κ_2, κ_3 by (5), C_{tail} and K as in lemma 5.14, and the factor $B \geq 2$ of theorem 3.20 for the format of the instance (the mixed box X , the degree d , and the monomial support of F_0 with n added linear terms);
- (2) $\rho = 1/(4B)$, $g_0 = \rho\sigma/(2W)$, $\tau = \rho\sigma/(2K)$ and $A = 2 + nL/g_0$;
- (3) the stopping depth J , the least $j \geq 0$ satisfying (16);
- (4) the grid size M , the least power of two with $M \geq \max\{2, 2^J, 4nC_{\text{tail}}/\rho, 2K/\rho\}$.

Every quantity in the schedule has encoding length $\text{poly}_d(I)$: the logarithms of B , C_{tail} , K , κ_2 , κ_3 , $1/\rho$, $1/g_0$, $1/\tau$ and A are polynomial in I , so $J \leq \text{poly}_d(I)$ and $\log_2 M \leq \text{poly}_d(I)$. The order matters. The factor B is fixed first; the thresholds and J depend on B but not on M ; and M is chosen last. The sampled coefficients have bit length $b = \log_2 M + O(\text{size } \sigma)$, which enters only the polynomial factor of the fallback.

Lemma 5.16 (Rare late closure). *Under the schedule of definition 5.15, the closure test succeeds at some level $j \leq J$ except on an event of probability at most $1/(2B)$.*

Proof. By lemma 5.14, $\mathbb{P}\{g_* < g_0\} \leq \rho/2 + 2nC_{\text{tail}}/M \leq \rho$, and the probability that $g_* \geq g_0$ while an active continuous derivative has magnitude at most τ is at most $K\tau/\sigma + K/M \leq \rho/2 + \rho/2$. Outside these two events proposition 5.13 applies at level J . The union has probability at most $2\rho = 1/(2B)$. $\square$

Definition 5.17 (Sparse cell algorithm). Given a rational input with fixed-degree polynomial factors on a mixed box and a tree decomposition:

- (0) Preprocess as in section 2.1. If no unfixed coordinate remains, draw γ once, return the unique feasible point, which is rational, and its sampled value, and stop. Otherwise the result is a sparse mixed polynomial instance with $n \geq 1$.
- (1) Compute the schedule of definition 5.15, convert the tree decomposition to maximum degree three, assign factors and noise terms to bags, and draw γ once from the grid law with this M .
- (2) Set $\mathcal{W}_0(\beta) = \{X_\beta\}$ for every bag. For $j = 0, 1, \dots, J$: form the row lists, run the dynamic program of lemma 5.5 for all min-marginals and a minimizer of F_γ on A_j , update the incumbent, apply the retention rule (10), and run the closure test of definition 5.11. If the test returns, stop with its output. Otherwise form $\mathcal{W}_{j+1}$ by generating children and removing duplicates.

- (3) If no closure test has returned by level J , run the exact fallback of [theorem 3.20](#) on F_γ over X and return its output.

In step (0) the law is $U_{\sigma,2}$; the draw only determines the reported value, which includes the sampled constants of the fixed coordinates.

Proof of [theorem 5.2](#). Exactness. A closure output is correct on every draw by [proposition 5.12](#), and the fallback is correct on every draw by [theorem 3.20](#). The fallback runs only if no closure test returns, which by [lemma 5.16](#) has probability at most $1/(2B)$. No draw is discarded or repeated.

Work of one level. Let $j \geq 1$. Generating $\mathcal{W}_j(\beta)$ costs $O(2^{|\beta|} |\mathcal{W}'_{j-1}(\beta)|)$ cell operations and a sort, the row list has at most $2^{|\beta|} |\mathcal{W}_j(\beta)|$ entries, the dynamic program costs $O(\sum_\beta |\mathcal{R}_j(\beta)|)$ operations by [lemma 5.5](#), and the retention rule and the hulls cost $O(2^{|\beta|} |\mathcal{W}_j(\beta)|)$ further operations; level 0 has one cell per bag. Every grid coordinate at level $j \leq J$ has bit length $O(I + j + \log s)$, every sampled coefficient $O(I + \log M)$, every factor value and message $\text{poly}_d(I)$, and a sort or dictionary operation on a list of R rows costs $O(\log R)$ comparisons with $\log R \leq \log |\mathbb{G}_j| \leq \text{poly}_d(I)$. The closure test costs $\text{poly}_d(I)$ plus the hull computation. Hence the work of level j is at most $8^p \sum_\beta (1 + |\mathcal{W}'_{j-1}(\beta)|) \text{poly}_d(I)$.

Expectation. By [proposition 5.9](#), with $M \geq 2^J$, $\mathbb{E} |\mathcal{W}'_{j-1}(\beta)| \leq 2^p \Theta_\beta$ for $1 \leq j \leq J$, and $\Theta_\beta \leq [4 + (1 + n/2)Lw_{\max}/(2\sigma)]^p$. The work is a sum of nonnegative terms, each bounded by a constant times a retained-list size, so its expectation is bounded by linearity; no product of random table sizes occurs. There are $J + 1 \leq \text{poly}_d(I)$ levels and polynomially many bags. The fallback costs $B \text{poly}_d(I + b)$ and runs with probability at most $1/(2B)$, so it contributes $\text{poly}_d(I)$ to the expectation ([lemma 3.18](#)). Computing the schedule and drawing $n \log_2 M$ random bits cost $\text{poly}_d(I)$. This proves (8) with $C_0 = 16$. The global proof record consists of quantities computed during the run, so its length is bounded by the work; it is verified by recomputing the dynamic programs of the levels, which costs the same.

Evaluation. Part (iii) is [proposition 5.18](#) below. □

5.7 Exact implicit output and arbitrary-precision evaluation

A patch output determines one point exactly, but the coordinates of that point are usually irrational algebraic numbers whose minimal polynomials can have large degree. The output contract does not include them. What it includes is evaluation to any requested precision in time polynomial in the precision.

Proposition 5.18 (Evaluation). *Let $q \geq 0$.*

- (a) *Given a patch output $\Pi = (S, \bar{x}_S, P, g_0)$ of [theorem 5.2](#) with optimizer $\hat{x}$, a rational $y \in X$ with $y_S = \bar{x}_S$ and a rational a with $a \leq f^* \leq F_\gamma(y) \leq a + 2^{-q}$ and $\|y - \hat{x}\|_2 \leq 2^{-q}$ are computed in $\text{poly}_d(I + q)$ bit operations. If $S = [n]$, then $y = \hat{x} = \bar{x}$ and $a = F_\gamma(\bar{x})$.*
- (b) *Given a fallback output, a rational $y \in X$ with the same integer labels as the returned optimizer x° , and a rational a with $a \leq f^* \leq F_\gamma(y) \leq a + 2^{-q}$ and $\|y - x^\circ\|_2 \leq 2^{-q}$, are computed in $B \text{poly}_d(I + q)$ bit operations.*

In particular the expected cost of a q -bit evaluation is $\text{poly}_d(I + q)$.

The proof is in [appendix C.2](#). A point output is rational and is evaluated exactly. For a positive-dimensional patch, part (a) applies weak convex optimization in the bit model [[24](#), Definition 2.1.10 and Corollary 4.2.7] to the capped epigraph of $F_\gamma(\bar{x}_S, \cdot)$ over P , with an explicit repair of the

approximately feasible output; the modulus g_0 converts value accuracy into distance, including at boundary minimizers, and enters the cost only through its encoding length. The cost does not depend on the numerical ratio L/g_0 , as a grid-based evaluation would. Evaluation does not decide exact comparisons of f^* with a rational threshold, and it does not identify the active set of $\hat{y}$. Without a verified modulus, value accuracy alone would not localize the optimizer: a function that is flat along a direction has points of equal value far from each other.

5.8 Quadratic and mixed-integer quadratic objectives

For $d = 2$ the patch and the fallback can both be solved in rational arithmetic, so the output is rational on every draw.

Corollary 5.19 (Sparse mixed-integer quadratic programs). *Let $F_0(x) = \frac{1}{2}x^\top Hx + b^\top x + c_0$ be rational on a mixed box X as in [definition 5.1](#), with a tree decomposition of bag size p containing $\{i, k\}$ whenever $H_{ik} \neq 0$, and let $L > 0$ be rational with $L \geq \max_i H_{ii}$ (verified). There is a power of two M , computed from the base instance with $\log_2 M \leq \text{poly}(I)$, such that for the ambient perturbation model with marginal law $U_{\sigma, M}$:*

- (i) *on every draw the algorithm returns a rational global minimizer of F_γ on X and the optimal value, of bit length $\text{poly}(I)$;*
- (ii) *the expected bit work is at most $C_0^p[4 + (1 + n/2)Lw_{\max}/(2\sigma)]^p \text{poly}(I)$.*

No growth, uniqueness, convexity, bound on negative inertia, bound on variable occurrences, or bound on the integer dimension is used.

The algorithm is that of [definition 5.17](#) with two changes. After a patch output, the strongly convex quadratic program $\min\{F_\gamma(\bar{x}_S, y) : y \in P\}$ is solved exactly; its unique solution is rational of polynomial bit length and is computed in polynomial time by the ellipsoid method for convex quadratic programming [\[31\]](#). The fallback is the elementary enumeration of integer assignments and continuous faces of [lemma C.3](#), with $B = 3^{n_c}N_Z$; it needs no real algebraic geometry. The proof is in [appendix C.3](#). For a quadratic κ_3 may be replaced by zero in [\(15\)](#), because the Hessian is constant.

The bound is fixed-width polynomial, as before. It does not contradict the strong NP-hardness of box-constrained quadratic programming at treewidth two [\[19\]](#). That result concerns the threshold question for the unperturbed objective, whereas the corollary solves the sampled instance, with a bound that is polynomial only when $Lw_{\max}/\sigma$ is polynomially bounded. For the treewidth-two reduction itself, [remark 9.8](#) exhibits no-instances, with r a size parameter, that have a point of value 2^{-2r-4} above the threshold with a coordinate equal to one. By [proposition 9.7](#), the sampled minimum of such an instance falls below the threshold with probability at least $\frac{1}{2} - 2^{-2r-5}/\sigma - 1/(2M)$, so on these instances the exact sampled optimum does not answer the original threshold question. We make no claim about other hardness reductions.

5.9 Relation to earlier work

The ingredients are classical. Dynamic programming over tree decompositions is nonserial dynamic programming [\[14\]](#). The allowance $Lw^2/8$ per coordinate is the maximal separation of the α BB underestimator [\[4\]](#), used here with the opposite sign as a lower-bound correction. For linear tilts of a polynomial objective on a regular compact semialgebraic set, uniqueness of the minimizer and a growth inequality hold for generic tilts [\[32\]](#); these statements are qualitative and give no probability

bound for a finite law. Smoothed analyses of discrete optimization use isolation of winners [11, 39]; a continuous box has no positive gap between distinct feasible values, which is why the count above uses near-optimal grid nodes instead. Treewidth-based linear programming approximations of sparse polynomial optimization [15] give accuracy-dependent formulations rather than exact optimizers. Exact quadratic optimization on a box is polynomial on forests and strongly NP-hard at treewidth two [19]. An unpublished companion manuscript [1] develops a deterministic counterpart for factored objectives on mixed boxes: corrected coordinate grids, dynamic programming of min-marginals on a tree decomposition, filtering with certificates whose validity does not use growth, the conditional-recourse filter of [proposition 5.10\(a\),\(b\)](#), and running-time bounds fixed-parameter in the width under a quadratic growth condition of the instance; the end of [section 5.4](#) explains why those bounds do not average to an expected bound under noise. This section re-proves the parts of that pipeline it needs. The new element is the composition under noise: an expected count of retained bag cells, for every base instance and without a growth premise, under one finite independent-noise law that does not grow with the mesh, combined with sound closure on a certified strongly convex patch, an exact same-draw fallback, and an output whose evaluation costs polynomial time in the requested precision.

6 Feasible sets beyond boxes

The proof of [theorem 5.2](#) used four properties of a mixed box.

- (P1) For every bag, the set of feasible values of the coordinates outside the bag does not depend on the bag’s values. The conditional value V_β of (12) is then a minimum over a fixed set, independent of the bag’s own noise, with the curvature of the objective.
- (P2) Every feasible point of a candidate cell can be rounded, with its mean preserved and a second-order error, to feasible corners that lie in the same candidate cells of every bag.
- (P3) Sound first-order tests identify the active constraints near an optimizer, after which a certified strongly convex patch contains every optimizer.
- (P4) The point-growth and margin events have finite-law tails, and an exact fallback exists on the same domain.

Arbitrary sparse affine constraints do not have these properties, and the method cannot be extended to them without further structure: [theorem 9.5](#) shows that bounded-width affine equalities with objective noise can encode SUBSET SUM, and [example 9.6](#) shows that even a single totally unimodular inequality destroys the coordinatewise count of [lemma 5.8](#). This section proves the method for four structured classes. Each supplies these four proof steps or replaces one of them by a proved substitute. Graph constraints and affine-state recurrences satisfy (P1) after a reduction to a box of retained or free coordinates, and products of simplices satisfy it because every bag contains whole blocks. Order constraints do not satisfy (P1): the feasible values outside a bag depend on the bag’s values. There, a transport of feasible points supplies the semiconcave upper bound on the conditional value that (P1) gives for boxes ([lemma D.8](#)). [Table 3](#) lists the classes.

Two reductions below are conditional applications of [theorem 5.2](#) to a family of objectives indexed by part of the noise. The constants of its schedule must then be chosen uniformly over the family, before any noise is drawn; choosing them after observing part of the noise would make the law depend on the draw. The following lemma records what this requires.

Table 3: Feasible-domain extensions. All use the ambient model with law $U_{\sigma,M}$ on every original coordinate, $\log_2 M \leq \text{poly}(I)$, exact output on every draw, and the exact output of [theorem 3.20](#) on rare draws. The expected work is $C_0^{p'} \Theta^{p'} \text{poly}(I)$, where p' is the bag size used by the algorithm; for order constraints it is the bound (26).

Result	Feasible set and structural premise	Curvature bound	Factor Θ
theorem 6.4(E)	graph of an explicit polynomial map of bounded depth over a mixed box; $p' \leq p \max\{1, k^{D_g}\}$	retained, after substitution	$4 + (1 + n/2)Lw_{\max}/(2\sigma)$
theorem 6.4(I)	graph of monotone scalar equations with global brackets; $p' \leq p \max\{1, k\}$	retained, after substitution	$4 + (1 + n)Lw_{\max}/(2\sigma)$
theorem 6.11	affine-state recurrence with polynomial actuators, any horizon; $p' = p$	reduced, uniform in state noise	$4 + (1 + n/2)Lw_{\max}/(2\sigma)$
theorem 6.15	disjoint unit simplices and integer intervals, whole-block bags; $p' = p$	block Hessian	$4 + (2 + n/2)L \max\{1, w_{\max}\}/(2\sigma)$
theorem 6.17	order constraints on continuous and binary coordinates	full Hessian	$2 + 3n_c \bar{L}/(4\sigma)$, power p_c

Lemma 6.1 (Uniform schedules). *Let X be a mixed box with $n \geq 1$ coordinates and a tree decomposition of bag size p , and let $\mathcal{H} \subseteq \mathbb{R}^m$ be a box. For every rational $\eta \in \mathcal{H}$ let G_η be an explicit rational polynomial of degree at most d' on X whose monomial support lies in a fixed set computed from the base instance and whose factor scopes lie in the bags. Suppose that there are polynomials P_1, P_2 , whose degrees depend only on fixed format constants such as degrees and depths, such that for every $b \geq 1$ and every rational $\eta \in \mathcal{H}$ with entries of bit length at most b , the coefficients of G_η have bit length at most $P_1(I + b)$ and G_η is computed from the base instance and η in at most $P_2(I + b)$ bit operations. Suppose also that*

- (a) *rationals $L > 0$, κ_2 and κ_3 , computed from the base instance, satisfy (3) and (5) for every G_η , $\eta \in \mathcal{H}$, and*
- (b) *a deterministic algorithm solves every problem $\min_X (G_\eta + \gamma^\top x)$ with rational $\eta \in \mathcal{H}$ and $\gamma \in \mathbb{Q}^n$ exactly, and refines its output to precision 2^{-q} , within $B \text{poly}(I + b + q)$ bit operations when the entries of η and γ have bit length at most b ; here $B \geq 2$ depends only on the base instance and $\log B \leq \text{poly}(I)$.*

Compute the schedule of [definition 5.15](#) from L , κ_2 , κ_3 , W , this B , and the constants C_{tail} and $K = \max\{1, n_c 3^{n_c} N_Z \max\{1, d' - 1\}^{n_c}\}$ of [lemma 5.14](#) for degree d' , and let b be the largest bit length of an atom of the resulting law $U_{\sigma,M}$. Let η be a random rational point of $\mathcal{H}$ whose entries have bit length at most b , with an arbitrary law, and let γ be independent of η with law $U_{\sigma,M}^{\otimes n}$. Then the algorithm of [definition 5.17](#), with the fallback (b), applied to $G_\eta + \gamma^\top x$ is correct on every draw, runs its fallback with probability at most $1/(2B)$, and has expected work at most $C_0^p [4 + (1 + n/2)Lw_{\max}/(2\sigma)]^p \text{poly}(I + b)$, where the degree of the polynomial factor depends on d' , P_1 and P_2 . Since $b \leq \text{poly}(I)$, this is $C_0^p [4 + (1 + n/2)Lw_{\max}/(2\sigma)]^p \text{poly}(I)$.

Proof. Every statement of [section 5](#) about one objective uses only the validity of L , κ_2 and κ_3 for that objective, the tail constants, the fallback, and the bit length of the objective's coefficients, which enters the cost of each arithmetic operation polynomially. The constant C_{tail} of [lemma 3.17](#) depends on the number of variables, the degree and the number of atoms of the domain formula, and K on n_c , N_Z and d' ; neither depends on the coefficients of G_η . All schedule quantities are therefore fixed

before η and γ are drawn, and $b = O(\log_2 M + \text{size } \sigma)$ is polynomial in I . Conditionally on η , the coefficients γ are independent with law $U_{\sigma, M}$, so [lemma 5.14](#), [proposition 5.9](#), [lemma 5.16](#) and the proof of [theorem 5.2](#) apply to G_η with the same schedule, after the cost $P_2(I + b)$ of computing G_η and with factor values of bit length polynomial in $P_1(I + b)$. Averaging over η gives the claim. $\square$

In the applications below, the fallback (b) is [theorem 3.20](#) applied to the *original* constrained problem: an explicit polynomial on a compact semialgebraic domain ([definition 3.19](#)) tilted by the ambient noise on all coordinates. Its minimizers correspond to those of the reduced problem, and its factor B is fixed by the base instance, while the sampled coefficients enter only through b .

6.1 Global graph parameterizations

Some coupled equality constraints can be eliminated by a global chart that keeps a set of *retained* coordinates ranging over a mixed box. The retained coordinates remain ambient coordinates with their own independent noise. Conditioning on the noise of the eliminated coordinates then leaves exactly the setting of [section 5](#).

Definition 6.2 (Graph instance). A graph instance consists of rational data as follows.

- (i) Retained coordinates x in a mixed box $X \subseteq \mathbb{R}^n$ as in [definition 5.1](#) (i), with $n \geq 1$, and dependent coordinates $y \in \mathbb{R}^m$ given by a chart $y = \psi(x)$ of one of two kinds.
 - (E) *Explicit*: $y_j = P_j(x, y_1, \dots, y_{j-1})$, $j = 1, \dots, m$, where each P_j is an explicit rational polynomial of degree at most e in at most k of the variables (its *parents*), and every chain of dependencies has length at most D_g ; e and D_g are fixed.
 - (I) *Implicit*: $q_j(x_{S_j}, y_j) = 0$ and $y_j \in V_j = [\ell'_j, u'_j]$, $j = 1, \dots, m$, where q_j is an explicit rational polynomial of degree at most e , $|S_j| \leq k$, and rationals $\mu_j > 0$ are given with

$$\partial_y q_j(x, y) \geq \mu_j, \quad q_j(x, \ell'_j) \leq 0 \leq q_j(x, u'_j) \quad (x \in \widehat{X}_{S_j}, y \in V_j). \quad (18)$$

The feasible set is $\mathfrak{X} = \{(x, \psi(x)) : x \in X\}$; there are no other constraints. Bounds on dependent coordinates in case (E) must be implied by the chart, and in case (I) they only select the branch.

- (ii) An objective $F_0(x, y)$ that is a sum of explicit rational polynomial factors of degree at most d .
- (iii) A tree decomposition of bag size p of the ambient variables whose bags contain every objective scope and every constraint scope, that is, $\{y_j\}$ with its parents in case (E) and $S_j \cup \{y_j\}$ in case (I).
- (iv) A noise half-width $\sigma > 0$ and a rational $L > 0$ such that

$$\partial_{ii} G_\eta(x) \leq L, \quad G_\eta(x) = F_0(x, \psi(x)) + \eta^\top \psi(x), \quad (19)$$

for all $x \in \widehat{X}$, all $i \in [n]$, and all $\eta \in [-\sigma, \sigma]^m$.

The noise is ambient: $\gamma \in \mathbb{Q}^n$ and $\eta \in \mathbb{Q}^m$ have independent coordinates with law $U_{\sigma, M}$, and the sampled objective is $F_{\gamma, \eta}(x, y) = F_0(x, y) + \gamma^\top x + \eta^\top y$.

In case (I), (18) gives every $x \in \widehat{X}$ exactly one root $\psi_j(x_{S_j}) \in V_j$ (lemma 6.6); the brackets and the derivative floor must hold on the whole real hull, including between integer labels, because the algorithm evaluates the chart at such points. Brackets at individual query points would not establish uniqueness, smoothness, or the validity of rounding between grid points. A dependent coordinate whose chart has no retained argument is a constant. It is substituted only when it is rational, that is, given by a constant polynomial in case (E) or by a one-point interval V_j in case (I). Other constant roots, such as the root of $y^2 = 2$ in $[1, 2]$, stay in the chart and in the original constrained problem used by the fallback, so that no algebraic constant enters the rational data. If preprocessing fixes every retained coordinate, the feasible set is one point; the algorithm returns it directly, in case (I) as the fixed retained values together with the chart, which determines the dependent roots.

Status of the premises. In case (E), ψ is an explicit polynomial map obtained by substitution, and (19) is verified by monomial bounds that hold uniformly for $|\eta_j| \leq \sigma$, or promised if sharper. In case (I), (18) and (19) are promised, or certified with their verification cost added to the work; arbitrary polynomial positivity is not tested. Curvature is measured after substitution. An ambient diagonal bound does not suffice (example 6.8), and the dependent noise η can contribute to it.

Lemma 6.3 (Expanded decomposition). *For a dependent coordinate y_j let $\Lambda(y_j)$ be the set of retained coordinates on which it depends: its retained ancestors in case (E) and S_j in case (I); let $\Lambda(x_i) = \{x_i\}$. Replacing every bag β by $\beta' = \bigcup_{v \in \beta} \Lambda(v)$ gives a tree decomposition, on the same tree, of the retained coordinates. Its bags contain the scope of every reduced factor $f_a(x, \psi(x))$ and of every term $\eta_j \psi_j(x)$, and its bag size is $p' \leq p \max\{1, k^{D_g}\}$ in case (E) and $p' \leq p \max\{1, k\}$ in case (I).*

The proof is in appendix D.1. The equality scopes must lie in the supplied bags: a decomposition of the objective alone does not give running intersection after substitution.

Theorem 6.4 (Sparse optimization over global graphs). *Fix d, e and, in case (E), D_g . For graph instances there is an algorithm and a power of two M , computed from the base instance with $\log_2 M \leq \text{poly}(I)$, such that for the ambient model with law $U_{\sigma, M}$ on all $n + m$ coordinates the following hold.*

- (i) *On every draw the algorithm returns an exact global optimizer of $F_{\gamma, \eta}$ on $\mathfrak{X}$. On ordinary draws the output is a charted patch output (definition 6.5) of length $\text{poly}(I)$ with its global proof record. On the remaining draws, of probability at most $1/(2B)$, it is the exact output of theorem 3.20 for $F_{\gamma, \eta}$ on the compact semialgebraic set $\mathfrak{X}$, computed on the same draw.*
- (ii) *The expected bit work is at most*

$$(E) \quad C_0^{p'} \left[4 + \frac{(1 + n/2)Lw_{\max}}{2\sigma} \right]^{p'} \text{poly}(I), \quad (I) \quad C_0^{p'} \left[4 + \frac{(1 + n)Lw_{\max}}{2\sigma} \right]^{p'} \text{poly}(I),$$

with p' from lemma 6.3; the global proof record obeys the same bounds in expectation.

- (iii) *A q -bit evaluation returns a rational retained point $\tilde{x} \in X$ within distance 2^{-q} of the retained part $\hat{x}$ of the optimizer and a rational interval $[a_q, u_q]$ of width at most 2^{-q} with $a_q \leq f^* \leq F_{\gamma, \eta}(\tilde{x}, \psi(\tilde{x})) \leq u_q$, where $f^* = \min_{\mathfrak{X}} F_{\gamma, \eta}$. In case (E) the lift $(\tilde{x}, \psi(\tilde{x}))$ is rational, satisfies every graph equation, and lies within distance 2^{-q} of the optimizer. In case (I) the lift $(\tilde{x}, \psi(\tilde{x}))$ is an exactly feasible point given by rational retained coordinates and isolating brackets of its dependent roots, and a rational point within distance 2^{-q} of the optimizer is also returned;*

this rational point need not satisfy the graph equations. The cost is $\text{poly}(I + q)$ on ordinary draws and $B \text{poly}(I + q)$ on fallback draws, and its expectation is $\text{poly}(I + q)$.

No growth, uniqueness or strict-complementarity premise is used.

The exponents of the polynomials depend on the fixed degrees. The bound is fixed-width polynomial in the expanded bag size p' ; it is not a fixed-parameter bound. Case (I) loses the factor $1 + n/2$ to $1 + n$ because its costs are algebraic and are computed with certified error (lemma 6.7). The two cases are not comparable: (E) allows chains of dependent coordinates of bounded depth and gives rational feasible points, whereas (I) allows charts that are not polynomial but only one layer of dependent coordinates. Even one cubic equation can force irrational dependent coordinates at rational retained points, so no rational feasible point need exist in case (I).

Definition 6.5 (Charted patch output). For a graph instance and a draw (γ, η) , a charted patch output (definition 2.5(c)) is a patch output $(S, \bar{x}_S, P, g_0)$ in the sense of section 5.1 for the reduced objective $G_\eta + \gamma^\top x$ on X , verified with (7), together with the chart: the polynomials P_j in case (E); the equations q_j , the intervals V_j and the bounds μ_j in case (I). It denotes the point $(\hat{x}, \psi(\hat{x}))$, where $\hat{x} = (\bar{x}_S, \hat{y})$ and $\hat{y}$ is the unique minimizer of $G_\eta(\bar{x}_S, \cdot) + \gamma^\top(\bar{x}_S, \cdot)$ on P . Equivalently, $\hat{x}$ is the unique solution of the box KKT system of this reduced problem, and in case (I) the dependent coordinates are the unique roots of $q_j(\hat{x}_{S_j}, y_j) = 0$ in V_j , with graph multipliers $\lambda_j = -(\partial_{y_j} F_0 + \eta_j) / \partial_{y_j} q_j$ at that point.

The descriptor determines one point because the reduced objective is strongly convex on P , and the global proof record shows that this point is the global optimizer. No positive definiteness of the ambient Hessian is claimed or needed.

The explicit case. By lemma 6.3, the reduced objective $G_\eta + \gamma^\top x$ is an explicit polynomial of degree at most de^{D_g} on X whose factor scopes lie in the bags β' . Substitution of a dependent-noise vector η with entries of bit length b produces coefficients of bit length polynomial in $I + b$, in time polynomial in $I + b$, with exponents that depend only on d, e and D_g ; repeated products of supplied coefficients can make these lengths exceed $I + b$, but only by a fixed polynomial. The pair $(x, \psi(x)) \mapsto x$ is a bijection from $\mathfrak{X}$ onto X that preserves objective values, so the sampled problem is equivalent to minimizing $G_\eta + \gamma^\top x$ on X . Conditionally on η , the coefficients γ are independent with law $U_{\sigma, M}$. Lemma 6.1 therefore applies, with L from (19), κ_2, κ_3 from monomial bounds uniform in $|\eta_j| \leq \sigma$, degree $d' = de^{D_g}$, and the fallback of theorem 3.20 on $\mathfrak{X}$, whose domain formula consists of the mixed box and the polynomial equations $y_j = P_j$. The rest of the proof is in appendix D.1.

The implicit case. Here the reduced objective is algebraic but not polynomial, and three ingredients replace exact rational arithmetic: a chart oracle (lemma 6.6), a dynamic program on certified lower costs (lemma 6.7), and closure tests with certified derivative errors. The tails use the original polynomial graph, and so does the fallback.

Lemma 6.6 (Chart). *In case (I), for every $x \in \hat{X}$ each equation has exactly one root $\psi_j(x_{S_j}) \in V_j$, and ψ_j is infinitely differentiable on $\hat{X}_{S_j}$, with local smooth extensions at its boundary. If $A_j \geq 1$ bounds all partial derivatives of q_j of orders one to three on $\hat{X}_{S_j} \times V_j$, then every partial derivative of ψ_j of order $\nu \in \{1, 2, 3\}$ is bounded on $\hat{X}_{S_j}$ by*

$$\Pi_{1,j} = \frac{A_j}{\mu_j}, \quad \Pi_{2,j} = \frac{A_j(1 + \Pi_{1,j})^2}{\mu_j}, \quad \Pi_{3,j} = \frac{A_j[(1 + \Pi_{1,j})^3 + 3(1 + \Pi_{1,j})\Pi_{2,j}]}{\mu_j}.$$

At a rational $x \in \widehat{X}$, rational brackets of width ϵ for all roots, and rational enclosures of width ϵ of $G_\eta(x)$, $\nabla G_\eta(x)$ and $\nabla^2 G_\eta(x)$, are computed in time polynomial in I , the length of x and η , and $\log(1/\epsilon)$.

The derivative bounds have polynomial encoding length; through the chain rule they give rationals κ_2, κ_3 satisfying (5) for every G_η , $|\eta_j| \leq \sigma$, and the chart Lipschitz bound

$$\|(x, \psi(x)) - (x', \psi(x'))\|_2 \leq K_\psi \|x - x'\|_2, \quad K_\psi = 1 + \sum_j |S_j| \Pi_{1,j} \quad (20)$$

on $\widehat{X}$. The proof of lemma 6.6 is in appendix D.1.

Lemma 6.7 (Certified lower costs). *In case (I), at level j compute for every row of every bag a rational lower cost ϕ_β with $\phi_\beta \leq \phi_\beta \leq \phi_\beta + \delta_j$, where ϕ_β is the sum of the reduced terms assigned to β and $\delta_j = E_j/N_\mathfrak{B}$ with $N_\mathfrak{B}$ the number of bags. Run the dynamic program on the lower costs, update the incumbent with $U_j = \min\{U_{j-1}, \underline{m}_j + E_j\}$, where $\underline{m}_j$ is the lower-cost minimum, initialize $U_{-1} = +\infty$, and store the feasible implicit lift associated with the chosen upper bound. Update this bound and point before pruning. Retain a cell when $\underline{q}_C - E_j \leq U_j$, with $\underline{q}_C$ the least lower-cost min-marginal over its corners. Then on every draw all global optimizers are retained, every point removed at level j has value above $U_j \geq f^*$, and every retained cell has a corner $v = y_\beta$ of an allowed grid point y with $G_\eta(y) + \gamma^\top y \leq f^* + 4E_j$.*

The proof (appendix D.1) is that of proposition 5.7 with the error $D_j = N_\mathfrak{B}\delta_j \leq E_j$ of the summed lower costs. The witness tolerance $4E_j$ replaces $2E_j$; lemma 5.8 then gives intervals of length $La_i + 8E_j/a_i \leq (1+n)La_i$, which explains the factor in case (I). Exact comparisons of algebraic values are never made: the dynamic program is exact rational arithmetic on certified rational costs, and different bags may use different brackets of the same root without affecting global consistency.

The closure test of definition 5.11 is used with certified derivatives. In step (3), a rational $\hat{g}$ with $\|\hat{g} - \nabla(G_\eta + \gamma^\top x)(c)\|_\infty \leq \epsilon_g$ replaces the exact gradient, and a continuous coordinate is recorded at its lower bound when $\hat{g}_i - \kappa_2 r - \epsilon_g > 0$ and at its upper bound when $\hat{g}_i + \kappa_2 r + \epsilon_g < 0$. In step (5), a rational symmetric $\hat{H}$ with $\|\hat{H} - \nabla_{\bar{S}\bar{S}}^2 G_\eta(\bar{x}_S, c^P)\|_2 \leq \epsilon_H$ replaces the exact Hessian; it is obtained from enclosures of width at most ϵ_H/n of the entries by taking midpoints and symmetrizing, since a symmetric error matrix with entries at most ϵ_H/n in absolute value has spectral norm at most ϵ_H . The test is $\hat{H} - (\kappa_3 r^P + \epsilon_H + g_0)I \succ 0$. With $\epsilon_g = \tau/8$ and $\epsilon_H = g_0/8$, these tests are sound on every draw and succeed under the hypotheses of proposition 5.13 (appendix D.1).

Example 6.8 (Two necessary premises). *Curvature after substitution.* Under the equality $y = x$, the ambient polynomial $F_0(x, y) = Hxy + \varepsilon(x^2 + y^2)$ has both ambient diagonal second derivatives equal to 2ε , while its pullback $(H + 2\varepsilon)x^2$ has second derivative $2H + 4\varepsilon$. The chart is linear and well conditioned, yet the ambient cross derivative becomes retained curvature. *Retained ambient coordinates.* Under the parameterization $(x_1, x_2) = (t_1 + t_2, t_1 + 2t_2)$, $t \in [0, 1]^2$, which has determinant one and a bounded inverse, independent ambient coefficients pull back to $c_1 = \gamma_1 + \gamma_2$ and $c_2 = \gamma_1 + 2\gamma_2$. Under the grid law, the event $c_2 = 3\sigma$ forces $\gamma_1 = \gamma_2 = \sigma$, so $c_1 = 2\sigma$ is conditionally deterministic: a singleton receives conditional probability one instead of at most $1/M$, and the conditional count of lemma 5.8 fails. This does not show that the parallelogram is hard; it shows that a bounded inverse alone does not preserve the noise model.

Example 6.9 (A connected chain of products). Let $x \in [0, 1]^n$, where some coordinates may be binary, and impose $y_i = x_i x_{i+1}$ for $i < n$, with $F_0 = \sum_{i < n} (y_i - a_i)^2 + \lambda \sum_i x_i^2$, $\lambda \geq 0$. The constraints

form a connected chain of triangles $\{x_i, x_{i+1}, y_i\}$, not a product of separate blocks. Substitution gives a quartic with path bags $\{x_i, x_{i+1}\}$, and $\partial_{ii}G_\eta = 2x_{i-1}^2 + 2x_{i+1}^2 + 2\lambda \leq 4 + 2\lambda$ for all targets a_i and all η , since the dependent noise terms $\eta_i x_i x_{i+1}$ have zero pure second derivatives. Thus $L = 4 + 2\lambda$, independently of n , and case (E) applies with $p' = 2$.

6.2 Affine-state recurrences with polynomial actuators

Bounded-depth substitution does not cover dynamics over a long horizon, because the states depend on all earlier controls. For affine state dependence and affine state costs, an adjoint recursion eliminates the states while adding only unary terms.

Definition 6.10 (Actuator instance). For a horizon $N \geq 1$ and $k = 0, \dots, N-1$, the states and controls satisfy

$$s_{k+1} = a_k s_k + \chi_k(u_k), \quad |a_k| \leq a < 1, \quad (21)$$

with rational a_k and a , explicit rational univariate polynomials χ_k of degree at most d , nonempty closed rational state intervals I_k , $k = 0, \dots, N$, and control domains U_k that are closed rational intervals or native integer intervals. Box invariance holds:

$$a_k I_k + \chi_k(\text{hull } U_k) \subseteq I_{k+1} \quad (k < N). \quad (22)$$

There are no other constraints. The free variables are $z = (s_0, u_0, \dots, u_{N-1})$; a fixed initial state is allowed. The objective is $F_0 = q(z) + \sum_{k=1}^N c_k s_k$, where q is a sum of explicit rational polynomial factors of degree at most d whose scopes lie in the bags of a supplied tree decomposition of the free variables with bag size p . The noise is ambient on all original coordinates, states and controls.

Condition (22) is verified in polynomial time at fixed degree, by comparing the rational interval endpoints with the extreme values of χ_k at the endpoints of $\text{hull } U_k$ and at the real roots of χ'_k , using univariate root isolation. It makes the state intervals redundant: every free point has exactly one feasible trajectory. If preprocessing fixes every free variable, that is, the initial state is fixed and every control domain is a single point, the unique feasible trajectory is rational and the algorithm returns it with its sampled value directly.

Condition on the state coefficients $\eta = (\gamma_{s_1}, \dots, \gamma_{s_N})$ and define the adjoints $\lambda_N = c_N + \eta_N$ and $\lambda_k = c_k + \eta_k + a_k \lambda_{k+1}$ for $k = N-1, \dots, 1$. Substituting (21) backward from $k = N$ gives the identity

$$\sum_{k=1}^N (c_k + \eta_k) s_k = a_0 \lambda_1 s_0 + \sum_{k=0}^{N-1} \lambda_{k+1} \chi_k(u_k) \quad (23)$$

on every trajectory. The sampled problem is therefore equivalent to minimizing

$$Q_\eta(z) + \xi^\top z, \quad Q_\eta(z) = q(z) + a_0 \lambda_1 s_0 + \sum_{k=0}^{N-1} \lambda_{k+1} \chi_k(u_k),$$

over the free box, where $\xi = (\gamma_{s_0}, \gamma_{u_0}, \dots, \gamma_{u_{N-1}})$ are independent with law $U_{\sigma, M}$ conditionally on η . The added terms are unary, so Q_η has the same degree and the same tree decomposition as q , for every horizon. With $C_s \geq \max_k |c_k|$, induction gives $|\lambda_k| \leq \Lambda := (C_s + \sigma)/(1 - a)$ for every conditioned draw. Let L_q be a rational upper bound on the pure second derivatives $\partial_{ii}q$ on the free hull; it may be negative. If $G_\nu \geq \max_k \sup_{\text{hull } U_k} |\chi_k^{(\nu)}|$, then

$$L = \max\{1, L_q + \Lambda G_2\} \quad (24)$$

is positive and bounds the coordinate curvature of every Q_η , because the added terms contribute at most $|\lambda_{k+1}| |\chi_k''| \leq \Lambda G_2$ to $\partial_{u_k u_k} Q_\eta$ and nothing to $\partial_{s_0 s_0} Q_\eta$. Analogous bounds with ΛG_2 and ΛG_3 give uniform κ_2 and κ_3 . The adjoints are computed by N affine steps, so their denominators divide products of the denominators of the a_k , c_k and η_k , and their bit lengths are bounded by sums of the lengths of these numbers plus the length of Λ ; the coefficients of Q_η therefore have bit length polynomial in $I + b$, as [lemma 6.1](#) requires.

Theorem 6.11 (Affine-state recurrences with polynomial actuators). *Fix d . For actuator instances, with L as in [\(24\)](#) or any uniform valid bound, there is an algorithm and a power of two M , computed from the base instance with $\log_2 M \leq \text{poly}_d(I)$, such that for the ambient model with law $U_{\sigma, M}$ the following hold, where n is the number of free variables after substituting fixed ones. The all-fixed branch returns its rational trajectory directly; the displayed width bound concerns the remaining nonempty free box.*

- (i) *On every draw the algorithm returns an exact global optimizer, as a patch output in the free variables together with the recurrence [\(21\)](#), or, with probability at most $1/(2B)$, as the exact output of [theorem 3.20](#) for the sampled objective on the set of feasible trajectories, described by [\(21\)](#) and the variable bounds.*
- (ii) *The expected bit work is at most $C_0^p[4 + (1 + n/2)Lw_{\max}/(2\sigma)]^p \text{poly}_d(I)$.*
- (iii) *A q -bit evaluation returns a rational free point and its exactly feasible rational trajectory, within distance 2^{-q} of the optimal trajectory, and a rational interval of width 2^{-q} containing the optimal value, in expected bit work $\text{poly}_d(I + q)$.*

The proof is in [appendix D.2](#). The bound holds for every horizon, because the reduced decomposition is the free-variable decomposition. The restrictions are substantive. Nonlinear state dependence or nonlinear state costs raise the degree or couple many controls under elimination; for instance $s_{k+1} = s_k/2 + u_k^2/4$ with terminal cost s_N^2 produces the monomial $u_i^2 u_l^2$ for every pair of controls. Additional path or terminal constraints destroy the free product domain. The next example shows why the equalities cannot simply be kept as constraints in the sparse algorithm.

Example 6.12 (Predecessor fibers). Let $s_1 = u_0^2$ and $s_2 = u_1^2$ with $s_0, s_1, s_2 \in [0, 1]$ and $u_0, u_1 \in [-1, 1]$, linear objective $\gamma^\top(s, u)$, and bags $\{s_0, u_0, s_1\}$ and $\{s_1, u_1, s_2\}$. Conditioning on the coefficients outside the second bag, its conditional value at $(s_1, u_1, s_2) = (t, 0, 0)$ is $\min\{0, \gamma_{s_0}\} + \gamma_{s_1} t - |\gamma_{u_0}| \sqrt{t}$, because $u_0 = \pm\sqrt{t}$. Its second difference at $0, h, 2h$ is $|\gamma_{u_0}|(2 - \sqrt{2})\sqrt{h}$, which is not $O(h^2)$. Under the grid law $\gamma_{u_0} \neq 0$ on every draw, since M is even. No finite coordinate curvature bound holds for this conditional value, so the count of [lemma 5.8](#) does not apply, although the dynamics are box invariant and smooth. The elimination of [\(23\)](#) avoids predecessor fibers altogether.

Example 6.13 (A stateful nonlinear actuator). On $[-1, 1]$, the transition $s_{k+1} = \frac{1}{4}s_k + \frac{1}{4}u_k + \frac{1}{8}u_k^2$ maps $[-1, 1] \times [-1, 1]$ into $[-\frac{3}{8}, \frac{5}{8}]$, so [\(22\)](#) holds. Continuous or integer controls in $[-1, 1]$, affine state costs, and any coupled nonconvex control cost $\sum_\beta q_\beta(s_0, u_\beta)$ with the supplied decomposition satisfy [definition 6.10](#).

6.3 Products of simplices

Resource constraints inside disjoint blocks are not eliminated by a chart: a stick-breaking map from a box onto a simplex makes the ambient linear noise nonlinear. Instead, feasible rounding and the count are adapted to the faces of the simplices.

Definition 6.14 (Simplex instance). The continuous coordinates are partitioned into blocks b ; each block is an *inequality block* with feasible set $\Delta_b = \{x_b \geq 0, \sum_{i \in b} x_i \leq 1\}$, or an *equality block* with $\Delta_b^- = \{x_b \geq 0, \sum_{i \in b} x_i = 1\}$ and $|b| \geq 2$. The remaining coordinates are native integer intervals as in [definition 5.1](#), and there is at least one coordinate. The objective F_0 is a sum of explicit factors of degree at most d , and a supplied tree decomposition contains every factor scope and *whole blocks*: a bag containing a coordinate of b contains all of b . Its largest number of scalar coordinates is p . A rational $L > 0$ satisfies

$$\nabla_{bb}^2 F_0 \preceq LI \text{ for every block } b, \quad \partial_{ii} F_0 \leq L \text{ for every integer coordinate } i, \quad (25)$$

on the continuous hull of the domain. The noise is ambient on all coordinates. Let $w_{\max}$ be the largest integer width, or 1 if there is none, and $W = n_c + \sum_{i \in I_Z} w_i$.

The bound (25) on the block Hessian is stronger than a bound on its diagonal entries, and it is needed: rounding within a block is correlated. It is verified, for example, by the row-sum bound $\max_{i \in b} \sum_{k \in b} \sup |\partial_{ik} F_0|$, or promised. The derivative bounds κ_2, κ_3 of (5) are taken on the enclosing coordinate box $[0, 1]^{n_c} \times \hat{X}_Z$, not only on the product of simplices, because the closure tests evaluate derivatives at midpoints of coordinate hulls, which can lie outside a simplex. Whole-block bags are essential: they make the domain outside a bag independent of the bag's values. A model with simplex constraints in the scalar primal graph satisfies this after enlarging bags by at most the largest block size; a single resource constraint over all variables gives no width advantage.

Theorem 6.15 (Products of simplices). *Fix d . For simplex instances there is an algorithm and a power of two M , computed from the base instance with $\log_2 M \leq \text{poly}_d(I)$, such that for the ambient model with law $U_{\sigma, M}$:*

- (i) *on every draw the algorithm returns an exact global optimizer: on ordinary draws a relative patch output of length $\text{poly}_d(I)$ (fixed integer labels, fixed coordinates, recorded tight budgets, a rational relative polytope P , and a verified modulus g_0 with $v^\top \nabla^2 F_\gamma v \geq g_0 \|v\|_2^2$ on P for all directions v of its affine hull), or a rational point, with its global proof record; on the remaining draws, of probability at most $1/(2B)$, the exact output of [theorem 3.20](#) on the same domain;*
- (ii) *the expected bit work is at most $C_0^p [4 + (2 + n/2)L \max\{1, w_{\max}\}/(2\sigma)]^p \text{poly}_d(I)$;*
- (iii) *a q -bit evaluation, with an exactly feasible rational point, costs $\text{poly}_d(I + q)$ on ordinary draws and has expectation $\text{poly}_d(I + q)$.*

The proof is in [appendix D.3](#). We describe its four ingredients.

Cells and rounding. The mesh $h_j = s2^{-j}$ of [section 5.2](#) is shared with the integer coordinates, so it can exceed one. While $h_j \geq 1$, each block has one cell, its simplex, whose corners are the vertices of the simplex. For $h = h_j = 1/m < 1$ the cells of a block are the nonempty sets $C = \Delta_b \cap \prod_{i \in b} [hk_i, h(k_i + 1)]$ with $k_i \in \{0, \dots, m - 1\}$, or the same with Δ_b^- , and the corners of C are its points in $(h\mathbb{Z})^b$. After scaling, C is the unit cube intersected with $\sum_i z_i \leq m - \sum_i k_i$ (or equality), whose right-hand side is an integer, so every vertex of C is a cube corner ([lemma D.3](#)). Every point of a cell is therefore a convex combination of its corners, which gives mean-preserving feasible rounding with $\mathbb{E} \|Y_b - x_b\|^2 \leq |b| h_j^2/4$; coordinates on grid hyperplanes stay fixed, so all candidate lists are preserved. Rounding blocks independently and using (25) gives the allowance $E_j = nLh_j^2/8$, and [proposition 5.7](#) holds verbatim.

Count by faces. Classify a block grid point v_b by its smallest face of the simplex: its positive support S and whether its budget is tight. On a slack face the directions $e_i, i \in S$, are feasible in

both signs; on a tight face, with the least index $\iota \in \mathbf{S}$ as anchor, the directions $e_i - e_\iota$, $i \in \mathbf{S} \setminus \{\iota\}$, are. Conditioning also on the anchor coefficients, each direction confines one remaining independent coefficient to an interval of length at most $Lh \|d\|^2 + 4E_j/h \leq (2 + n/2)Lh$. A face of dimension r has at most h^{-r} relative-interior grid points, which cancels the probability h^r (lemma D.4). The face classification, not a coordinate classification, is what makes the count independent of h .

Closure on original faces. For an inequality block, a positive partial derivative throughout the hull forces $x_i = 0$; a negative one forces the budget to be tight; and a positive difference $\partial_i F - \partial_l F$ forces $x_i = 0$. For an equality block only the last test is valid: decreasing one coordinate alone is infeasible, and a strictly convex objective can have positive partial derivatives at an interior minimizer of a probability simplex. The margins concern *active inequalities* only: zero coordinates, and tight budgets of inequality blocks. The budget of an equality block is imposed structurally; its multiplier is unrestricted in sign and no margin is required of it. For $F_0 = 0$ on $x_1 + x_2 = 1$, $x \geq 0$, the multiplier of the budget in the sign convention of lemma D.5 is $-\min\{\gamma_1, \gamma_2\}$, which is negative whenever both coefficients are positive, although the point growth is positive whenever $\gamma_1 \neq \gamma_2$; a margin requirement on it would fail with probability close to $1/4$. At an optimizer whose active inequalities have multipliers larger than τ , the tests detect every active inequality without a lower bound on positive coordinates: in an inequality block with a tight budget, the derivative of a positive coordinate is minus the budget multiplier, and in every block with a tight or equality budget, the multiplier of a zero coordinate is its derivative difference from a positive coordinate, in which the budget multiplier cancels. The patch test is $Z^\top \nabla^2 F_\gamma(c^P)Z - (\kappa_3 r^P + g_0)Z^\top Z \succ 0$ for a rational tangent basis Z of the patch, a rational point c^P of it, and the largest hull width r^P ; subtracting a multiple of $Z^\top Z$ rather than of I keeps the curvature in the metric of the original coordinates (lemma D.5).

Margins. The face stationarity equations depend only on tangent noise, the differences $\gamma_i - \gamma_\iota$, and the multiplier of each active inequality depends with slope ± 1 on one further original coefficient. The transformed coefficients are not independent under the grid law, and the proof does not treat them as independent: it counts transformed tuples, with at most $2M - 1$ values per difference (lemma D.6).

6.4 Order constraints with continuous and binary coordinates

Order constraints $x_i \leq x_l$ may couple the whole instance; they are not confined to blocks, so (P1) fails. Instead, the conditional value is controlled by transporting feasible points, and the active constraints are identified by linear programming.

Definition 6.16 (Order instance). Let $[n] = I_C \cup I_Z$, $n_c = |I_C| \geq 1$, $n_z = |I_Z|$, and let E be a set of m directed edges on $[n]$; cycles are allowed. The feasible set is

$$X = \{x \in [0, 1]^n : x_i \leq x_l \text{ for } (i, l) \in E, x_i \in \{0, 1\} \text{ for } i \in I_Z\}.$$

The objective F_0 is a sum of explicit factors of degree at most d , and a supplied tree decomposition contains every factor scope and every edge. Let p be its bag size and p_c the largest number of continuous coordinates in a bag. A rational $\bar{L} \geq 1$ satisfies $\nabla^2 F_0 \preceq \bar{L}I$ on $[0, 1]^n$. The noise is ambient on all coordinates.

The zero vector is feasible. If $n_c = 0$, ordinary finite-state dynamic programming over binary bag rows solves the instance exactly for every γ in $2^{O(p)} \text{poly}_d(I)$ bit work, and no noise is needed. The full Hessian bound $\bar{L}$ is needed because feasible rounding of ordered coordinates is correlated; it is verified by a row-sum bound or promised.

Theorem 6.17 (Order constraints). *Fix d . For order instances with $n_c \geq 1$ there is an algorithm and a power of two M , computed from the base instance with $\log_2 M \leq \text{poly}_d(I)$, such that for the ambient model with law $U_{\sigma, M}$:*

- (i) *on every draw the algorithm returns an exact global optimizer: on ordinary draws a binary assignment together with a relative patch output on the continuous slice, consisting of fixed coordinates, a block structure $x_{I_C} = Dy + \bar{x}$ with disjoint zero-one columns, a rational box and the remaining order inequalities in y , and a modulus g_0 with verified $\nabla_y^2 F_\gamma(Dy + \bar{x}) \succeq g_0 D^\top D$ on the patch, or a rational point if no block remains; on the remaining draws, of probability at most $1/(2B)$, the algebraic fallback output;*
- (ii) *the expected bit work is at most*

$$C_0^p p_c! (p_c + 1) \left[2 + \frac{3n_c \bar{L}}{4\sigma} \right]^{p_c} \text{poly}_d(I); \quad (26)$$

- (iii) *a q -bit evaluation, with an exactly feasible rational point, costs $\text{poly}_d(I + q)$ on ordinary draws and has expectation $\text{poly}_d(I + q)$.*

The number of binary coordinates is unrestricted, and they do not enter the bracket. For $n_z = 0$ the bound is $C_0^p p! (p + 1) [2 + 3n\bar{L}/(4\sigma)]^p \text{poly}_d(I)$. The factorial is a function of p alone; the bound is fixed-width polynomial, not fixed-parameter, because the bracket contains n_c . Only binary integer coordinates are covered: transporting points with arbitrary integer levels would need fixed interior knots and a different count. The proof is in [appendix D.4](#); its ingredients are as follows.

Rounding. Continuous coordinates use dyadic cells of width $h_j = 2^{-j}$, and binary coordinates the cell $\{0, 1\}$ at level 0 and singletons afterwards. Rounding all continuous coordinates by one common threshold, $Y_i = h(\lfloor x_i/h \rfloor + \mathbf{1}\{U < \text{frac}(x_i/h)\})$ with U uniform on $[0, 1]$, is a nondecreasing map for each value of U . It therefore preserves every order constraint and every binary value, fixes grid nodes, and stays in every candidate cell. It is mean preserving with $\mathbb{E} \|Y - x\|^2 \leq n_c h^2/4$, and the full Hessian bound gives the allowance $E_j = n_c \bar{L} h_j^2/8$. The threshold is a proof device; the algorithm does not sample it. A similar mean-preserving coupling of ordered coordinates by one quantile appears in isotonic optimization [\[8\]](#).

Transport and count. Fix a bag, a binary bag row, and the noise outside the continuous bag coordinates. For a feasible continuous bag tuple v in the relative interior of a face of a standard order simplex $\{0 \leq v_{\pi(1)} \leq \dots \leq v_{\pi(c)} \leq 1\}$, take a conditional minimizer and apply to all its coordinates the nondecreasing piecewise-affine map that moves the distinct interior values of v to those of a target w in the same closed face while fixing 0 and 1. The result is feasible, keeps every binary value, has bag part w , and is affine in w ; along the edge directions of the face every coordinate moves at rate in $[0, 1]$, so the transported objective is an upper support for the conditional value with directional curvature at most $n_c \bar{L}$ ([lemma D.8](#)). The transported fiber need not equal the true fiber, and the projected mixed domain need not be convex; an upper support is what the comparison needs. Counting relative-interior grid points of the faces of the $p_c!$ order simplices, as in [lemma 5.8](#), gives the bracket of (26) ([proposition D.9](#)).

Closure by linear programming. When every binary hull is a singleton, all optimizers lie in one continuous slice, a polytope with zero-one vertices. With g the gradient at the hull midpoint and $\delta = \kappa_2 r$ a bound on its error, a proposed equality $x_i = x_l$ is certified when the least value of $g^\top x$ on the slice face $\{x_i = 0, x_l = 1\}$ exceeds the least value on the slice by more than $n_c \delta$; bound equalities use the faces $\{x_i = 1\}$ and $\{x_i = 0\}$. These are rational linear programs. At every optimizer the true slice gradient exposes a face containing it, and if the equality failed some zero-one vertex of

that face would lie in the opposite face, which bounds the computed gap by $n_c\delta$ (lemma D.10). The certified equalities merge coordinates into blocks, and all other order inequalities are kept in the patch, so no lower bound on inactive slacks is needed.

Example 6.18 (Binary labels must be fixed before exposure). On $0 \leq x \leq z \leq 1$ with z binary, the objective $(x - \frac{1}{2})^2 + 10z^2 - 11z$ has the unique mixed optimizer $(\frac{1}{2}, 1)$ with value -1 and point-growth modulus one. Its full gradient there is $(0, 9)$. Over the continuous relaxation, the least value of $9z$ is 0, while on the face $\{x = 0, z = 1\}$ opposite to the equality $x = z$ it is 9. The relaxed test would certify $x = z$, which is false. After $z = 1$ is fixed, the slice gradient is zero and no equality is certified. The algorithm therefore applies the linear-programming test only after all binary hulls are singletons.

6.5 What is not covered

General totally unimodular constraints on the searched coordinates are not covered by the smoothed guarantees of this paper. This differs from theorem 8.14, where a totally unimodular system constrains only the integer recourse of a small perturbed core. An unpublished companion manuscript [1] treats them deterministically with a coupling algorithm. For fixed-degree polynomial objectives it returns approximation certificates without a growth premise, with table sizes that then depend on the accuracy. For rational quadratic objectives its exact version terminates without a growth or uniqueness premise, and a set-growth condition bounds its number of levels. In both cases the stated accuracy-independent operation bounds also need set growth and a bound r on the number of distinct values that each coordinate takes over the optimal set, which is a separate premise. These deterministic bounds do not by themselves give expected work bounds under the noise; section 5.4 discusses the limits of integrating the available growth tail. For $P = \{x \in [0, 1]^n : Ax \leq b\}$ with A integral and totally unimodular and b integral, aligned feasible rounding holds: after scaling a cell of width $h = 2^{-j}$, the cell polytope $\{z \in [0, 1]^n : Az \leq b/h - Ak\}$ has an integral right-hand side and hence integral vertices [26], so every feasible cell point is a mean of feasible corners, and a full Hessian bound $\bar{L}$ gives the allowance $n\bar{L}h^2/8$. With every constraint inside a bag, the pruning of proposition 5.7 is then sound. This deterministic rounding, allowance and filtering layer specializes results of the companion manuscript [1]. What is missing is a count of retained cells whose constants are controlled by the input: example 9.6 shows that coordinate comparisons fail for a single totally unimodular inequality, and the arguments of this paper provide neither such a count nor closure and tail estimates for general totally unimodular systems. Order constraints are the totally unimodular case in which the transport count, the linear-programming closure and the tails are all complete. Overlapping resource constraints, affine images of simplices, general integer ranges in order systems, extra constraints that cut a graph chart, and nonlinear state recurrences are also outside the theorems of this section.

Relation to earlier work. The geometric ingredients are classical: order polytopes have zero-one vertices and are triangulated by order simplices [43]; isotonic optimization uses the common-quantile coupling [8]; totally unimodular systems have integral vertices [26]; and implicit elimination of states inside validated global search for dynamic models is established [47]. The new element is again the composition: an expected near-optimal row count under one finite independent-noise law for each domain class, sound closure tests adapted to the domain, and an exact same-draw fallback with a polynomial-time evaluable output.

7 Conditional recourse: a small nonconvex core

Conditional optimization can reduce a global search to a few coordinates. We call these coordinates $v \in [0, 1]^k$ the *core*, and write z for the *residual* variables. At each core point an oracle optimizes over one fixed residual feasible set Y , returning an exact value or a certified interval. The core changes the residual costs, while Y stays the same. A few design or control parameters that enter the operating costs of a large convex or combinatorial model are an example. A two-stage problem has this form only if its first-stage decisions change second-stage costs and leave second-stage constraints unchanged; when first-stage decisions enter second-stage right-hand sides or bounds, the method below fails, as [example 7.3](#) shows. Of the three structural routes in this paper, a small nonconvex factor, sparse bags, and a small core with tractable conditional recourse, these two sections develop the third. Its parameter is the core size k together with the ratio L/σ of core curvature to noise. The number of residual variables, their coupling, and their curvature enter only through the input length.

The mechanism has four parts.

- (i) Minimizing over a residual set that does not depend on the core preserves upper coordinate curvature. The conditional value $V(v) = \min_z F_\gamma(v, z)$ therefore obeys the corrected-corner inequality of [lemma 3.7](#) on the core cube, and no residual coordinate is rounded.
- (ii) Independent linear noise on the core coefficients bounds the expected number of near-optimal core grid points by a product of k one-dimensional factors. Residual noise, when present, is conditioned on.
- (iii) Residual coordinates are never refined, so closure needs a separate global certificate. A few restricted recourse calls prove that every global optimizer lies in a small product patch (this section) or uses one residual integer label ([section 8](#)).
- (iv) A base-selected finite law makes failure of these certificates rare enough to pay for an exact fallback on the same draw ([section 2.3](#)).

[Table 4](#) lists the results of this section. Throughout,

$$Q_{\text{ex}} = \left[3 + \frac{(1 + k/2)L}{2\sigma} \right]^k, \quad Q_{\text{ap}} = \left[3 + \frac{(1 + k)L}{2\sigma} \right]^k. \quad (27)$$

The first factor applies when conditional values are computed exactly, the second when they are computed with certified error. Both are numerical: they depend on the magnitude of L/σ , not on its encoding length.

7.1 The conditional value function

Definition 7.1 (Core-recourse instance). A *core-recourse instance* consists of integers $k, n_R \geq 0$ with $n = k + n_R \geq 1$, a nonempty compact *residual set* $Y \subseteq \mathbb{R}^{n_R}$, an explicit rational polynomial $F_0(v, z)$ of total degree at most d in the core variables $v \in \mathbb{R}^k$ and residual variables $z \in \mathbb{R}^{n_R}$, a rational $L > 0$ with

$$\frac{\partial^2 F_0}{\partial v_i^2}(v, z) \leq L \quad (i = 1, \dots, k, (v, z) \in [0, 1]^k \times \text{conv } Y), \quad (28)$$

and a rational noise half-width $\sigma > 0$. The feasible set is $X = [0, 1]^k \times Y$. We write C and R for the core and residual index sets. In this section $Y = [0, 1]^{n_R}$; [section 8](#) uses integer sets.

Table 4: Continuous conditional recourse. Here $b = \log_2 M$ is the number of sampling bits per perturbed coordinate, and “implicit” is the patch output of [definition 2.5\(b\)](#). On rare draws each result returns the exact output of a fallback on the same draw. For the two quadratic results this output is rational, so their output is rational on every draw. For the other two it is the algebraic output ([definition 2.5\(d\)](#)) of [theorem 3.20](#): one isolated real root and polynomial maps for all coordinates of one global minimizer and for the optimal value.

Result	Noise; residual premise and oracle	b	Expected work	Ordinary output
theorem 7.9	ambient; quadratic, exact recourse on every residual subbox	$\text{poly}(I)$	$8^k Q_{\text{ex}} \text{poly}(I)$	rational
corollary 7.10	ambient; quadratic, residual interaction graph a forest	$\text{poly}(I)$	$8^k Q_{\text{ex}} \text{poly}(I)$	rational
theorem 7.13	ambient; certified values on every residual subbox, e.g. residual convexity	$\text{poly}_d(I)$	$8^k Q_{\text{ap}} \text{poly}_d(I)$	implicit
theorem 7.15	core-only; residual Hessian $\succeq \mu I$	$\text{poly}_d(I)$	$8^k Q_{\text{ap}} \text{poly}_d(I)$	implicit

The essential restriction is that the residual feasible set does not depend on the core: the core enters only the objective. Core coordinates with other rational ranges can be mapped affinely onto $[0, 1]$; this rescales L and the physical noise, and the rescaled data are part of the input. A valid L can always be computed from monomial coefficients, but the numerical factors below use the supplied value, which may be much smaller. A sharper value is a certified premise: the encoding of its proof is part of I , and the work of checking the proof is part of the stated work bound. For the sampled objective $F_\gamma(v, z) = F_0(v, z) + \gamma_C^\top v + \gamma_R^\top z$ (with $\gamma_R = 0$ under core-only noise) define, for a nonempty closed $Y' \subseteq Y$,

$$V_{Y'}(v) = \min_{z \in Y'} F_\gamma(v, z), \quad V = V_Y, \quad f^* = \min_X F_\gamma = \min_{[0,1]^k} V. \quad (29)$$

Lemma 7.2 (Conditional values). *Let $G \geq \sup\{\sum_{i \in C} |\partial_{v_i} F_\gamma(v, z)| : v \in [0, 1]^k, z \in \text{conv } Y\}$.*

- (a) *The function $V(v) - \gamma_C^\top v$ depends on γ_R but not on γ_C .*
- (b) *For every $i \in C$ and every v , the function $t \mapsto V(v + te_i) - Lt^2/2$ is concave on the interval where $v + te_i \in [0, 1]^k$.*
- (c) *For every nonempty closed $Y' \subseteq Y$, the function $V_{Y'}$ is G -Lipschitz with respect to the maximum norm on the core: $|V_{Y'}(v) - V_{Y'}(w)| \leq G \|v - w\|_\infty$.*

None of these statements uses convexity of the residual problem.

Part (b) is the curvature part of the value-function lemma of an unpublished companion manuscript [1], which also assumes a fixed recourse feasible set; we include its proof for completeness. This part is the reason for fixed residual feasibility. It makes V satisfy the curvature hypothesis of [theorem 3.4](#) in dimension k : for a dyadic cell of side $h_j = 2^{-j}$ in the core cube,

$$\min_{v \in C} V(v) \geq \min_{v \text{ corner of } C} V(v) - E_j, \quad E_j = \frac{kLh_j^2}{8}. \quad (30)$$

Only core coordinates contribute to E_j . When residual feasibility depends on the core, (30) fails, even for linear data.

Example 7.3 (Core-dependent feasibility). Let $k = n_R = 1$, $F_0(v, z) = z$, and replace the fixed residual set by $Y(v) = \{z \in [0, 1] : z \geq v - \frac{1}{3}, z \geq \frac{1}{3} - v\}$. Then $\partial_{vv}F_0 = 0$ and $V(v) = \left\lfloor v - \frac{1}{3} \right\rfloor$. At every level j the cell containing $\frac{1}{3}$ has corners at distances $h_j/3$ and $2h_j/3$ from $\frac{1}{3}$, because $2^j \not\equiv 0 \pmod{3}$. Its minimum corner value is $h_j/3$, while $\min_C V = 0$. Thus (30) fails with $L = 0$, and for every $L > 0$ it fails at all levels with $h_j < 8/(3kL)$. A search based on it would report the value interval $[U_j - E_j, U_j]$, which excludes $f^* = 0$.

The corrected-corner inequality also shows why conditional values must be exact or certified, rather than computed by rounding the residual variables to a grid. Rounding residual variables reintroduces a discretization error whose variation over core states grows with the number of residual variables, and normalizing grid messages does not remove it.

Example 7.4 (Rounded residuals). Let $m = 32$, $\varepsilon = \frac{1}{16}$, and on $[0, 1]^{m+1}$ let

$$F_0(v, z) = v^2 - \frac{v}{2} \sum_{i=1}^m z_i + \sum_{i=1}^m z_i^2 + \left(\frac{m}{16} - 1 - \varepsilon\right)v.$$

Its interaction graph is a star, every coordinate second derivative equals 2, and $V(v) = (1 - \frac{m}{16})v^2 + (\frac{m}{16} - 1 - \varepsilon)v$ is strictly concave with $V(0) = 0 > V(1) = -\varepsilon$. The unique optimizer is $(1, \frac{1}{4}, \dots, \frac{1}{4})$ with value $-\frac{1}{16}$. With $h = \frac{1}{2}$, let $V_h(v)$ minimize over residual points of the grid $\{0, \frac{1}{2}, 1\}^m$. Then $V_h(0) = 0$, $V_h(\frac{1}{2}) = \frac{23}{32}$, and $V_h(1) = \frac{31}{16}$, and $V_h(v) - V(v) = m \text{dist}(v/4, \{0, \frac{1}{2}, 1\})^2$. The bag cell $[\frac{1}{2}, 1] \times [0, \frac{1}{2}]$ of $\{v, z_1\}$ contains the optimizer's bag projection, its smallest grid min-marginal is $\frac{23}{32}$, the grid incumbent is 0, and the bag-local correction $|B|Lh^2/8 = \frac{1}{8}$ discards it. The global correction $(m+1)Lh^2/8 = \frac{33}{16}$ is valid. With exact conditional values V in place of V_h , the corner correction is $Lh^2/8$ for the single core coordinate, independently of m .

This deterministic example is a member of the first star family of the companion manuscript [1]. Proposition 9.2 extends it to every draw of small noise and shows that no closure test succeeds at level 0, so the search reaches the level at which the cell is deleted.

The proof of example 7.4 (appendix E.7) also gives a uniformly positive definite variant on a star with $m = d^2$ leaves, $d \geq 8$, in which the variation of $V_h - V$ between two adjacent core states is $(d-1)h^2$. This variant is the case $h = \frac{1}{4}$ of the companion's second, positive definite star family [1]. Positive definiteness and bounded coloring number therefore do not make a bag-local budget valid for grid min-marginals. These examples concern the retention rule; they are not hardness results.

The core search. A *conditional oracle* receives a rational core point v . In *exact mode* it returns $V(v)$ and a residual point $z \in Y$ with $F_\gamma(v, z) = V(v)$. In *certified mode* it also receives a rational accuracy $\eta > 0$ and returns rationals $\ell \leq V(v) \leq u$ with $u - \ell \leq \eta$ and a residual point $z \in Y$ with $F_\gamma(v, z) = u$. For uniform notation, exact mode returns $\ell = u = V(v)$.

The *core search* processes levels $j = 0, 1, \dots$. At level 0 the only generated cell is $[0, 1]^k$. At level $j \geq 1$ the generated cells are the 2^k children of every cell retained at level $j-1$; the full level- j grid is never enumerated. At every level the search queries the oracle once at every distinct corner of a generated cell. In certified mode the accuracy is

$$\eta_j = E_j \quad (k \geq 1), \quad \eta_j = 4^{-j} \quad (k = 0), \quad (31)$$

and in exact mode we put $\eta_j = 0$. Let U_j be the least upper value u returned at this level, attained at a corner c_j with completion z_j . A generated cell is retained if and only if

$$\min_{v \text{ corner of } C} \ell(v) - E_j \leq U_j. \quad (32)$$

The *retained hull* $\mathcal{H}_j$ is the smallest box containing all retained cells. For $k = 0$ there is one empty core cell, which is always retained, $E_j = 0$, each level makes one query, and $\mathcal{H}_j$ is a point. In that case a certified answer of width 4^{-j} gives only an incumbent within 4^{-j} of f^* , never an exact one; the positive allowance η_j is what the stopping analysis uses.

Proposition 7.5 (Core search). *Let $\varepsilon_j = E_j + \eta_j$; thus $\varepsilon_j = E_j$ in exact mode, $\varepsilon_j = 2E_j$ in certified mode with $k \geq 1$, and $\varepsilon_j = 4^{-j}$ in certified mode with $k = 0$. On every draw and for every admissible oracle answer:*

- (a) *for every global minimizer (a, z^*) of F_γ on X , a retained cell contains a , so $a \in \mathcal{H}_j$;*
- (b) *$f^* \leq U_j \leq f^* + \varepsilon_j$, and c_j is a corner of a retained cell, so $c_j \in \mathcal{H}_j$;*
- (c) *every retained cell has a corner v with $V(v) \leq f^* + 2\varepsilon_j$.*

Suppose the core coefficients γ_C are independent with law $U_{\sigma, M}$ and independent of γ_R , and $M \geq 2^J$. Conditionally on γ_R , for every $j \leq J$ the expected number of points v of the full grid $(h_j \mathbb{Z})^k \cap [0, 1]^k$ with $V(v) \leq f^* + 2\varepsilon_j$ is at most Q_{ex} in exact mode and Q_{ap} in certified mode. The expected number of oracle calls through level J is at most $2^k + 8^k JQ$, with $Q = Q_{\text{ex}}$ or $Q = Q_{\text{ap}}$ respectively.

The expectation statement uses a deterministic grid fixed before sampling. Each retained cell is charged to a near-optimal corner, and near-optimality of a fixed grid point is a condition on the true function V , so arbitrary noise-dependent choices of the oracle within its error bounds do not affect the count. The bound is [theorem 3.4](#) applied, conditionally on the residual noise, to $V - \gamma_C^\top v$, which has upper coordinate curvature L and does not depend on γ_C . With exact values the search is a variant of the corrected-corner search of [theorem 3.9](#) on the core, with the incumbent taken over the current level. The proof is in [appendix E.2](#). With an exact oracle, parts (a)–(c) are the retention, value-interval and witness guarantees of the deterministic core search in the companion manuscript [1], whose separate query bound uses point growth. Here we also allow certified oracle errors and bound the expected number of queries without a growth premise.

7.2 Global containment and closure certificates

The core search alone does not finish the problem: it localizes the core but says nothing about residual coordinates. The next two lemmas supply sound certificates. Here $Y = \prod_{i \in R} [\ell_i, u_i]$ is a box.

Let c be a point of a core box $\mathcal{H}$, let $z_c \in Y$, and let $r > 0$. The *residual patch* is

$$P = \prod_{i \in R} [\max\{\ell_i, z_{c,i} - r\}, \min\{u_i, z_{c,i} + r\}].$$

For each $i \in R$ with $\ell_i < z_{c,i} - r$ form the *lower slab* $Y_i^- = \{z \in Y : z_i \leq z_{c,i} - r\}$, and for each $i \in R$ with $z_{c,i} + r < u_i$ the *upper slab* $Y_i^+ = \{z \in Y : z_i \geq z_{c,i} + r\}$. No slab is formed on a side where the patch reaches an original bound; a singleton slab at an original bound could contain an optimizer and destroy the certificate.

Lemma 7.6 (Excluded slabs). *The slabs cover $Y \setminus P$. Let λ_{out} be at most the least conditional value $V_{Y_i^\pm}(c)$ over all slabs ($\lambda_{\text{out}} = +\infty$ if there are none), let $U \geq V(c)$, and let $w_\infty(\mathcal{H})$ be the largest coordinate width of $\mathcal{H}$. If*

$$\lambda_{\text{out}} - U > 2G w_\infty(\mathcal{H}), \tag{33}$$

then for every $v \in \mathcal{H}$ every minimizer of $F_\gamma(v, \cdot)$ on Y lies in P . If moreover $\mathcal{H}$ contains the core of every global minimizer, then every global minimizer of F_γ on X lies in $\mathcal{H} \times P$.

The certificate needs a certified *lower* bound on every slab. A feasible value on a slab, which is all an approximate solver returns by itself, does not suffice: a better competitor could be hidden in the slab. It also needs restricted recourse on slabs, so the conditional oracle must accept arbitrary residual subboxes. This is the *box-stability* requirement of the following sections.

Once global containment is certified, two sound local tests finish the problem. Let $Q \subseteq [0, 1]^n$ be a box containing every global minimizer, with midpoint m and maximum-norm radius r_Q , and put

$$K_2 \geq \max\left\{1, \max_i \sum_j \sup_{[0,1]^n} |\partial_{ij} F_0|\right\}, \quad K_3 \geq \max\left\{1, \max_i \sum_{j,l} \sup_{[0,1]^n} |\partial_{ijl} F_0|\right\}, \quad (34)$$

rational bounds computed from monomial coefficients.

Lemma 7.7 (Sign fixing and convexity tests).

- (a) *If the original lower bound of coordinate i lies in the i th interval of Q and $\partial_i F_\gamma > 0$ on Q , every global minimizer has x_i at that bound; symmetrically for an upper bound and $\partial_i F_\gamma < 0$. The midpoint test $\pm \partial_i F_\gamma(m) > K_2 r_Q$ is a sufficient condition for the sign; for quadratics the sign can be decided exactly. Every global minimizer lies in the box Q' obtained by fixing all coordinates that pass.*
- (b) *Let m' and $r_{Q'}$ be the midpoint and radius of Q' and let $g_0 > 0$. If the Hessian of F_0 in the unfixed coordinates satisfies $\nabla^2 F_0(m') - (K_3 r_{Q'} + g_0)I \succeq 0$, then F_γ is g_0 -strongly convex on Q' , its unique minimizer on Q' is the unique global minimizer of F_γ on X , and (Q', g_0) is an implicit patch output ([definition 2.5](#)).*
- (c) *If F_0 is quadratic and its Hessian in the unfixed coordinates is positive semidefinite, then every exact minimizer of the convex quadratic program $\min_{Q'} F_\gamma$ is a global minimizer of F_γ on X .*

The tests may fix only *original* bounds, and only bounds that lie in the current interval. Fixing an artificial patch endpoint would be unsound. A coordinate whose derivative vanishes at a contained optimizer never passes a strict sign test. All tests in this section are exact rational computations and are sound on every draw; the probabilistic analysis only shows that they pass early.

7.3 Exact recourse for quadratic objectives

Definition 7.8 (Box-stable exact recourse). Let $F_0(x) = \frac{1}{2}x^\top A x + a^\top x + c_0$ be a rational quadratic on $[0, 1]^n$ with core C and residual R . A *box-stable exact recourse oracle* receives a rational core point $v \in [0, 1]^k$, a rational box $\prod_{i \in R} [\ell'_i, u'_i] \subseteq [0, 1]^{n_R}$, possibly with degenerate intervals, and rational linear coefficients γ , and returns an exact rational global minimizer of $z \mapsto F_\gamma(v, z)$ on that box and its value. Its bit work is bounded by a polynomial in the query length whose exponent does not depend on k .

An oracle that solves only the unrestricted residual problem does not satisfy this definition, and the slab certificate cannot be run with it.

Theorem 7.9 (Exact quadratic recourse). *Let $F_0(x) = \frac{1}{2}x^\top A x + a^\top x + c_0$ be a rational quadratic on $X = [0, 1]^n$, let C be a supplied core of size $k \geq 1$, and let $L > 0$ be rational with $A_{ii} \leq L$ for $i \in C$ (verified). Assume a box-stable exact recourse oracle. There is a power of two M , computed from the base instance with $\log_2 M \leq \text{poly}(I)$, such that for the ambient perturbation model with marginal law $U_{\sigma, M}$ the following hold.*

- (i) On every draw the algorithm returns a rational global minimizer of F_γ on X and the optimal value, of bit length $\text{poly}(I)$.
- (ii) The expected bit work, including oracle calls and the computation of M , is at most $8^k Q_{\text{ex}} \text{poly}(I)$.

No growth, uniqueness, nondegeneracy, or residual convexity premise is used.

The polynomial factor depends on the oracle's bound but has an exponent independent of k . The algorithm runs the core search in exact mode; at each level it tests lemma 7.6 with at most $2n_R$ additional oracle calls and then the tests of lemma 7.7. If they have not passed by a base-computed level J , it enumerates the 3^n faces of the box on the same draw (lemma C.3). The proof in appendix E.5 shows that under point growth g_0 and active-gradient margin τ , both chosen from base data, all tests pass by level J , and that the growth and active-gradient tails make the fallback rare. The stopping depth depends only on base data: no rational-reconstruction bound that grows with the sampling precision is needed.

Corollary 7.10 (Residual forests). *In theorem 7.9, suppose that deleting the core C from the interaction graph of A (vertices $[n]$, an edge ij when $i \neq j$ and $A_{ij} \neq 0$) leaves a forest. Then the box-stable exact recourse oracle exists, and the expected bit work is $8^k Q_{\text{ex}} \text{poly}(I)$. This bound is fixed-parameter in the size k of the supplied feedback vertex set and in L/σ .*

The oracle is the exact algorithm of Del Pia and Khajavirad [19] for box-constrained quadratic programs whose interaction graph is a forest: fixing core coordinates and adding linear terms change only linear coefficients, and restricting a residual box and rescaling its nondegenerate intervals keeps the residual graph a forest. The residual quadratic need not be convex, have bounded negative inertia, or be separable. The corollary is not fixed-parameter in treewidth alone: a graph of treewidth two can require a large feedback vertex set, and box-constrained quadratic programming is strongly NP-hard at treewidth two [19].

7.4 Certified convex recourse for polynomial objectives

For nonlinear recourse the conditional values are usually irrational, and exact comparisons between them are unavailable. The theorem below uses only certified intervals for conditional values and feasible rational completions.

Definition 7.11 (Box-stable certified recourse). For an explicit rational polynomial F_0 on $[0, 1]^n$ with core C , a *box-stable certified recourse oracle* receives a rational core point v , a rational residual box $Y' \subseteq [0, 1]^{n_R}$, rational linear coefficients γ , and a rational $\eta > 0$. It returns a rational ℓ and a rational $z \in Y'$ with $\ell \leq V_{Y'}(v) \leq F_\gamma(v, z) \leq \ell + \eta$. Its bit work is polynomial in the query length, including the length of η , with an exponent independent of k .

Lemma 7.12 (Certified convex recourse). *Fix d . If $\nabla_{RR}^2 F_0 \succeq 0$ on $[0, 1]^n$, a box-stable certified recourse oracle exists with bit work poly_d in the query length. Moreover, the returned lower bound can be taken as the tangent bound*

$$\ell = f(z) + \min_{y \in Y'} \nabla f(z)^\top (y - z), \quad f = F_\gamma(v, \cdot), \quad (35)$$

which a verifier checks by evaluating f and ∇f at the rational point z and choosing one endpoint per coordinate.

The proof (appendix E.4) uses weak convex optimization over a capped epigraph [24, Definition 2.1.10 and Corollary 4.2.7] to find a feasible point of small objective error; convexity alone makes (35) a global lower bound, and a smoothness bound along a feasible segment makes its gap small. No strong convexity modulus is used. The premise $\nabla_{RR}^2 F_0 \succeq 0$ says that the polynomial $q(x, y) = y^\top \nabla_{RR}^2 F_0(x) y$ is nonnegative on $[0, 1]^n \times \mathbb{R}^{n_R}$. The algorithms do not decide it; it is certified. One certificate writes $q = \sigma_0 + \sum_i \sigma_i x_i (1 - x_i)$ with sum-of-squares polynomials σ_0, σ_i in (x, y) given by rational Gram matrices; it is checked by a polynomial identity and exact rational semidefiniteness tests. A structural certificate writes F_0 as a sum of a polynomial in v , terms that are affine in z for fixed v , and even powers $(a^\top z + b(v))^{2p}$ with rational vectors a and polynomials b ; each term has a positive semidefinite residual Hessian. A sum-of-squares representation of F_0 itself is not a certificate: a nonnegative polynomial need not be convex.

Theorem 7.13 (Polynomial recourse with certified conditional values). *Fix d . Let F_0 be an explicit rational polynomial of degree at most d on $X = [0, 1]^n$, let C be a supplied core of size $k \geq 0$, and let $L > 0$ be rational with (28) (verified by monomial bounds or certified). Assume a box-stable certified recourse oracle; by lemma 7.12 one exists when $\nabla_{RR}^2 F_0 \succeq 0$ on X (certified). There is a power of two M , computed from the base instance with $\log_2 M \leq \text{poly}_d(I)$, such that for the ambient perturbation model with marginal law $U_{\sigma, M}$ the following hold.*

- (i) *On every draw the algorithm returns an exact global optimizer of F_γ on X . On ordinary draws the output is an implicit patch output (S, z_S, Q, g_0) of length $\text{poly}_d(I)$, with a global proof record. On the remaining draws it is the output of the exact fallback of theorem 3.20 on the same draw: an algebraic output (definition 2.5(d)) of the lexicographically least global minimizer and of the optimal value, whose length is at most $B \text{poly}_d(I)$ for a base-computable $B = 2^{\text{poly}_d(I)}$.*
- (ii) *The expected bit work and the expected size of the global proof record are at most $8^k Q_{\text{ap}} \text{poly}_d(I)$.*
- (iii) *A q -bit evaluation of an implicit patch output costs $\text{poly}_d(I + q)$ bit operations; the expected cost of a q -bit evaluation of the output, over all draws, is $\text{poly}_d(I + q)$.*

No growth, uniqueness, or nondegeneracy premise is used.

The ordinary branch decides no exact comparison between irrational conditional values. The patch descriptor determines one point because F_γ is strongly convex on Q , and the global proof record shows that this point is the global optimizer; the record contains the pruning trace and has only an expected size bound. An expanded algebraic representation of the optimizer can have large degree even on an ordinary draw; the output contract does not include one. For $k = 0$ there is no core, and each level makes one certified call on the whole residual box with accuracy 4^{-j} (31); the problem need not be convex, since the oracle is an assumption of the theorem. In the convex specialization of lemma 7.12 it is a convex polynomial on a box with ambient noise, and the patch certificate replaces any need to decide uniqueness or the active set.

The residual objective may be nonlinear, nonseparable, dense, and not strongly convex. For example, $F_0(v, z) = \psi(v) + \sum_i z_i^4 + \|Dz - Ev\|^2$ has convex recourse for any rational matrices D, E and any polynomial ψ . The core curvature bound need not reflect the size of the core-residual coupling, and a small core need not arise from a low-rank quadratic convexifier.

Proposition 7.14 (A core not given by a low-rank convexifier). *For integers $m \geq 1$ and rationals $\lambda, T > 0$, $t_0 \in [0, 1]$, $\beta \in \mathbb{Q}^m$, let*

$$F(t, z) = \frac{\lambda}{2}(t - t_0)^2 + T t P(z) - \beta^\top z, \quad P(z) = \sum_{i=1}^m z_i^4 + \left(\sum_{i=1}^m z_i \right)^4,$$

on $[0, 1] \times [0, 1]^m$. Then $\partial_{tt}F = \lambda$, the residual problem is convex for every t , and the core-residual cross derivatives grow linearly with T . If a symmetric positive semidefinite matrix K makes $F + \frac{1}{2}(t, z)^\top K(t, z)$ convex on the box, then $K_{zz} \succ 0$; in particular $\text{rank } K \geq m$.

[Theorem 7.13](#) applies to this family with one core coordinate and $L = \lambda$, so its numerical factor does not depend on T ; the coefficient T enters only the polynomial factor through its bit length. By the proposition, no fixed positive semidefinite quadratic correction of rank independent of m makes the objective convex. The proposition excludes only fixed quadratic corrections. A one-coordinate correction $Ct \log t$ with C of order $T \max P$ does convexify the family, so the example separates the theorem from small-rank quadratic convexifiers, not from all difference-of-convex representations. It is not a hardness result.

7.5 Core-only noise and strongly convex recourse

[Theorem 7.13](#) perturbs every coordinate. Residual noise is what produces full point growth and nonzero active gradients at residual bounds with high probability. Under core-only noise neither holds in general: the residual optimal set can be a segment on every draw, and residual bound multipliers can vanish identically. A residual modulus restores the patch closure developed here.

It is one sufficient condition for that closure, not a necessary condition for evaluating every full optimizer. The exact-arithmetic companion [\[2\]](#) already proves a full selected-point evaluator for residual-convex cubics on product boxes without a residual modulus or a joint convexifier. Its selector is the lexicographically least optimal core followed by the least-norm point in its optimal residual fiber. Under one base-selected finite core-noise law, point and value evaluation has expected cost $k^{O(k)}(1 + L/\sigma)^k \text{poly}(I + q)$. For general residual-convex quartics, the same companion gives the core-noise point obstruction reproduced in [theorem 9.10\(c\)](#). Thus the degree-four obstruction does not rule out the degree-three result. The theorem below treats fixed degree with a uniform residual modulus and returns an exact patch descriptor.

Theorem 7.15 (Core-only noise with strongly convex recourse). *Fix d . Let F_0 , $X = [0, 1]^n$, C , and L be as in [theorem 7.13](#), with $k \geq 1$ and $n_R \geq 1$, and let $\mu > 0$ be rational with*

$$\nabla_{RR}^2 F_0(x) \succeq \mu I \quad (x \in X) \quad (36)$$

(certified). There is a power of two M , computed from the base instance with $\log_2 M \leq \text{poly}_d(I)$, such that for the core-only perturbation model with marginal law $U_{\sigma, M}$ (all residual coefficients unperturbed) the conclusions (i)–(iii) of [theorem 7.13](#) hold. The ordinary output is an implicit patch output in which some original core and residual bounds may be fixed.

No residual interiority, strict complementarity, active-set, or growth premise is used, and residual bound multipliers may vanish identically. The numerical factor contains the core curvature L only. The modulus μ and the size of mixed derivatives enter the polynomial factor through the bit lengths of the derived constants. By [proposition 2.6](#), the exact sampled optimizer is within $k\sigma$ of $\min_X F_0$ in original objective value, independently of the number of residual variables. With $\sigma = \varepsilon/(2k)$ and an evaluation of accuracy $\varepsilon/2$, this certifies an ε -approximate solution of F_0 on every draw, with expected work $8^k[3 + (1 + k)kL/\varepsilon]^k$ times a polynomial in the input length. That is polynomial in $1/\varepsilon$ for fixed k and L , and polynomial in I and $1/\varepsilon$ only when L is polynomially bounded; a binary-encoded L can be exponentially large.

The class admits a fixed rank- k quadratic convexifier: the Schur complement of the residual block and [\(34\)](#) give $\nabla^2 F_0 + \alpha \text{diag}(I_C, 0) \succeq 0$ for $\alpha = K_2 + K_2^2/\mu$. Its curvature can be much larger than L . The contribution of [theorem 7.15](#) is that the numerical factor keeps L/σ under

core-only noise, despite changing and weakly active residual bounds; it is not that the class lacks a low-rank convexifier. For example, the Hessian of $F_0(v, z) = Lv^2/2 + T vz + \mu z^2/2$ is $\begin{pmatrix} L & T \\ T & \mu \end{pmatrix}$, so a core quadratic convexifier must have scale $\alpha \geq \max\{0, T^2/\mu - L\}$. The direct count uses L/σ even when this scale grows as T^2/μ . The companion's full selected-point theorem under a supplied convexifier instead uses $k^{O(k)}(1 + \alpha/\sigma)^k \text{poly}(I + q)$; it applies to cubic objectives convexified on the feasible polytope, or to fixed-degree objectives whose convexification is convex on all of $\mathbb{R}^n$. These output and convexity premises differ from the patch theorem here.

The proof (appendix E.6) has three new ingredients. First, the residual selector $s(v) = \arg \min_{z \in Y} F_0(v, z)$ is unique and (K_2/μ) -Lipschitz, the conditional value is continuously differentiable, and projected growth g_V of V at its optimal core lifts to full point growth $\min\{g_V/(1 + 2K_2^2/\mu^2), \mu/4\}$. The growth tail of theorem 3.16 applies to V on the k -dimensional core box. Second, core noise keeps the optimal core away from changes of the residual active set, with a coefficient-independent probability bound. Third, small residual multipliers can be released without losing a positive Hessian modulus. We state the last two ingredients because they are the reason the theorem needs no strict complementarity.

For a face K of $[0, 1]^k$ with $q \geq 1$ free coordinates, identified with a q -dimensional box, let $A_i^\ell = \{v \in K : s_i(v) = \ell_i\}$ and $A_i^u = \{v \in K : s_i(v) = u_i\}$, and define

$$S_K = \partial K \cup \bigcup_{i \in R} (\text{bd}_K A_i^\ell \cup \text{bd}_K A_i^u), \quad \mathcal{E}_K = -\nabla_K V_0(S_K) \subseteq \mathbb{R}^q, \quad (37)$$

where bd_K is the boundary relative to K , V_0 is the conditional value without noise, and ∇_K is the gradient in the free coordinates of K .

Proposition 7.16 (Active-set changes are rarely near the optimum). *In the setting of theorem 7.15, let K be a face of $[0, 1]^k$ with $q \geq 1$ free coordinates.*

- (a) S_K and $\mathcal{E}_K$ are compact semialgebraic sets of dimension at most $q - 1$. If $a \in \text{relint } K$ and $0 < \eta' < \text{dist}(a, S_K)$, the relative ball of radius η' about a lies in $\text{relint } K$, and the bound status of each residual coordinate of $s(v)$ is constant on it.
- (b) There is a base-computable D_* with $\log D_* \leq \text{poly}_d(I)$ such that $\mathcal{E}_K$ lies in the zero set of a nonzero polynomial of degree at most D_* .
- (c) If γ_K has independent coordinates with law $U_{\sigma, M}$, then for every $\delta \geq 0$

$$\Pr\{\text{dist}(\gamma_K, \mathcal{E}_K) \leq \delta\} \leq \min\left\{1, C(q, D_*)\left(\frac{\delta}{\sigma} + \frac{\sqrt{q}}{M}\right)\right\}, \quad C(q, D) = 16q^{q+1}D(4D + 2)^{q-1}.$$

At an optimal core $a \in \text{relint } K$ the free coordinates of the noise satisfy $\gamma_K = -\nabla_K V_0(a)$, so (c) bounds the probability that the optimal core is close to a change of the residual active set. Part (b) uses fixed-block quantifier elimination only as an analysis device: the algorithm never computes S_K or the polynomial. Part (c) combines a tube estimate for singular algebraic sets [9] with a jitter argument that covers every atom of the grid, including atoms on $\mathcal{E}_K$. The conclusion is a stable residual active set near the optimum; it gives no positive lower bound on active multipliers.

Proposition 7.17 (Releasing small residual multipliers). *In the setting of theorem 7.15, let $x^* = (a, s(a))$ with a in the relative interior of a face K with $q \geq 1$ free coordinates. Suppose the residual bound status is constant on the relative ball of radius $\eta > 0$ about a , $\nabla_K V_\gamma(a) = 0$, and*

$\nabla_K^2 V_\gamma(a) \succeq 2gI$ with $g > 0$. For an active residual coordinate i let $\lambda_i = \pm \partial_{z_i} F_0(x^*) \geq 0$ be its inward multiplier. Put $H = K_2/\mu$, $K_\lambda = \max\{1, K_3(1+H)^3\}$, and

$$\theta = \min\left\{\frac{K_\lambda \eta^2}{2}, \frac{\mu g}{4n_R K_\lambda}\right\}, \quad \nu = \min\left\{\frac{\mu}{2}, \frac{\min\{g, \mu/2\}}{1+2H^2}\right\}.$$

Let $\mathcal{J}$ be any set of active residual coordinates with $\lambda_i \leq \theta$. Then the Hessian of F_0 at x^* , restricted to the free coordinates of K , the residual coordinates strictly inside their bounds, and $\mathcal{J}$, is at least νI . The same holds for every principal submatrix.

The bound uses the two-sided ball in the free core directions: nonnegativity of a multiplier on both sides of a and a Hessian bound force a small multiplier to have a small gradient, which controls the cross terms between released coordinates and the core. The algorithm never identifies the active set or classifies multipliers. It fixes the coordinates whose derivative signs are certified, which include all active residual coordinates with $\lambda_i > \theta$ once the patch is small, and tests the Hessian on the rest.

Remark 7.18 (Interior recourse). If, in addition, every conditional residual minimizer is interior to the residual box (certified, for example by strict inward derivative signs on the residual facets), then no residual coordinate is active. After the active core coordinates are fixed, point growth alone gives a positive Hessian on all remaining coordinates, and the proof of [theorem 7.15](#) needs neither [proposition 7.16](#) nor [proposition 7.17](#). No quantitative distance from the residual bounds is used, and an accepted patch may touch them.

The following examples show which hypotheses cannot be dropped. Proofs are in [appendix E.7](#).

Example 7.19 (Flat residual fibers). On $[0, 1]^3$ with core v and residual (z_1, z_2) let $F_\gamma(v, z) = (v - \frac{1}{2})^2 + (z_1 - vz_2)^2 + \gamma v$. The residual Hessian is positive semidefinite, $\partial_{vv} F_0 \leq 4$, and for every $\gamma \in (-1, 1)$ the optimal core $a_\gamma = \frac{1}{2} - \frac{\gamma}{2}$ is unique with projected growth constant one. The optimal set $\{(a_\gamma, a_\gamma s, s) : s \in [0, 1]\}$ is a segment, and the full Hessian is indefinite at every point with $z_1 \neq vz_2$. Hence no full-dimensional box in X carries a convex restriction of F_γ . Residual convexity without a modulus does not support the closure of [theorem 7.15](#). If the residual term is replaced by $(z_1 - \frac{1}{2} - v(z_2 - \frac{1}{2}))^2 + (z_2 - \frac{1}{2})^4$, the residual problem is strictly convex with the unique interior minimizer $(\frac{1}{2}, \frac{1}{2})$ for every v , the optimizer $(a_\gamma, \frac{1}{2}, \frac{1}{2})$ is unique, and still the full Hessian is indefinite at points arbitrarily close to it. Strict residual convexity and interior uniqueness therefore do not replace the uniform modulus μ .

Example 7.20 (Vanishing multipliers). On $[0, 1]^2$ let $F_0(v, y) = \phi(v) + y^2$, with residual y . The residual modulus is 2, the selector is $s \equiv 0$, and the multiplier of the active bound $y = 0$ vanishes identically, on every draw of core-only noise. The coupled variant $F_0 = (v - \frac{1}{2})^2 + (y + (v - \frac{1}{2})^2)^2$ has multiplier $2(v - \frac{1}{2})^2$, which vanishes at an interior core point while the conditional value has quadratic growth. A closure that requires residual multipliers bounded away from zero fails here; [proposition 7.17](#) keeps such coordinates free.

Example 7.21 (Why active core coordinates are fixed first). On $[0, 1]^2$ let $F_\gamma(v, y) = \frac{1}{4}v^2 + vy + \frac{1}{2}y^2 + \gamma v$ with residual y . The selector is $y = 0$, $V_\gamma(v) = \frac{1}{4}v^2 + \gamma v$, the residual modulus is one, and the full Hessian $\begin{pmatrix} 1/2 & 1 \\ 1 & 1 \end{pmatrix}$ is indefinite. The multiplier $\lambda(v) = v$ is nonnegative on the feasible side only; at the core vertex $v = 0$ it vanishes while its derivative is one. For $\gamma > 0$ the optimum is $(0, 0)$, the core derivative there is γ , and the closure succeeds only because the sign test fixes $v = 0$ first. [Proposition 7.17](#) applies only in the free directions of the optimal core face.

7.6 Scope

All results in this section optimize the sampled objective exactly; none recovers an optimizer of F_0 . The core, the curvature bound L , the residual convexity or modulus, and the box-stable oracle are inputs. The results do not recognize a good core, and they do not decide residual convexity of an arbitrary polynomial. The residual set must not depend on the core (example 7.3), and the oracle must handle every residual subbox. The numerical factors depend on the magnitude of L/σ , so these are expected polynomial bounds only when that ratio is polynomially bounded. The certificates are exact and sound on every draw, while their early success, the stopping depth, and the size of the sampling grid rest on growth and active-gradient tails for the base-selected finite law (theorem 3.16 and lemma 3.17). All four results have polynomial sampling precision. For a fixed oracle implementation, corollary 3.21 therefore replaces the instance-selected grid size by one depending only on the input length, with the same output contracts and the same numerical factors. Under core-only noise, residual convexity without a modulus is not enough for the closure used here (example 7.19). The companion’s residual-convex cubic theorem shows that other completion certificates can evaluate full points without such a modulus; its general quartic obstruction prevents an unrestricted extension with the same polynomial evaluation guarantee unless the stated complexity consequences hold. For integer residual variables the patch and Hessian tests are replaced by label certificates; section 8 treats that case, together with core-only noise for integer flows.

8 Integer recourse, flows, and random components

This section treats residual variables that are native integers. The core search of section 7 applies unchanged, because it uses only the conditional value function on a fixed residual set. Closure changes. A patch and a Hessian test are meaningless for integer coordinates; instead, a few bound-restricted recourse calls certify that one residual integer label is optimal throughout the retained core hull, and an exact solver for the small continuous core completes the problem. Under core-only noise residual ties can persist on every draw, and for integer flows and totally unimodular systems the label certificate is replaced by one that works with the whole set of tied optimal labels. The section ends with two results of a different kind: exact optimization of mixed boxes under strong noise through random components, and an exact lattice argument for separable integer problems with a low-rank concave part.

Table 5 lists the results. The factor Q_{ex} is defined in (27), and c_d is the constant of theorem 8.1.

8.1 An exact solver for a small continuous core

All label certificates end with the same task: minimize an explicit polynomial exactly over a box of small dimension. A general exact method returns separate defining polynomials for the coordinates; pairing their roots costs a product of degrees, and it does not by itself say which coordinate roots belong to the same optimizer. The following solver returns all coordinates and the optimal value as polynomials in one common real algebraic number, with a base that is exponential only in the dimension.

Theorem 8.1 (Exact box solver through joint critical limits). *Fix $d \geq 1$, and let D be the least even integer greater than d . There are a constant c_d , depending only on d , and a deterministic algorithm with the following properties. The input is an explicit rational polynomial f of degree at most d in k variables and a nonempty closed rational box $B \subseteq \mathbb{R}^k$; let H be its length. The algorithm returns a global minimizer x^* of f on B and the value $f(x^*)$ in the algebraic format of definition 2.5(d): one*

Table 5: Integer recourse and random components. “Label + algebraic” is an integer residual label together with the algebraic output of [definition 2.5\(d\)](#) for the core and the optimal value; its size bound holds on every draw. The function $f_d(k)$ is effective.

Result	Noise; residual structure	$\log_2 M$	Expected work	Output
theorem 8.4	ambient; exact integer oracle on tightened bounds	$\text{poly}_d(I)$	$[8^k Q_{\text{ex}} + c_d^k] \text{poly}_d(I)$	label + algebraic
corollary 8.5	ambient; as above	$\text{poly}_d(I)$	$8^k Q_{\text{ex}} \text{poly}_d(I)$	label + implicit
theorem 8.9	core-only; convex integer flow, interior optimal cores	$\text{poly}_d(I)$	$[8^k Q_{\text{ex}} + c_d^k] \text{poly}_d(I)$	label + algebraic
theorem 8.12	core-only; convex integer flow	$f_d(k) \text{poly}_d(I)$	$f_d(k) Q_{\text{ex}} \text{poly}_d(I)$	label + algebraic
corollary 8.13	core-only; flow, bilinear coupling	$\text{poly}_d(I)$	$[8^k Q_{\text{ex}} + c_d^k] \text{poly}_d(I)$	label + algebraic
theorem 8.14	core-only; totally unimodular	as for flows	as for flows	label + algebraic
theorem 8.16	ambient, strong; any mixed box	any	see (40)	component
theorem 8.17	aligned; convex piecewise-polynomial unaries on an integer box	$\text{poly}(I)$	$8^k H_{\text{rat}} \text{poly}(I)$	rational, no fallback

real root θ of a squarefree integer polynomial P of degree at most $(D-1)^k$, given with a rational isolating interval, and polynomials $r_1, \dots, r_k, r_f \in \mathbb{Q}[T]$ of degree less than $\deg P$ with

$$x_i^* = r_i(\theta) \quad (i = 1, \dots, k), \quad f(x^*) = r_f(\theta), \quad r_f = f(r_1, \dots, r_k) \bmod P.$$

Coordinates whose interval is a point have constant maps. As an auxiliary representation, used only for comparisons, the algorithm also returns a nonzero integer polynomial with the root $f(x^*)$ and an isolating interval of that root. The bit work and the output length are at most $c_d^k \text{poly}_d(H)$. For every $q \geq 0$, a rational point within Euclidean distance 2^{-q} of x^* , a feasible rational point with certified objective gap at most 2^{-q} , and a rational interval of width at most 2^{-q} containing $f(x^*)$ are computed within $c_d^k \text{poly}_d(H+q)$. No genericity is assumed: ties, singular critical points, and positive-dimensional sets of minimizers are allowed.

For each face of the box, the solver deforms the restricted objective by $\varepsilon \sum_i x_i^D$. Written in $z = 1/\varepsilon$, the deformed stationary equations have the pure-power leading monomials x_i^{D-1} over the field of rational functions in z and after every finite specialization of z . In particular this proves the claim for symbolic ε and for every $\varepsilon \neq 0$. They therefore form a Gröbner basis, and the quotient algebra has the fixed dimension $(D-1)^r$ on a face with r free coordinates. At $\varepsilon = 0$ the deformation disappears and the original stationary set can be positive dimensional; the limit $\varepsilon \rightarrow 0$ is handled by a branch analysis, not by specializing. View the complex critical points of the deformed problem as branches parametrized by ε . Call a linear form $\lambda^\top x$ *good* if it does not vanish on the leading coefficient vector of any branch that escapes to infinity as $\varepsilon \rightarrow 0$, and if it separates the distinct finite limits of the bounded branches. For a good form, the leading coefficient in z of the characteristic polynomial of $\lambda^\top x$ in the quotient algebra vanishes exactly at the projections of these finite limits, and its derivatives with respect to independent coefficients of λ , divided by a greatest common divisor, recover all coordinates of each limit as polynomials in its projection. For other forms the leading coefficient can retain bounded projections of escaping branches, so its roots need not be projections of limits. A deterministic list of $O(r(D-1)^{2r})$ moment-curve

forms contains a good form. Every listed form is screened by degree and divisibility tests, and a point reconstructed from any form is kept only after an exact test that it lies in the box. A bad form that passes the screening can therefore add only feasible candidates, whose values are at least the minimum. The deformed minimizers have a convergent subsequence whose limit is a global minimizer of f , and a good form recovers it, so the kept candidate of least value is optimal. The proof is in [appendix F.1](#); it uses finite-dimensional quotient algebras, simultaneous triangularization, Puiseux series, and exact univariate algorithms, and [lemma F.9](#) gives the explicit choice $c_d = 2^{10000D}$ for one elementary implementation. That constant is a proved bound used to define noise laws below, not a practical estimate. The ingredients are classical. Recovering all coordinates from one univariate polynomial and derivatives of a characteristic polynomial is the mechanism of the rational univariate representation [\[40\]](#), and deformations of stationary systems are standard in critical-point methods [\[10\]](#). We give a self-contained proof for boxes, including singular and positive-dimensional critical sets, because the recourse theorems need its exact contract and its constant-base bound; we do not claim a new general algebraic complexity result.

Corollary 8.2 (Mixed boxes and exact comparisons). *For a mixed box with k continuous coordinates and native integer coordinates with $N_1, \dots, N_t$ labels, enumerating the labels and applying [theorem 8.1](#) to each gives an exact global minimizer and value, in the algebraic format with the integer labels listed, in $c_d^k \prod_i N_i \text{poly}_d(H)$ bit operations. Values of two algebraic outputs of [theorem 8.1](#) can be compared exactly, including equality, within $c_d^k \text{poly}_d(H)$; the comparison uses the two auxiliary value polynomials and forms no field containing both.*

8.2 Native integer recourse

Definition 8.3 (Native integer recourse). A *native integer recourse instance* is a core-recourse instance ([definition 7.1](#)) whose residual set is

$$Y = \{z \in \mathbb{Z}^{n_R} : Az \leq b, \ell \leq z \leq u\},$$

with a rational matrix A , a rational vector b , and integer bounds encoded in binary; the curvature bound [\(28\)](#) is required on $[0, 1]^k \times \prod_i [\ell_i, u_i]$ (verified by monomial bounds or certified). Put $N_Z = \prod_i (u_i - \ell_i + 1)$ and $S = k + \sum_i (u_i - \ell_i)$. A *box-stable exact integer oracle* receives a rational core point v , rational linear coefficients, and integer bounds $\ell \leq \ell' \leq u' \leq u$, and returns the exact optimal value of $\min\{F_\gamma(v, z) : z \in Y, \ell' \leq z \leq u'\}$ and an attaining integer point, or reports infeasibility, in bit work polynomial in the query length with an exponent independent of k .

The core changes costs, never feasibility: supplies, right-hand sides, and bounds do not depend on v . The residual set is nonempty by [definition 7.1](#); one oracle call with the original bounds decides this before sampling, and an empty Y is reported as such. Exact values are rational because the core point is rational and residual points are integral. The oracle must handle arbitrary tightened coordinate bounds, which is how competing labels are excluded.

Theorem 8.4 (Native integer recourse). *Fix d . Let a native integer recourse instance with $k \geq 1$ be given, together with a box-stable exact integer oracle. There is a power of two M , computed from the base instance with $\log_2 M \leq \text{poly}_d(I)$, such that for the ambient perturbation model with marginal law $U_{\sigma, M}$ the following hold.*

- (i) *On every draw the algorithm returns a global minimizer (v^*, z^*) of F_γ on $X = [0, 1]^k \times Y$ and the optimal value: the label z^* exactly, and v^* and the value in the algebraic format of [theorem 8.1](#). Its length is at most $c_d^k \text{poly}_d(I)$ on every draw, and a q -bit refinement costs $c_d^k \text{poly}_d(I + q)$.*

- (ii) The expected bit work and the expected size of the global proof record are at most $[8^k Q_{\text{ex}} + c_d^k] \text{poly}_d(I)$.
- (iii) If F_0 is quadratic, the output is rational on every draw and the expected work is $[8^k Q_{\text{ex}} + 3^k] \text{poly}(I)$.

No growth, label-gap, uniqueness, or residual convexity premise is used.

For $k = 0$ there is no core, and one oracle call with the original bounds solves the sampled problem exactly. For $k \geq 1$ the certificate is the following. At the incumbent corner c_j with label z_j , the $2n_R$ restrictions $z_i \leq z_{j,i} - 1$ and $z_i \geq z_{j,i} + 1$, each together with the original constraints, have union exactly $Y \setminus \{z_j\}$, however the residual coordinates are coupled. A restriction that leaves the original bounds ($z_{j,i} = \ell_i$ in the first case, $z_{j,i} = u_i$ in the second) is empty and receives the value $+\infty$ without an oracle call, as does a restriction that the oracle reports infeasible. Let $V_{\text{other}}(c_j)$ be the least of the $2n_R$ values; it is $+\infty$ when all restrictions are empty, that is, when z_j is the only feasible label; the minimum of an empty restriction list is also $+\infty$. If

$$V_{\text{other}}(c_j) - V(c_j) > 2G w_{\infty}(\mathcal{H}_j), \quad (38)$$

then z_j is the unique optimal label at every core point of the retained hull $\mathcal{H}_j$ (lemma 7.2(c)), so every global minimizer lies in $\mathcal{H}_j \times \{z_j\}$. The whole slice $[0, 1]^k \times \{z_j\}$ is feasible and contains a global minimizer, so theorem 8.1 applied to $F_{\gamma}(\cdot, z_j)$ on $[0, 1]^k$ finishes the problem, whether the core minimizers are unique, tied, or form a continuum. Feasible values for the restricted problems would not suffice; (38) needs their exact optimal values. Distinct integer labels are at unit distance, so point growth forces (38) at a base-computed level; no label gap is supplied or charged separately. On the remaining draws the algorithm enumerates the N_Z labels on the same draw, solves each core slice, and keeps the best by exact comparisons (corollary 8.2). It returns only the winning label and its core representation, so the output bound does not depend on how many labels were examined. The proof is in appendix F.2.

Corollary 8.5 (Implicit core output). *Under the hypotheses of theorem 8.4, an algorithm with expected bit work $8^k Q_{\text{ex}} \text{poly}_d(I)$ returns on ordinary draws the exact label z^* and an implicit patch output for the core slice $F_{\gamma}(\cdot, z^*)$, and on the remaining draws the output of theorem 8.4. A q -bit evaluation of the ordinary output costs $\text{poly}_d(I + q)$, and its expected cost over all draws is $\text{poly}_d(I + q)$.*

The variant replaces the exact core solve by the sign and Hessian tests of lemma 7.7 on the core coordinates of the certified slice; it pays for an additional active-gradient event instead of the term c_d^k .

Separable convex recourse over totally unimodular systems. The main source of exact integer oracles in this paper is the following classical result.

Proposition 8.6 (Separable convex integer oracles). *Let $Y = \{z \in \mathbb{Z}^{n_R} : Az = b, \ell \leq z \leq u\}$, where A is an integer totally unimodular matrix (promised; automatic for network matrices), b is integral, and the bounds are integral and finite. Let $F_0(v, z) = \phi(v) + \sum_i f_i(v, z_i)$ with explicit rational polynomials of fixed degree, and suppose $t \mapsto f_i(v, t)$ is convex on $[\ell_i, u_i]$ for every $v \in [0, 1]^k$ (promised, or certified). Then a box-stable exact integer oracle exists for every rational linear perturbation, with bit work polynomial in the query length and logarithmic dependence on the numerical bounds. This includes integer flows on a directed network with integral supplies and arc bounds; inequality systems $Cz \leq b$ with C totally unimodular reduce to this form (corollary 8.15).*

The oracle is the proximity-scaling algorithm of Hochbaum and Shanthikumar [25, Theorem 4.3], which solves separable convex integer minimization over totally unimodular systems with logarithmically many linear programs in the numerical bounds and function evaluations at prescribed points. At a rational core point all evaluations are rational of polynomial length; tightening coordinate bounds and adding linear terms keep the class; and convex tangent extensions beyond the native intervals supply evaluations if the algorithm requests them (appendix F.2). The convexity premise concerns each scalar cost on its whole real interval. For networks, optimality also has a short certificate.

Lemma 8.7 (Marginal potentials). *Let Y be the integer flows of a directed network with s nodes, fixed integral supplies, and arc bounds, where parallel arcs and loops are allowed, and let each $t \mapsto f_a(t)$ be convex on $[\ell_a, u_a]$. A feasible z minimizes $\sum_a f_a(z_a)$ over Y if and only if there are node potentials π for which every arc of the residual network of z has nonnegative reduced cost, where a forward arc $a = (p, p')$ with $z_a < u_a$ costs $f_a(z_a + 1) - f_a(z_a)$ and a backward arc with $z_a > \ell_a$ costs $f_a(z_a - 1) - f_a(z_a)$, and the reduced cost adds $\pi_p - \pi_{p'}$ along the residual arc from p to p' . If the residual arc costs are rational numbers of bit length at most H_m , shortest-path distances from an added zero-cost source supply such potentials, of bit length polynomial in s and H_m .*

This is a certificate, not an algorithm: repeated unit augmentation is not claimed to run in polynomial time. The characterization holds for arbitrary real convex costs. The bit-length statement needs rational marginal costs of polynomial length; explicit rational polynomials of fixed degree have them at rational core points, while an arbitrary convex real function need not. With proposition 8.6, theorem 8.4 applies to separable convex integer flows and totally unimodular systems whose costs depend arbitrarily on the core. For instance, $F_0(v, z) = \phi(v) + \sum_a \psi_a(z_a) + v^\top Wz$ with convex ψ_a allows arbitrary core-to-arc coupling W ; the curvature bound L is that of ϕ , and W enters only through its bit length. Convexity of a fixed-degree univariate ψ_a on its interval can be verified by univariate sign determination.

8.3 Core-only noise for integer flows

Under core-only noise the residual costs are not perturbed, and many optimal labels can persist on every draw. Full point growth then fails, and the competing-label certificate cannot pass when two labels are optimal at the optimal core. Throughout this subsection the instance is a core-recourse instance (definition 7.1) in which Y is the set of integer flows of a directed network with s nodes and n_R arcs (parallel arcs and loops allowed), integral supplies, and integral arc bounds $[\ell_a, u_a]$ in binary; in particular Y is nonempty. Nonemptiness is decided before sampling by one exact feasibility linear program, whose vertices are integral by total unimodularity, and an infeasible network is reported as such. The sampled objective is

$$F_\gamma(v, z) = \phi(v) + \sum_a f_a(v, z_a) + \gamma^\top v, \quad v \in [0, 1]^k, \quad z \in Y, \quad (39)$$

with explicit rational polynomials of degree at most $d \geq 1$ in expanded form, each $f_a(v, \cdot)$ convex on $[\ell_a, u_a]$ for every $v \in [0, 1]^k$ (promised or certified), and a curvature bound L as in (28) (verified or certified). Only the k core coefficients are perturbed. By proposition 2.6 the sampled optimizer is within $k\sigma$ of $\min_X F_0$ in original objective value, whatever the size of the network.

A deterministic baseline shows what the noise adds. It uses no flow structure: exact or certified conditional values on a fixed core grid give an additive approximation for every core-recourse instance.

Corollary 8.8 (A deterministic core grid). *Let a core-recourse instance (definition 7.1) with $k \geq 1$ and a fixed rational linear term γ , for instance $\gamma = 0$, be given, together with a conditional oracle (section 7.1) that answers either exactly, in which case put $\eta = 0$, or in certified mode with a fixed accuracy $\eta > 0$. For an integer $m \geq 1$, query the oracle at all $(m+1)^k$ points of $G_m = \{0, \frac{1}{m}, \dots, 1\}^k$, let $(\hat{v}, \hat{z})$ be a returned completion of least upper value, and put $\underline{f} = \min_{v \in G_m} \ell(v) - kL/(8m^2)$. Then $(\hat{v}, \hat{z}) \in X$ and*

$$\underline{f} \leq f^* \leq F_\gamma(\hat{v}, \hat{z}) \leq \underline{f} + \frac{kL}{8m^2} + \eta.$$

For every $\varepsilon > 0$, $m = \lceil \sqrt{kL/(8\varepsilon)} \rceil$ gives an additive gap at most $\varepsilon + \eta$ with $(m+1)^k$ oracle calls. For integer flows, proposition 8.6 supplies exact values.

The grid rounds only the k core coordinates. The error and the number of calls do not depend on the number of residual variables, which the oracle optimizes exactly; rounding them as well would reintroduce a dimension-dependent error (example 7.4). The corollary is deterministic and uses no noise; its proof is in appendix F.3. It is an approximation, not an exact method. With $\sigma = \varepsilon/(2k)$ and an evaluation of accuracy $\varepsilon/2$, the core-only theorems below also certify an ε -approximate solution of F_0 on every draw (proposition 2.6), but their expected work contains the factor $[3 + (1 + k/2)kL/\varepsilon]^k$, whose dependence on $1/\varepsilon$ is worse than the $(2 + \sqrt{kL/(8\varepsilon)})^k$ calls of the grid. The smoothed theorems do not improve additive approximation of F_0 ; what they add is exact optimization of the sampled objective, with exact output, on every draw.

A natural exact certificate selects one flow and proves it optimal on the retained hull by polynomial potentials. Boundary core optima defeat it: proposition 9.11 gives two parallel arcs and a two-dimensional core on which, with probability $\frac{1}{16}$ for every grid size, the optimal core is a vertex with projected growth constant at least one, both flows are optimal there, and no single flow is optimal on any box at that vertex, while the core search retains such a box at every level. With m stages, an algorithm that insists on a single-flow certificate and enumerates labels when it fails does work at least $2^m/16$ in expectation. The obstruction concerns that certificate; the instance itself is easy. Two remedies follow: a premise that all optimal cores are interior, and a certificate that uses all tied flows at once.

Theorem 8.9 (Core-only flow recourse with interior optimal cores). *Fix d , and assume, in addition to the model (39) with $k \geq 1$, that for every $\gamma \in [-\sigma, \sigma]^k$ every optimal core point of F_γ lies in $(0, 1)^k$ (promised, or certified, for instance by $\partial_i F_0 \leq -\sigma - \tau'$ on $\{v_i = 0\}$ and $\partial_i F_0 \geq \sigma + \tau'$ on $\{v_i = 1\}$ for all z in the bounding box $\prod_a [\ell_a, u_a]$ and some rational $\tau' > 0$). There is a power of two M , computed from the base instance with $\log_2 M \leq \text{poly}_d(I)$, such that for core-only noise with marginal law $U_{\sigma, M}$ the algorithm returns on every draw an optimal integer flow and an algebraic output of one optimal core and the optimal value, of length $c_d^k \text{poly}_d(I)$, with q -bit refinement in $c_d^k \text{poly}_d(I + q)$, and with expected bit work $[8^k Q_{\text{ex}} + c_d^k] \text{poly}_d(I)$. For quadratic objectives the output is rational.*

The interiority premise is used only in the running-time analysis; correctness holds on every draw without it, and its sufficient certificate enters no numerical factor. The premise is not vacuous because Y is nonempty. At the incumbent corner the algorithm computes an optimal flow, a tight shortest-path tree of its residual network, and the corresponding polynomial potentials, and it tests exactly whether every residual reduced cost is nonnegative on the whole retained hull; identically zero reduced costs pass, so persistent ties are allowed. If the test passes, the flow is optimal on the hull, and the exact solver finishes its whole core slice. The analysis uses a finite family of reduced-cost polynomials over all labels and trees, fixed before sampling: an optimal core in the interior is a stationary point of every attaining fixed-flow slice, so the noise equals a negative gradient

there, and the event that the optimal core is near a zero of a chart polynomial has small probability by an algebraic tube estimate.

All tied flows at once. At a core point c on a face $\mathcal{A}$ of the core cube, let z^0 be an optimal flow and π potentials certifying it (lemma 8.7). The adjusted costs $g_a(c, t) = f_a(c, t) + (\pi_p - \pi_{p'})t$ of the arcs $a = (p, p')$ are convex, and their integer minimizers form intervals $I_a \ni z_a^0$, found by binary search on monotone first differences. The set $Y_0 = \{z \in Y : z_a \in I_a \text{ for all } a\}$ is exactly the set of optimal flows at c : the potential terms sum to a constant over Y , and the objective gap is a sum of nonnegative scalar gaps. Minimizing an inward core derivative over Y_0 is again a convex flow problem, because on an interval with at least three integers the adjusted cost is constant, so the original arc cost is affine there and its inward derivative is convex; two-point intervals use linear interpolation, and singleton intervals are substituted. Finally, every feasible flow is close to Y_0 .

Lemma 8.10 (Proximity to the optimal set). *For every $z \in Y$ there is $\bar{z} \in Y_0$ with $\|z - \bar{z}\|_1 \leq n_R \sum_a \text{dist}(z_a, I_a)$.*

The deterministic certificate uses these facts on a rational core box $\mathcal{D}$ whose intervals in the normal coordinates of $\mathcal{A}$ contain the corresponding bounds, with projection $\mathcal{D}_{\mathcal{A}}$ onto the face and $c \in \mathcal{D}_{\mathcal{A}}$. Let $\varsigma_i = +1$ for a normal coordinate fixed at 0 and $\varsigma_i = -1$ for one fixed at 1, let T_1 and T_∞ be the sum and the maximum of the normal widths of $\mathcal{D}$, let $R_{\mathcal{A}}$ bound $\|w - c\|$ on $\mathcal{D}_{\mathcal{A}}$, and let $H \geq 1$ and $K \geq 0$ be rational bounds on $\|\nabla_v^2 F_0\|$ and on $|\partial_{v_i} \partial_{z_a} f_a|$ on the real bounding domain.

Theorem 8.11 (An optimal-face certificate). *Suppose that, with the potentials taken as polynomials of the core obtained from a tight shortest-path tree at c :*

- (a) *for every arc, $g_a(w, t) = g_a(w, z_a^0)$ for all $w \in \mathcal{D}_{\mathcal{A}}$ and all integers $t \in I_a$;*
- (b) *every first adjusted marginal outside I_a that stays in the native interval is at least $\mu_{\text{out}} = n_R K T_1$ on $\mathcal{D}_{\mathcal{A}}$; and*
- (c) *if $\mathcal{A}$ has normal coordinates, $\beta = \min_i \beta_i$ satisfies $\beta - H R_{\mathcal{A}} - \frac{H}{2} T_\infty > 0$, where $\beta_i = \min_{z \in Y_0} \varsigma_i \partial_{v_i} F_\gamma(c, z)$.*

Then every flow of Y_0 is optimal at every point of $\mathcal{D}_{\mathcal{A}}$, and every point of $\mathcal{D} \times Y$ off the face is strictly worse than some feasible point on it. If $\mathcal{D}$ contains the core of every global minimizer, the slice $[0, 1]^k \times \{z^0\}$ contains a global minimizer. Conditions (a)–(c) are decided exactly, (a) by at most $d + 1$ polynomial identities per arc, (b) by exact polynomial minimization on a rational box (theorem 8.1), and (c) by one convex flow problem per normal coordinate.

The certificate is sound on every draw and needs neither uniqueness of the optimal flow nor a noise assumption. Both (b) and (c) are needed: (c) uses the minimum over *all* optimal flows, since a single tied flow can have the wrong inward sign, and without (b) a flow that loses at c can become optimal off the face. In the instance of proposition 9.11 the certificate with the origin face closes every box of maximum width less than $2 \min_i \gamma_i$.

Theorem 8.12 (Core-only flow recourse at arbitrary core faces). *Fix d . For the model (39) with $k \geq 1$ there are an effective function f_d and a power of two M , computed from the base instance with*

$$\log_2 M \leq f_d(k) \text{poly}_d(I),$$

such that for core-only noise with marginal law $U_{\sigma, M}$ the algorithm returns on every draw an optimal integer flow and an algebraic output of one optimal core and the optimal value, of

length $f_d(k) \text{poly}_d(I)$, with q -bit refinement in $f_d(k) \text{poly}_d(I + q)$, and with expected bit work $f_d(k) Q_{\text{ex}} \text{poly}_d(I)$. No interiority, growth, or uniqueness premise is used, and arbitrarily many optimal flows may persist.

At each level the algorithm tries the certificate of [theorem 8.11](#) on every face of the core cube compatible with the retained hull, at the midpoint of the projected hull, and solves the whole core slice of z^0 when one passes. The analysis needs three probability bounds: projected growth, distance of the optimal core from the zeros of a base-only family of chart polynomials (tube estimate), and a margin τ for the minimal inward derivative over all optimal flows. The last is obtained without conditioning on a random label: on each fixed face, the optimal core and its whole optimal flow set do not depend on the normal noise coefficients. A tube controls the distance from a zero set, not the value of a polynomial away from it. Converting distance into a value margin uses fixed-block quantifier elimination with its coefficient-height bound, and costs a precision of $f_d(k) \text{poly}_d(I)$ bits. This is why the sampling precision in [theorem 8.12](#) depends on k .

Corollary 8.13 (Bilinear coupling). *If $F_\gamma(v, z) = \phi(v) + \sum_a \psi_a(z_a) + v^\top Wz + \gamma^\top v$ with a rational matrix W , fixed degree $d \geq 2$, convex ψ_a on $[\ell_a, u_a]$, and $\partial_{ii}\phi \leq L$ on $[0, 1]^k$, then the conclusions of [theorem 8.12](#) hold with $\log_2 M \leq \text{poly}_d(I)$, output length $c_d^k \text{poly}_d(I)$, refinement $c_d^k \text{poly}_d(I + q)$, and expected bit work $[8^k Q_{\text{ex}} + c_d^k] \text{poly}_d(I)$. For quadratic ϕ the output is rational.*

All chart polynomials are then affine in the core, and their value margins have polynomial length. Let p be such a chart, nonzero, with coefficients of bit length at most H_0 , on a face identified with $[0, 1]^q$. If its zero set Z in the cube is nonempty, then p is not constant and $|p(x)| \geq b_{\min} \text{dist}(x, Z)$, where $b_{\min} \geq 2^{-H_0}$ is the least absolute value of a nonzero linear coefficient. If p is a nonzero constant or has no zero in the cube, then $|p|$ is at least its least absolute vertex value, a positive rational at least $2^{-(q+1)H_0}$; this case includes vertex faces. Identically zero charts are discarded from the exceptional family, and they pass the non-strict certificate tests. The curvature in the numerical factor is that of ϕ alone. This would fail for nonlinear coupling: a term $v_i^2 z_a^2$ makes the core curvature grow quadratically with the capacity of arc a .

8.4 Totally unimodular systems

Theorem 8.14 (Totally unimodular recourse). *Fix d and $k \geq 1$. Let $Y = \{z \in \mathbb{Z}^{n_R} : Az = b, \ell \leq z \leq u\}$ be nonempty, where A is an integer totally unimodular matrix (promised), b is integral, and the bounds are finite integers encoded in binary. Let*

$$F_\gamma(v, z) = \phi(v) + \sum_i f_i(v, z_i) + \gamma^\top v \quad (v \in [0, 1]^k, z \in Y),$$

where ϕ and the f_i are explicit rational polynomials of degree at most d in expanded form, each $f_i(v, \cdot)$ is convex on $[\ell_i, u_i]$ for every $v \in [0, 1]^k$ (promised or certified), $L > 0$ is a rational bound [\(28\)](#) on $[0, 1]^k \times \prod_i [\ell_i, u_i]$ (verified or certified), and only the k core coefficients are perturbed, with law $U_{\sigma, M}$. Then the conclusions of [theorem 8.12](#) hold. If moreover $F_\gamma = \phi(v) + \sum_i \psi_i(z_i) + v^\top Wz + \gamma^\top v$ with the premises of [corollary 8.13](#), its conclusions hold.

These are the premises of the flow model, with the network replaced by a totally unimodular system; certificates are part of the input length, and nonemptiness is decided by one linear program. The premises are used: the proof evaluates and differentiates explicit rational polynomials at rational core points, needs rational unit marginals of polynomial length for its charts, and uses L for the count. Total unimodularity with arbitrary convex costs, or with succinctly encoded costs, does not provide these interfaces. Two further ingredients replace network structure. Shortest-path potentials

become a dual vector satisfying at most $2n_R$ adjacent-slope inequalities, which exist because the piecewise-linear interpolation of the separable costs has an integral optimum over the real totally unimodular polytope; a dual vertex in a pre-draw box has inverse-basis entries in $\{0, \pm 1\}$, which bounds the chart coefficients. Directed cycles become conformal circuits of $\ker A$ with entries in $\{0, \pm 1\}$, which gives [lemma 8.10](#) with the same factor n_R . Both ingredients use total unimodularity; neither holds for general integer matrices.

Corollary 8.15 (Bounded totally unimodular inequalities). *Under the premises of [theorem 8.14](#) on the core and the costs, its conclusions hold for a nonempty $Y = \{z \in \mathbb{Z}^{n_R} : Cz \leq b, \ell \leq z \leq u\}$ with C totally unimodular, b integral, and finite integral bounds.*

Add the slack $s_i = b_i - C_i z$ with the explicit bounds $0 \leq s_i \leq b_i - \min_{\ell \leq z \leq u} C_i z$; the matrix $[C \ I]$ is totally unimodular, projection is a bijection onto Y , and slacks have zero cost, so the curvature bound and all objective values are unchanged. A negative slack bound proves infeasibility.

8.5 Strong noise and random components

The previous results allow every noise scale $\sigma > 0$, with numerical ratios such as L/σ in their bounds, and they need a small core. The following result needs neither a core nor a decomposition, but it requires a lower bound on the noise: noise that dominates the variation of every coordinate derivative. Let $X = \prod_i X_i$ be a mixed box ([section 2.1](#)) with n coordinates after preprocessing, and let F_0 be an explicit rational polynomial of degree at most d . If $n = 0$ the objective is a constant, which is returned directly; assume $n \geq 1$. The *primal graph* joins two coordinates when they occur together in a nonzero monomial, so each monomial support is a clique; let $\Delta_+ = \max\{1, \text{maximum degree}\}$. Before sampling, compute rational enclosures $\partial_i F_0 \in [\underline{\partial}_i, \bar{\partial}_i]$ on the continuous hull, by multiplying exact interval ranges of univariate powers over each derivative monomial. Put $R_i = \bar{\partial}_i - \underline{\partial}_i$,

$$a_i = \begin{cases} c_d, & i \text{ continuous,} \\ u_i - \ell_i + 1, & i \text{ native integer,} \end{cases} \quad q_i = U_{\sigma, M}([- \bar{\partial}_i, - \underline{\partial}_i]), \quad \beta = \max_i a_i q_i,$$

where c_d is the constant of [theorem 8.1](#); for quadratic objectives one may take $a_i = 3$ for continuous coordinates. Each q_i is an exactly computable rational number, which depends on M .

Theorem 8.16 (Strong fields and random components). *Fix d , and let $M \geq 2$ be any power of two. Under the ambient perturbation model with marginal law $U_{\sigma, M}$, a deterministic algorithm returns on every draw an exact global minimizer of F_γ on X in the component format of [definition 2.5\(e\)](#): rational values for the pinned coordinates, and for each remaining component its integer labels and one common-root algebraic representation of its continuous coordinates and value. If $4\Delta_+\beta < 1$, its expected bit work is at most*

$$\text{poly}_d(I + \log_2 M) \left[1 + \frac{\sum_i a_i q_i}{1 - 4\Delta_+\beta} \right]. \quad (40)$$

A q -bit evaluation of the output, that is, a rational point within Euclidean distance 2^{-q} of the represented optimizer and a rational interval of width at most 2^{-q} containing the optimal value, together with a feasible rational point whose objective gap is certified to be at most 2^{-q} , has expected cost bounded by (40) with $\text{poly}_d(I + \log_2 M + q)$ in place of $\text{poly}_d(I + \log_2 M)$. For quadratic objectives the output is an exact rational optimizer and value, with $a_i = 3$ for continuous coordinates. A sufficient condition for a bounded denominator is

$$\sigma \geq 8\Delta_+ \max_i (a_i R_i), \quad M \geq 16\Delta_+ \max_i a_i, \quad (41)$$

which gives $a_i q_i \leq 1/(8\Delta_+)$ and $4\Delta_+\beta \leq \frac{1}{2}$.

Outside the closed interval in the definition of q_i , the sampled derivative $\partial_i F_\gamma$ has a strict sign throughout the continuous hull, so every global minimizer has x_i at the corresponding bound; this holds for integer coordinates as well, by monotonicity between labels. Call a coordinate *bad* when its coefficient lies inside that closed interval, so this sign test leaves it unpinned. These bad events depend on one coefficient each, so they are independent Bernoulli events with the exact probabilities q_i ; recomputing the enclosures after pinning would destroy this independence and is not needed. After pinning, every surviving monomial lies in one connected component of the graph induced by the bad coordinates, so the objective splits into independent component problems, each solved exactly by [corollary 8.2](#) or, for quadratics, by face enumeration. At most $(4\Delta_+)^{t-1}$ connected sets of size t contain a given vertex, and a subcritical branching bound gives (40). The q -bit evaluation refines each of the at most n components with $q + \lceil \log_2(n+1) \rceil$ bits, for its point, its value, and its certified objective gap separately. Combining their point errors in Euclidean norm and adding their value widths and gaps gives the stated joint contract; an objective gap alone would not bound the distance to the optimizer on a flat or singular component. The proof is in [appendix F.5](#).

The algorithm is exact on every draw, also when $4\Delta_+\beta \geq 1$, but the theorem then gives no useful bound. The probabilities q_i are not monotone in M : grids of sizes M and $2M$ are not nested, so an interval without atoms of one grid can contain atoms of the other. A value of β computed for one grid size therefore does not transfer to another; q_i and β must be computed for the M actually used. The sufficient condition (41), in contrast, holds for every larger power of two. This sufficient regime scales with the full derivative variation, with the graph degree, and, for integer coordinates, with the numerical number of labels; for continuous coordinates of nonquadratic objectives it scales with the large effective constant c_d . The total value is reported as a rational constant plus a sum of component values. Expanding that sum into one defining polynomial can have degree exponential in the number of components, and exact sign or equality tests for sums of unrelated algebraic numbers are not claimed; rational enclosures of the total are obtained by refining each component. The regret bound of [proposition 2.6](#) is $\sigma \sum_i w_i$, which is large in this regime.

8.6 Separable integer problems with a low-rank concave part

The last result uses the row-scaled aligned extension described after [definition 2.2](#) and an exact argument of a different kind: the original objective values of the sampled problem lie on a lattice whose spacing is controlled by the same grid that controls the expected count. Let $X = \prod_{j=1}^n X_j$ with $X_j = \{L_j, \dots, U_j\}$ and integer bounds $L_j \leq U_j$ in binary, and

$$F(x) = \sum_{j=1}^n g_j(x_j) - \frac{\alpha}{2} \|Tx\|^2, \quad (42)$$

with a rational $\alpha > 0$ and an arbitrary supplied $T \in \mathbb{Q}^{k \times n}$, whose rows may be zero, linearly dependent, or constant on X . Fix a degree bound d_* . Each g_j is a continuous convex function on $[L_j, U_j]$, given by an explicit list of polynomial pieces of degree at most d_* , in expanded monomial form with rational coefficients, on consecutive intervals with rational breakpoints that cover $[L_j, U_j]$; a single polynomial is a list with one piece. Fixed-degree shifted or factored pieces can first be expanded in polynomial time; their expanded coefficients define the denominator budget. Convexity is verified: continuity and the order of the one-sided derivatives at each breakpoint are rational evaluations, and nonnegativity of the second derivative of each piece on its interval is a univariate sign determination. Smoothness at the breakpoints is not assumed. For $k \geq 1$ choose a rational $\sigma_i > 0$ for each row, let $\ell_i = \min_X (Tx)_i$, $u_i = \max_X (Tx)_i$, $w_i = u_i - \ell_i + 2\sigma_i/\alpha$, and $s = \max_i w_i$. Let Q_{den} be the product of the denominators of all base rational data, with the pieces in expanded

form, put $D_0 = 2Q_{\text{den}}^3$, and let M be the least power of two with $M \geq \max\{2, k\alpha s^2 D_0\}$. The sampled objective is $F_\xi(x) = F(x) + \xi^\top Tx$, where $\xi_1, \dots, \xi_k$ are independent and ξ_i has law $U_{\sigma_i, M}$; all rows use the same M .

Theorem 8.17 (Exact low-rank separable integer optimization). *If $k = 0$, a deterministic algorithm minimizes F on X exactly in polynomial time. If $k \geq 1$, then for every draw a deterministic algorithm returns an exact integer minimizer of F_ξ on X and the exact rational optimal value, both of polynomial length. Its expected bit work is at most*

$$8^k H_{\text{rat}} P(I), \quad H_{\text{rat}} = \prod_{i=1}^k \left[3 + \frac{(1+2k)\alpha w_i}{2\sigma_i} \right],$$

where P is a polynomial with an absolute exponent and coefficients depending only on d_* . No growth, uniqueness, or separation premise is used, and there is no fallback. The number of integer coordinates, the numbers of pieces, and the numerical lengths of the integer intervals do not enter H_{rat} except through the row ranges $u_i - \ell_i$.

Completing the square gives the auxiliary value $W(a) = \frac{\alpha}{2} \|a\|^2 + \min_{x \in X} [G(x) - \alpha a^\top Tx]$ on the fixed box $\prod_i [\ell_i - \sigma_i/\alpha, u_i + \sigma_i/\alpha]$, with $G = \sum_j g_j$. It has upper coordinate curvature α , because $W - \frac{\alpha}{2} \|a\|^2$ is a minimum of affine functions. Its inner problem separates into scalar problems over consecutive integers. Real convexity of g_j makes the forward differences of $g_j(t) - \lambda t$ nondecreasing, also between different pieces, so binary search on them finds an exact integer minimizer with $O(1 + \log(U_j - L_j + 1))$ evaluations, without listing labels. Moreover $\min_a W(a) + \xi^\top a$ equals $\min_X F_\xi$ up to the known constant $\|\xi\|^2/(2\alpha)$. The corrected-corner search of [theorem 3.9](#) on the k auxiliary coordinates gives an integer witness whose original gap is below $E_J < 1/(D_0(M-1))$ at the predetermined level $J = \log_2 M$, while every original value $F_\xi(x)$ lies in $\frac{1}{D_0(M-1)}\mathbb{Z}$. At an integer argument every expanded monomial is an integer, so the unary degree and the pieces add no denominator, and one common grid denominator suffices because the noise enters the original objective linearly. The auxiliary values may have denominators involving $(M-1)^2$, but no lattice property of them is used. The proof is in [appendix F.6](#).

The argument uses discrete convexity of the unaries, their exact evaluation, and a common denominator of their values, and nothing else about them. The fixed degree and the explicit piece lists guarantee exact evaluation in polynomial time with values of polynomial length. A succinct encoding would not: evaluating a monomial t^{2^b} with a binary exponent at $t = 2$ produces an integer of length 2^b . Continuous coordinates and coupled integer constraints are outside the theorem: continuous minima can be irrational, and coupling destroys the separable scalar recourse.

The numerical ratios $\alpha w_i/\sigma_i$ are explicit; binary-encoded bounds alone do not make them small. For binary variables, and more generally for explicitly listed labels, deterministic fixed-rank algorithms already exist. On $\{0, 1\}^n$ each g_j agrees with the affine function $g_j(0) + c_j x_j$, $c_j = g_j(1) - g_j(0)$, so F is a concave function of $(Tx, c^\top x)$; it attains its minimum at a vertex of the zonotope that is the image of the cube, and this zonotope in dimension $k+1$ has $O(n^k)$ vertices. With explicitly listed labels, the planar polytopes $\text{conv}\{(T_j t, g_j(t)) : t \in X_j\}$ give a Minkowski-sum enumeration that, for fixed k , is polynomial in the total number of labels. The theorem does not improve these cases; it allows binary-encoded intervals without listing their labels.

8.7 Scope

The residual feasible set must not depend on the core, and the exact integer oracle must handle all tightened coordinate bounds. The flow and totally unimodular results need nonempty residual sets

and separable costs that are convex in each residual coordinate for every core point; they do not cover nonseparable integer convexity, arbitrary integer matrices, or core-dependent supplies and right-hand sides. Under core-only noise, single-flow certificates fail at boundary cores ([proposition 9.11](#)); the face certificate needs a parameter-dependent sampling precision for nonlinear coupling, which bilinear coupling avoids. The strong-field theorem is a strong-noise statement, and the low-rank theorem uses aligned rather than ambient noise. All results optimize the sampled objective exactly; none optimizes F_0 .

The native, interior-flow, and bilinear results, including the bilinear totally unimodular case, and the lattice theorem have polynomial sampling precision. By [corollary 3.21](#), with a fixed oracle implementation where an oracle is used, their grid size can therefore be replaced by one depending only on the input length; in the lattice theorem the level cap is reset to $\log_2 M$. For the nonlinear results at arbitrary core faces ([theorem 8.12](#) and the corresponding case of [theorem 8.14](#)), the level cap and the grid exponent are bounded by $F_d(k)P_d(I)$ for an effective nondecreasing F_d . A common resolution then exists for the instances with a supplied bound $K \geq k$, and $F_d(K)$ enters their sampling, output, and work bounds; taking $K = I$ gives a law that depends only on I , but its sampling cost need not stay within $f_d(k) \text{poly}_d(I)$ for the actual k . The strong-noise theorem holds for every M , with q_i and β computed for the M used.

The integer oracles, fixed-block elimination, tube estimates, and connected-set counting are established tools; what this section proves is their composition with one base-selected finite law, the label and face certificates, and the bit and output bounds, including exact algebraic output in which all coordinates and the value within one completion or component share one root.

9 Boundaries

The positive results combine four ingredients: retention of grid cells by a rounding allowance, objective-only linear noise, a choice of which coefficients are perturbed, and exact closure certificates of specific kinds. This section shows by explicit families how each ingredient limits the scope. [Table 6](#) lists the results, including the examples of [section 7](#) that concern the same questions.

None of these results is an unconditional lower bound for all algorithms. Each is either a limitation of a specified mechanism or a conditional complexity implication. Some mechanism bounds hold on every draw; others concern an expected count or an event of positive probability. Several of the instances are easy for a different method, and the text says so. The purpose is to show that the restrictions of the positive theorems are not artifacts of their proofs: removing them requires a different algorithmic idea, and in some cases it would resolve an open complexity question.

9.1 The width exponent of global-error retention

The sparse theorems have bounds of the form $C^p[c + (1 + n/2)Lw_{\max}/(2\sigma)]^p \text{poly}_d(I)$. The factor n inside the bracket comes from the rounding allowance: rounding a feasible point to the corners of its cell changes the objective by up to $nLh^2/8$ in expectation, and every coordinate contributes ([section 5.4](#)). We show that, for the retention rule as specified, the resulting power of n is real.

The global-error search. We recall the search of [section 5](#) in the special case of a continuous box $X = [0, 1]^n$, with a tree decomposition whose bags $\beta \subseteq [n]$ have at most p coordinates and contain every monomial scope of F_0 , and with a rational $L \geq \max_i \sup_X \partial_{ii} F_0$. At level $j = 0, 1, \dots$ the mesh is $h_j = 2^{-j}$ and the grid is $\mathbb{G}_j = (h_j \mathbb{Z} \cap [0, 1])^n$. A cell of bag β at level j is a product of intervals $[t_i h_j, (t_i + 1)h_j]$, $i \in \beta$. At level 0 each bag has the single candidate cell $[0, 1]^\beta$; at level

Result	What it limits	Kind
Theorem 9.1	Global-error retention in the sparse route: more than $(5n/(6p))^{p/2}$ retained cells	Mechanism-specific, every draw
Proposition 9.2 , example 7.4	Replacing the global allowance by a bag-local one	The modified rule deletes all optimizers
Corollary 9.3	Certified conditional values remove the ambient dimension from the count	Sufficient interface
Theorem 9.4	Dimension-free counts under independent uniform noise in the low-rank route	Count surrogate, in expectation
Theorem 9.5	Objective noise with arbitrary sparse affine constraints	Would give $\text{NP} \subseteq \text{ZPP}$
Example 9.6 , example 7.3	Counting and corner bounds when feasibility depends on the bag or core	Counting argument fails
Proposition 9.7	Reading original threshold answers from sampled instances	Probability bound
Example 7.19	Full-dimensional convex closure under core-only noise	Mechanism-specific, every draw
Proposition 9.9	Converting value accuracy into residual distance; Tikhonov selection	Unconditional, method-specific
Theorem 9.10	Convex quartic residual point output under core-only noise; treewidth two and bounded coefficients for Square Root Sum	Would give Las Vegas algorithms for Square Root Sum and PosSLP
Proposition 9.11	One-flow certificates at boundary core optima	Mechanism-specific, probability 1/16
Proposition 7.14	Reducing small-core recourse to a fixed quadratic convexifier	Structural separation

Table 6: Boundary results. “Mechanism-specific” results show that a stated rule or certificate fails; they do not bound the work of other algorithms.

$j \geq 1$ the candidate cells are the children of the cells retained at level $j - 1$. The *allowed grid* A_j is the set of points of $\mathbb{G}_j$ whose projection onto every bag is a corner of a candidate cell of that bag. The search computes the exact bag min-marginals $m_\beta(v) = \min\{F_\gamma(y) : y \in A_j, y_\beta = v\}$, updates the incumbent value U_j , the least value of a feasible point found so far, and retains a candidate cell C of bag β if and only if

$$q_C - E_j \leq U_j, \quad q_C = \min_{v \text{ a corner of } C} m_\beta(v), \quad E_j = \frac{nLh_j^2}{8}. \quad (43)$$

Closure is attempted on the hull $Q_j = \prod_i [a_i, b_i]$, where $[a_i, b_i]$ is the intersection, over the bags containing i , of the smallest interval containing the i th projections of their retained cells. A coordinate is fixed at its original lower (upper) bound only if a sound enclosure of $\partial_i F_\gamma$ over Q_j is strictly positive (negative). On the remaining box a convexity test certifies a positive lower bound on the Hessian of the restricted objective. If no closure succeeds by the stopping depth J , the exact fallback runs. This is [definition 5.17](#) for a continuous box. The statements below hold for every implementation of these rules, every sound gradient enclosure, and every sound convexity test.

Theorem 9.1 (Global-error retention forces a width-dependent exponent). *Let $n \geq p \geq 2$. Let $\mathcal{G}$ be the graph on $[n]$ formed by a clique on $\{1, \dots, p\}$ and the path $p, p+1, \dots, n$, with edge set $\mathcal{E}$. Put $\phi(t) = t^2(1-t)^2$, $\eta = 1/(100n^2)$, and*

$$F_0(x) = \sum_{i=1}^n \phi(x_i) + \eta \sum_{\{i,l\} \in \mathcal{E}} x_i x_l \quad \text{on } X = [0, 1]^n.$$

- (a) F_0 is a quartic whose interaction graph is $\mathcal{G}$, which has treewidth $p - 1$. The bags $\{1, \dots, p\}$ and $\{i, i+1\}$, $p \leq i < n$, form a tree decomposition of bag size p containing every monomial scope. For every $\gamma \in \mathbb{R}^n$, $\partial_{ii} F_\gamma \leq 2$ on X , so $L = 2$ is valid. All widths are one, and the base length is $I = O((n + p^2) \log n)$.
- (b) Let $\gamma \in [-\frac{1}{100}, \frac{1}{100}]^n$ be arbitrary, and run the global-error search on F_γ with $L = 2$. At every level j with $h_j^2 \geq 3p/(10n)$, every candidate cell of every bag is retained, the hull Q_j is $[0, 1]^n$, no sound gradient-sign test fixes a coordinate, and no sound convexity test succeeds.
- (c) Let j_* be the last level with $h_{j_*}^2 \geq 3p/(10n)$. If the stopping depth satisfies $J > j_*$, the search generates more than $(5n/(6p))^{p/2}$ retained cells of the clique bag at level j_* , on every such draw. The stopping depth of [definition 5.15](#) with $\sigma \leq \frac{1}{100}$ satisfies $J > j_*$.
- (d) Consequently, for every constant c there is no function f such that the work of the global-error search is at most $f(p)I^c$ on all instances of this family.

Proof. (a) Expanding $\phi(x_i) = x_i^2 - 2x_i^3 + x_i^4$ shows that the only monomials involving two variables are the products $x_i x_l$ with $\{i, l\} \in \mathcal{E}$, each with coefficient $\eta \neq 0$. Hence the interaction graph is $\mathcal{G}$. Every tree decomposition of a graph containing a p -clique has a bag containing that clique, so the treewidth is at least $p - 1$. The displayed bags, arranged as the path $\{1, \dots, p\}, \{p, p+1\}, \{p+1, p+2\}, \dots$, cover every edge and every variable, and each variable occurs in one bag or in two adjacent bags; so they form a tree decomposition of width $p - 1$. The linear and bilinear terms have zero diagonal second derivatives, so

$$\partial_{ii} F_\gamma(x) = \phi''(x_i) = 2 - 12x_i(1 - x_i) \leq 2.$$

There are $O(n + p^2)$ monomials, each with indices and coefficients of $O(\log n)$ bits, and the decomposition has $O(n + p)$ entries; this gives the bound on I .

(b) Fix γ with $\|\gamma\|_\infty \leq 1/100$. The function $P_\gamma(x) = \eta \sum_{\mathcal{E}} x_i x_l + \gamma^\top x$ is affine in each coordinate separately, so moving one coordinate at a time to a better endpoint shows that its minimum over X is attained at a vertex. Since $\phi \geq 0$ and $\phi(0) = \phi(1) = 0$,

$$f^* := \min_X F_\gamma \geq \min_X P_\gamma = \min_{\{0,1\}^n} P_\gamma = \min_{\{0,1\}^n} F_\gamma \geq f^*,$$

so f^* is attained at a vertex a . Let β be a bag and $v \in [0, 1]^\beta$, and put $y = (v, a_{-\beta})$. Using $\phi \leq 1/16$, $|\gamma_i(v_i - a_i)| \leq 1/100$, $|y_i y_l - a_i a_l| \leq 1$, $|\mathcal{E}| \leq n(n-1)/2$, and $p \geq 2$,

$$F_\gamma(y) - f^* \leq |\beta| \left(\frac{1}{16} + \frac{1}{100} \right) + \eta |\mathcal{E}| \leq \frac{29p}{400} + \frac{1}{200} \leq \frac{3p}{40}. \quad (44)$$

We show by induction that at every level j with $h_j^2 \geq 3p/(10n)$ all cells are candidates and all are retained. At level 0 every cell is a candidate. Suppose all level- j cells are candidates. Then $A_j = \mathbb{G}_j$. For a cell C of bag β and a corner v of C , the point $(v, a_{-\beta})$ lies in $\mathbb{G}_j$ because a is a vertex, so (44) gives $q_C \leq m_\beta(v) \leq f^* + 3p/40$. Here $E_j = nh_j^2/4 \geq 3p/40$, and $U_j \geq f^*$ because U_j is the value of a feasible point. Therefore $q_C - E_j \leq f^* \leq U_j$, and C is retained. As every level- j cell is retained, every level- $(j+1)$ cell is a candidate.

Since all cells are retained, every coordinate projection of every bag covers $[0, 1]$, and $Q_j = [0, 1]^n$. Set all coordinates other than i to zero. Then $\partial_i F_\gamma = \phi'(x_i) + \gamma_i$, and $\phi'(t) = 2t(1-t)(1-2t)$ gives the values $\gamma_i + 3/16 > 0$ at $x_i = 1/4$ and $\gamma_i - 3/16 < 0$ at $x_i = 3/4$. A sound enclosure contains both values, so it is neither strictly positive nor strictly negative, and no coordinate is fixed. The remaining box is therefore $[0, 1]^n$. At its midpoint $c = (\frac{1}{2}, \dots, \frac{1}{2})$,

$$\nabla^2 F_\gamma(c) = -I + \eta A_G,$$

where A_G is the adjacency matrix. Its largest eigenvalue is at most $-1 + \eta(n-1) < 0$. A function convex on a box is positive semidefinite at interior points, so no sound convexity test can succeed.

(c) Since $h_0 = 1$ and $3p/(10n) < 1$, the level j_* exists, and $h_{j_*+1}^2 = h_{j_*}^2/4 < 3p/(10n)$ gives $h_{j_*}^2 < 6p/(5n)$. The clique bag has $h_{j_*}^{-p} = (h_{j_*}^{-2})^{p/2} > (5n/(6p))^{p/2}$ cells at level j_* , all retained by (b). No closure succeeds at levels up to j_* , so if $J > j_*$ the fallback does not run before level j_* , and the search generates all these cells. The stopping depth of definition 5.15 satisfies $h_J \leq 1/(4A)$ with $A \geq nL/g_0$ and $g_0 = \rho\sigma/(2W) \leq \sigma/(8W)$, where $W = \sum_i w_i = n$. Here $L = 2$ and $\sigma \leq 1/100$, so $h_J \leq \sigma/(64n^2) < \sqrt{3p/(10n)} \leq h_{j_*}$, that is, $J > j_*$.

(d) Fix $p > 2c$. For this p , the family has $I = O(n \log n)$ and work at least $(5n/(6p))^{p/2}$, which exceeds $f(p)I^c$ for all sufficiently large n . $\square$

The lower bound holds on every draw in the noise box, so it also bounds the expected work for every law supported there, including $U_{1/100, M}$ for every M . It does not contradict theorem 5.2, whose bound has the same form $n^{O(p)}$ for fixed numerical scales; it shows that this exponent is necessary for the retention rule as specified. It is not a lower bound for sparse optimization: the instances are easy, since every optimizer is a vertex and binary dynamic programming with $O(2^p)$ states per bag solves them. Tests on individual cells, rather than on the whole hull, could also prune these instances much earlier.

The natural repair is to replace E_j by an allowance for the bag alone, $|\beta| L h_j^2/8$. Example 7.4 shows that this is unsound for the unperturbed star quadratic: the error comes from minimizing over the grid outside the bag, and it varies with the bag's own coordinates. That deterministic example comes from an unpublished companion manuscript [1]. The next proposition adds two points: the failure persists on every draw of small noise, and no closure succeeds at level 0, so the search reaches the failing level. It is thus a failure of the smoothed algorithm and not of one exceptional draw.

Proposition 9.2 (A bag-local allowance is unsound on every draw). *Let $m = 32$ and let F_0 be the star quadratic of [example 7.4](#) on $[0, 1]^{1+m}$, with center v , leaves $z_1, \dots, z_m$, bags $\{v, z_i\}$, and $L = 2$. Let $\gamma \in [-\sigma, \sigma]^{1+m}$ with $\sigma \leq 1/1000$, and run the search of [\(43\)](#) with E_j replaced by $|\beta| Lh_j^2/8$ and with a stopping depth $J \geq 1$, for instance that of [definition 5.15](#). Then no closure succeeds at level 0, so the search reaches level 1 ($h_1 = 1/2$). There every global optimizer of F_γ projects into the cell $C = [\frac{1}{2}, 1] \times [0, \frac{1}{2}]$ of the bag $\{v, z_1\}$, and C is deleted. The global allowance $(1+m)Lh_1^2/8 = 33/16$ retains C .*

Proof. Every F_γ differs from F_0 by at most $(1+m)\sigma = 33\sigma$ on the box, and $F_0(v, z) = v^2 - \frac{v}{2} \sum_i z_i + \sum_i z_i^2 + \frac{15}{16}v$. The schedule of [definition 5.15](#) gives $h_J \leq 1/(4A) < 1 = h_0$, so $J \geq 1$ there.

Location of the optimizers. For fixed v , the coordinate z_i minimizes $z_i^2 - (v/2 - \gamma_i)z_i$, a strictly convex function whose minimizer on $[0, 1]$ is at most $(1/2 + \sigma)/2 < 1/2$. Let $V_\gamma(v)$ be the minimum of F_γ over z . Each inner minimum is at least $-(v/2 - \gamma_i)^2/4$, so for $v \in [0, \frac{1}{2}]$

$$V_\gamma(v) \geq v^2 + \left(\frac{15}{16} - \sigma\right)v - 8\left(\frac{v}{2} + \sigma\right)^2 = -v^2 + \left(\frac{15}{16} - 9\sigma\right)v - 8\sigma^2 \geq -8\sigma^2,$$

the last step because the concave middle expression is at least its endpoint values $-8\sigma^2$ and $7/32 - 9\sigma/2 - 8\sigma^2 > 0$. The feasible point $v = 1$, $z_i = 1/4$ has $F_0 = -1/16$, so $V_\gamma(1) \leq -1/16 + 9\sigma < -8\sigma^2$. Hence every optimizer has $v > 1/2$ and $z_1 < 1/2$.

Level 0. At level 0 each bag has one cell, and its value q is the minimum of F_γ over the vertices of the box, which is at most $F_\gamma(0) = 0$. The bag-local allowance at $h_0 = 1$ is $1/2$, and the incumbent is at least $\min F_\gamma \geq -1/16 - 33\sigma > -1/2$. So every cell is retained, and the hull is the whole box. No closure succeeds there. Each leaf derivative $\partial_{z_i} F_\gamma = 2z_i - v/2 + \gamma_i$ is negative at $(v, z_i) = (1, 0)$ and positive at $(0, 1)$; the center derivative $\partial_v F_\gamma = 2v - \frac{1}{2} \sum_i z_i + \frac{15}{16} + \gamma_v$ is negative at $(v, z) = (0, 1)$ and positive at $(1, 0)$. So no sound sign test fixes a coordinate. The Hessian is the constant matrix with diagonal 2 and center-leaf entries $-\frac{1}{2}$, and its quadratic form at $(1, \frac{1}{4}\mathbf{1})$ is $2 - 8 + 4 = -2$; so no sound convexity test succeeds. Hence the search reaches level 1, every cell of level 1 is a candidate, and the allowed grid is $\{0, \frac{1}{2}, 1\}^{1+m}$.

The level-1 test. For $\gamma = 0$, [example 7.4](#) computes the smallest grid min-marginal of C as $23/32$; in general it is at least $23/32 - 33\sigma$. The incumbent satisfies $U_1 \leq F_\gamma(0) = 0$, because the origin is an allowed grid point. The bag-local allowance is $2 \cdot 2 \cdot \frac{1}{4}/8 = \frac{1}{8}$, and

$$q_C - \frac{1}{8} \geq \frac{19}{32} - 33\sigma > 0 \geq U_1,$$

so C is deleted. With the global allowance, $q_C - \frac{33}{16} \leq \frac{23}{32} + 33\sigma - \frac{33}{16} < -1 < U_1$, so C is retained. $\square$

Certified values of the true conditional function remove this error, and with it the factor n in the count. The following statement makes this precise. It is a sufficient interface, not an algorithm: computing such values requires optimizing over all variables outside the bag. In the notation of [section 5.3](#), let $W_\beta(v) = V_\beta(v) + \gamma_\beta^\top v$ be the exact conditional value of a bag tuple, a minimum over the original outside domain. Parts (a) and (b) below restate deterministic guarantees of the companion's certified bag filter [\[1\]](#); part (c) is the finite-law count of [proposition 5.10\(c\)](#).

Corollary 9.3 (Certified conditional values remove the dimension factor). *Let the modified search and certified conditional oracle of [proposition 5.10](#) be given, with an error allowance ae_j , $a \geq 0$, and $e_j = pLh_j^2/8$. Then:*

(a) *every global optimizer is retained on every draw at every level;*

- (b) the incumbent satisfies $U_j \leq f^* + (1+a)e_j$, and every retained bag cell has a corner with true conditional value at most $f^* + 2(1+a)e_j$;
- (c) for $j \leq J$ and $M \geq 2^J$, the expected number of such tuples in the complete bag grid is bounded by (14), with error parameter a .

Thus, for fixed a , the count is FPT jointly in p and $Lw_{\max}/\sigma$.

Proof. These are the retention, witness, and counting conclusions of [proposition 5.10](#). Its incumbent includes the current oracle answers before pruning, and its error includes the entire outside problem. $\square$

With a fixed, the per-bag factor depends on the bag size p but not on the dimension n , so the count would be fixed-parameter in p and the numerical ratio. What is missing for a general sparse algorithm is the oracle: the grid min-marginals of [section 5](#) do not satisfy the interface ([proposition 9.2](#)), and an exact conditional value can require solving almost the whole instance. The recourse route of [section 7](#) realizes the interface for one search block, a small core, by assuming a tractable residual problem; its count is the case of a single bag of size k . For general sparse instances no such oracle is available, and whether another method achieves a bound FPT jointly in width and numerical ratios under linear noise remains open.

9.2 Ambient uniform noise and dimension-free counts

The low-rank route has bounds FPT jointly in structure and numerical ratios under aligned noise ([theorem 4.21](#)) and under Gaussian-like noise on all coefficients ([theorem 4.19](#)), whereas under independent uniform noise on all coefficients its count contains powers of the dimension ([theorem 4.20](#)). The reason is that the factor component and the residual component of a uniform cube vector are dependent: conditionally on the residual component, the factor component can be concentrated much more than its variance suggests. For a Gaussian vector the two components are independent. [Proposition 4.25](#) states that the dimension dependence of the uniform count is real for the counted quantities, even for perfectly conditioned factors and fixed projected widths. We prove it here, in a slightly more general form.

We use the notation of [section 4.2](#) with $\Lambda = \alpha I$. For a factor $T \in \mathbb{Q}^{k \times N}$ with $TT^\top = I_k$, write $\gamma = T^\top \xi + \zeta$ with $\xi = T\gamma$ and $T\zeta = 0$, and

$$V(a) = \min_{x \in P} \left[F_0(x) + \zeta^\top x + \frac{\alpha}{2} \|a - Tx\|^2 \right] + \xi^\top a, \quad a \in \mathbb{R}^k. \quad (45)$$

The search of [theorem 4.20](#) refines dyadic meshes on a box that contains the projected ranges enlarged by the largest possible factor noise. When all its widths are equal, the allowance at mesh h is $E_h = \alpha kh^2/8$, and the count of [theorem 4.20](#) bounds the expected number of grid nodes v satisfying the local event $V(v) \leq V(v \pm he_l) + 2E_h$ for every coordinate l that is interior at v .

Theorem 9.4 (Uniform ambient counts depend on the dimension). *Let $k \geq 1$, let $m \geq 256$ be even, let $\alpha \in \{1, m\}$, and put $N = km^2$. There are a symmetric rational matrix A , a rational factor T with $TT^\top = I_k$ and $A + \alpha T^\top T \succeq 0$, and a bounded rational polytope $P \subset \mathbb{R}^N$ with $O(N)$ inequalities, all of encoding length polynomial in N , with the following properties for $F_0(x) = \frac{1}{2}x^\top Ax$.*

- (a) A has exactly k negative eigenvalues, each equal to $-\nu$ with $\nu = \alpha(\sqrt{17/8} - 1)$, so $\alpha < 4\nu$, and each coordinate of Tx ranges over an interval of width 8 as x ranges over P .

(b) Let the coordinates of γ be independent, either uniform on $[-1, 1]$ or with law $U_{1,M}$ for a power of two $M \geq 32$. For every grid of spacing h on an interval containing $[-\frac{3}{2}, \frac{3}{2}]$, with product grid $\mathbb{G}_h$ in $\mathbb{R}^k$:

- (i) if $8/\sqrt{\alpha m} \leq h \leq 16/\sqrt{\alpha m}$, then $\mathbb{E} \#\{v \in \mathbb{G}_h : V(v) \leq \min V + 2E_h\} \geq (\sqrt{\alpha m}/64)^k$;
- (ii) if $16/(\alpha m) \leq h \leq 32/(\alpha m)$, then the expected number of nodes $v \in \mathbb{G}_h \cap [-1, 1]^k$ with $V(v) \leq V(v \pm he_l) + 2E_h$ for every l and both signs is at least $(\alpha m/128)^k$.

The dyadic meshes of the search of [theorem 4.20](#) include values of h in both ranges.

(c) For $\alpha = 1$ these bounds are $64^{-k}(N/k)^{k/4}$ and $128^{-k}(N/k)^{k/2}$. Consequently no bound of the form $f(k, R)I^c$, with an absolute constant c , holds for either expected count, where R records the frame $TT^\top$, the projected widths, the ratio α/ν , and the noise scale, all of which are fixed here.

The proof is in [appendix G.1](#). Each block of the construction has a negative direction coupled to a thin coordinate, and a simplex of total mass $4m$ whose two halves carry the signs of the factor row. With constant probability the smallest noise coefficients of the two halves are both close to -1 ; then minimization over the simplex cancels the factor noise up to $O(1/m)$, the auxiliary objective is flat to within $O(1/m)$ on a long interval, and many grid nodes pass every comparison.

The theorem bounds the count surrogates used in the analysis, not the work of the algorithm: the closure step could resolve these instances before the counted nodes are visited. It also leaves room for bounds in terms of the ambient diameter, which grows like m here. What it excludes is a proof of a fixed-parameter bound for uniform ambient noise that bounds these counts in terms of projected widths and conditioning alone. [Theorem 4.19](#) avoids the obstruction because the factor and residual components of a Gaussian vector are independent.

9.3 Constraints and decision thresholds

The sparse results allow mixed boxes, products of simplices, order constraints, and graph equalities ([section 6](#)). The next result shows that objective noise cannot handle arbitrary sparse affine constraints: their feasibility problem can encode an NP-complete question that no small objective perturbation changes.

Theorem 9.5 (Objective noise does not remove sparse feasibility hardness). *There is a polynomial-time map from instances $(a_1, \dots, a_n; B)$ of SUBSET SUM, with positive integers, to the following instances. Put $A = \sum_i a_i + B$. The variables are $t, z_1, \dots, z_n, w_1, \dots, w_n \in \{0, 1\}$ and $s_1, \dots, s_n \in [0, 1]$; with $s_0 = 0$ the constraints are*

$$z_i + w_i = t, \quad s_i = s_{i-1} + \frac{a_i}{A} z_i \quad (i = 1, \dots, n), \quad s_n = \frac{B}{A} t,$$

and the objective is $F_0 = -t$. Every constraint has at most three variables and coefficients of absolute value at most one, the constraint and objective scopes have a tree decomposition of bag size four, and the all-zero point is feasible. For every $\gamma \in \mathbb{R}^{3n+1}$ with $\|\gamma\|_\infty \leq \sigma := 1/(10(3n+2))$, every global optimizer of $F_\gamma = -t + \gamma^\top x$ has $t = 1$ if the instance is a yes-instance and $t = 0$ otherwise.

Consequently, suppose an exact smoothed algorithm handles mixed boxes with additional affine equality constraints whose scopes fit a supplied tree decomposition of bag size four, under ambient noise drawn from a base-chosen finite rational law supported in $[-\sigma, \sigma]$ with polynomially many sampling bits. If its expected work is polynomial in I and $1/\sigma$, then SUBSET SUM has a Las Vegas algorithm with expected polynomial running time, and $\text{NP} \subseteq \text{ZPP}$.

Proof. If $t = 0$, then $z_i + w_i = 0$ forces $z_i = w_i = 0$, the recurrence forces $s_i = 0$, and $s_n = 0$ holds; so the all-zero point is the only feasible point with $t = 0$. If $t = 1$, then $w_i = 1 - z_i$ and $s_i = A^{-1} \sum_{l \leq i} a_l z_l \in [0, 1]$, and $s_n = B/A$ holds exactly when $\sum_l a_l z_l = B$. So a feasible point with $t = 1$ exists exactly for yes-instances.

The bags $P_i = \{t, s_{i-1}, s_i, z_i\}$ (with s_0 omitted) form a path, and attaching $\{t, z_i, w_i\}$ to P_i gives a tree. Every constraint lies in a bag, t lies in every bag, and every other variable lies in one bag or in two adjacent bags; so this is a tree decomposition of bag size four.

There are $3n + 1$ variables in $[0, 1]$, so $|\gamma^\top x| \leq (3n + 1)\sigma < 1/10$. The zero point has value 0, and every feasible point with $t = 1$ has value below $-9/10$. Hence the optimizers have $t = 1$ exactly on yes-instances.

For the consequence, construct the instance, compute and sample the law, run the algorithm, and read t from its exact output. The answer is correct on every draw, and the expected running time is polynomial because $1/\sigma = O(n)$ and the instance has length polynomial in the SUBSET SUM input. $\square$

The objective is linear, so any positive L is a valid curvature bound, all widths are one, and $1/\sigma = O(n)$. A constrained analogue of the sparse bounds with these parameters would therefore be polynomial on this family. Bounded coefficient magnitudes do not bound denominators, Hoffman constants, or the constants of feasible repair, and the reduction makes no claim about them. A constrained theorem can avoid the barrier by assuming structure that preserves feasibility under rounding, as the simplex, order, and graph results of [section 6](#) do, or by charging parameters that are large on this family.

Constraints also affect the counting argument itself. The box count uses the fact that a conditional value, obtained by minimizing over the variables outside a bag or a core, has bounded upper curvature in each coordinate. This requires the outside feasible set to be independent of the bag or core values. [Example 7.3](#) shows that otherwise the corrected-corner lower bound itself fails. The following example shows that the expected count fails as well, even for a single totally unimodular row.

Example 9.6 (Core-dependent feasibility defeats the coordinatewise count). Let $v_1, v_2, y \in [0, 1]$ with the single constraint $y \geq v_1 - v_2$, which together with the bounds is totally unimodular. Minimize $2y + \gamma_1 v_1 + \gamma_2 v_2 + \gamma_y y$, with independent coefficients either uniform on $[-\frac{1}{2}, \frac{1}{2}]$ or with law $U_{1/2, M}$. Regard (v_1, v_2) as a core, or as a bag, and y as the residual variable. For fixed γ_y the conditional value is

$$V(v) = \kappa \max\{0, v_1 - v_2\}, \quad \kappa = 2 + \gamma_y \in [\frac{3}{2}, \frac{5}{2}],$$

which has a convex kink along the diagonal. Let $h = 1/r$ and consider the diagonal grid nodes (kh, kh) , $1 \leq k \leq r - 1$. Their four coordinate neighbors increase $V + \gamma_1 v_1 + \gamma_2 v_2$ by

$$h(\kappa + \gamma_1), \quad -h\gamma_1, \quad h\gamma_2, \quad h(\kappa - \gamma_2).$$

On the event $\{\gamma_1 < 0 < \gamma_2\}$, which has probability $1/4$ under both laws because 0 is not an atom of $U_{1/2, M}$, all four increments are positive. So every one of the $r - 1$ diagonal nodes passes every coordinate-neighbor comparison, even with tolerance zero, and the expected number of passing nodes is at least $(r - 1)/4$, unbounded as $h \rightarrow 0$. Under a coordinate upper-curvature bound L , the two increments in one coordinate sum to at most Lh^2 , which confines the coefficient to an interval of length $O(Lh)$; here they sum to κh .

The example does not bound the number of truly near-optimal nodes: along the diagonal the objective is $kh(\gamma_1 + \gamma_2)$, and comparisons in the diagonal direction remove the spurious candidates.

The constrained sparse results therefore use feasible rounding and comparisons along the faces of their polytopes, and the recourse results require a residual feasible set that does not depend on the core.

Finally, the answer to a threshold question about the base instance cannot in general be read off from the exact optimum of a sampled instance when the threshold gap is smaller than the noise. Worst-case hardness results for structured nonconvex optimization, such as strong NP-hardness of box-constrained quadratic programming at treewidth two [19], concern such threshold questions, and bounded coefficients do not by themselves give a lower bound on the distance of a no-instance's minimum from the threshold.

Proposition 9.7 (Small gaps do not survive the noise). *Let Ψ be continuous on a compact set X , let $\delta \geq 0$, and let $y \in X$ satisfy $\Psi(y) \leq \delta$ and $|y_i| \geq 1$ for some coordinate i . Let γ have independent coordinates with laws symmetric about zero, and let γ_i be uniform on $[-\sigma, \sigma]$. Then*

$$\Pr\left\{\min_X(\Psi + \gamma^\top x) < 0\right\} \geq \frac{1}{2} - \frac{\delta}{2\sigma}.$$

If instead γ_i has law $U_{\sigma, M}$, the bound holds with the additional term $-1/(2M)$.

Proof. Put $Z = \gamma^\top y$. It is a sum of independent symmetric terms, so Z is symmetric and $\Pr\{Z < -\delta\} = \frac{1}{2}(1 - \Pr\{|Z| \leq \delta\})$. Conditionally on $(\gamma_l)_{l \neq i}$, the event $|Z| \leq \delta$ confines γ_i to an interval of length $2\delta/|y_i| \leq 2\delta$, which has probability at most δ/σ , or at most $\delta/\sigma + 1/M$ by (2). On the event $Z < -\delta$, $\min_X(\Psi + \gamma^\top x) \leq \Psi(y) + Z < 0$. $\square$

If $\Psi \geq 0$ and $0 < \min_X \Psi \leq \delta \ll \sigma$, the base instance is a no-instance of the question “is $\min \Psi \leq 0$?”, while the sampled instance has a negative minimum with probability close to $1/2$. Reading the original answer directly from the sampled optimum therefore fails on such instances, and expected polynomial bounds for sampled instances are compatible with strong NP-hardness of the corresponding threshold questions. These guarantees do not by themselves provide a polynomial-time algorithm for the original threshold questions. The proposition does not exclude other uses of a smoothed solver, for instance reductions with a robust gap or repeated queries at chosen noise scales.

Remark 9.8 (The treewidth-two reduction). The strong NP-hardness proof of Del Pia and Khajavirad [19] reduces SUBSET SUM to the question whether a nonnegative box quadratic Ψ of treewidth two with bounded integral coefficients has minimum zero. In that construction Ψ combines nonnegative binary penalties $x_i(1 - x_i)$ with squared residuals for bit copies, bit-serial and partial-sum recurrences, and a final square that compares the selected partial sum with the target. Take $r \geq 1$, $A = 2^r$, items A and $A + 2$, and target $A + 1$, a no-instance. Selecting the first item and following every recurrence exactly gives a point of the unit box at which every term vanishes except the final square, which equals 2^{-2r-4} , and the indicator of the selected item equals one. Proposition 9.7 with $\delta = 2^{-2r-4}$ shows that, under independent noise of half-width σ with law $U_{\sigma, M}$, the sampled minimum is negative with probability at least $\frac{1}{2} - 2^{-2r-5}/\sigma - 1/(2M)$, although the base instance is a no-instance. The number of variables of the construction is linear in r , so the gap is exponentially small in the instance size.

9.4 Core-only noise and residual outputs

Core-only noise perturbs the few coordinates that carry the nonconvexity and leaves the residual costs unchanged. Ties and flat directions in the residual variables then persist on every draw. The continuous core-only theorem (theorem 7.15) assumes a certified uniform strong-convexity modulus of the residual problem, and the integer core-only results of section 8 use a certificate that represents

all tied optimal flows at once. These are sufficient structures, rather than necessary conditions for every residual problem. The exact-arithmetic companion [2] also gives full selected-optimizer oracles under core-only noise for residual-convex cubics, without a residual modulus. Its supplied-convexifier theorem also gives point oracles when the transformed objective is cubic on the feasible polytope or is a fixed-degree polynomial convex on all of $\mathbb{R}^n$. The general arithmetic obstruction below begins at degree four. The examples explain the limits of the certificates used here.

Example 7.19 concerns the closure step used under ambient noise: a full-dimensional box around the optimal set on which the objective is convex. Under core-only noise such a box need not exist, even when the core has projected growth one, the residual problem is convex or strictly convex, and the optimizer is unique and interior. The example has a short exact optimality certificate, so it limits one certificate, not exact optimization. Two further results concern the output itself. The first separates value accuracy from distance to an optimizer. It shows that no value tolerance with polynomially many bits localizes the residual coordinates of a convex quartic with constant-size coefficients, and that a noisy core does not change this.

Proposition 9.9 (Value accuracy does not localize residual points). *For $n \geq 1$ and $x_0 = \frac{1}{2}$, let*

$$G_n(x) = \sum_{i=1}^n (8x_i^2 - x_{i-1}^2 x_i), \quad \Phi_\gamma(v, x, y) = \left(v - \frac{1}{2}\right)^2 + \gamma v + G_n(x) + x_n^2(y - 1)^2$$

on $[0, 1] \times [0, 1]^n \times [0, 1]$, with $|\gamma| \leq \frac{1}{2}$.

- (a) Φ_γ is a convex quartic with $O(n)$ monomials and constant-size coefficients other than γ , and $\nabla^2 G_n \succeq 10I$ on $[0, 1]^n$.
- (b) Φ_γ has the unique minimizer $(v_\gamma, x^*, 1)$, where $v_\gamma = (1 - \gamma)/2$, x^* is the interior minimizer of G_n , and $0 < x_n^* \leq 2^{-2^{n+1}}$.
- (c) The feasible point $(v_\gamma, x^*, 0)$ is at distance one from the minimizer, and its objective gap is $(x_n^*)^2 \leq 2^{-2^{n+2}}$.
- (d) For $\lambda > 0$ let (x_λ, y_λ) minimize $G_n(x) + x_n^2(y - 1)^2 + \lambda(\|x\|^2 + y^2)$ on $[0, 1]^{n+1}$. If $y_\lambda \geq \frac{1}{2}$, then $\lambda \leq 2^{-2^{n+2}}$.

Proof. (a) The Hessian of G_n is tridiagonal, with diagonal entries $16 - 2x_{i+1}$ for $i < n$ and 16 for $i = n$, and off-diagonal entries $-2x_i$. On the box each diagonal entry is at least 14 and each row has off-diagonal absolute sum at most 4, so $\nabla^2 G_n \succeq 10I$ by Gershgorin's theorem. For a direction (p, q) in (x, y) and $u = 1 - y \in [0, 1]$, direct differentiation gives

$$(p, q)^\top \nabla^2 [x_n^2(y - 1)^2] (p, q) = 2(x_n q - 2u p_n)^2 - 6u^2 p_n^2,$$

so the Hessian of $G_n + x_n^2(y - 1)^2$ has quadratic form at least $10\|p\|^2 - 6p_n^2 \geq 0$. The core term is convex and separate.

(b) The derivative $\partial_i G_n = 16x_i - x_{i-1}^2 - 2x_i x_{i+1}$ (the last term absent for $i = n$) equals $-x_{i-1}^2$ at $x_i = 0$ and is at least 13 at $x_i = 1$. Since $x_0 > 0$, induction on i shows that the strongly convex function G_n has its minimizer x^* in the interior, with $x_i^* = (x_{i-1}^*)^2 / (16 - 2x_{i+1}^*) \leq (x_{i-1}^*)^2 / 14$, where $x_{n+1}^* := 0$. Writing $x_i^* = 14e_i$ gives $e_i \leq e_{i-1}^2$ and $e_0 = 1/28$, so $x_n^* \leq 14 \cdot 28^{-2^n} \leq 4^{-2^n}$. Since $\Phi_\gamma \geq (v - \frac{1}{2})^2 + \gamma v + G_n(x)$, with equality exactly when $x_n^2(y - 1)^2 = 0$, and $x_n^* > 0$, the unique minimizer is $(v_\gamma, x^*, 1)$.

(c) is immediate from (b).

(d) The regularized objective is strongly convex, so its minimizer is unique. For fixed x , minimizing $x_n^2(y-1)^2 + \lambda y^2$ over y gives $y_\lambda = x_{n,\lambda}^2/(x_{n,\lambda}^2 + \lambda)$. The boundary argument of (b) applies unchanged to the regularized problem, and its stationarity equations give $x_{i,\lambda} = x_{i-1,\lambda}^2/(16 + 2\lambda - 2x_{i+1,\lambda})$ for $i < n$ and $x_{n,\lambda} = x_{n-1,\lambda}^2/(16 + 2\lambda + 2(1 - y_\lambda)^2)$. Hence $x_{n,\lambda} \leq 4^{-2^n}$ as before. If $y_\lambda \geq \frac{1}{2}$, then $\lambda \leq x_{n,\lambda}^2 \leq 2^{-2^{n+2}}$. $\square$

The input has length $O(n \log n)$, so part (c) says that a value accuracy with fewer than 2^{n+2} bits, which is superpolynomial in I , certifies no residual accuracy below one, on every draw. Part (d) shows that isotropic Tikhonov selection, the usual way to pick a point from a convex optimal fiber, needs a regularization parameter of exponentially many bits here. The instance is nonetheless easy: setting $y = 1$ leaves a uniformly strongly convex problem. So the proposition limits methods that convert values into points, not residual optimization in general.

The next result concerns every method. It encodes two arithmetic decision problems in the residual optimizer of a convex quartic. *Square Root Sum* asks, for positive integers $a_1, \dots, a_n$ and an integer B , whether $\sum_i \sqrt{a_i} \leq B$. A *straight-line program* computes a sequence of integers, starting from the constants 0 and 1, each a sum, difference, or product of two earlier ones; its output is the last integer A , and PosSLP asks whether $A > 0$. Both problems lie in the counting hierarchy, and Square Root Sum is decidable in polynomial time with a PosSLP oracle [5]; whether either is in P is open.

Theorem 9.10 (Residual points under core-only noise).

- (a) *There is a polynomial-time algorithm that maps every Square Root Sum instance either to its answer or to an explicit rational quartic Φ on a unit box Y , convex on Y , with a unique minimizer $\hat{x}$ whose designated coordinate $\hat{x}_y$ is 1 if $\sum_i \sqrt{a_i} > B$ and 0 if $\sum_i \sqrt{a_i} < B$; equality does not occur in the produced instances. The interaction graph of Φ has treewidth at most two, every nonconstant coefficient has absolute value bounded by an absolute constant, and $\partial_{ii}\Phi \leq 20$ on Y .*
- (b) *There is a polynomial-time algorithm that maps every straight-line program with output A to an explicit rational quartic Φ on a rational box Y , convex on Y , with a unique minimizer $\hat{x}$ whose designated coordinate is 1 if $A > 0$ and 0 if $A \leq 0$.*
- (c) *For Φ from (a) or (b), consider*

$$F_\gamma(v, x) = \left(v - \frac{1}{2}\right)^2 + \gamma v + \Phi(x) \quad \text{on } [0, 1] \times Y,$$

with one noisy core coordinate v , no residual noise, and $|\gamma| \leq \frac{1}{2}$. The core optimizer is $v_\gamma = (1 - \gamma)/2$, the projected growth is one, $\partial_{vv}F_\gamma = 2$, the residual objective is convex, and the unique optimizer is $(v_\gamma, \hat{x})$ on every draw. Consequently, if an exact smoothed algorithm for this class, under a base-chosen finite rational core-noise law with polynomially many sampling bits, returns an output from which a point within distance $1/4$ of the optimizer can be computed with expected polynomial total work, then Square Root Sum and PosSLP have Las Vegas algorithms with expected polynomial running time; for Square Root Sum the instances of (a), with treewidth two and bounded coefficients, suffice. If a deterministic algorithm returns, in polynomial time, a point within distance $1/4$ of the minimizer of every explicit rational quartic that is convex on a rational box and has a unique minimizer, then Square Root Sum and PosSLP are in P.

The proof is in [appendix G.2](#). All three parts, including the core-only-noise and deterministic consequences, also appear in the unpublished companion manuscript [2]; the constructions are reproduced here to keep the paper self-contained. The two parts have different structure: part (a) has treewidth two and bounded coefficients, while the construction of part (b) bounds neither its treewidth nor its coefficient magnitudes. The theorem is a conditional implication, not an NP-hardness result. Values remain easy: convex optimization approximates $\min \Phi$ in polynomial time, and the program $\min_{\gamma} \Phi$ is itself a compact exact description of $\hat{x}$. What the theorem limits is the efficient evaluation of residual coordinates. The residual problem has no positive uniform strong-convexity modulus on its box: along the designated coordinate its curvature vanishes on the face where both activation coordinates are zero, and at $\hat{x}$ it can be extremely small. This is consistent with the certified modulus assumed in [theorem 7.15](#). Part (a) is also relevant to the sparse route: its residual problems have treewidth two and bounded coefficients, so sparsity of the residual problem does not by itself make residual points efficiently computable under core-only noise. Noise on the residual coordinates is what supplies point growth in the ambient theorems.

For integer recourse under core-only noise, the analogue of a convex patch is a certificate that one residual label, for instance one flow, is optimal throughout a small core box. Boundary core optima defeat this certificate on an event whose probability does not shrink as the noise grid is refined.

Proposition 9.11 (One-flow certificates fail at boundary core optima). *Let $v \in [0, 1]^2$, let one integral unit be routed through two parallel arcs, so that the residual labels are $Y = \{(1, 0), (0, 1)\}$, and let*

$$F_{\gamma}(v, z) = \frac{1}{2} \|v\|^2 + \gamma^{\top} v + v_1 z_1 + v_2 z_2,$$

with core-only noise γ_1, γ_2 independent with law $U_{1,M}$, $M \geq 4$. Then $L = 1$ is valid and the event $\mathcal{E} = \{\gamma_1 \geq \frac{1}{2}, \gamma_2 \geq \frac{1}{2}\}$ has probability exactly $1/16$. On $\mathcal{E}$:

- (a) *the projected value $V_{\gamma}(v) = \frac{1}{2} \|v\|^2 + \gamma^{\top} v + \min\{v_1, v_2\}$ satisfies $V_{\gamma}(v) \geq (\frac{1}{2} + \min_i \gamma_i) \|v\|^2$, so $v = 0$ is the unique optimal core, with growth at least one, and both labels are optimal there;*
- (b) *on every box $[0, a] \times [0, b]$ with $a, b > 0$ no single label is optimal throughout, so no certificate of optimality of a single label on such a box exists;*
- (c) *the corrected-grid core search with allowance $kLh^2/8 = h^2/4$ retains the cell $[0, h]^2$ at every level.*

If the two arcs are followed by $m - 1$ further stages of two parallel arcs with zero cost, there are 2^m feasible flows and the same conclusions hold. A strategy that requires a one-label certificate on the retained hull and otherwise enumerates all flows then has expected work at least $2^m/16$, while the input length is $O(m \log m)$ plus the sampling bits.

The proof is in [appendix G.3](#). The instance is easy: on $\mathcal{E}$ the bound in (a) certifies the origin and any flow. The integer core-only theorem of [section 8](#) avoids the obstruction with a certificate that represents every optimal flow at once and fixes the core face instead of one label.

9.5 Relations between the structural routes

The low-negative-inertia route uses a quadratic correction $\alpha \|Tx\|^2/2$ of small rank k that makes the objective convex. The recourse route uses a small core whose fixing makes the residual problem tractable. [Proposition 7.14](#) gives a family with a core of size one and convex polynomial recourse for which every fixed positive semidefinite quadratic correction that makes the objective convex

has rank at least the residual dimension. [Theorem 7.13](#) applies to it with the core curvature as numerical parameter, independently of the coupling scale, while a quadratic convexification would need rank equal to the residual dimension. The two routes therefore measure different structure. The separation concerns the broader class with merely convex recourse. A uniformly strongly convex residual problem, as in [theorem 7.15](#), does admit a quadratic convexifier supported on the core, of rank at most k , by a Schur-complement bound; for that class the contribution of the recourse theorem is that its numerical factor keeps the original core curvature L/σ rather than the scale of such a convexifier. If the full Hessian blocks are bounded in operator norm by M and the residual modulus is μ , a sufficient convexifier scale is of order $M + M^2/\mu$. The direct theorem retains the original L/σ in its count. Neither statement is a hardness result.

The sparse and recourse routes are related in the opposite way. A bag of the sparse route plays the role of a core, and the sparse count contains the factor n only because it lacks exact conditional values for the variables outside the bag ([proposition 9.2](#) and [corollary 9.3](#)). Where such values are available, as in the recourse theorems, the factor n disappears. The strong-noise route of [section 8](#) uses no structural parameter at all, and its noise regime is different from the other three.

10 Discussion

10.1 What the results permit

The theorems concern structured nonconvex instances under the specific finite laws constructed before sampling. Independence is in the coordinates of the perturbation model: original coefficients for ambient noise, core coefficients for core-only noise, and factor coefficients for aligned noise. Within these contracts they give three kinds of guarantees.

First, they return exact global solutions of sampled instances, with expected bit work bounded by the structural parameters and displayed numerical ratios. If uncertain or estimated costs follow the actual law and hypotheses of a theorem, the sampled instance can itself be the problem of interest. An exogenous cost distribution is covered only when it matches that contract; arbitrary uncertainty distributions do not inherit these work bounds.

Second, for routes valid at every positive noise scale, [proposition 2.6](#) and [section 2.6](#) give an exactly feasible point and an original-objective certificate of any prescribed accuracy ε . The recipe uses widths in the perturbation coordinates and the exact Gaussian-like support factor $b + 20$. The expected work is polynomial in $1/\varepsilon$ for fixed structure and fixed other numerical parameters; it is polynomial in I and $1/\varepsilon$ only when those numerical parameters are polynomially bounded. Implicit graphs use exact algebraic lifts of rational free points. The strong-field theorem retains the regret bound, but its lower noise-scale condition prevents this calibration in general.

Third, the proofs give certificates checked exactly and sound on every draw: corrected-corner bounds, critical regions of parametric convex programs, strongly convex patches after sound coordinate fixing, excluded-region and competing-label gaps, and optimal-face certificates for integer flow and TU recourse. Probability bounds control when these certificates pass. They could serve as termination tests inside other global solvers after branching has localized the optimum; their practical effectiveness is unestablished.

The structures have natural application interpretations. A small core with convex, network-flow, or TU recourse describes two-stage problems in which a few nonconvex first-stage decisions enter residual costs while residual feasibility stays fixed. General coupling to second-stage constraints is excluded. Separable convex costs plus a low-rank concave term can describe economies of scale in aggregate quantities with many integer variables. Orders with binary variables describe isotonic fitting with switching; simplex products describe allocation; graph equalities describe actuator

models. Each application must meet the actual law, curvature, domain, and oracle hypotheses. These examples provide no performance evidence.

Core-only noise locates the core, while a residual modulus or an optimal-face certificate supplies point output in our theorems. These are sufficient structures, rather than necessary conditions for every problem. The exact-arithmetic companion also gives full selected-point oracles for residual-convex cubics without a modulus, and supplied-convexifier point oracles under its stated degree and global-convexity hypotheses [2]. The general arithmetic obstruction of [theorem 9.10](#) starts at degree four.

10.2 Limitations

Numerical ratios such as L/σ and $\nu \text{diam}(X)/\sigma$ enter through their magnitudes. The FPT claims parameterize them jointly with structure; for a fixed bounded ratio they are FPT in structure. A polynomially growing ratio in $(1+R)^{O(k)} \text{poly}(I)$ gives $I^{O(k)}$, rather than FPT in k alone. The sparse bounds are polynomial for each fixed width, with a dimension factor necessary for the specified retention rule ([theorem 9.1](#)).

Finite laws need enough resolution for the base-only format, stopping-depth, and fallback budgets. Polynomial-precision routes admit one resolution per input length, scaled by the supplied noise scale ([proposition 2.3](#)). General nonlinear core-only flow and TU recourse retain a budget depending on a supplied core-size bound. Arbitrary coarse laws are not covered. Most closure routes handle ties with a rare same-draw fallback; lattice and strong-field methods finish directly.

Several premises are supplied: sharper curvature bounds, cores, tree decompositions, convexifiers, residual convexity or moduli, and recourse oracles valid under every required restriction. Certificate encoding length belongs in I and verification work in the runtime. A general polynomial inequality test is not hidden in the input model.

Exact output also has a representation contract. Ordinary implicit or charted descriptors are compact and support evaluation to any precision; they do not provide exact threshold comparisons or active-set decisions. Fallback outputs have base-exponential bounds on every draw, and global proof records can also be large; their expected lengths enter the work bounds. Small-core algebraic outputs retain their per-draw parameter bounds. Component outputs give common-root data for each component and a symbolic sum of values; exact comparisons of unrelated algebraic sums are not claimed. The companion instead gives Cauchy names of a canonical selected point with one per-draw work factor controlling all precision queries under one base-selected law. Our finite descriptor and global proof record are an output distinction, rather than by themselves an improvement in optimization time.

The constants are large. In particular, the nonquadratic strong-field sufficient condition contains the conservative component-solver base $c_d = 2^{10000D}$, with D the least even integer greater than d , in both the derivative-variation and grid-size requirements. The quadratic case uses $a_i = 3$ for continuous coordinates and has a smaller sufficient scale. These conditions define proved regimes, but their conservative size and the absence of computational evidence leave practical performance unsettled.

10.3 Open questions

Width. Can exact sparse polynomial optimization under independent linear noise have expected work FPT jointly in width and numerical ratios? [Theorem 9.1](#) and [proposition 9.2](#) rule out the analyzed global-error retention rule and its naive bag-local replacement. Certified conditional values are one sufficient way to remove the ambient dimension from this count ([proposition 5.10](#)); the

current grid min-marginals do not supply them. They are not a proved necessary ingredient of every FPT algorithm. The deterministic growth-conditioned graded-grid companion gives another relevant starting point [1]. A linear small-growth tail alone does not integrate a generic inverse-growth work factor κ^r , $r > 1$, to an FPT expectation without a cutoff-dependent factor: its truncated-moment estimate retains $K^{1-1/r}$. This limits that integration argument, rather than other algorithms (section 5.4 and remark 4.10).

Uniform ambient noise. Can the low-rank route be FPT jointly in negative inertia and numerical ratios under independent uniform ambient noise? Theorem 9.4 excludes a dimension-free bound for the specified auxiliary counts; it does not exclude another algorithm.

Constraints. Which sparse linear constraints beyond the proved graph, simplex, and order domains admit exact smoothed algorithms? The affine hardness construction needs restrictions beyond sparsity (theorem 9.5), and the TU diagonal example calls for counts adapted to feasible faces (example 9.6).

Residual points. Which structures beyond the positive cubic, convexifier, modulus, and integer-face results permit efficient residual point evaluation under core-only noise? Convex quartics already carry the Square Root Sum and PosSLP implications of theorem 9.10. A continuous optimal-face certificate is one possible direction.

Formats and precision. Which further classes admit short common-root algebraic outputs on every draw? Can general nonlinear core-only flow and TU recourse use polynomial sampling precision while retaining the current parameter dependence, as bilinear coupling does?

Computation. Implementations with certified convex solvers, compact proof records, and smaller constants are needed to assess the certificates as practical termination tests.

A Proofs for Section 3

This appendix proves the results of Section 3. Standard facts from measure theory and convex analysis are cited where they are used.

A.1 Local counts

Proof of Lemma 3.2. In $F_c(v) \leq F_c(v + he_i) + \delta$ the terms $c_l v_l$ with $l \neq i$ cancel, leaving $f(v) \leq f(v + he_i) + c_i h + \delta$, which is the lower bound on c_i . The comparison with $v - he_i$ gives $f(v) \leq f(v - he_i) - c_i h + \delta$, which is the upper bound. The difference of the endpoints is $(f(v + he_i) + f(v - he_i) - 2f(v))/h + 2\delta/h$. If $g(t) = f(v + te_i) - L_i t^2/2$ is concave, then $g(h) + g(-h) \leq 2g(0)$, that is, $f(v + he_i) + f(v - he_i) - 2f(v) \leq L_i h^2$. $\square$

The three concentration bounds stated after Definition 3.3 are elementary. A density bounded by ϕ gives $\mathbb{P}(Y \in J) \leq \phi|J|$. An interval of length ℓ contains at most $\ell(M-1)/(2\sigma) + 1$ points of the endpoint grid with spacing $2\sigma/(M-1)$, so its probability under $U_{\sigma,M}$ is at most $\ell(M-1)/(2\sigma M) + 1/M \leq \ell/(2\sigma) + 1/M$. If $|\mathbb{P}(Y \leq t) - \nu(-\infty, t]| \leq \delta_K$ for all t , the same bound holds for left limits, and the probability of every interval, which is a difference of two such values, differs from its ν -probability by at most $2\delta_K$.

Proof of Theorem 3.4. Fix $v \in G$. By Lemma 3.2, $c \in E_v^\delta$ implies $c_i \in I_i(v, h_i, \delta)$ for every $i \in Q_v$. These intervals do not depend on c and have length at most $L_i h_i + 2\delta/h_i$. By independence,

$$\mathbb{P}(c \in E_v^\delta) \leq \prod_{i \in Q_v} \mathbb{P}(c_i \in I_i(v, h_i, \delta)) \leq \prod_{i \in Q_v} \min\{1, \pi_i(L_i h_i + 2\delta/h_i)\}.$$

Define $\psi_i(t) = 1$ for the two endpoints $t \in \{a_i, b_i\}$ and $\psi_i(t) = \min\{1, \pi_i(L_i h_i + 2\delta/h_i)\}$ for the $m_i - 1$ interior points of G_i . The bound reads $\mathbb{P}(c \in E_v^\delta) \leq \prod_i \psi_i(v_i)$, and summing over the tensor grid gives $\sum_{v \in G} \prod_i \psi_i(v_i) = \prod_i \sum_{t \in G_i} \psi_i(t)$, which is the claim. If f depends on a random element independent of c , apply this argument for each fixed value of that element and integrate. $\square$

Proof of Corollary 3.6. Since $\rho_i h_{j+1} = \rho_i h_j/2$, the least admissible power of two at level $j+1$ is m_{ij} or $2m_{ij}$. Equal division into m_{ij} parts is refined by equal division into $2m_{ij}$ parts, so the partitions are nested. At level 0, $w_i \leq \rho_i S_\rho = \rho_i h_0$, so $m_{i0} = 1$. Since $w_i/2^j \leq \rho_i h_j$, $m_{ij} \leq 2^j$. If $m_{ij} \geq 2$, minimality gives $w_i/(m_{ij}/2) > \rho_i h_j$, that is, $h_{ij} > \rho_i h_j/2$.

Let i satisfy $m_{ij} \geq 2$. For every l , including unrefined coordinates, $L_l h_{lj}^2 \leq L_l \rho_l^2 h_j^2 \leq c_{\text{bal}} L_i \rho_i^2 h_j^2 < 4c_{\text{bal}} L_i h_{ij}^2$. Summing over l ,

$$\frac{4E_j}{h_{ij}} = \frac{1}{2h_{ij}} \sum_{l=1}^k L_l h_{lj}^2 < 2c_{\text{bal}} k L_i h_{ij},$$

which proves the displayed inequality.

For the counts apply Theorem 3.4 with $\delta = 2E_j$, so that $L_i h_{ij} + 2\delta/h_{ij} \leq C_k L_i h_{ij}$ for refined coordinates; an unrefined coordinate contributes the factor 2. With densities, $(m_{ij} - 1)\phi_i C_k L_i h_{ij} \leq \phi_i C_k L_i w_i$. With $c_i \sim U_{\sigma_i, M}$ and $M \geq m_{ij}$, $(m_{ij} - 1)(C_k L_i h_{ij}/(2\sigma_i) + 1/M) \leq C_k L_i w_i/(2\sigma_i) + 1$. $\square$

A.2 Corrected-corner search

Proof of Lemma 3.7. Fix $x \in C$. Let $R_1, \dots, R_k$ be independent random variables with $R_i \in \{l_i, l_i + \eta_i\}$ and $\mathbb{E}R_i = x_i$; if $\eta_i = 0$ put $R_i = l_i$. Let $Y^{(0)} = x$, and let $Y^{(i)}$ be obtained from $Y^{(i-1)}$ by replacing its i th coordinate with R_i . Every $Y^{(i)}$ lies in C . Conditional on $Y^{(i-1)}$, the function $t \mapsto F_c(Y^{(i-1)} + (t - x_i)e_i) - L_i t^2/2$ is concave on $[l_i, l_i + \eta_i]$, because $F_c - f$ is linear. Jensen's inequality gives

$$\mathbb{E}[F_c(Y^{(i)}) \mid Y^{(i-1)}] \leq F_c(Y^{(i-1)}) + \frac{L_i}{2}(\mathbb{E}R_i^2 - x_i^2) = F_c(Y^{(i-1)}) + \frac{L_i}{2} \text{Var } R_i \leq F_c(Y^{(i-1)}) + \frac{L_i \eta_i^2}{8}.$$

Iterating, $\mathbb{E}F_c(Y^{(k)}) \leq F_c(x) + \sum_i L_i \eta_i^2/8$. The point $Y^{(k)}$ is a corner of C , so the least corner value is at most this expectation. $\square$

Proof of Theorem 3.9. Item (i) is Lemma 3.7, since every level- j cell has side lengths h_{ij} and hence correction E_j .

Let x^* be a global minimizer, which exists because f is lower semicontinuous and X is compact. We claim that at every processed level j , either $U_j = F^*$, or some processed cell $C_j \ni x^*$ of level j is not closed and is retained. At level 0 the cell X contains x^* . Suppose that a processed cell $C_{j'}$ of level j' contains x^* . If it is closed, the closure step finds a value at most $F_c(x^*) = F^*$, so $U_j = F^*$ for all $j \geq j'$, because the incumbent never increases and never falls below F^* . Otherwise $\underline{F}(C_{j'}) \leq F^* \leq U_{j'}$ by (i), so $C_{j'}$ is retained and one of its children contains x^* and is processed at level $j' + 1$. This proves the claim.

For (iii), if $U_j \neq F^*$, the cell C_j of the claim has a corner with value at most $F^* + E_j$ by Lemma 3.7, so $U_j \leq F^* + E_j$. For (ii), if $U_j \neq F^*$, the claim provides a retained cell. For (iv), a retained cell satisfies $\underline{F}(C) \leq U_j$, so its least corner value is at most $U_j + E_j \leq F^* + 2E_j$. Such a corner v satisfies $F_c(v) \leq \inf_X F_c + 2E_j$ and hence $c \in E_v^{2E_j}$. Assign to every retained cell one such corner. A grid node is a corner of at most 2^k cells of one level, which gives the bound on the number of retained cells. For (v), every retained cell has $\underline{F}(C) \geq U_j - E_j$, because all its corners were evaluated and U_j is at most their values; hence $U_j - \underline{U}_j \leq E_j$. If $U_j = F^*$, then $\underline{U}_j \leq U_j = F^*$;

otherwise the retained cell C_j of the claim satisfies $\underline{F}(C_j) \leq F^*$. The expectation bound follows from (iv) and Corollary 3.6. $\square$

Proof of Corollary 3.10. With $\rho_i = 1$, $S_\rho = s$ and $h_{iJ} \leq h_J = s2^{-J}$, so $E_J \leq kLs^24^{-J}/8 \leq \varepsilon$ by the choice of J . If the search stops before level J , Theorem 3.9(ii) gives an exact minimizer. Otherwise item (v) at level J gives the stated certificate with $\underline{U} = \underline{U}_J$ and x the stored incumbent point. Level 0 uses 2^k evaluations. A level $j \geq 1$ processes at most 2^k children of each cell retained at level $j-1$, and each child has 2^k corners. By Theorem 3.9 the expected number of cells retained at level $j-1$ is at most $2^k\mathcal{H}$, with $\mathcal{H}$ from Corollary 3.6; for the grid law, $M \geq 2^J \geq m_{ij}$ for all $j \leq J$. Hence each level $j \geq 1$ costs at most $8^k\mathcal{H}$ expected evaluations. $\square$

A.3 Finite laws

Proof of Lemma 3.13. (a) Let $x_j = -\sigma + 2\sigma j/(M-1)$. For $t \in [x_j, x_{j+1})$ with $0 \leq j \leq M-2$, the grid distribution function equals $(j+1)/M$, and the continuous one lies in $[j/(M-1), (j+1)/(M-1))$. The differences are at most $(j+1)/M - j/(M-1) = (M-1-j)/(M(M-1)) \leq 1/M$ in one direction and less than $(j+1)/(M-1) - (j+1)/M \leq 1/M$ in the other. Outside $[-\sigma, \sigma)$ both functions are 0 or 1. The second claim is Lemma 3.14(b), since scaling by σ preserves the Kolmogorov distance.

(b) For $0 \leq r \leq n$ let $\lambda^{(r)} = \nu_1 \otimes \cdots \otimes \nu_r \otimes \mu_{r+1} \otimes \cdots \otimes \mu_n$. The laws $\lambda^{(r-1)}$ and $\lambda^{(r)}$ differ only in the r th factor. By Fubini's theorem,

$$\lambda^{(r)}(E) - \lambda^{(r-1)}(E) = \int [\nu_r(E_{x_{-r}}) - \mu_r(E_{x_{-r}})] d\left(\bigotimes_{l < r} \nu_l \otimes \bigotimes_{l > r} \mu_l\right)(x_{-r}),$$

where $E_{x_{-r}}$ is the section of E . By hypothesis every section is a union of at most C intervals, hence a disjoint union of at most C intervals. The probability of an interval is a difference of two values of the distribution function or of its left limits, and both differ between μ_r and ν_r by at most δ_r . The integrand is therefore at most $2C\delta_r$ in absolute value. Summing over r proves the claim. $\square$

Construction and proof of Lemma 3.14. Put $K = b + 20$ and $e = 2^{-K}$. Let N be the least power of two with $\Delta = 2K/(N-1) \leq e/K$, so $\log_2 N = O(b)$, and let $z_j = -K + j\Delta$ for $0 \leq j < N$. Write $\varphi(z) = \exp(-z^2/2)$. The weight rule below assigns to every j a number $w_j \in [0, 1]$ with a common dyadic denominator 2^q , $q = O(b)$, and $|w_j - \varphi(z_j)| \leq e/K$.

Sampler. Repeat at most $T = \lceil 4K^2 \rceil$ times: draw j uniformly from $\{0, \dots, N-1\}$ using $\log_2 N$ fair bits, draw U uniformly from $\{0, \dots, 2^q - 1\}$ using q fair bits, and return z_j if $U < 2^q w_j$. If no attempt succeeds, return 0. Let $a = N^{-1} \sum_j w_j$ be the success probability of one attempt. The output equals z_j with probability $(w_j/(Na))(1 - (1-a)^T)$, so conditional on success its law π is proportional to the weights; the failure atom at 0 has mass $(1-a)^T$.

Failure mass. At least $2/\Delta - 1$ grid points lie in $[-1, 1]$, a fraction of at least $(2-\Delta)/(2K+\Delta) \geq 1/(2K)$ of all points, and there $w_j \geq \varphi(1) - e/K \geq 1/2$. Hence $a \geq 1/(4K)$ and $(1-a)^T \leq \exp(-T/(4K)) \leq \exp(-K) \leq e$.

Kolmogorov distance. Let Φ be the standard normal distribution function, and compare in four steps. First, the output law differs from π by at most the failure mass, hence by at most e in Kolmogorov distance. Second, let π^0 be the law proportional to $\varphi(z_j)$. With $W = \sum_j w_j$ and $W_0 = \sum_j \varphi(z_j)$, every set A of grid points satisfies

$$\left| \frac{\sum_A w_j}{W} - \frac{\sum_A \varphi(z_j)}{W_0} \right| \leq \frac{2 \sum_j |w_j - \varphi(z_j)|}{W} \leq \frac{2Ne}{KW}.$$

The Riemann estimate below gives $\Delta W_0 \geq \int_{-1}^1 \varphi - (K+2)\Delta \geq 1$, so $\Delta W \geq 1 - N\Delta e/K \geq 1/2$ and the displayed quantity is at most $4N\Delta e/K = 4(2K+\Delta)e/K \leq 9e$. Third, φ is bounded by 1 and 1-Lipschitz. For $t \in [-K, K]$, comparing $\Delta \sum_{z_j \leq t} \varphi(z_j)$ cell by cell with $\int_{-K}^t \varphi$ gives an error at most $(2K/\Delta)\Delta^2/2 + 2\Delta = (K+2)\Delta$, and the same holds for the total mass. Normalizing by masses at least 1, the distribution function of π^0 differs from that of the normal law truncated to $[-K, K]$ by at most $2(K+2)\Delta \leq 3e$. Fourth, truncation to $[-K, K]$ changes Φ by at most $3\Phi(-K) \leq 3\exp(-K^2/2) \leq e$. The total is at most $14e < 2^{-b}$.

Weights. Fix j and put $t = z_j^2/2 \leq K^2/2$ and $s = \lceil \log_2 K^2 \rceil + 1$, so that $u = t/2^s \leq 1/4$. Let $P = s + 3 + \lceil \log_2(K/e) \rceil$. Compute y_0 by rounding $\sum_{r=0}^P (-u)^r/r!$ to a multiple of 2^{-P} and clamping to $[0, 1]$; the alternating series gives $|y_0 - e^{-u}| \leq 4^{-P-1} + 2^{-P} \leq 2^{1-P}$. For $r = 0, \dots, s-1$, let y_{r+1} be y_r^2 rounded to a multiple of 2^{-P} and clamped to $[0, 1]$. Since $|\alpha^2 - \beta^2| \leq 2|\alpha - \beta|$ on $[0, 1]$ and clamping cannot increase the distance to a point of $[0, 1]$, the error at most doubles and gains 2^{-P} per step, so $|y_s - \varphi(z_j)| \leq 2^s(2^{1-P} + 2^{-P}) \leq e/K$. Put $w_j = y_s$ and $q = P$. The rounded weights have $O(b)$ bits. The exact Taylor sum before rounding has polynomial bit length: u has $O(b)$ bits and $P = O(b)$, so its powers and the factorial denominators require $O(b^2 + b \log(b+1))$ bits. Thus a weight still costs $\text{poly}(b)$ bit operations.

Cost and support. Each attempt uses $O(b)$ fair bits and $\text{poly}(b)$ bit operations, and there are $O(b^2)$ attempts. The atoms are 0 and the rationals $z_j = -K + 2Kj/(N-1)$, of bit length $O(b)$, in $[-(b+20), b+20]$. $\square$

A.4 The growth tail

We use three standard facts. A Lipschitz map $\mathbb{R}^n \rightarrow \mathbb{R}^n$ is differentiable almost everywhere (Rademacher's theorem) [21]. For a Lipschitz map $P : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and a Lebesgue measurable set A , the area formula gives $\int_A |\det DP(z)| dz = \int_{\mathbb{R}^n} \#(A \cap P^{-1}\{y\}) dy$ [21, 22]. A finite convex function on $\mathbb{R}^n$ is differentiable outside a Lebesgue null set [38].

Proof of Theorem 3.16. Measurability. Let $\varepsilon > 0$. If $g_*(c) \geq \varepsilon$, the minimizer is unique and satisfies the growth inequality with the finite coefficient ε , by taking limits of admissible coefficients if necessary. For a singleton the inequality is vacuous for every finite coefficient. Let $c_m \rightarrow c$ with $g_*(c_m) \geq \varepsilon$, and let x_m be the corresponding minimizers, so that $f(x_m) \leq f(x) + c_m^\top(x - x_m) - \varepsilon \|x - x_m\|^2$ for all $x \in X$. Along a subsequence $x_m \rightarrow \bar{x} \in X$, and lower semicontinuity gives $f(\bar{x}) \leq f(x) + c^\top(x - \bar{x}) - \varepsilon \|x - \bar{x}\|^2$. Hence $g_*(c) \geq \varepsilon$. The set $\{g_* \geq \varepsilon\}$ is closed and $\{g_* < \varepsilon\}$ is open.

A conjugate in coefficient space. Put $z = -c$, $\lambda = 2\varepsilon$, $\ell_i = \min_X x_i$, $r_i = \max_X x_i$, and

$$H(a) = \max_{x \in X} \{a^\top x - f(x) + \varepsilon \|x\|^2\}, \quad a \in \mathbb{R}^n.$$

The maximum is attained because $-f$ is upper semicontinuous and X is compact. H is convex and finite. Every maximizer x_a is a subgradient: $H(a') \geq a'^\top x_a - f(x_a) + \varepsilon \|x_a\|^2 = H(a) + (a' - a)^\top x_a$. Every subgradient $u \in \partial H(a)$ satisfies $u_i \in [\ell_i, r_i]$: for $t > 0$, $H(a + te_i) \leq H(a) + tr_i$ and $H(a - te_i) \leq H(a) - t\ell_i$, while $H(a \pm te_i) \geq H(a) \pm tu_i$.

Proximal map. Let $P(z) = \arg \min_a \{H(a) + \|a - z\|^2/(2\lambda)\}$, which is unique by strong convexity, and $Q(z) = z - P(z)$. Optimality gives $Q(z)/\lambda \in \partial H(P(z))$, so $Q_i(z) \in [\lambda\ell_i, \lambda r_i]$. Monotonicity of ∂H gives $\langle P(z) - P(z'), Q(z) - Q(z') \rangle \geq 0$. Writing $z - z'$ as the sum of the two differences shows that P and Q are monotone and 1-Lipschitz. In particular, for fixed z_{-i} , the function $t \mapsto Q_i(t, z_{-i})$ is nondecreasing, Lipschitz, and has values in an interval of length λw_i .

Regular points certify growth. Suppose H is differentiable at $a = P(z)$. Then $\partial H(a) = \{\nabla H(a)\}$, so every maximizer in the definition of $H(a)$ equals $x^* = \nabla H(a) = Q(z)/\lambda$, and $x^* \in X$. Maximality

gives, for all $x \in X$, $f(x) - f(x^*) - a^\top(x - x^*) \geq \varepsilon(\|x\|^2 - \|x^*\|^2)$. Substituting $a = z - 2\varepsilon x^*$ yields

$$f(x) - z^\top x - (f(x^*) - z^\top x^*) \geq \varepsilon \|x - x^*\|^2.$$

With $c = -z$ this is the growth inequality, so $g_*(c) \geq \varepsilon$.

The bad event has singular Jacobian. Let N_0 be a Borel null set containing the points where H is not differentiable, and let $E = P^{-1}(N_0)$, a Borel set. By the previous step, $\{z : g_*(-z) < \varepsilon\} \subseteq E$. For every m , the area formula applied to $A = E \cap [-m, m]^n$ gives $\int_A |\det DP| = \int \#(A \cap P^{-1}\{y\}) dy = 0$, because the integrand vanishes outside the null set N_0 . Hence $\det DP = 0$ almost everywhere on E .

Trace bound. At a point where P is differentiable, $DP + DQ = I$, and monotonicity of P and Q makes the symmetric parts of DP and DQ positive semidefinite. If $\det DP = 0$, choose $v \neq 0$ with $DPv = 0$; then $DQv = v$, and the trace of DQ , which equals the trace of its positive semidefinite symmetric part, is at least the Rayleigh quotient $v^\top DQv / \|v\|^2 = 1$. Therefore

$$\mathbf{1}_E(z) \leq \operatorname{tr} DQ(z) = \sum_{i=1}^n \partial_i Q_i(z) \quad \text{for almost every } z,$$

where outside E the right side is nonnegative. Here $\partial_i Q_i$ is the (i, i) entry of DQ ; by Fubini's theorem it coincides almost everywhere with the derivative of the section $t \mapsto Q_i(t, z_{-i})$.

Integration. The coordinates of $z = -c$ are independent with densities $p_i \leq \phi_i$, so null sets have probability zero. For fixed z_{-i} , the section of Q_i is absolutely continuous and nondecreasing with total increase at most λw_i , so $\int_{\mathbb{R}} \partial_i Q_i(t, z_{-i}) p_i(t) dt \leq \phi_i \lambda w_i$. Integrating over z_{-i} and summing over i ,

$$\mathbb{P}\{g_*(c) < \varepsilon\} \leq \mathbb{P}(z \in E) \leq \sum_i \mathbb{E} \partial_i Q_i(z) \leq 2\varepsilon \sum_i \phi_i w_i.$$

Consequences and sharpness. For uniform laws on $[-\sigma, \sigma]$, $\phi_i = 1/(2\sigma)$. Taking $\varepsilon = \rho/(2\sum_i \phi_i w_i)$ gives the high-probability statement; $g_* > 0$ forces a unique minimizer. For the example, $c \neq 0$ has the unique minimizer $x^* = -\operatorname{sgn}(c)w/2$ and $F_c(x) - F_c(x^*) = |c| |x - x^*| \geq (|c|/w) |x - x^*|^2$, with equality at the other endpoint. Hence $g_* = |c|/w$ and $\mathbb{P}\{g_* < \varepsilon\} = \mathbb{P}\{|c| < \varepsilon w\} = \varepsilon w/\sigma$ for $\varepsilon w \leq \sigma$. $\square$

A.5 Finite-law tails and the fallback budget

We use the following block-sensitive quantifier-elimination theorem.

Theorem A.1 (Renegar [37, Theorem 1.1]). *Consider a formula*

$$Q_1 x^{(1)} \dots Q_\omega x^{(\omega)} \Psi(y, x^{(1)}, \dots, x^{(\omega)})$$

with alternating quantifier blocks of positive sizes $n_1, \dots, n_\omega$, $\ell \geq 1$ free variables y , and a quantifier-free matrix Ψ with $m \geq 1$ atomic polynomial sign conditions of degree at most $d \geq 2$. There is an equivalent quantifier-free formula in y , a disjunction of conjunctions of polynomial sign conditions, in which the number of conjunctions, the number of atoms in each conjunction, and their degrees are at most $(md)^{2^{O(\omega)\ell}} \prod_k n_k$. This holds over the reals for arbitrary real coefficients. If the coefficients are integers of bit length at most L , the output coefficients have bit length at most $(L+1)(md)^{2^{O(\omega)\ell}} \prod_k n_k$, and the formula is computed using $(L+1) \log(L+2) \log \log(L+3) (md)^{2^{O(\omega)\ell}} \prod_k n_k$ bit operations and $(md)^{O(\ell + \sum_k n_k)}$ evaluations of Ψ on sign vectors.

The constant a_R of Lemma 3.17 is chosen so that, for $\omega = 2$, $\ell \leq 2$, and $n_1 = n_2 = N$, the format bound is at most $(md)^{a_R(N+1)^2}$.

Proof of Lemma 3.17. Let Φ be the formula describing $\mathcal{D}$. If $n + m = 0$, the domain is a singleton and the growth assertion is immediate; there are no active coordinates. In the remaining argument $n + m \geq 1$, so the two quantified blocks have positive sizes.

(a) Let $\gamma(t)$ have i th coordinate t and the fixed values elsewhere. We claim that $g_*(\gamma(t)) \geq \varepsilon$ if and only if

$$\exists(x, y) \forall(x', y') : \quad \Phi(x, y) \wedge \left[-\Phi(x', y') \vee F(x', y') + \gamma(t)^\top x' - F(x, y) - \gamma(t)^\top x - \varepsilon \|x' - x\|^2 \geq 0 \right].$$

If $g_* \geq \varepsilon$ with minimizer x^* , take y^* with $F(x^*, y^*) = f(x^*)$; then every feasible (x', y') satisfies $F(x', y') + \gamma^\top x' \geq f(x') + \gamma^\top x' \geq f(x^*) + \gamma^\top x^* + \varepsilon \|x' - x^*\|^2$. Conversely, if (x, y) satisfies the formula, then taking $x' = x$ shows $F(x, y) = f(x)$, and taking y' optimal for each x' gives the growth inequality at x . The formula has two blocks of $n + m$ variables, one free variable t , $2s + 1$ atoms, and degree at most $\max\{d, 2\}$, because the terms $t(x'_i - x_i)$ and $\varepsilon \|x' - x\|^2$ have degree two. Theorem A.1 gives an equivalent quantifier-free formula in t with at most H_s^2 polynomial occurrences of degree at most H_s . Discard the identically zero polynomials, whose signs are constant. The others have at most H_s^3 real roots in total. Between consecutive roots and at each root every atom has constant truth value, so the good set and its complement are unions of at most $2H_s^3 + 1$ intervals. The bound does not depend on ε or on the fixed values, which enter only as real coefficients.

(b) Under independent continuous uniform laws on $[-\sigma, \sigma]$, Theorem 3.16 gives $\mathbb{P}\{g_* < \varepsilon\} \leq S\varepsilon/\sigma$; the reduced objective f is lower semicontinuous because $\mathcal{D}$ is compact and F is continuous. The event $\{g_* < \varepsilon\}$ is open. By (a) and Lemma 3.13 with $\delta_i = 1/M$, the probability changes by at most $2nC_{\text{tail}}/M$. Since $\{g_* = 0\} = \bigcap_{\varepsilon > 0} \{g_* < \varepsilon\}$, letting $\varepsilon \downarrow 0$ gives the second bound.

(c) The same argument with the density bound $1/(\sigma\sqrt{2\pi})$ and $\delta_i = \delta_K$.

(d) Fix an integer assignment, a face of the continuous box (each continuous coordinate fixed at its lower bound, fixed at its upper bound, or free), and a coordinate i fixed at a bound on that face. Let u be the k' free coordinates. Condition on all coefficients other than γ_i . The free stationarity system $\nabla_u F(u, x_{\text{fixed}}) + \gamma_u = 0$ does not involve γ_i , and its equations have degree at most D . By Lemma C.1, it has at most $D^{k'}$ nonsingular complex solutions; for $k' = 0$ there is one candidate, and if $d \leq 1$ and $k' > 0$ no solution is nonsingular. At each such solution the active gradient $\partial_i F_\gamma$ equals γ_i plus a number that is fixed under the conditioning, so the event that its absolute value is at most τ confines γ_i to an interval of length 2τ . Its conditional probability is at most $\tau/\sigma + 1/M$ under $U_{\sigma, M}$ and at most $\sqrt{2/\pi} \tau/\sigma + 2\delta_K$ under the laws of (c).

If $g_* > 0$, the minimizer x^* is unique. Let its integer assignment and its smallest continuous face be the ones fixed above. Its free coordinates lie in the open intervals, so $\nabla_u F_\gamma(x^*) = 0$. For every free direction v and small t , $x^* \pm tv$ is feasible, and the growth inequality with a second-order expansion gives $v^\top \nabla_{uu}^2 F(x^*) v \geq 2g_* \|v\|^2$. The free coordinates of x^* therefore form one of the counted nonsingular solutions. A union over at most R_Z integer assignments, 3^{n_c} faces, n_c choices of i , and D^{n_c} solutions proves the bound. $\square$

Proof of Lemma 3.18. (a) $\mathbb{E}[1_{\mathcal{E}} W_1] \leq \mathbb{P}(\mathcal{E}) \sup W_1 \leq B^{-1} B P(L_{\max})$.

(b) Under $U_{\sigma, M}$, Lemma 3.17(b) gives $\mathbb{P}\{g_* < g_0\} \leq \rho + 2nC_{\text{tail}}/M$, and part (d) gives a probability at most $K_{\text{act}}(\tau/\sigma + 1/M) \leq \rho + K_{\text{act}}/M$ for the active-gradient event, which contains the second part of $\mathcal{E}_{\text{bad}}$. The sum is at most $2\rho + (2nC_{\text{tail}} + K_{\text{act}})/M \leq 3\rho$. Under the Gaussian-like condition the two bounds are $\sqrt{2/\pi} \rho + 2nC_{\text{tail}}\delta_K$ and $\sqrt{2/\pi} \rho + 2K_{\text{act}}\delta_K$, with sum at most 3ρ . $\square$

A.6 The exact fallback

Lemma A.2 (One shared root for a specified tuple). *Let $m \geq 1$. For $i = 1, \dots, m$, let $p_i \in \mathbb{Z}[t]$ have positive degree δ_i , and let a rational interval J_i isolate a specified real root a_i of p_i . Point*

intervals are allowed. Let $H \geq 1$ bound the coefficient and endpoint bit lengths and put $D = \prod_i \delta_i$. A deterministic algorithm constructs a squarefree $P \in \mathbb{Z}[t]$ of degree at most D , an interval isolating one real root θ , and rational polynomials r_i of degree less than $\deg P$ such that $a_i = r_i(\theta)$. Construction work and all coefficient and endpoint lengths are polynomial in $D, m, H + 1$, with absolute exponents. Refinement to scalar error 2^{-q} costs $\text{poly}(D, m, H + q + 1)$; the same bound gives Euclidean tuple error 2^{-q} after an additional $O(\log(m + 1))$ bits. Exact signs of explicitly listed fixed-degree rational polynomials in the tuple are decidable within a polynomial in these bounds and in the polynomial's encoding length.

Proof. A reduced tensor algebra. Replace each p_i by its primitive squarefree part, with positive leading coefficient. Rational gcd and exact division preserve its distinct roots and take polynomial degree/height work. Write d_i for the new degree and $N = \prod_i d_i \leq D$. Divide by the leading coefficients to work with monic rational polynomials. The algebra

$$\mathcal{A} = \mathbb{Q}[X_1, \dots, X_m] / (p_1(X_1), \dots, p_m(X_m))$$

has the N -element basis of monomials $X_1^{e_1} \dots X_m^{e_m}$ with $0 \leq e_i < d_i$. Multiplication by X_i is the tensor product of its univariate companion matrix with the other identity matrices; denote it by M_i . Over $\mathbb{C}$, evaluation identifies this algebra with $\prod_{\zeta \in \mathcal{T}} \mathbb{C}$, where $\mathcal{T} = \prod_i Z_{\mathbb{C}}(p_i)$ consists of exactly N distinct tuples. Thus the M_i are simultaneously diagonalizable with joint eigenvalues $\zeta \in \mathcal{T}$.

A separating form with an exact test. For integers

$$0 \leq t \leq (m - 1) \binom{N}{2}, \quad \lambda(t) = (1, t, \dots, t^{m-1}),$$

use $t^0 = 1$, also when $t = 0$. Two distinct tuples ζ, η have equal projections only when the nonzero polynomial $\sum_i (\zeta_i - \eta_i) t^{i-1}$ vanishes. Each pair excludes at most $m - 1$ integers, so the displayed list contains a form separating every complex tuple. For each form compute

$$M_\lambda = \sum_i \lambda_i M_i, \quad \chi_\lambda(T) = \det(TI - M_\lambda) = \prod_{\zeta \in \mathcal{T}} (T - \lambda^\top \zeta).$$

The form separates precisely when $\gcd(\chi_\lambda, \chi'_\lambda) = 1$. Choose the first form passing this rational gcd test. This includes $m = 1$ and $N = 1$.

Coordinate maps. For the chosen form compute independently for each i

$$q_i(T) = - \frac{\partial}{\partial u} \det(TI - M_\lambda - u M_i) \Big|_{u=0}.$$

The product formula gives

$$q_i(T) = \sum_{\zeta \in \mathcal{T}} \zeta_i \prod_{\eta \neq \zeta} (T - \lambda^\top \eta), \quad q_i(\lambda^\top \zeta) = \zeta_i \chi'_\lambda(\lambda^\top \zeta).$$

Since χ'_λ is invertible modulo χ_λ , define $r_i = q_i(\chi'_\lambda)^{-1} \pmod{\chi_\lambda}$, taking the representative of degree less than N . These identities recover every coordinate of every Cartesian tuple. Clearing a positive denominator and removing the content of χ_λ gives a squarefree integer polynomial P with the same roots. It need not be irreducible or minimal.

Selection of the designated root. The target root is $\theta = \sum_i \lambda_i a_i$. Every real root of P corresponds to an all-real tuple: otherwise its distinct complex conjugate would have the same real projection, contradicting separation. Let 2^{-S} be an effective strict lower bound on distances between distinct

roots of P . Integer polynomial separation bounds give $S \leq \text{poly}(N, H_P + 1)$, where H_P is the coefficient length of P [10]. Put $\Lambda_0 = \sum_i |\lambda_i| \geq 1$. Refine each original isolator J_i to width at most $2^{-S-2}/\Lambda_0$, preserving its chosen root. The rational interval $J_\theta = \sum_i \lambda_i J_i$ then contains θ and has width at most 2^{-S-2} , so it contains no other root of P . Endpoint tests and closed-interval Sturm counting certify its root count. This step selects the tuple specified by the original isolators; separation of all Cartesian tuples alone would not select it.

Heights and work. Companion entries have heights polynomial in D, H . One common denominator is the product of the leading coefficients of the squarefree p_i , of bit length polynomial in D, m, H . The number of forms is $O(mN^2 + 1)$; tested integers t have $O(\log(m+1) + \log(N+1))$ bits, and each $\lambda_i = t^{i-1}$ has $O(m \log((m+1)(N+1)))$ bits. Hence matrix entry lengths stay polynomial. If $c > 0$ clears their denominators, put $A_\lambda = cM_\lambda$ and $A_i = cM_i$. The integer polynomials

$$\tilde{\chi}(T) = \det(cTI - A_\lambda) = c^N \chi_\lambda(T), \quad \tilde{q}_i(T) = -[u^1] \det(cTI - A_\lambda - uA_i) = c^N q_i(T)$$

have coefficient lengths $O(N(H_{\text{mat}} + \log(N+1)))$ by determinant bounds. They can be computed by interpolation on $0, \dots, N$ in each of the two variables T, u , using polynomially many rational determinants. Only two-variable interpolation is involved; sample values, interpolation denominators, and intermediate heights are polynomial in D, m, H . Subresultant or Sylvester-minor bounds likewise control gcds, modular inverses, and the r_i by a polynomial with absolute exponents. The same univariate bounds control root isolation and refinement. In particular one may take $S = CN^2(H_P + \log(N+1) + 1)$ for an effective absolute C , so the source precision used to select θ is polynomially bounded.

For later evaluation, a Cauchy bound gives $|\theta| \leq 2^{H_P+1}$. Termwise derivative bounds on this interval give rational $K_i \geq \max\{1, |r'_i|\}$ of polynomial logarithmic length. Refine θ with $q + \lceil \log_2 \max_i K_i \rceil + O(1)$ bits and evaluate the maps by rational interval arithmetic. An extra $\lceil \frac{1}{2} \log_2 \max\{1, m\} \rceil + O(1)$ bits gives the Euclidean bound. Finally substitute these maps in any explicitly listed fixed-degree rational polynomial, clear positive denominators, and reduce modulo P . Univariate sign determination at the selected isolated root decides its sign, including zero, with polynomial work. This proves every assertion. $\square$

Proof of Theorem 3.20. A canonical minimizer. The set $\mathcal{O}$ of global minimizers of F_γ on the compact set $\mathcal{D}$ is nonempty and compact. Put $\mathcal{O}_0 = \mathcal{O}$ and, for $i = 1, \dots, N$, let $\mathcal{O}_i$ be the set of points of $\mathcal{O}_{i-1}$ whose i th coordinate is minimal. Each $\mathcal{O}_i$ is nonempty and compact, and $\mathcal{O}_N$ is a single point x^{lex} , the lexicographically least global minimizer. It exists even when $\mathcal{O}$ is a continuum.

Singleton formulas. Let Φ be Φ_0 conjoined with the disjunctions $\bigvee_{t=\ell_j}^{u_j} (x_j = t)$ for $j \in \mathcal{Z}$, and let

$$\text{LexGE}(y, x) = \left(\bigwedge_j y_j = x_j \right) \vee \bigvee_{j=1}^N \left[\left(\bigwedge_{i < j} y_i = x_i \right) \wedge y_j > x_j \right],$$

which has $O(N^2)$ atoms and expresses $y \geq_{\text{lex}} x$. For $1 \leq i \leq N$ consider

$$\Theta_i(t) : \quad \exists x \forall y : \Phi(x) \wedge x_i = t \wedge \left[\neg \Phi(y) \vee F_\gamma(y) > F_\gamma(x) \vee (F_\gamma(y) = F_\gamma(x) \wedge \text{LexGE}(y, x)) \right],$$

and let $\Theta_0(t)$ be the same formula with $x_i = t$ replaced by $F_\gamma(x) = t$. If x witnesses $\Theta_i(t)$, then x is feasible, no feasible point has a smaller value, and every other minimizer is lexicographically larger; hence $x = x^{\text{lex}}$ and $t = x_i^{\text{lex}}$. Conversely, $x = x^{\text{lex}}$ witnesses $\Theta_i(x_i^{\text{lex}})$. Thus the truth set of Θ_i is $\{x_i^{\text{lex}}\}$, and that of Θ_0 is $\{f^*\}$. All $N+1$ formulas refer to the same point.

Elimination. Each formula has two blocks of N variables, one free variable, at most $2s + O(N^2)$ atoms, where s is the number of atoms of Φ , and degree at most $\max\{d, 2\}$. Here $\log(s+1)$ is

polynomial in I , because the integer disjunctions contribute $\sum_j (u_j - \ell_j + 1)$ atoms with binary-encoded bounds. Multiply every atom by a positive integer that clears its denominators. The resulting integer coefficients have bit length $L_c \leq \text{poly}_d(I + b)$, because each atom involves the coefficients of F_0 , Φ_0 , and at most N coefficients γ_i . By Theorem A.1, an equivalent quantifier-free formula in t is computed within $(L_c + 1) \log(L_c + 2) \log \log(L_c + 3) \cdot 2^{\text{poly}_d(I)}$ bit operations plus $2^{\text{poly}_d(I)}$ evaluations of the matrix, each costing time polynomial in its explicit length $2^{\text{poly}_d(I)}$. The output has at most $2^{\text{poly}_d(I)}$ polynomials of degree at most $2^{\text{poly}_d(I)}$ with integer coefficients of bit length at most $(L_c + 1)2^{\text{poly}_d(I)}$. All bounds on the exponential factors depend only on the base data.

Univariate recovery. Replace constant atoms by their truth values. The singleton θ in the truth set is a root of some remaining nonconstant polynomial: otherwise all atoms have locally constant signs near θ , and the formula would hold on a neighborhood. Let P be the squarefree part of the product of the nonconstant polynomials. Isolate the real roots of P . At each root decide the sign of every atom polynomial: a polynomial vanishes at the root exactly when the root is a root of its greatest common divisor with P , which is decided by counting roots in the isolating interval; a nonvanishing polynomial has no root in that interval, so its sign is the sign at an endpoint. Evaluate the formula at each root; exactly one root satisfies it. Squarefree decomposition, real-root isolation, sign determination, and refinement of an isolating interval to width 2^{-q} for an integer polynomial of degree D and coefficient bit length τ take $\text{poly}(D, \tau, q)$ bit operations [10]. There is therefore a base-computable $A_0 = 2^{\text{poly}_d(I)}$ bounding every scalar degree independently of b , while scalar coefficient and endpoint lengths are at most $A_0(I + b + 1)^{e_0}$ for a fixed exponent e_0 . These are compatible scalar representations of the one canonical tuple, not yet the shared-root output.

Shared-root conversion and its base budget. Apply Lemma A.2 to the $N + 1$ scalars consisting of the coordinates and the value. Their product degree is at most A_0^{N+1} , whose logarithm is $\text{poly}_d(I)$ and does not depend on b . The lemma's absolute polynomial work and height bounds are consequently at most $B(I + b + q + 1)^{e_d}$ after enlarging the base-only integer $B = 2^{\text{poly}_d(I)}$ and a fixed exponent e_d . This proves the claimed separation of degree from coefficient height: no power of b or q depending on N occurs. The enlarged B , including elimination, all scalar recoveries, and conversion, is computed before rare-event thresholds or sampling precision are chosen.

The source isolators select the tuple (x^{lex}, f^*) , so at the selected root $F_\gamma(r_1(\theta), \dots, r_N(\theta)) = r_f(\theta)$. This equality can be verified by exact sign determination there. It need not hold modulo P : other Cartesian tuples may pair coordinate roots with an unrelated value root. For an integer coordinate, refine its scalar isolator to width less than $1/2$ and extract its unique possible integer; the domain premise guarantees that integer exists. List it explicitly and discard its map from the submitted representation. All other maps and the value share θ , as required by the model. Refinement with the additional $O(\log(N + 1))$ coordinate precision allowance gives (ii).

Rational feasible approximation. Let $\mathcal{D}$ be a mixed box with continuous coordinates in $[\ell_i, u_i]$, and let G be the rational number obtained by bounding every monomial derivative termwise on the box $\prod_i [\ell_i, u_i]$; then $G \geq \max\{1, \sup \|\nabla F_\gamma\|_1\}$ over the box, and $\log G = \text{poly}_d(I + b)$. Put $\delta = 2^{-q}$. Refine every continuous coordinate of x^{lex} to an interval of width at most $\delta/(2G \max\{1, N\})$, take its rational midpoint, and clip it to $[\ell_i, u_i]$; clipping does not increase the distance to $x_i^{\text{lex}} \in [\ell_i, u_i]$. Refine every integer coordinate to an interval of width less than 1 and take the integer in it. The resulting rational point y lies in $\mathcal{D}$ and has the integer coordinates of x^{lex} , so the segment between them lies in the box and $0 \leq F_\gamma(y) - f^* \leq G \|y - x^{\text{lex}}\|_\infty \leq \delta/2$. Refine f^* to an interval of width at most $\delta/2$ and let ℓ be its lower endpoint. Then $\ell \leq f^*$ and $F_\gamma(y) - \ell \leq \delta$. These refinements also give $\|y - x^{\text{lex}}\| \leq \delta$; they need $q + O(\log(N + 1)) + \text{poly}_d(I + b)$ bits and fit in the bound of (ii). $\square$

A.7 Resolution uniform over an input-length class

Proof of Corollary 3.21. Grid schedules. If $I_\Pi \leq I$, monotonicity of P_0 gives $M(I) \geq M_\Pi$. The schedule hypothesis supplies every-draw correctness, the same output format and same-draw fallback, and the original structural and numerical work factor. Only its fixed bit polynomial is now evaluated at $I_\Pi + \log_2 M(I)$, which is polynomial in I . The algorithm evaluates the fixed envelope and samples by its bounded sampler; it does not enumerate inputs, verify all promises in the class, or materialize atoms. Certificate verification and any explicit oracle factor remain charged.

Gaussian support and precision together. Put $H = A(I) + D(I) + 21$ and $t = \lceil 4 \log_2 H \rceil$. Since $H \geq 3$, $t \leq 4 \log_2 H + 1$ and $\log_2 H \leq H$. With $b = A(I) + D(I)t$,

$$b + 20 \leq H + 4H \log_2 H \leq H + 4H^2 \leq H^4 \leq 2^t, \quad 2^t \leq 2H^4.$$

The integer t is computed by comparing powers of two with H^4 , so the construction needs no real logarithm oracle. Also $b \geq A(I_\Pi) + D(I_\Pi)t \geq b_\Pi(t)$. Thus both schedule requirements hold for the same pair (t, b) : the sampler support fits the trial box and its cap is computed for that box before drawing noise. The integers t, b are polynomially bounded in I , so the bit factor stays polynomial. The schedule preserves the original structural and numerical factor. Later q -bit evaluations refine the same exact output descriptor; they neither change the law nor redraw the instance.

A supplied structural bound. In (c) first compute the integer envelope $F(K)$, charging its stated evaluation cost. Since the encoding of K is part of I , this cost is at most $(1 + F(K))^{c_F} \text{poly}(I)$. The common resolution then dominates each required resolution with parameter at most K . The sampled lengths are bounded by $\text{poly}(I) + (1 + F(K))P_0(I)$ rather than by $\text{poly}(I)$ alone. Substituting this in a fixed polynomial work or output bound gives an extra factor $(1 + F(K))^c$ for a fixed c , enlarged to cover all charged costs. This proves (c) while retaining its explicit dependence on K . $\square$

We verify the schedule premises for the stated routes next. They do not follow from fixed polynomial degree alone.

Effective format and height envelopes. Fix the degree and other format constants of a route. Explicit dimensions, monomial lists, supplied factors, scales, curvature bounds, brackets, and certificates are counted in its base length. Positive rationals of length at most I have logarithmic magnitudes and reciprocal magnitudes $O(I)$. Rational matrix operations, fixed-degree derivative bounds, and rational LP vertex bounds have polynomial bit lengths. On the domains used by the polynomial-bit routes, this bounds the logarithms of their widths, inverse frame bounds, curvature bounds, and reciprocal scales by effective polynomials in I . These logarithmic estimates choose the law; they do not remove large numerical ratios from expected counts.

The fallback proof separates format degree from added coefficient height. It gives $B_\Pi \leq 2^{P_B(I_\Pi)}$ and work $B_\Pi \text{poly}_d(I_\Pi + b + q)$ with a fixed exponent, including shared-root conversion. Integer disjunctions may have exponentially many atoms, but their count has polynomial logarithm. Hence $\log H_s$ and $\log C_{\text{tail}}$ in Lemma 3.17, and the mixed-box active constant K_{act} , have polynomial envelopes. The fixed-block elimination formats for recourse tube constants have the same property in the routes stated with polynomial precision. These are effective estimates from the fixed algorithms. Finitely many estimates can be dominated by one nondecreasing polynomial with effective integer coefficients and degree independent of the structural parameters.

The geometric qualification matters. The compact degree-two domain $1 \leq x_0 \leq 2$, $x_i = x_{i-1}^2$ for $1 \leq i \leq N$, has description length $O(N \log N)$, but final width $2^{2^N} - 1$. Format and coefficient height alone do not give a polynomial logarithmic width bound there. This example concerns the

budget inference, not the difficulty of optimizing on that particular chain. The polynomial-bit routes use boxes, rational polytopes, fixed-depth explicit graphs, or supplied brackets and curvature data.

Uniform grids and lattice termination. The grid closure routes select M above finitely many base quantities such as 2^J , inverse failure budgets, fallback factors, and tail or tube constants. The preceding envelopes bound the logarithm of each by a fixed polynomial. In the proofs, increasing M decreases the atom term $1/M$, product-replacement errors, and finite isolation errors, while support stays $[-\sigma, \sigma]$. Their thresholds, search box, and cap remain valid. This verifies (a) for sparse, constrained, continuous and native-integer recourse, and interior and bilinear flow/TU. Aligned row noise uses the same $M(I)$ in each row, rescaled by that row's supplied σ_i .

For Theorem 4.9, $M \geq 2nDB$ has polynomial logarithm, and the capped-moment proof improves with larger M . Its Gaussian version similarly permits larger polynomial-bit accuracy. For Theorem 8.17, reset $J = \log_2 M$: terminal curvature error is $O(M^{-2})$, whereas value spacing is $1/[D_0(M-1)]$. Their ratio improves with M once the base inequality holds. Counts remain valid through the new polynomial cap. Using one M in every row retains the common value denominator.

Gaussian-like quadratic routes. In Appendices B.6 and B.8, trial radius $2^t \sigma$ gives nominal box width at most 2^t times its initial width. The mixed-gap schedule enforces $h_J \leq 1$, making its gap constant support independent. Thus $J_\Pi(t) \leq J_\Pi(0) + t$; the weaker $J_\Pi(t) \leq J_\Pi(0) + 2t$ would also suffice. The deterministic node budget $(J+1)(2^J+1)^k$ and bad-event requirements give

$$b_{\text{req}, \Pi}(t) \leq A(I_\Pi) + D(I_\Pi)t$$

for effective nondecreasing integer polynomials $A, D \geq 1$. Indeed $k \leq \text{poly}(I_\Pi)$, $J_\Pi(0)$ has a polynomial envelope, and $\log(J+1)$ is bounded by a polynomial base term plus t . Rational normalization increases encoding length only polynomially. The two-direction requirement $2^b \geq 2nDB$ can be included by enlarging A and is independent of support. The Gaussian-weighted counts use original projected widths, so the common trial support adds depth and bit costs without a new numerical width factor. This verifies (b), including anisotropic separable normalization. Original-objective regret uses the new support $(b(I) + 20)\sigma$; a calibration made with smaller support must be recomputed.

Strong fields and nonlinear boundary flow. Theorem 8.16 holds at every resolution with its actual interval probabilities $q_i(M)$ and derived $\beta(M)$. These probabilities need not decrease under refinement: endpoint grids of sizes M and $2M$ are not nested. Hence (a) does not preserve arbitrary distribution-dependent subcriticality. In the sufficient regime (41), however, the scale premise is retained and the lower bound on M has polynomial logarithm. A common grid preserves that sufficient regime, with q_i, β recomputed for the new law.

For nonlinear boundary flow and its TU transfer, the available distance-to-chart-margin proof gives $J_\Pi, \log_2 M_\Pi \leq F_d(k)P_d(I_\Pi)$. This parameter function can enter coefficient bit length, not just an analysis-only degree. The explicit parameter powers admit a nondecreasing integer envelope in supplied $K \geq k$ whose evaluation costs a fixed polynomial in $1 + \text{size}(K) + F_d(K)$. Choose this envelope for (c). Sampling, output, and work then carry a factor depending on K , with polynomial exponents independent of K . Setting $K = I$ supplies a computable input-length law but may charge that precision even when the actual core is small. Thus the proof gives no polynomial-bit law independent of k preserving the original actual-parameter bound in this family. This is a limitation of the available budget, not an impossibility theorem; interior and bilinear cases belong to (a).

Finally, each oracle or curvature premise must hold on the new support. The contracts here cover the requisite coefficient cubes or all rational queries, rather than a single old grid. Fixing the cost model is essential: unrelated oracle programs with arbitrary polynomial exponents do not share a uniform work constant. These arguments standardize a scalar family rescaled by the supplied noise scales in the theorem's perturbation coordinates. They neither interchange law families nor permit unbounded raw-grid refinement under a fixed atomic law.

B Proofs for Section 4

This appendix proves the results of Section 4. The expected-work theorems for quadratic programs, mixed-integer quadratic programs, and separable recourse share one argument, given in Appendix B.6; the separable and anisotropic results are derived from it in Appendix B.8 by listing the inputs that change.

B.1 Rational normalization

Proof of Lemma 4.4. Preliminaries. Test $A \succeq 0$ exactly by symmetric elimination; if it holds, stop. Otherwise $A \neq 0$. Let $H = \max_i \sum_j |A_{ij}| \geq \|A\|$, let $d = \text{rank } A \geq 1$, and let q_A be the product of the denominators of the entries of A . The coefficient of t^{n-d} in $\det(tI - q_A A)$ is a nonzero integer equal, up to sign, to q_A^d times the product of the d nonzero eigenvalues of A . Since the other nonzero eigenvalues have absolute value at most H , every nonzero eigenvalue has absolute value at least

$$\mu = q_A^{-d} H^{-(d-1)}.$$

In particular $\mu \leq \nu$. Start from $\beta = H$ and halve β while $A + (\beta/2)I \succeq 0$. Each successful test is equivalent to $\beta/2 \geq \nu$, so the final value satisfies $\nu \leq \beta < 2\nu$, after at most $1 + \log_2(H/\mu)$ tests. Compute the orthogonal projector R onto the range of A as $R = C(C^\top C)^{-1}C^\top$ for a maximal set C of linearly independent columns of A . All these numbers have polynomial bit length.

Rational Jacobi rotations. Put $\epsilon = \mu^2/(16n\beta)$ and $\tau = \epsilon/n$. If $n = 1$, no rotation is needed. Starting from $Q = I$ and $\bar{A} = A$, repeat while some off-diagonal entry of $\bar{A}$ exceeds τ in absolute value: choose a largest one, $\bar{A}_{pq} = b'$, with $a' = \bar{A}_{pp}$ and $c' = \bar{A}_{qq}$; for a rational t let $G(t)$ be the rotation in the (p, q) plane with entries $c_t = (1 - t^2)/(1 + t^2)$ and $s_t = 2t/(1 + t^2)$; replace Q by $QG(t)$ and $\bar{A}$ by $G(t)^\top \bar{A} G(t)$. The new (p, q) entry is $\varphi(t)/(1 + t^2)^2$ with $\varphi(t) = b'(1 - 6t^2 + t^4) + 2(c' - a')t(1 - t^2)$. Here $\varphi(0) = b'$, and at $t_{\text{end}} = -\text{sgn}(b'(c' - a'))/2$ (or $1/2$ if $c' = a'$), $\varphi(t_{\text{end}}) = -\frac{7}{16}b' - \frac{3}{4}|c' - a'|\text{sgn}(b')$ has the opposite sign. Bisection on the sign of φ finds a dyadic t within $\tau/(8H)$ of a root. The rotated block is a principal block of an orthogonal conjugate of A , so its norm is at most H ; the angle $2\arctan t$ has derivative at most 2 in t , so the new entry has derivative at most $4H$ in t , and its absolute value is at most $\tau/2$.

A rotation in the (p, q) plane preserves $\bar{A}_{pr}^2 + \bar{A}_{qr}^2$ for every $r \notin \{p, q\}$, so the off-diagonal energy $\text{Off} = \sum_{p \neq q} \bar{A}_{pq}^2$ changes by $2(b''^2 - b'^2) \leq -\frac{3}{2}b'^2$, where $|b''| \leq \tau/2 < |b'|/2$ is the new entry. Since $b'^2 \geq \text{Off}/(n(n-1))$, each rotation multiplies Off by at most $1 - 3/(2n(n-1))$. Initially $\text{Off} \leq \|A\|_F^2 \leq nH^2$, and the loop can continue only while $\text{Off} > \tau^2$. Hence there are $O(n^2 \log(nH/\tau))$ rotations. A rotation with $t = m'/2^\theta$ has common denominator $2^{2\theta} + m'^2$; the common denominators of the products multiply, so their bit lengths add, and exact orthogonality bounds all entries of Q by 1. All numbers therefore have polynomial bit length, and $\log(H/\tau)$ is polynomial because $\log \mu$ is. At termination $Q^\top A Q = \text{diag}(a_1, \dots, a_n) + E$ with zero diagonal in E and $\|E\| \leq \|E\|_F \leq n\tau = \epsilon$.

Selecting the negative directions. The eigenvalues of A are at most $-\mu$ (k of them) or nonnegative. By Weyl's inequality and $\epsilon \leq \mu/16$, exactly k of the a_j are below $-\mu/2$. Let B be the $n \times k$ matrix of the corresponding columns q_j of Q , so $B^\top B = I_k$, and let Π be the orthogonal projector onto the span of the eigenvectors of A with negative eigenvalues. From $Aq_j = a_j q_j + QEe_j$ we get $\|(A - a_j I)q_j\| \leq \epsilon$. On the range of $I - \Pi$, the matrix $A - a_j I$ has eigenvalues at least $\mu/2$, so $\|(I - \Pi)q_j\| \leq 2\epsilon/\mu$. For projectors of equal rank, $\|BB^\top - \Pi\|_F^2 = 2\|(I - \Pi)B\|_F^2$, hence $\|BB^\top - \Pi\| \leq 2\sqrt{2k}\epsilon/\mu \leq 3n\epsilon/\mu$.

The correction. Put $U = RB$ and $T = U^\top$. Since the range of Π lies in the range of R , $R\Pi R = \Pi$, and $\|UU^\top - \Pi\| = \|R(BB^\top - \Pi)R\| \leq 3n\epsilon/\mu$. On the range of A , the matrix $A + 2\beta\Pi$ has eigenvalues at least μ : a negative eigenvalue $\lambda \geq -\nu$ becomes $\lambda + 2\beta \geq \beta \geq \mu$, and a positive one is at least μ . Replacing Π by $UU^\top$ changes the matrix by at most $2\beta \cdot 3n\epsilon/\mu = 3\mu/8$, so $A + 2\beta UU^\top \succeq \frac{5}{8}\mu I$ on the range of A . On $\ker A$, both A and $U^\top = B^\top R$ vanish. Hence $A + 2\beta T^\top T \succeq 0$, and the rows of T lie in the range of A .

Frame. $\|U\| \leq \|R\| \|B\| \leq 1$, so $TT^\top \preceq I$. Moreover $TT^\top = B^\top RB = I - B^\top(I - R)B$, and $\|(I - R)B\|_F^2 \leq \|(I - \Pi)B\|_F^2 \leq 4k\epsilon^2/\mu^2 \leq k/(64n^2) \leq 1/64$, using $\mu \leq \beta$. Thus $TT^\top \succeq \frac{63}{64}I$. $\square$

B.2 The auxiliary value function and face candidates

Proof of Lemma 4.5. (a) For every x , expanding the square gives

$$F_\gamma(x) + \frac{1}{2}(a - Tx + \Lambda^{-1}\xi)^\top \Lambda(a - Tx + \Lambda^{-1}\xi) - \frac{1}{2}\xi^\top \Lambda^{-1}\xi = F(x) + \zeta^\top x + \frac{1}{2}(a - Tx)^\top \Lambda(a - Tx) + \xi^\top a,$$

because $\gamma^\top x = \xi^\top Tx + \zeta^\top x$. Minimizing over x gives the identity. The right side of the identity is at least $f^* - \frac{1}{2}\xi^\top \Lambda^{-1}\xi$, with equality at $a = Tx^* - \Lambda^{-1}\xi$ and $x = x^*$. For an attaining witness x_a , dropping the nonnegative square gives $V(a) \geq F_\gamma(x_a) - \frac{1}{2}\xi^\top \Lambda^{-1}\xi$.

(b) $W_\zeta(a) - \frac{1}{2}a^\top \Lambda a = \min_x [F(x) + \zeta^\top x + \frac{1}{2}(Tx)^\top \Lambda Tx - a^\top \Lambda Tx]$ is the infimum of a compactly indexed family of affine functions of a , and is therefore concave. Compactness and continuity separately give attainment; the family need not be finite.

(c) For every z , the witness x_a is feasible in the problem defining $W_\zeta(z)$, so $W_\zeta(z) \leq W_\zeta(a) + (a - Tx_a)^\top \Lambda(z - a) + \frac{1}{2}(z - a)^\top \Lambda(z - a)$. Adding $\xi^\top z$ gives the upper model. Evaluating it at $z = a - \Lambda^{-1}g$ and using $V(z) \geq V^*$ gives the inequality.

(d) The minimizer $Tx^* - \Lambda^{-1}\xi$ from (a) lies in $\mathcal{A}$. $\square$

Proof of Lemma 4.11. (a) A global minimizer exists by compactness. Among all global minimizers choose $\hat{x}$ whose minimal face Φ (the face containing $\hat{x}$ in its relative interior) has least dimension. Let $\mathcal{S}$ be the set of rows active at $\hat{x}$; the affine hull of Φ is $\{x : D_S x = e_S\}$, with tangent space $\ker D_S$. For $v \in \ker D_S$ and small $|t|$, $\hat{x} + tv \in \mathcal{Q}$, so $\nabla G(\hat{x})^\top v = 0$ and $v^\top H v \geq 0$. If $v \neq 0$ and $v^\top H v = 0$, then G is constant on the line $\hat{x} + tv$, which leaves the bounded face Φ through its relative boundary at a global minimizer on a face of smaller dimension, a contradiction. Hence H is positive definite on $\ker D_S$, or $\ker D_S = \{0\}$. Let $S \subseteq \mathcal{S}$ index a maximal linearly independent set of rows, so $\ker D_S = \ker D_{\mathcal{S}}$. If $K_S(v; \mu) = 0$, then $D_S v = 0$ and $v^\top H v = -v^\top D_S^\top \mu = 0$, so $v = 0$, and then $D_S^\top \mu = 0$ gives $\mu = 0$. Finally $\nabla G(\hat{x})$ is orthogonal to $\ker D_S$, hence equals $-D_S^\top \mu$ for some μ , and $D_S \hat{x} = e_S$.

(b) Every retained candidate is feasible, so the least retained value is at least the minimum, and by (a) the minimizer $\hat{x}$ is retained. Each subset costs polynomial rational work, and Cramer's rule bounds the bit lengths of the solutions.

(c) Each nonempty slice is a bounded polytope on which the objective is a quadratic in the continuous variables; apply (b) to every slice and compare.

(d) The faces of a box are obtained by fixing each coordinate at its lower bound, fixing it at its upper bound, or leaving it free, and the rows active on a face form an independent set unless a coordinate has equal bounds. $\square$

Proof of Lemma 4.8. Fix ε , the coordinate i , and the other coefficients, and write F_t for the objective with $\gamma_i = t$. Call a *candidate type* a pair $\omega = (z, S)$ of a label and a set of linearly independent rows of its slice. For a fixed Hessian, whether the stationarity matrix of ω is nonsingular does not depend on t ; there are at most F types. For every type with a nonsingular matrix for the Hessian A , the candidate $x_\omega(t)$ is affine in t , and the set of t where it is feasible is an interval. For such ω , the modified quadratic $\tilde{F}_{\omega,t}(y) = F_t(y) - \varepsilon \|y - x_\omega(t)\|^2$ has Hessian $A - 2\varepsilon I$ and a linear coefficient affine in t . Its candidates $y_{\omega,\omega'}(t)$, for types ω' with nonsingular matrices for $A - 2\varepsilon I$, are affine in t with interval feasibility sets, and $d_{\omega,\omega'}(t) = \tilde{F}_{\omega,t}(y_{\omega,\omega'}(t)) - F_t(x_\omega(t))$ is a polynomial of degree at most two.

We claim that $g_* \geq \varepsilon$ if and only if some $x_\omega(t)$ is feasible and $d_{\omega,\omega'}(t) \geq 0$ for every ω' with $y_{\omega,\omega'}(t)$ feasible. If this holds, then by Lemma 4.11(c) applied to $\tilde{F}_{\omega,t}$ the minimum of $\tilde{F}_{\omega,t}$ over X is attained at a feasible modified candidate, so $\min_X \tilde{F}_{\omega,t} \geq F_t(x_\omega(t)) = \tilde{F}_{\omega,t}(x_\omega(t))$, which is the growth inequality at $x_\omega(t)$. Conversely, if $g_* \geq \varepsilon$, the minimizer x^* is unique, so by Lemma 4.11 it is a candidate $x_\omega(t)$, and the growth inequality makes x^* a minimizer of $\tilde{F}_{\omega,t}$, so all $d \geq 0$.

The truth value of this condition can change only at the at most $2F$ endpoints of feasibility intervals of the x_ω , the at most $2F^2$ endpoints for the $y_{\omega,\omega'}$, and the at most $2F^2$ roots of nonzero polynomials $d_{\omega,\omega'}$. With at most $K_b = 2F + 4F^2$ breakpoints, the good set and the bad set are unions of at most $2K_b + 1 \leq 8(F + 1)^2$ intervals. The probability bounds follow from Theorem 3.16 and Lemma 3.13(b), with Kolmogorov distances $1/M$ and 2^{-b} and density bounds $1/(2\sigma)$ and $1/(\sigma\sqrt{2\pi})$. $\square$

B.3 The growth-conditioned algorithm and two negative directions

Proof of Theorem 4.6. Algorithm. Compute β and T by Lemma 4.4 and put $\alpha = 2\beta$, $\Lambda = \alpha I$, $\zeta = 0$, $\xi = 0$, so that $V = W_0$ and $V^* = f^*$. Compute the ranges $[\ell_i, u_i]$ of $(Tx)_i$ over X by linear programming, and remove the coordinates with $\ell_i = u_i$ only from the mesh. Every reduced auxiliary query is embedded in the full auxiliary space with these coordinates fixed at their original values; all rows of T and the full certified PSD matrix P remain in the inner problem. If no mesh coordinate remains, one exact convex solve at the constant value of Tx returns a minimizer by Lemma 4.5(a). By Lemma 4.11(a) and Cramer's rule, there are integers V and R , with polynomially many bits and computable from the data, such that f^* has a denominator at most V and, if the minimizer x^* is unique, every coordinate of Tx^* has a denominator at most R . Run the search of Definition 3.8 on W_0 over the box of ranges with isotropic meshes and curvature α , storing an exact attaining witness at every corner. At the end of every level j do the following.

- (1) If no cell is retained, return the witness of the incumbent.
- (2) If $E_j < 1/(2V^2)$, find the unique rational $\hat{f}$ with denominator at most V in the interval $[\underline{U}_j, \underline{U}_j]$ of Theorem 3.9(v) by continued fractions [24]. If a stored witness x has $F(x) = \hat{f}$, return x .
- (3) Only after step (2) has reconstructed $\hat{f}$, with v_j a corner attaining $\underline{U}_j$, look in each remaining mesh coordinate for a rational with denominator at most R within $1/(4R^2)$ of $(v_j)_i$. If all exist, solve the inner problem exactly at the resulting point $\hat{a}$, embedded with all removed coordinates fixed, and return its witness $\hat{x}$ if $F(\hat{x}) = \hat{f}$.

(a) *Correctness.* In (1), Theorem 3.9(ii) and Lemma 4.5(a) show that the witness is a minimizer. In (2) and (3), the interval $[U_j, U_j]$ contains f^* and has length less than $1/V^2$, while two distinct rationals with denominators at most V differ by at least $1/V^2$; hence $\hat{f} = f^*$, and any returned feasible point of value $\hat{f}$ is a minimizer. This holds on every instance.

(b) *Termination.* Let $g > 0$ and $t^* = Tx^*$. For every a and feasible x , $\|a - t^*\| \leq \|a - Tx\| + \|x - x^*\|$ since $\|T\| \leq 1$, and the Cauchy–Schwarz inequality gives $\frac{\alpha}{2} \|a - Tx\|^2 + g \|x - x^*\|^2 \geq g_W \|a - t^*\|^2$ with $g_W = \alpha g / (2g + \alpha)$. Hence $W_0(a) - f^* \geq g_W \|a - t^*\|^2$ and $\alpha/g_W = 2 + \alpha/g < \kappa$. By Theorem 3.9(iv), every cell retained at level j has a corner v with $W_0(v) \leq f^* + 2E_j$, so $\|v - t^*\|^2 \leq 2E_j/g_W \leq k\kappa h_j^2/4$. In a refined coordinate the grid spacing exceeds $h_j/2$, so the box of half-width $h_j\sqrt{k\kappa}/2$ around t^* contains at most $2\sqrt{k\kappa} + 2$ grid values per coordinate. Hence at most $2^k(2\sqrt{k\kappa} + 2)^k$ cells are retained and at most $4^k(2\sqrt{k\kappa} + 2)^k$ are processed at the next level. Step (2) applies once $E_j \leq k\alpha s^2 4^{-j}/8 < 1/(2V^2)$, after $\text{poly}(L_{\text{in}})$ levels. If no stored witness attains f^* , then $W_0(v_j) = U_j \leq f^* + E_j$ by Theorem 3.9(iii), so $\|v_j - t^*\|^2 \leq E_j/g_W \leq k\kappa s^2 4^{-j}/8$. Once this is less than $1/(16R^4)$, which happens after $\text{poly}(L_{\text{in}}) + \frac{1}{2} \log_2 \kappa$ levels, each coordinate of t^* is the unique rational with denominator at most R within $1/(4R^2)$ of $(v_j)_i$, so $\hat{a} = t^*$. At t^* every inner minimizer $\hat{x}$ satisfies $F(\hat{x}) + \frac{\alpha}{2} \|t^* - T\hat{x}\|^2 = W_0(t^*) = f^*$, so $F(\hat{x}) = f^*$ and step (3) returns. Level- j corners have bit length polynomial in $L_{\text{in}} + j$, so each exact convex solve, reconstruction, and comparison costs $\text{poly}(L_{\text{in}} + j)$. Summing over the levels proves (b).

(c) For $k \leq 2$ and $Z \geq 1$, $\kappa \leq 2 + 4Z \leq 6Z$, so $(1 + \sqrt{\kappa})^k \leq (1 + \sqrt{6})^2 Z^{k/2}$ and $J_g \leq \text{poly}(L_{\text{in}}) + 3 + \log_2 Z$. Substituting in (b) gives the bound. $\square$

Proof of Corollary 4.7. Use the same full convexifying factorization and auxiliary objective, but solve every inner query by Proposition 4.2(b) and polish its returned integer label by part (a). Compute auxiliary coordinate ranges over the rational relaxation $\mathcal{P}$, so they enclose TX and are available by polynomial-time LP. Remove singleton auxiliary ranges only from the mesh, with the same full-space embedding as above. The mixed smallest-face argument of Lemma 4.11(c) and the LP bounds on integer labels give precomputable polynomial-height denominator bounds for f^* and, under uniqueness, Tx^* . The growth-transfer inequality holds for every $x \in X$ without using convexity of X . Upper coordinate curvature, corner rounding, the near-optimal-corner packing, and the two guarded rational reconstructions therefore apply verbatim. At Tx^* every exact mixed recourse witness has original objective value f^* by square completion. All later queried and polished data have polynomial length in $L_{\text{in}} + j$; only the inner solve costs the additional factor $f(n_z)$. This proves the stated work and correctness contracts. $\square$

Proof of Theorem 4.9. Every sampled coefficient has bit length $O(\log M)$ in case (a) and $O(b)$ in case (b), plus the bit length of σ . The least sufficient choices, or the stated polynomial-bit enlargements, therefore give $L_{\text{in}} \leq \text{poly}(I)$, including the sampler’s polynomial cost. If $S = 0$, LP fixes every original coordinate and direct evaluation returns the rational point and value; no growth threshold or mesh is used. Below assume $S > 0$ and $k \geq 1$; $k = 0$ uses the convex primitive. Interleave, one bit operation at a time, the algorithm of Theorem 4.6 and the fallback of Lemma 4.11(b) on the same sampled objective, and stop when either returns. Both are exact, so the output is exact on every draw. The work is at most twice the smaller of the two running times, up to a polynomial simulation overhead. The fallback costs at most $B \text{poly}(L_{\text{in}})$. If $g_* > 0$, the conditioned algorithm costs at most $\text{poly}(L_{\text{in}}) Z^p (1 + \log Z)^{d_0}$ with $p = k/2$ and $Z = \max\{1, \nu/g_*\}$. Put $Z = \infty$ if $g_* = 0$. If $Z^p \leq B$, then $\log Z \leq p^{-1} \log B$; otherwise the minimum below is at most B . Hence

$$\min\{B, Z^p(1 + \log Z)^{d_0}\} \leq (1 + p^{-1} \log B)^{d_0} \min\{B, Z^p\}.$$

For $t \geq 1$, $\{Z > t\} = \{g_* < \nu/t\}$. By Lemma 4.8 with $F = 2^r$, $\mathbb{P}\{Z > t\} \leq r_0/t + \beta$ with $r_0 = \nu S/\sigma$ and $\beta = 2nD/M \leq 1/B$ in case (a), and with $r_0 = \sqrt{2/\pi} \nu S/\sigma$ and $\beta = 2nD2^{-b} \leq 1/B$ in case (b). Since $\min\{B, Z^p\} \geq 1$,

$$\mathbb{E} \min\{B, Z^p\} = 1 + \int_1^B \mathbb{P}\{Z > t^{1/p}\} dt \leq 1 + r_0 \int_1^B t^{-1/p} dt + \beta B,$$

which is at most $2 + r_0$ for $p = 1/2$ and at most $2 + r_0 \log B$ for $p = 1$. Since $\log B = O(r + 1)$, the expected work is at most $\text{poly}(I)(1 + \nu S/\sigma)$. For $k = 0$, $A \succeq 0$ and one exact convex solve suffices. $\square$

B.4 Pieces, gaps, isolation, and tubes

Proof of Lemma 4.12. Write $x = (z, x_c)$, and let P_{cc} , D_c , T_c be the blocks of P , D , T for the continuous variables. For fixed (a, ζ) the slice objective is $\frac{1}{2}x_c^\top P_{cc}x_c + p(a, \zeta)^\top x_c$ plus a constant, where $p(a, \zeta) = P_{cz}z + b_c + \zeta_c - \alpha T_c^\top a$ is affine, and the slice is $\mathcal{Q}_z = \{x_c : D_c x_c \leq e - D_z z\}$. Since $P \succeq 0$, $P_{cc} \succeq 0$.

Extraction. Solve the slice exactly (Proposition 4.2) and let x^0 be the optimal point and $g^0 = P_{cc}x^0 + p(a, \zeta)$. For $y \in \mathcal{Q}_z$ the objective gap equals $g^{0\top}(y - x^0) + \frac{1}{2}(y - x^0)^\top P_{cc}(y - x^0)$, a sum of two nonnegative terms, so the optimal set is the polytope $\mathcal{O} = \mathcal{Q}_z \cap \{P_{cc}(y - x^0) = 0, g^{0\top}(y - x^0) = 0\}$. Find a vertex y of $\mathcal{O}$ by linear programming. Let $\mathcal{S}$ be the rows active at y . If $v \in \ker D_{c,S}$ with $v^\top P_{cc}v = 0$, then $P_{cc}v = 0$, $y \pm tv \in \mathcal{Q}_z$ for small t , optimality gives $g^{0\top}v = 0$ (the gradient at y equals g^0 because $P_{cc}(y - x^0) = 0$), and $y \pm tv \in \mathcal{O}$, contradicting that y is a vertex. Hence P_{cc} is positive definite on $\ker D_{c,S}$. By the normal-cone formula for polyhedra there are multipliers $\mu \geq 0$ with $-(P_{cc}y + p) = D_{c,S}^\top \mu$; find them by linear programming. While the rows in the support of μ are dependent, move μ along a dependency until a support coefficient vanishes, keeping $\mu \geq 0$. Extend the final independent support to a maximal independent set $S \subseteq \mathcal{S}$, with zero multipliers on the added rows. Then $\ker D_{c,S} = \ker D_{c,S}$, so $K_S = \begin{pmatrix} P_{cc} & D_{c,S}^\top \\ D_{c,S} & 0 \end{pmatrix}$ is nonsingular, as in the proof of Lemma 4.11.

(a)–(b). The system $K_S(x_c; \lambda) = (-p(a', \zeta'); e_S - D_{z,S}z)$ has an affine solution in (a', ζ') ; the derivative in a' is the first block Ξ_S of $K_S^{-1}(\alpha T_c^\top; 0)$, which depends on neither ζ' , z , b , nor e . At the query, (y, μ_S) solves the system, so a lies in the region for the queried ζ . The same vertex and multiplier existence argument applies to every real (a, ζ) for each feasible fixed label; it requires no rational computation for this existence assertion. Thus the finitely many regions cover that label's full real parameter space, including the conditioned lines used in the section-count proof. On the region the point is feasible and satisfies the Karush–Kuhn–Tucker conditions with nonnegative multipliers on the active rows S ; for a convex quadratic program these conditions are sufficient. The region is described by the r feasibility rows and the $|S| \leq n$ sign conditions, with normals formed from Ξ_S and the a' -derivative of λ .

(c). Let $\Phi(a', x) = F(x) + \zeta'^\top x + \frac{\alpha}{2} \|a' - Tx\|^2$. Then $\partial_{a'}\Phi = \alpha(a' - Tx)$ and $\nabla_{x_c}\Phi = P_{cc}x_c + p(a', \zeta')$, which equals $-D_{c,S}^\top \lambda$ along the solution. Since $D_{c,S}x_{z,S}(a', \zeta') = e_S - D_{z,S}z$ for all a' , $D_{c,S}\Xi_S = 0$, and the chain rule gives $\nabla_{a'}q_{z,S} = \alpha(a' - Tx_{z,S}) - (D_{c,S}\Xi_S)^\top \lambda = \alpha(a' - Tx_{z,S})$. Hence $H_S = \alpha(I - T_c\Xi_S)$. Multiply K_S and the right side $(\alpha T_c^\top; 0)$ by a common denominator of P , D , and $\alpha T^\top$. By Cramer's rule every entry of Ξ_S is a quotient of two integer determinants of order at most $2n$, the denominator nonzero, with entries bounded by C_0 ; hence its absolute value is at most U_0 . Every entry of $T_c\Xi_S$ is then at most $n\tau_T U_0$, and $\|T_c\Xi_S\| \leq \|T_c\Xi_S\|_F \leq kn\tau_T U_0$. There are at most Z labels and 2^r row sets. $\square$

Proof of Lemma 4.13. (a) Distinct integer vectors differ in some coordinate by at least one, so the union of the $2n_z$ restricted problems consists exactly of the points of X with a label other than z_* . Each restricted problem is a convex MIQP with the same integer variables and one more row.

(b) For a label z put $m_z(a) = W_\zeta^z(a) - \frac{\alpha}{2} \|a\|^2$, the minimum over the points x of label z of affine functions of a with slopes $- \alpha T x$, $T x \in T\mathcal{P}$. For a and v , using the minimizer at v as a test point and the bound on all slopes,

$$- \alpha \max_{y \in T\mathcal{P}} (a - v)^\top y \leq m_z(a) - m_z(v) \leq - \alpha \min_{y \in T\mathcal{P}} (a - v)^\top y.$$

The interval is the same for every label and has length at most $\alpha \|a - v\| \text{diam}(T\mathcal{P}) \leq \alpha \sqrt{k} h \bar{w}$ for $a \in C$. Hence, for $z \neq z_*$ and $a \in C$, $W_\zeta^z(a) - W_\zeta^{z_*}(a) \geq \Delta(v) - \alpha \sqrt{k} h \bar{w} \geq 0$. The label z_* is therefore optimal on C , and on $C \subseteq \mathcal{R}_{z_*, S}(\zeta)$ its value is $q_{z_*, S}$ with attaining point $x_{z_*, S}$ by Lemma 4.12.

(c) For every feasible x and every a , Lemma 4.5(a) gives $F_\gamma(x) - f^* \leq [F(x) + \zeta^\top x + \frac{\alpha}{2} \|a - T x\|^2 + \xi^\top a] - V^*$. For the winning witness at v the bracket is $V(v)$, and for the best witness of another label it is $V(v) + \Delta(v)$. Both are at most $V^* + 2E + \theta_{\text{gap}} h \leq V^* + (k\alpha/4 + \theta_{\text{gap}})h$, using $2E \leq k\alpha h^2/4 \leq k\alpha h/4$. $\square$

Proof of Lemma 4.14. Fix i and all coefficients except γ_i . For each value t taken by z_i on $\mathcal{Z}$, put $a_t = \min\{h(z) + \sum_{l \neq i} \gamma_l z_l : z \in \mathcal{Z}, z_i = t\}$. As a function of $u = \gamma_i$, the least value is the lower envelope of the lines $a_t + tu$, whose slopes are distinct integers in $[L_i, U_i]$. The active slope is nonincreasing in u and no slope is active on two separated intervals, so the envelope has at most $U_i - L_i$ breakpoints, which are fixed under the conditioning. Suppose that at u some line $a_t + tu$ attains the envelope and another line $a_{t'} + t'u$ lies within ε of it. The two lines cross within distance $\varepsilon/|t - t'| \leq \varepsilon$ of u . Moving from u towards the crossing, the envelope either changes its active line before the crossing or has both lines active at the crossing, which is a breakpoint because their slopes differ. Hence u lies within ε of a breakpoint, an event of conditional probability at most $(U_i - L_i)\pi_i(2\varepsilon)$.

If $\Delta \leq \varepsilon$, a best label and a different second-best label differ in some coordinate i , and their groups satisfy the preceding event for that i . A union over i , followed by integration over the other coefficients, proves the bound. $\square$

Proof of Lemma 4.17. The family. For every pair $\omega = (z, S)$ write $\nabla_a q_\omega(a, \zeta) = H_\omega a + p_\omega(\zeta)$, with p_ω affine. If H_ω is nonsingular, then for every region row $N a \leq n_0 + \Gamma \zeta$ with $N \neq 0$ let $u = H_\omega^{-\top} N^\top / \|H_\omega^{-\top} N^\top\|$ and

$$\ell(\xi, \zeta) = \frac{N H_\omega^{-1} (\xi + p_\omega(\zeta)) + n_0 + \Gamma \zeta}{\|H_\omega^{-\top} N^\top\|}.$$

For fixed ζ , $|\ell(\xi, \zeta)|$ is the distance from ξ to the image of the hyperplane $\{N a = n_0 + \Gamma \zeta\}$ under the affine map $\Phi_\zeta(a) = -(H_\omega a + p_\omega(\zeta))$. If H_ω is singular, choose a unit vector $u \in \ker H_\omega$ (a Hessian is symmetric, so $u^\top H_\omega = 0$), and let $\ell(\xi, \zeta) = u^\top (\xi + p_\omega(\zeta))$. Each pair contributes at most $r + n + 1$ functionals.

The estimate. Let $\omega = (z_v, S_v)$ and $g = \nabla_a q_\omega(v, \zeta) + \xi$. By Lemma 4.12(c), $g = \alpha(v - T x_\omega(v, \zeta)) + \xi$, where $x_\omega(v, \zeta)$ is an attaining witness of $W_\zeta(v)$ because z_v wins. Lemma 4.5(c) gives $\|g\|^2 \leq 2\alpha(V(v) - V^*) \leq 4\alpha E_j \leq k\alpha^2 h_j^2/2$. If H_ω is singular, $u^\top \nabla_a q_\omega(v, \zeta) = u^\top p_\omega(\zeta)$, so $|\ell(\xi, \zeta)| = |u^\top g| \leq \|g\|$. If H_ω is nonsingular, some corner a' of the cell violates a region row that v satisfies. The normal of that row is nonzero, since a row with zero normal holds at v exactly when

it holds everywhere. The segment from v to a' meets the boundary hyperplane at a point a_Π with $\|a_\Pi - v\| \leq \text{diam } C \leq \sqrt{k}h_j$. Then

$$|\ell(\xi, \zeta)| \leq \|\xi - \Phi_\zeta(a_\Pi)\| \leq \|\xi - \Phi_\zeta(v)\| + \|H_\omega(v - a_\Pi)\| \leq \|g\| + \Theta\sqrt{k}h_j,$$

since $\xi - \Phi_\zeta(v) = g$. In both cases $|\ell| \leq \sqrt{k}(\alpha/\sqrt{2} + \Theta)h_j$.

Ambient form. With $\xi = \Psi\gamma$ and $\zeta = \Pi_T\gamma$, $\ell = L^\top\gamma + c$ with $L = \Psi^\top u + \Pi_T w$, where w collects the coefficients of ζ . Since $T\Psi^\top = TT^\top(TT^\top)^{-1} = I$ and $T\Pi_T = 0$, $TL = u$, so $\|L\| \geq \|u\| / \|T\| \geq 1$. $\square$

Lemma B.1 (Tube probabilities). *Let $\ell(\xi, \zeta) = u^\top\xi + w^\top\zeta + c$ be one of the functionals of Lemma 4.17, and $\eta > 0$.*

- (a) *If $\zeta = 0$ and $\xi_1, \dots, \xi_k$ are independent with law $U_{\sigma, M}$, then $\mathbb{P}\{|\ell| \leq \eta\} \leq \sqrt{k}\eta/\sigma + 1/M$.*
- (b) *If T has full row rank, $\|T\| \leq 1$, and $\gamma_1, \dots, \gamma_n$ are independent with law $U_{\sigma, M}$, then $\mathbb{P}\{|\ell| \leq \eta\} \leq \sqrt{n}\eta/\sigma + 1/M$.*
- (c) *Under the conditions of (b) with independent γ_i satisfying $d_K(\gamma_i, N(0, \sigma^2)) \leq \delta_K$, $\mathbb{P}\{|\ell| \leq \eta\} \leq \eta/\sigma + 2n\delta_K$.*

Proof. (a) Some coordinate has $|u_i| \geq k^{-1/2}$. Conditional on the others, the event confines ξ_i to an interval of length $2\eta/|u_i| \leq 2\sqrt{k}\eta$. (b) In the form $|L^\top\gamma + c| \leq \eta$ with $\|L\| \geq 1$, some coordinate has $|L_i| \geq n^{-1/2}$, and the same argument applies. (c) Under the proxy $N(0, \sigma^2 I_n)$, $L^\top\gamma$ is normal with standard deviation at least σ , so the probability is at most $2\eta/(\sigma\sqrt{2\pi}) \leq \eta/\sigma$. Every coordinate section of the event is an interval, possibly empty or the whole line, so Lemma 3.13(b) with $C = 1$ adds at most $2n\delta_K$. $\square$

B.5 Section counts and local counts under the three laws

In the following lemma, a *closure structure* is one of: continuous pieces ($n_z = 0$) or mixed pieces of Lemma 4.12, with $R = Z2^r$; or separable pieces of Section 4.7, with $R = R_{\text{sep}}$. Put $P_{\text{sec}} = 2R$ for continuous and separable pieces and $P_{\text{sec}} = 2R^2$ for mixed pieces.

Lemma B.2 (Section counts). *Fix a grid node v of any level, a coordinate i of γ , and real values of the other coordinates. The set of values of γ_i for which the local event E_v holds is a union of at most $C_{\text{sec}} = [(2k+1)P_{\text{sec}} + 1](8k+2)$ intervals.*

Proof. Write $\gamma_i = t$. Then $\xi = \Psi\gamma$ and $\zeta = \Pi_T\gamma$ are affine in t . Fix a query a . For every piece, membership of a in its region is a finite set of affine inequalities in t , hence an interval of t , and on that interval the piece's value is a quadratic polynomial in t . For continuous and separable pieces, every region certifies $W_\zeta(a)$ itself, the regions cover the line (Lemmas 4.12 and 4.28), and valid formulas agree in value where intervals overlap. Hence between consecutive interval endpoints, of which there are at most $2R$, a single quadratic gives $W_\zeta(a)$. For mixed pieces, $W_\zeta(a)$ is the least value of the pieces whose interval contains t : the pieces of one label give that label's value, and the minimum over labels is $W_\zeta(a)$. Between consecutive points among the at most $2R$ endpoints and the at most $R(R-1)$ crossings of distinct quadratics, the available pieces and their order are constant, so again one quadratic gives the value; there are at most $R^2 + R \leq 2R^2$ such points. Gradients of overlapping formulas are not used.

The event E_v involves at most $2k+1$ queries, giving at most $(2k+1)P_{\text{sec}}$ breakpoints. On each open interval between breakpoints, the event is a conjunction of at most $2k$ inequalities between

quadratics in t (the term $\xi^\top a$ is affine in t). Each nonzero difference has at most two roots, and identically zero differences have constant sign, so the interval splits into at most $8k + 1$ pieces on which the event is constant. Counting the breakpoints themselves as possible isolated components gives at most $[(2k + 1)P_{\text{sec}} + 1](8k + 1) + (2k + 1)P_{\text{sec}} \leq C_{\text{sec}}$ intervals. $\square$

Proof of Lemma 4.23. Let the columns of V_0 form a basis of $\ker T$ and $S = [T^\top V_0]$, which is invertible. Then $\gamma = S(\xi, \eta)$ with $\zeta = V_0\eta$. The vector (ξ, η) has density $|\det S|(2\sigma)^{-n}$ on $S^{-1}[-\sigma, \sigma]^n$. For fixed values of the coordinates other than ξ_Q , the allowed set of ξ_Q is contained in a box of volume $\prod_{i \in Q} \ell_i$. The remaining coordinates range over the projection of the parallelotope $S^{-1}[-\sigma, \sigma]^n$, a zonotope whose volume is $(2\sigma)^{n-t} \sum_{|J|=n-t} |\det(S^{-1})_{Q^c, J}|$. By Jacobi's complementary-minor identity, $|\det S| |\det(S^{-1})_{Q^c, J}| = |\det S_{J^c, Q}| = |\det T_{Q, J^c}|$. Multiplying gives the first bound, with $K = J^c$. By Cauchy–Schwarz and the Cauchy–Binet formula, $\sum_{|K|=t} |\det T_{Q, K}| \leq \binom{n}{t}^{1/2} \det(T_Q T_Q^\top)^{1/2} \leq n^{t/2}$, since $\|T_Q\| \leq \|T\| \leq 1$. $\square$

Lemma B.3 (Local counts under uniform ambient noise). *Let T have full row rank and $\|T\| \leq 1$, let $\Lambda = \alpha I$, use isotropic meshes on a box with widths w_i , and let γ be continuous uniform on $[-\sigma, \sigma]^n$. At every level, $\sum_{v \in G_j} \mathbb{P}(E_v) \leq \mathcal{H}_{\text{amb}} = \prod_i [2 + (1 + 2k)\alpha\sqrt{n}w_i/(2\sigma)]$.*

Proof. For fixed ζ , apply Lemma 3.2 to W_ζ , which has upper coordinate curvature α by Lemma 4.5(b). In the comparisons of E_v the term $\xi^\top a$ contributes only $\pm h_{ij}\xi_i$, so E_v confines each ξ_i , $i \in Q_v$, to an interval of length at most $(1 + 2k)\alpha h_{ij}$ (Corollary 3.6) that depends on ζ only. Lemma 4.23, applied to the coordinates of Q_v whose factor $\sqrt{n}(1 + 2k)\alpha h_{ij}/(2\sigma)$ is less than one, gives $\mathbb{P}(E_v) \leq \prod_{i \in Q_v} \min\{1, \sqrt{n}(1 + 2k)\alpha h_{ij}/(2\sigma)\}$. Summing over the tensor grid as in the proof of Theorem 3.4 proves the claim. $\square$

Proof of Lemma 4.26. Independence and a density majorant. Under the proxy, $\xi = \Psi\gamma$ and $\zeta = \Pi_T\gamma$ are jointly Gaussian with $\text{Cov}(\xi, \zeta) = \sigma^2\Psi\Pi_T = \sigma^2(\Psi - \Psi T^\top\Psi) = 0$, because $\Psi T^\top = I$ and Π_T is symmetric. They are therefore independent, and $\text{Cov}\xi = \sigma^2(TT^\top)^{-1}$ has eigenvalues in $[\sigma^2, \sigma^2/c_{\text{fr}}]$. Every principal submatrix of order t has eigenvalues in the same interval, so the density p_Q of ξ_Q satisfies

$$p_Q(y) \leq (2\pi\sigma^2)^{-t/2} \exp(-c_{\text{fr}} \|y\|^2 / (2\sigma^2)) = c_{\text{fr}}^{-t/2} \prod_{i \in Q} \phi_s(y_i), \quad s = \sigma/\sqrt{c_{\text{fr}}},$$

where ϕ_s is the centered normal density with standard deviation s .

Location. Fix v and ζ , and let x_v be an attaining witness of $W_\zeta(v)$, chosen as a function of ζ . With $g = \Lambda(v - Tx_v) + \xi$, Lemma 4.5(c) gives $V(v \pm h_{ij}e_i) \leq V(v) \pm h_{ij}g_i + \lambda_i h_{ij}^2/2$. Together with the comparisons of E_v this yields $|g_i| \leq \lambda_i h_{ij}/2 + 2E_j/h_{ij} \leq C_k \lambda_i h_{ij}/2$ by Corollary 3.6. Since $(Tx_v)_i \in [\ell_i, u_i]$, the event E_v implies $\xi_i \in J_i(v) = \lambda_i[\ell_i - v_i, u_i - v_i] + [-C_k \lambda_i h_{ij}/2, C_k \lambda_i h_{ij}/2]$, a deterministic interval. As in Lemma B.3, E_v also confines ξ_i to an interval $I_i(\zeta)$ of length at most $C_k \lambda_i h_{ij}$. Only the event needs to lie in the intersection of the two intervals.

Probability of one node. By independence of ξ and ζ and the density majorant,

$$\mathbb{P}(E_v) \leq c_{\text{fr}}^{-|Q_v|/2} \prod_{i \in Q_v} \min\left\{1, C_k \lambda_i h_{ij} \sup_{y \in J_i(v)} \phi_s(y)\right\}.$$

The factor $c_{\text{fr}}^{-|Q_v|/2}$ stays outside the minima.

A weighted lattice sum. Let $\Delta > 0$, $C > 0$, and $A \leq B'$ with $W' = B' - A$. We claim

$$\sum_{y \in y_0 + \Delta\mathbb{Z}} \min\left\{1, C\Delta \sup_{[A-y-C\Delta/2, B'-y+C\Delta/2]} \phi_s\right\} \leq CW' \phi_s(0) + 2C + 4.$$

At most $W'/\Delta + C + 2$ lattice points satisfy $y \in [A - C\Delta/2, B' + C\Delta/2]$, and each contributes at most $\min\{1, C\Delta\phi_s(0)\}$, which gives at most $CW'\phi_s(0) + C + 2$. For the points beyond $B' + C\Delta/2$, the summand is $\min\{1, C\Delta\phi_s(\theta)\}$, where θ is the distance to that point. The summand decreases with θ , so the tail sum is at most its first term, bounded by 1, plus its integral divided by Δ . This integral term is at most $C \int_0^\infty \phi_s = C/2$. The tail contributes at most $1 + C/2$; the same holds below $A - C\Delta/2$.

Summation. Apply the claim with $\Delta = \lambda_i h_{ij}$, the lattice of $\lambda_i v_i$ (the interior nodes of coordinate i form a subset), core $[\lambda_i \ell_i, \lambda_i u_i]$, and $C = C_k$. The two endpoint nodes contribute one each. Expanding the product over the tensor grid,

$$\sum_{v \in G_j} \mathbb{P}(E_v) \leq \prod_{i=1}^k \left[2 + c_{\text{fr}}^{-1/2} (C_k \lambda_i (u_i - \ell_i) \phi_s(0) + 2C_k + 4) \right],$$

and $c_{\text{fr}}^{-1/2} \phi_s(0) = 1/(\sigma\sqrt{2\pi})$. Unrefined coordinates have only their two endpoints. Nothing depends on the box. $\square$

B.6 Proofs of the main theorems

We prove Theorems 4.19, 4.20, and 4.21 together.

Base quantities. If nonemptiness is not promised, first use Proposition 4.2(b), or (a) when $n_z = 0$, with zero objective to decide feasibility. If $X = \emptyset$, return that report before constructing any thresholds, meshes, or law. On the remaining nonempty instances, compute from the base data, in polynomial time, the factorization (Lemma 4.4 or the supplied one, checked), the ranges $[\ell_i, u_i]$, the bound Θ of Lemma 4.12, the integer bounds $L_i^z = \lceil \min_{\mathcal{P}} x_i \rceil$ and $U_i^z = \lfloor \max_{\mathcal{P}} x_i \rfloor$ for $i \leq n_z$, and

$$Z = \prod_i (U_i^z - L_i^z + 1), \quad G = \sum_i (U_i^z - L_i^z), \quad R = Z2^r, \quad K = R(r + n + 1), \quad B = \max\{2, Z2^r\},$$

with $Z = 1$ and $G = 0$ if $n_z = 0$; further C_{sec} from Lemma B.2, $\bar{w}$, θ_{gap} , and C_{gap} from Lemma 4.13, and $\Psi = (TT^\top)^{-1}T$ for the ambient laws. All these numbers have bit length polynomial in I , with absolute exponents; for example $\log \Theta = O(n \log n + n \log C_0)$.

Sampling parameters. Write $h_j = s2^{-j}$ for the nominal mesh on the box $\mathcal{A}$ with largest width s , and $Q_J = (J + 1)(2^J + 1)^k$, which bounds the number of nodes of all grids of levels $0, \dots, J$.

- (A) *Aligned law* ($n_z = 0$, $\zeta = 0$): $\mathcal{A} = \prod_i [\ell_i - \sigma/\alpha, u_i + \sigma/\alpha]$; J is the least integer with $s2^{-J} \leq \sigma/(2kKB(\alpha + \Theta))$; M is the least power of two with $M \geq \max\{2, 2^J, 2KB\}$.
- (B) *Uniform ambient law*: $\mathcal{A} = \prod_i [\ell_i - s_i/\alpha, u_i + s_i/\alpha]$ with $s_i = \sigma \|\Psi_{i,:}\|_1$; J is the least integer with $s2^{-J} \leq \min\{1, \sigma/(2B[nkK(\alpha + \Theta) + GC_{\text{gap}}])\}$; M is the least power of two with $M \geq \max\{2, 2B(K + G), 2nC_{\text{sec}}Q_J\}$.
- (C) *Gaussian-like law*: for $t = 0, 1, 2, \dots$ let $s_i(t) = 2^t \sigma \|\Psi_{i,:}\|_1$, let $\mathcal{A}_t$ and $s(t)$ be the corresponding box and largest width, let $J(t)$ be the least integer with $s(t)2^{-J} \leq \min\{1, \sigma/(4B[kK(\alpha + \Theta) + GC_{\text{gap}}])\}$, and let $b(t)$ be the least positive integer with $2^b \geq \max\{8B(nK + G), 2nC_{\text{sec}}Q_{J(t)}\}$. Stop at the first t with $2^t \geq b(t) + 20$, and use $\mathcal{A} = \mathcal{A}_t$, $J = J(t)$, $b = b(t)$.

In (C), $s(t) \leq 2^t s(0)$, so $J(t) \leq J(0) + t$, and $\log_2 Q_{J(t)} \leq k(J(0) + t + 1) + \log_2(J(0) + t + 1)$. Since $J(0)$ and the other base quantities are polynomial in I , $b(t)$ grows at most linearly in t up to an additive polynomial, and the loop stops after polynomially many steps with J, b polynomial in I . In every case the sampled coefficients have bit length polynomial in I , and the parameters are fixed before the draw.

Containment and correctness. On every draw the box $\mathcal{A}$ contains a global minimizer of V by Lemma 4.5(d): in (A), $|\xi_i| \leq \sigma$; in (B), $|\xi_i| \leq \sigma \|\Psi_{i,:}\|_1$; in (C), the support of $\mathcal{G}_{b,\sigma}$ lies in $[-\sigma(b+20), \sigma(b+20)] \subseteq [-2^t\sigma, 2^t\sigma]$. By Lemmas 4.12 and 4.13(b), every closure step of the search of Definition 4.16 computes $\min_C V$ exactly with an attaining witness. Theorem 3.9 therefore applies to V on $\mathcal{A}$; with Lemma 4.5(a) and Lemma 4.11(c), the output is exact on every draw, and its bit length is polynomial.

Unresolved cells are rare. Suppose a cell is retained and unclosed at level J . By Theorem 3.9(iv) it has a corner v with $V(v) - V^* \leq 2E_J$, and the closure test failed at v . Either some corner of the cell lies outside $\mathcal{R}_{z_v, S_v}(\zeta)$, and then by Lemma 4.17 one of K fixed functionals satisfies $|\ell_\kappa| \leq \eta_J = \sqrt{k}(\alpha + \Theta)h_J$; or $n_z \geq 1$ and $\Delta(v) < \theta_{\text{gap}}h_J$, and then by Lemma 4.13(c), with $h_J \leq 1$, the label gap of the sampled problem is at most $C_{\text{gap}}h_J$. This is one event about the sampled problem, not a union over visited cells. By Lemmas B.1 and 4.14, the latter applied conditionally on the continuous coefficients with $h(z)$ the slice optimum, the probability that the fallback runs is at most

$$\begin{aligned} \text{(A)} \quad & K(k(\alpha + \Theta)h_J/\sigma + 1/M) \leq \frac{1}{2B} + \frac{1}{2B}, \\ \text{(B)} \quad & K(\sqrt{nk}(\alpha + \Theta)h_J/\sigma + 1/M) + G(C_{\text{gap}}h_J/\sigma + 1/M) \leq \frac{1}{2B} + \frac{1}{2B}, \\ \text{(C)} \quad & K(\eta_J/\sigma + 2n2^{-b}) + G(C_{\text{gap}}h_J/\sigma + 2 \cdot 2^{-b}) \leq \frac{1}{4B} + \frac{1}{4B}. \end{aligned}$$

In (C) the concentration of $\mathcal{G}_{b,\sigma}$ is at most $2\varepsilon/(\sigma\sqrt{2\pi}) + 2 \cdot 2^{-b} \leq \varepsilon/\sigma + 2 \cdot 2^{-b}$ on intervals of length 2ε . The fallback costs $B \text{poly}(I)$, so by Lemma 3.18(a) its expected contribution is $\text{poly}(I)$.

Expected local counts. Let N_j be the number of nodes of G_j at which the local event E_v holds. By Theorem 3.9(iv), at most $2^k N_j$ cells are retained at level j , on every draw. In (A), Corollary 3.6 with $M \geq 2^J \geq m_{ij}$ gives $\mathbb{E}N_j \leq \mathcal{H}_{\text{al}}$ for every $j \leq J$. In (B), Lemma B.3 bounds the sum of the continuous probabilities by $\mathcal{H}_{\text{amb}}$ at every level. By Lemma B.2 and Lemma 3.13(b) with $\delta_i = 1/M$, each node's probability changes by at most $2nC_{\text{sec}}/M$, and the choice of M makes the total change over all levels at most one. Hence $\sum_{j \leq J} \mathbb{E}N_j \leq (J+1)\mathcal{H}_{\text{amb}} + 1$. In (C), Lemma 4.26 and the same transfer with $\delta_i = 2^{-b}$ give $\sum_{j \leq J} \mathbb{E}N_j \leq (J+1)\mathcal{H}_G + 1$.

Work. Level 0 processes one cell, and level $j+1$ processes at most 2^k children of each cell retained at level j . A processed cell needs 2^k corner queries, each one exact solve (convex QP, or $2n_z + 1$ convex MIQP solves and a polishing solve) with polynomial bit length, a containment test of at most $4^k(r+n)$ affine inequalities, and at most one closure by 3^k face systems of order at most k . Its cost is at most $f(n_z)6^k \text{poly}(I)$. Summing, the expected work before the fallback is at most $f(n_z)6^k(1 + 4^k \sum_{j \leq J} \mathbb{E}N_j) \text{poly}(I)$, and with $J \leq \text{poly}(I)$ this gives the bounds of the three theorems.

Intrinsic bounds. For the factorization of Lemma 4.4, $u_i - \ell_i \leq \|T\| \text{diam}(\mathcal{P}) \leq \text{diam}(\mathcal{P})$ and $\alpha < 4\nu$. In Theorem 4.19, $c_{\text{fr}}^{-1/2} = (64/63)^{1/2}$ gives $2 + c_{\text{fr}}^{-1/2}(4k+6) \leq 9 + 5k$, and $4/\sqrt{2\pi} \leq 2$. In Theorem 4.20, $\|(TT^\top)^{-1}\| \leq 64/63$ gives $\|\Psi_{i,:}\|_1 \leq \sqrt{n}\|\Psi\| \leq 2\sqrt{n}$, hence $w_i \leq \text{diam}(\mathcal{P}) + 4\sigma\sqrt{n}/\alpha$. $\square$

B.7 The fiber estimate

Proof of Proposition 4.24. Here $\xi = m\bar{\gamma}$, where $\bar{\gamma}$ is the mean of the coordinates, and $\zeta = \gamma - \bar{\gamma}\mathbf{1}$. Given ζ , the law of $\bar{\gamma}$ is uniform on $[-\sigma - \min_l \zeta_l, \sigma - \max_l \zeta_l]$, and $\xi - c(\zeta) = m(\min_l \gamma_l + \max_l \gamma_l)/2$ for the midpoint $c(\zeta)$ of the conditional support of ξ . Write $\gamma_l = \sigma(2U_l - 1)$ with independent uniform U_l on $[0, 1]$, and let A', B' be their minimum and maximum. The event is $|A' + B' - 1| \leq q$ with $q = \ell/(2\sigma m)$. Using the joint density $n(n-1)(B' - A')^{n-2}$ for $n \geq 2$ and integrating over the midpoint for a fixed range ϱ gives $n(n-1) \int_0^1 \varrho^{n-2} \min\{q, 1 - \varrho\} d\varrho = 1 - (1-q)^n$. For $n = 1$ we

have $A' = B' = U_1$ and the interval $|2U_1 - 1| \leq q$ has probability $q = 1 - (1 - q)^1$, giving the same formula. As $\ell \rightarrow 0$ this is $nq(1 + o(1)) = \ell\sqrt{n}/(2\sigma)(1 + o(1))$. Also $\text{Var } \xi = m^2\sigma^2/(3n) = \sigma^2/3$. For k disjoint blocks the events are independent, and each contributes the factor $m = (N/k)^{1/2}$, where $N = km^2$ is the total dimension. $\square$

Proposition 4.25 is the case $\alpha = 1$ of Theorem 9.4; its proof is in Appendix G.1.

B.8 Separable recourse

Proof of Lemma 4.28. (a) Convexity of φ_j makes δ_j nondecreasing. If $\delta_j(t - 1) \leq \lambda \leq \delta_j(t)$, then for $t' > t$, $\varphi_j(t') - \varphi_j(t) = \sum_{s=t}^{t'-1} \delta_j(s) \geq (t' - t)\lambda$, and symmetrically for $t' < t$; the converse follows by comparing t with $t \pm 1$. Binary search for the least t with $\delta_j(t) \geq \lambda$ (or the upper endpoint if none exists) returns such a point. Each evaluation locates the listed piece containing the integer and evaluates a rational quadratic.

(b) A point t minimizes $\varphi_j(t) - \lambda t$ on the interval exactly when λ lies in the subdifferential of φ_j at t , with the normal cone included at the endpoints. In the interior of a piece with $c_{jm} > 0$ the subdifferential is $\{c_{jm}t + \mu_{jm}\}$; at a breakpoint it is the interval between the one-sided derivatives. For every λ a minimizer exists. If it is a breakpoint or interior to a piece with positive curvature, a listed state applies. If it is interior to a piece with $c_{jm} = 0$, then $\lambda = \mu_{jm}$ equals both one-sided derivatives at the ends of that piece, and the breakpoint state at its left end applies. $\square$

Proof of the separable piece properties and Theorem 4.29. Pieces. The inner objective is $\frac{\alpha}{2} \|a\|^2 + \sum_j [\varphi_j(x_j) - \lambda_j(a, \zeta)x_j]$ with $\lambda_j = \alpha t_j^\top a - \zeta_j$, so the scalar problems are independent, and a choice ω of one state per coordinate whose intervals contain all λ_j certifies a global minimizer $x_\omega(a, \zeta)$, affine in (a, ζ) . Define q_ω by replacing each φ_j by the listed polynomial of its state (the constant $\varphi_j(t)$ for an integer state). For a free coordinate, the first-order condition $\psi'_{jm}(x_j) = \lambda_j$ cancels the derivative of the response, and constant coordinates have zero response derivative; hence $\nabla_a q_\omega = \alpha(a - Tx_\omega)$ identically, and the Hessian is $\alpha I - \alpha^2 \sum_{\text{free}} t_j t_j^\top / c_j$, bounded by Θ_{sep} . The regions have at most $2n$ rows with normals $\pm \alpha t_j^\top$ and offsets affine in ζ , and they cover $\mathbb{R}^k$ for every ζ by Lemma 4.28. Every region certifies W_ζ itself, so a cell inside a region is closed with no gap test, and $P_{\text{sec}} = 2R_{\text{sep}}$ in Lemma B.2.

Fallback. Each combination of an integer assignment and a listed piece for every continuous coordinate gives a box on which F_γ coincides with a rational quadratic; minimize it by Lemma 4.11(d) with 3^{n_c} faces and compare. The combinations cover X , so this is exact, at cost $B_{\text{sep}} \text{poly}(I + L_{\text{in}})$.

The theorem. The proof of Appendix B.6 applies with the following replacements: the exact recourse is the scalar procedure, of polynomial cost and no factor $f(n_z)$; $R = R_{\text{sep}}$, $K = R_{\text{sep}}(2n + 1)$, $B = B_{\text{sep}}$, $\Theta = \Theta_{\text{sep}}$, $G = 0$, and no gap test. In Lemma 4.17 the region rows are the scalar state inequalities, and every step of its proof uses only the gradient identity, the attaining witness at the query, and fixed normals with affine offsets. Part (a) uses law (A); integer variables are allowed there because no gap event occurs. Part (b) uses law (B). $\square$

Proof of Lemma 4.30. Let $\Gamma = TT^\top \succ 0$, let $H' \geq \|\Gamma\|$ be its largest row sum, and let q_Γ clear the denominators of Γ . As in the proof of Lemma 4.4, $\mu' = q_\Gamma^{-k} H'^{-(k-1)} \leq \lambda_{\min}(\Gamma)$. Rational Jacobi rotations with entry threshold $\mu'/(64k)$ give an exactly orthogonal rational Q with $Q\Gamma Q^\top = \text{diag}(a_i) + E$, E with zero diagonal and $\|E\| \leq \mu'/64$, in polynomial time. Each a_i is a Rayleigh quotient, so $\mu' \leq a_i \leq \|T\|^2$. Choose dyadic d_i with $2a_i \leq d_i^2 < 8a_i$, which exists because consecutive squared powers of two differ by the factor four. With $D = \text{diag}(d_i)$ and $U = D^{-1}QT$, $UU^\top = D^{-1} \text{diag}(a_i) D^{-1} + D^{-1} E D^{-1}$; the first term has diagonal entries in $(1/8, 1/2]$ and the second has norm

at most $(\mu'/64)/(2\mu') = 1/128$, so $\frac{1}{16}I \preceq UU^\top \preceq I$. Finally $U^\top \Lambda U = \alpha T^\top Q^\top D^{-1} D^2 D^{-1} Q T = \alpha T^\top T$ because Q is exactly orthogonal, and $\lambda_i = \alpha d_i^2 < 8\alpha a_i \leq 8\beta$. $\square$

Proof of Theorem 4.31. By Lemma 4.30, $F(x) = \sum_j \varphi_j(x_j) - \frac{1}{2}(Ux)^\top \Lambda(Ux)$ exactly, so Lemma 4.5 applies with (U, Λ) in place of (T, Λ) , and the scalar recourse applies with $\lambda_j = (\Lambda u_j)^\top a - \zeta_j$, where u_j is column j of U . The pieces satisfy $\nabla_a q_\omega = \Lambda(a - Ux_\omega)$, with Hessians $\Lambda - \sum_{\text{free}} (\Lambda u_j)(\Lambda u_j)^\top / c_j$ bounded by $\Theta_\Lambda = \lambda_{\max} + \sum_j \text{continuous } r_j^+ \|\Lambda u_j\|^2$.

Choose dyadic scales ρ_i with $1 \leq \lambda_i \rho_i^2 < 4$ and use the balanced nested meshes of Definition 3.5 with curvature λ ; then $c_{\text{bal}} < 4$, $C_k = 1 + 8k$, and $E_j \leq \frac{1}{8} \sum_i \lambda_i \rho_i^2 h_j^2 < k h_j^2 / 2$. In the proof of Lemma 4.17, Lemma 4.5(c) now gives $g^\top \Lambda^{-1} g \leq 4E_j < 2k h_j^2$, so $\|g\| \leq \sqrt{2k \lambda_{\max}} h_j$, and the cell diameter is at most $h_j \sum_i \rho_i$. The tube radius is therefore at most $C_{\text{tube}} h_j$ with the rational constant $C_{\text{tube}} = k(1 + \lambda_{\max}) + \Theta_\Lambda \sum_i \rho_i$. The ambient normals have norm at least one because $\|U\| \leq 1$. Lemma 4.26 applies with $c_{\text{fr}} = 1/16$ and gives $\mathcal{H}_\Lambda$; its bound uses $\lambda_i < 8\beta$ and $u_i - \ell_i \leq \|U\| \text{diam}(X) \leq \text{diam}(X)$.

For the law, use step (C) of Appendix B.6 with $s_i(t) = 2^t \sigma \|(A_U)_{i,:}\|_1$, $A_U = (UU^\top)^{-1}U$, the box $\prod_i [\ell_i - s_i(t)/\lambda_i, u_i + s_i(t)/\lambda_i]$, the nominal scale $S_\rho(t) = \max_i (u_i - \ell_i + 2s_i(t)/\lambda_i) / \rho_i \leq 2^t S_\rho(0)$, the cutoff $S_\rho(t) 2^{-J} \leq \sigma / (4KBC_{\text{tube}})$, and $2^b \geq \max\{8nKB, 2nC_{\text{sec}}Q_J\}$, where K , B , and C_{sec} are the separable quantities and $G = 0$. Then $J(t) \leq J(0) + t$, the loop stops after polynomially many steps, the probability of the fallback is at most $K(C_{\text{tube}} h_J / \sigma + 2n2^{-b}) \leq 1/(2B)$, and the rest of the proof is unchanged. $\square$

C Proofs for the sparse polynomial section

This appendix proves the evaluation statement [proposition 5.18](#) and [corollary 5.19](#). The evaluation lemma [lemma C.2](#) is stated for general polytopes and approximate oracles, because [section 6](#) uses it for simplices, order polytopes and implicit graphs. The margin bounds of [appendix D](#) count nonsingular stationary points with the following form of the refined Bézout inequality, which is also used in the proof of [lemma 3.17\(d\)](#).

C.1 Nonsingular common zeros

Lemma C.1 (Nonsingular common zeros). *Let $k \geq 1$ and $D \geq 1$, and let $P_1, \dots, P_k \in \mathbb{C}[u_1, \dots, u_k]$ have degree at most D . At most D^k points $u \in \mathbb{C}^k$ satisfy $P(u) = 0$ and $\det(\partial P / \partial u)(u) \neq 0$.*

Proof. Take any finite set of N distinct nonsingular common zeros. We show $N \leq D^k$, which also rules out a larger or infinite set. Consider

$$G_i(t, u) = (1 - t)P_i(u) + t(u_i^D - 1), \quad i = 1, \dots, k.$$

By the holomorphic implicit function theorem, each of the selected zeros continues to a zero of $G(t, \cdot)$ for all sufficiently small complex t . Choose disjoint neighborhoods of the original zeros. Shrinking the common parameter neighborhood keeps these N continued zeros distinct.

Homogenize each G_i to degree D in the variables $x_0, u_1, \dots, u_k$, and call the resulting polynomial $H_i(t)$. Choose an integer $r \geq \max\{D, k(D-1), N-1\}$. Let $\mathcal{V}_r$ be the vector space of homogeneous polynomials of degree r in these variables, and let

$$\mathcal{I}_r(t) = \sum_{i=1}^k H_i(t) \mathcal{V}_{r-D}.$$

The multiplication map from k copies of $\mathcal{V}_{r-D}$ to $\mathcal{V}_r$ has matrix entries polynomial in t . At $t = 1$ its generators are $u_i^D - x_0^D$. Successive division by these monic polynomials gives a unique remainder whose exponent in each u_i is below D . Equivalently, the quotient ring is a free $\mathbb{C}[x_0]$ -module with basis $\prod_i u_i^{a_i}$, $0 \leq a_i < D$. Since $r \geq k(D-1)$, its degree- r part has exactly D^k basis monomials, obtained by multiplying each such product by $x_0^{r-\sum_i a_i}$. Thus

$$\dim(\mathcal{V}_r/\mathcal{I}_r(1)) = D^k.$$

A matrix minor attaining the rank at $t = 1$ is a nonzero polynomial in t . Away from its finitely many zeros, the rank is at least that value and $\dim(\mathcal{V}_r/\mathcal{I}_r(t)) \leq D^k$.

Choose such a t sufficiently close to zero that all N continued zeros exist and remain distinct. Evaluation at their affine representatives $(1, u)$ maps $\mathcal{V}_r$ onto $\mathbb{C}^N$: products of $N-1$ affine linear factors give Lagrange polynomials for any N distinct affine points, and homogenizing them to degree r preserves their values at $x_0 = 1$. Every element of $\mathcal{I}_r(t)$ vanishes at these points. Hence the evaluation map factors through the quotient, giving $N \leq \dim(\mathcal{V}_r/\mathcal{I}_r(t)) \leq D^k$. $\square$

This is the weak Bézout bound for nonsingular zeros [10, Theorem 4.106]; Fulton [23] gives the intersection-theoretic background. The proof above does not need a degree estimate for the other components of the common zero set.

The lemma counts only nonsingular zeros. Stationarity equations of a face can have singular zeros and positive-dimensional components; these are neither excluded nor counted. In every application, positive point growth is what places the optimizer among the nonsingular zeros.

C.2 Weak convex optimization on a certified patch

We use two problems of Grötschel et al. [24, Section 2.1] for a convex body $\mathcal{K} \subseteq \mathbb{R}^m$, with $S(\mathcal{K}, \varepsilon)$ the closed ε -neighborhood of $\mathcal{K}$ and $S(\mathcal{K}, -\varepsilon)$ the set of points whose ε -ball lies in $\mathcal{K}$. The *weak separation problem* asks, for a rational w and a rational $\delta > 0$, either to assert $w \in S(\mathcal{K}, \delta)$ or to find a rational π with $\|\pi\|_\infty = 1$ and $\pi^\top x \leq \pi^\top w + \delta$ for all $x \in S(\mathcal{K}, -\delta)$. The *weak optimization problem* asks, for a rational objective π and a rational $\varepsilon > 0$, either to find a rational $w \in S(\mathcal{K}, \varepsilon)$ with $\pi^\top x \leq \pi^\top w + \varepsilon$ for all $x \in S(\mathcal{K}, -\varepsilon)$, or to assert that $S(\mathcal{K}, -\varepsilon)$ is empty. By Grötschel et al. [24, Corollary 4.2.7], for a convex body contained in the ball of radius R about the origin and given by a weak separation oracle, weak optimization is solved in oracle-polynomial time in the dimension, the encoding lengths of R , π and ε ; by their Section 4.1 conventions, a polynomial-time oracle with polynomial-length answers makes this ordinary polynomial bit time. The returned point need not lie in $\mathcal{K}$, and its objective is compared with $S(\mathcal{K}, -\varepsilon)$ only; the proof below repairs both.

Lemma C.2 (Evaluation of a certified convex patch). *Let $P \subseteq \mathbb{R}^N$ be a nonempty polytope given by rational linear equations and inequalities, and let rational data of total encoding length Σ be given as follows.*

- (a) *A point $c \in P$; a matrix $Z \in \mathbb{Q}^{N \times k}$, $k \geq 1$, whose columns span the direction space of the affine hull of P ; and rationals $r \in (0, \frac{1}{2}]$, $R \geq 1$ and $\zeta \geq \max\{1, \|Z\|_2\}$ such that $c + Zt \in P$ whenever $\|t\|_2 \leq r$, and $\|t\|_2 \leq R$ whenever $c + Zt \in P$.*
- (b) *A function f , continuously differentiable on an open set containing P and convex on P ; a norm ν on $\mathbb{R}^N$ equal to $\|\cdot\|_2$ or $\|\cdot\|_\infty$, with dual norm ν_* ; and rationals $V \geq \max\{1, \sup_P |f|\}$ and $G \geq \max\{1, \sup_P \nu_*(\nabla f)\}$.*

(c) An oracle that, for a rational $y \in P$ and a rational $\beta > 0$, returns rationals $l \leq f(y) \leq u$ with $u - l \leq \beta$ and a rational $\hat{g} \in \mathbb{Q}^k$ with $\|\hat{g} - Z^\top \nabla f(y)\|_\infty \leq \beta$, in time polynomial in Σ , the length of y , and $\log(1/\beta)$.

(d) A repair map ϱ , computable in time polynomial in Σ and the length of its argument, that sends every $w \in \mathbb{Q}^N$ to a rational point of P with $\nu(\varrho(w) - \bar{y}) \leq \|w - \bar{y}\|_2$ for every $\bar{y} \in P$.

Then for every rational $\eta \in (0, 1]$ one computes a rational $y \in P$ and rationals $a \leq U$ with

$$a \leq \min_P f \leq f(y) \leq U \leq a + \eta$$

in time polynomial in Σ and $\log(1/\eta)$. If moreover f is twice continuously differentiable and $v^\top \nabla^2 f(x) v \geq g \|v\|_2^2$ for all $x \in P$, all v in the column space of Z , and a rational $g > 0$, then f has a unique minimizer y^* on P , and $\|y - y^*\|_2 \leq \sqrt{2\eta/g}$.

Proof. Let $P' = \{t \in \mathbb{R}^k : c + Zt \in P\}$, $\tilde{f}(t) = f(c + Zt)$, and

$$\mathcal{K} = \{(t, \theta) \in \mathbb{R}^{k+1} : t \in P', \tilde{f}(t) \leq \theta \leq V + 2\}.$$

This is a compact convex set. It contains the ball of radius r about $o = (0, V + 1)$: a point of this ball has $t \in P'$ and $V + \frac{1}{2} \leq \theta \leq V + \frac{3}{2}$, while $\tilde{f}(t) \leq V$. It lies in the ball of radius $R_{\mathcal{K}} = R + 2V + 1$ about o , because $-V \leq \theta \leq V + 2$. Translating by o gives a circumscribed body in the sense of the cited corollary.

Weak separation. Let $D_1 = 2kR$, so that $\|t' - t\|_1 \leq D_1$ for $t, t' \in P'$. Given a rational (t, θ) and $\delta > 0$: if $c + Zt$ violates a defining constraint $\alpha^\top x \leq \alpha_0$ of P (equations being treated as two inequalities), then $Z^\top \alpha \neq 0$, because $\alpha^\top (c + Zt) = \alpha^\top c \leq \alpha_0$ would hold otherwise, and $(Z^\top \alpha, 0)$ separates strictly; if $\theta > V + 2$, the vector $(0, 1)$ separates strictly. Otherwise call the oracle at $y = c + Zt$ with $\beta = \delta/(2D_1 + 2)$. If $\theta < l - \beta D_1$, then for every $(t', \theta') \in \mathcal{K}$, convexity gives

$$\theta' \geq \tilde{f}(t') \geq \tilde{f}(t) + \nabla \tilde{f}(t)^\top (t' - t) \geq l + \hat{g}^\top (t' - t) - \beta D_1 > \theta + \hat{g}^\top (t' - t),$$

because $\nabla \tilde{f}(t) = Z^\top \nabla f(y)$; so $(\hat{g}, -1)$ separates strictly. Otherwise $\tilde{f}(t) - \theta \leq u - l + \beta D_1 \leq \delta$, so either $(t, \theta) \in \mathcal{K}$ or $(t, \tilde{f}(t)) \in \mathcal{K}$ lies within distance δ ; in both cases $(t, \theta) \in S(\mathcal{K}, \delta)$. A strictly separating vector, divided by its ∞ -norm, is a valid answer for every δ . All answers are rational of polynomial length.

Weak optimization. Let $\Xi = G\zeta + 2 + (2V + 1)/r$ and $\varepsilon = \min\{r/2, \eta/(2\Xi)\}$, and apply weak optimization with objective $(0, -1)$. Let t^* minimize $\tilde{f}$ on P' , put $z^* = (t^*, \tilde{f}(t^*))$ and $\lambda = \varepsilon/r \leq \frac{1}{2}$. The point $z_\lambda = (1 - \lambda)z^* + \lambda o$ lies in $S(\mathcal{K}, -\varepsilon)$: for $\|w\| \leq \varepsilon$, $z_\lambda + w = (1 - \lambda)z^* + \lambda(o + w/\lambda)$ with $\|w/\lambda\| \leq r$, a convex combination of points of $\mathcal{K}$. So $S(\mathcal{K}, -\varepsilon) \neq \emptyset$, and the algorithm returns $(t_o, \theta_o) \in S(\mathcal{K}, \varepsilon)$ with

$$\theta_o \leq (z_\lambda)_\theta + \varepsilon \leq \min_P f + \lambda(V + 1 - \min_P f) + \varepsilon \leq \min_P f + \varepsilon \left(1 + \frac{2V + 1}{r}\right).$$

Put $a = \theta_o - \varepsilon(1 + (2V + 1)/r)$, so $a \leq \min_P f$.

Repair. Some $(\bar{t}, \bar{\theta}) \in \mathcal{K}$ satisfies $\|(t_o, \theta_o) - (\bar{t}, \bar{\theta})\|_2 \leq \varepsilon$. Let $w = c + Zt_o$, $\bar{y} = c + Z\bar{t} \in P$ and $y = \varrho(w) \in P$. Then $\|w - \bar{y}\|_2 \leq \zeta \varepsilon$ and $\nu(y - \bar{y}) \leq \zeta \varepsilon$. Along the segment from $\bar{y}$ to y , which lies in P , the mean value theorem gives

$$f(y) \leq f(\bar{y}) + G\nu(y - \bar{y}) \leq \bar{\theta} + G\zeta \varepsilon \leq \theta_o + \varepsilon + G\zeta \varepsilon.$$

Call the oracle at y with $\beta = \eta/2$ and let $U = u$. Then $f(y) \leq U \leq f(y) + \eta/2$ and

$$U - a \leq \frac{\eta}{2} + \varepsilon \left(G\zeta + 2 + \frac{2V+1}{r} \right) \leq \eta.$$

All tolerances have polynomial encoding length, so the running time is polynomial.

Distance. Under the curvature hypothesis, f is strictly convex on P , so y^* is unique. For $y \in P$ the function $\varphi(\theta) = f(y^* + \theta(y - y^*))$ on $[0, 1]$ has $\varphi'(0) \geq 0$ by first-order optimality on the convex set P , and $\varphi'' \geq g \|y - y^*\|_2^2$ because $y - y^*$ lies in the column space of Z . Hence $f(y) - f(y^*) \geq \frac{g}{2} \|y - y^*\|_2^2$, and $\|y - y^*\|_2^2 \leq 2(U - a)/g \leq 2\eta/g$. $\square$

No inner-radius assumption on the location of the minimizer is used: the known inner ball is centered at c , and boundary minimizers cause no loss. The tolerance ε is exponentially small when r or V^{-1} is, but its encoding length is polynomial; the cost depends on $\log(1/r)$ and $\log V$, not on their magnitudes.

The lemma concerns positive tangent dimension. A patch consisting of a single rational point is handled before invoking it: evaluate a polynomial objective exactly, or use the proved value oracle for an algebraic chart. In the latter case fixed retained coordinates still require refinement of the dependent roots and the objective, as in [appendix D.1](#).

Proof of [proposition 5.18](#). (a) If $\bar{S} = \emptyset$ the point is rational and there is nothing to do. Otherwise apply [lemma C.2](#) with $N = |\bar{S}|$, the box P of the patch, $f = F_\gamma(\bar{x}_S, \cdot)$, c its midpoint, $Z = I$, $\zeta = 1$, $r = \min\{\frac{1}{2}, \min_{i \in \bar{S}}(b_i - a_i)/2\}$ and $R = \max\{1, \sum_{i \in \bar{S}}(b_i - a_i)\}$. Take $\nu = \|\cdot\|_2$; termwise monomial bounds on $\hat{X}$ give rationals $V \geq \max\{1, \sup |F_\gamma|\}$ and $G \geq \max\{1, \sup \|\nabla F_\gamma\|_1\} \geq \sup \|\nabla F_\gamma\|_2$ of polynomial length. The oracle evaluates f and its gradient exactly at rational points, and the repair map is coordinatewise clipping to P , the Euclidean projection onto a box, which is nonexpansive and fixes P . By (7) the curvature hypothesis holds with $g = g_0$. With $\eta = \min\{2^{-q}, \frac{g_0}{2} 2^{-2q}\}$, the lemma gives $y \in P$ and $a \leq f^* \leq F_\gamma(\bar{x}_S, y) \leq a + 2^{-q}$, where $\min_P f = f^*$ by [proposition 5.12](#), and $\|y - \hat{y}\|_2 \leq 2^{-q}$. The encoding length of all data is $\text{poly}_d(I + q)$.

(b) Let x° be the fallback optimizer and let $G' \geq \max\{1, \sup_{\hat{X}} \|\nabla F_\gamma\|_1\}$ be a rational monomial bound. Using the refinement of [theorem 3.20](#), compute for each $i \in I_C$ a rational $\tilde{x}_i$ with $|\tilde{x}_i - x_i^\circ| \leq \epsilon := 2^{-q-1}/(nG')$, and let y_i be $\tilde{x}_i$ clipped to $[\ell_i, u_i]$; clipping does not increase the error, since $x_i^\circ \in [\ell_i, u_i]$. For $i \in I_Z$ refine until the isolating interval contains exactly one integer, which is x_i° , and put $y_i = x_i^\circ$. Then $y \in X$, $\|y - x^\circ\|_\infty \leq \epsilon$, $\|y - x^\circ\|_2 \leq \sqrt{n}\epsilon \leq 2^{-q}$, and the segment from x° to y lies in $\hat{X}$, so $0 \leq F_\gamma(y) - f^* \leq G'\epsilon \leq 2^{-q-1}$. Refine the value to an interval of width at most 2^{-q-1} and let a be its lower endpoint. Then $a \leq f^* \leq F_\gamma(y) \leq a + 2^{-q}$. All refinements use $q + \text{poly}_d(I)$ bits of precision, so the cost is $B \text{poly}_d(I + q)$ by [theorem 3.20](#), after enlarging the base factor if necessary.

The expectation follows because the fallback output occurs with probability at most $1/(2B)$. $\square$

C.3 The quadratic case

Lemma C.3 (Face enumeration for quadratics). *Let $F_\gamma(x) = \frac{1}{2}x^\top Hx + (b + \gamma)^\top x + c_0$ be rational on a mixed box X . For every integer assignment z and every partition $I_C = L \cup U \cup F$, if H_{FF} is positive definite (or $F = \emptyset$), solve the linear system $\nabla_F F_\gamma(\tilde{x}) = 0$ for the point $\tilde{x}$ with integer part z , values ℓ_L , u_U on the fixed sets and unknowns on F , and keep $\tilde{x}$ if it lies in X . Some kept point is a global minimizer of F_γ on X . The enumeration costs $3^{n_c} N_Z \text{poly}(I + b)$ bit operations, where b bounds the bit length of the coefficients of γ , and every kept point is rational of bit length $\text{poly}(I + b)$.*

Proof. Let z^* be the integer part of some global minimizer. Among the global minimizers with integer part z^* choose x^* whose set F of continuous coordinates strictly inside their bounds is smallest. If $F = \emptyset$, x^* is a kept point. Otherwise $\nabla_F F_\gamma(x^*) = 0$ and $H_{FF} \succeq 0$ by the first- and second-order conditions in the open box. If $H_{FF}v = 0$ for some nonzero v supported on F , then $F_\gamma(x^* + \theta v) = F_\gamma(x^*)$ for all θ , because the gradient term and the quadratic term along v vanish; moving θ until a coordinate of F reaches a bound gives a global minimizer with integer part z^* and fewer interior coordinates, a contradiction. So H_{FF} is positive definite and x^* is the unique solution of its system, hence kept. Every kept point is feasible, so a kept point of least value is a global minimizer. There are $N_Z 3^{n_c}$ systems; each is a rational linear system whose solution has polynomial bit length by Cramer’s rule and Hadamard’s inequality. $\square$

The argument uses that a quadratic is constant along a null direction of its Hessian at a stationary point. It does not extend to polynomials of higher degree, for which [theorem 3.20](#) is used instead.

Proof of corollary 5.19. Run [definition 5.17](#) with the factor $B = 3^{n_c} N_Z$ of [lemma C.3](#) in the schedule and with this enumeration as the fallback; the scalar-section count C_{tail} of [lemma 5.14](#) is the one for $d = 2$. All statements about pruning, counting, closure and the schedule are those of the case $d = 2$ of [theorem 5.2](#); κ_3 may be replaced by 0 because the Hessian is constant. When the closure test returns a point, it is rational. When it returns $(S, \bar{x}_S, P, g_0)$, the problem $\min\{F_\gamma(\bar{x}_S, y) : y \in P\}$ is a convex quadratic program with a positive definite Hessian, whose unique solution is the global minimizer by [proposition 5.12](#); it is rational of polynomial bit length and is computed in polynomial time [\[31\]](#). On fallback draws, [lemma C.3](#) returns a rational global minimizer at cost $B \text{poly}(I + \log M)$, with probability at most $1/(2B)$ by [lemma 5.16](#). The work bound is that of [theorem 5.2](#). $\square$

D Proofs for the feasible-domain extensions

Throughout, $N_{\mathcal{B}}$ denotes the number of bags of the tree decomposition used by the algorithm, and C_0 an absolute constant that may change between occurrences.

D.1 Global graphs

Proof of lemma 6.3. Substitute only explicit rational constants and implicit coordinates with rational singleton brackets. A constant implicit root need not be rational: the root of $y^2 = 2$ in $[1, 2]$ has no parents. Such coordinates remain in the chart and in the rational original-domain formula used by the fallback. Their ancestor sets are empty and do not affect scope containment or running intersection. In case (E), induction along the dependency order shows that ψ_j depends only on $\Lambda(y_j)$ and that $|\Lambda(y_j)| \leq k^\delta$ for a coordinate at dependency depth $\delta \leq D_g$: $\Lambda(y_j)$ is the union of the sets $\Lambda(\pi)$ over its at most k parents π . In case (I), ψ_j depends on x_{S_j} and $|S_j| \leq k$. Hence $|\beta'| \leq \sum_{v \in \beta} |\Lambda(v)| \leq p \max\{1, k^{D_g}\}$, respectively $p \max\{1, k\}$.

Every retained coordinate lies in a bag of the original decomposition and hence in the corresponding expanded bag. A reduced factor $f_a(x, \psi(x))$ depends only on $\bigcup_{v \in S_a} \Lambda(v)$, which lies in β' for a bag $\beta \supseteq S_a$. The term $\eta_j \psi_j$ depends on $\Lambda(y_j)$, which lies in β' for every bag β containing y_j .

For running intersection, fix a retained coordinate x_i and let $\mathcal{D}_i$ be the set of dependent coordinates y with $x_i \in \Lambda(y)$. The expanded bags containing x_i are those of the original bags that contain x_i or a member of $\mathcal{D}_i$, that is, the union of the subtrees $\mathcal{T}_v$ of bags containing v , over $v \in \{x_i\} \cup \mathcal{D}_i$. Each $\mathcal{T}_v$ is connected. In case (I), $y_j \in \mathcal{D}_i$ means $i \in S_j$, and the bag containing $S_j \cup \{y_j\}$ lies in $\mathcal{T}_{x_i}$ and in $\mathcal{T}_{y_j}$. In case (E), a member of $\mathcal{D}_i$ is the end of a dependency path starting at x_i , consecutive elements of which are a parent and its child; the bag containing the

child's constraint scope lies in both of their subtrees. In both cases every $\mathcal{T}_y$, $y \in \mathcal{D}_i$, is connected to $\mathcal{T}_{x_i}$ within the union, which is therefore connected. $\square$

Proof of theorem 6.4, case (E). Reduction. Substitution in the dependency order computes every ψ_j as an explicit polynomial of degree at most e^{D_g} in at most $\min\{n, k^{D_g}\}$ retained variables, so it has $\text{poly}(I)$ monomials, and similarly for the reduced factors, whose degree is at most $d' = de^{D_g}$; the base part $F_0(x, \psi(x))$ is computed once, and the sampled part $\eta^\top \psi$ adds coefficients of length $\text{poly}(I + b)$. The production cost is also $\text{poly}(I + b)$, with exponents depending only on d , e and D_g , as required by lemma 6.1. The map $(x, y) \mapsto x$ is a bijection from $\mathfrak{X}$ onto X with inverse $x \mapsto (x, \psi(x))$, and $F_{\gamma, \eta}(x, \psi(x)) = G_\eta(x) + \gamma^\top x$. Hence minimizers correspond, and the sampled problem is that of lemma 6.1 for the family $\{G_\eta\}$ with the expanded decomposition of lemma 6.3.

Hypotheses of lemma 6.1. The bound L of (19) is valid for every η in the noise box. Monomial bounds for the derivatives of $F_0(x, \psi(x))$ and of the ψ_j on $\hat{X}$, the latter multiplied by $\sigma \geq |\eta_j|$, give uniform κ_2 and κ_3 of polynomial encoding length. For (b), apply theorem 3.20 to the base instance whose domain formula consists of the bounds of X , the integer constraints, and the equations $y_j - P_j(x, y_1, \dots, y_{j-1}) = 0$, with objective F_0 tilted by (γ, η) . It returns the lexicographically least minimizer (x°, y°) , and x° minimizes $G_\eta + \gamma^\top x$ on X . Its factor B depends only on the base instance. Lemma 6.1 gives (i) and (ii); the charted patch output consists of the patch of the reduced problem and the polynomials P_j .

Evaluation. If the retained patch is a point, substitute it into the explicit chart and evaluate the sampled polynomial exactly; no convex optimization is needed. For a positive-dimensional patch, let $K_\psi = 1 + \sum_j G_{\psi_j}$, where G_{ψ_j} is a rational monomial bound for $\sup_{\hat{X}} \|\nabla \psi_j\|_1$. For $x, x' \in \hat{X}$, $\|(x, \psi(x)) - (x', \psi(x'))\|_2 \leq \|x - x'\|_2 + \sum_j |\psi_j(x) - \psi_j(x')| \leq K_\psi \|x - x'\|_2$. On ordinary draws, proposition 5.18(a) applied to the reduced polynomial gives a rational $\tilde{x}$ in the patch with $\|\tilde{x} - \hat{x}\|_2 \leq 2^{-q}/K_\psi$ and a value interval of width at most 2^{-q} ; the lift $(\tilde{x}, \psi(\tilde{x}))$ is rational, satisfies the equations, and lies within 2^{-q} of the optimizer, and its value $G_\eta(\tilde{x}) + \gamma^\top \tilde{x}$ is the upper endpoint. On fallback draws, proposition 5.18(b) applied to the retained coordinates of (x°, y°) , with the chart map, gives the same. The extra precision $\log K_\psi$ is polynomial in I . $\square$

Proof of lemma 6.6. Roots. For $x \in \hat{X}_{S_j}$, the function $y \mapsto q_j(x, y)$ is strictly increasing on V_j and has a sign change there by (18), so it has exactly one root $\psi_j(x)$ in V_j . If $\ell'_j = u'_j$, then $q_j(x, \ell'_j) = 0$ for all x , and the coordinate is substituted.

Smoothness. Fix $x^0 \in \hat{X}_{S_j}$ and $y^0 = \psi_j(x^0)$. Since $\partial_y q_j(x^0, y^0) \geq \mu_j > 0$, the implicit function theorem gives an open neighborhood $\mathcal{N}$ of x^0 , a number $\epsilon > 0$, and an infinitely differentiable φ on $\mathcal{N}$ such that $\varphi(x)$ is the only root of $q_j(x, \cdot)$ in $(y^0 - \epsilon, y^0 + \epsilon)$ for $x \in \mathcal{N}$. For $y \in V_j$ with $|y - y^0| \geq \epsilon$, monotonicity gives $|q_j(x^0, y)| \geq \mu_j \epsilon$, so by uniform continuity $q_j(x, y) \neq 0$ for all such y and all $x \in \hat{X}_{S_j}$ close to x^0 . For these x , $\psi_j(x)$ lies in $(y^0 - \epsilon, y^0 + \epsilon)$ and therefore equals $\varphi(x)$. Thus ψ_j agrees near every point of $\hat{X}_{S_j}$ with an infinitely differentiable function, and its derivatives are obtained by differentiating $q_j(x, \psi_j(x)) = 0$.

Derivative bounds. Write q for q_j , subscripts for partial derivatives, and evaluate at $(x, \psi(x))$, where $q_y \geq \mu_j$ and all partial derivatives of orders one to three are bounded by A_j . Then $\psi_i = -q_i/q_y$, so $|\psi_i| \leq \Pi_{1,j}$, and

$$\psi_{ik} = -\frac{N_{ik}}{q_y}, \quad N_{ik} = q_{ik} + q_{iy}\psi_k + q_{ky}\psi_i + q_{yy}\psi_i\psi_k,$$

so $|\psi_{ik}| \leq A_j(1 + \Pi_{1,j})^2/\mu_j = \Pi_{2,j}$. Differentiating $q_y\psi_{ik} = -N_{ik}$ with respect to x_l gives $q_y\psi_{ikl} = -\frac{d}{dx_l}N_{ik} - \psi_{ik}\frac{d}{dx_l}q_y$. Here $\left|\frac{d}{dx_l}q_y\right| = |q_{yl} + q_{yy}\psi_l| \leq A_j(1 + \Pi_{1,j})$, and the total derivative of N_{ik} is a

sum of the terms $q_{ikl} + q_{iky}\psi_l$, $(q_{iyl} + q_{iyy}\psi_l)\psi_k + q_{iy}\psi_{kl}$, the analogous term with i and k exchanged, and $(q_{yy}l + q_{yyy}\psi_l)\psi_i\psi_k + q_{yy}(\psi_{il}\psi_k + \psi_i\psi_{kl})$. Bounding each factor gives $A_j[(1 + \Pi_{1,j})^3 + 2(1 + \Pi_{1,j})\Pi_{2,j}]$, and adding $\Pi_{2,j}A_j(1 + \Pi_{1,j})$ and dividing by μ_j gives $\Pi_{3,j}$.

Oracle. If there are no dependent coordinates, evaluate the rational polynomial and its derivatives directly. Otherwise, at a rational x , bisection on V_j with exact rational signs of $q_j(x, \cdot)$ keeps a bracket $[l_j, u_j] \subseteq V_j$ with $q_j(x, l_j) \leq 0 \leq q_j(x, u_j)$; monotonicity shows that it contains $\psi_j(x)$, and a zero sign identifies the root exactly. Width ϵ takes $O(\log((u'_j - \ell'_j)/\epsilon))$ steps, each a fixed-degree evaluation at a rational point whose length grows by one bit per step. By the chain rule and the formulas above, $G_\eta(x)$ and every entry of $\nabla G_\eta(x)$ and $\nabla^2 G_\eta(x)$ equal $\Phi(x, \psi(x))$ for an explicit expression $\Phi(x, y)$: a polynomial in η and in partial derivatives of F_0 and of the q_j at (x, y) , divided by powers of the $\partial_y q_j(x, y_j)$. On $\hat{X} \times \prod_j V_j$ these denominators are at least μ_j , so $|\partial_{y_j} \Phi|$ is bounded by a rational Λ_Φ of polynomial encoding length, obtained from monomial bounds. With brackets of widths ϵ_j and their rational midpoints $\tilde{y}$, $|\Phi(x, \psi(x)) - \Phi(x, \tilde{y})| \leq \Lambda_\Phi \sum_j \epsilon_j/2$, because the segment from $\psi(x)$ to $\tilde{y}$ stays in $\prod_j V_j$. Choosing $\epsilon_j \leq \epsilon/(m\Lambda_\Phi)$ gives the enclosures, and $\Phi(x, \tilde{y})$ is an exact rational evaluation. The number of expressions is polynomial, and so is the cost. The same chain-rule expressions and the bounds $\Pi_{\nu,j}$ give κ_2 , κ_3 and K_ψ , uniformly for $|\eta_j| \leq \sigma$. $\square$

Proof of lemma 6.7. Write $G = G_\eta + \gamma^\top x$ and, for an allowed grid point y , $\underline{H}(y) = \sum_\beta \phi_\beta(y_\beta)$. Each reduced term is assigned to exactly one bag, so $G(y) - D_j \leq \underline{H}(y) \leq G(y)$ with $D_j = N_{\mathfrak{B}}\delta_j = E_j$. Lemma 5.6 applies to G , whose coordinate curvature is at most L on $\hat{X}$ by (19). For $x \in \Omega_j$ in a candidate cell C of β , the rounded point Y satisfies $G(Y) \geq \underline{H}(Y) \geq \underline{q}_C$, so $\underline{q}_C - E_j \leq G(x)$. The dynamic-programming minimizer $y^{(j)}$ has $G(y^{(j)}) \leq \underline{m}_j + D_j \leq \underline{m}_j + E_j$, so U_j is an upper bound for the value of a feasible point and $U_j \geq f^*$; rounding a global minimizer gives $\underline{m}_j \leq f^* + E_j$ and $U_j \leq f^* + 2E_j$. As in proposition 5.7, every minimizer is retained and every removed point has value above U_j . A retained cell has $\underline{q}_C \leq U_j + E_j$, attained at an allowed y with y_β a corner, and $G(y) \leq \underline{H}(y) + D_j \leq f^* + 4E_j$. Initialize $U_{-1} = +\infty$, and store the point associated with the smallest certified upper value: replace the stored point by $y^{(j)}$ when $\underline{m}_j + E_j < U_{j-1}$, and otherwise keep the earlier point. Its implicit lift is feasible and has value at most U_j ; its exact value is never compared. Update this pair before pruning. $\square$

The certified rows at level j need precision $\delta_j = nLh_j^2/(8N_{\mathfrak{B}})$, of encoding length $\text{poly}(I + j)$, so the cost per row is polynomial by lemma 6.6. Rows reused from an earlier level are recomputed at the new precision. With the witness tolerance $4E_j$, the proof of proposition 5.9 applies to the reduced objective conditionally on η , with interval lengths $La_i + 8E_j/a_i \leq (1 + n)La_i$; the expected number of retained cells of a bag is at most $2^{|\beta|} \prod_{i \in \beta} [4 + (1 + n)Lw_i/(2\sigma)]$.

Lemma D.1 (Certified closure). *In case (I), the closure test with certified derivatives described in section 6.1 is sound on every draw. If $g_* \geq g_0$, the unique minimizer a has $|\partial_i G(a)| > \tau$ at every active continuous coordinate, $\epsilon_g \leq \tau/8$, $\epsilon_H \leq g_0/8$, $A = 2 + nL/g_0$ and $h_j \leq \min\{1/(4A), \tau/(4\kappa_2 A), g_0/(4\kappa_3 A)\}$, then the test of level j succeeds.*

Proof. For x in the hull box, $|\partial_i G(x) - \partial_i G(c)| \leq \kappa_2 r$ and $|\hat{g}_i - \partial_i G(c)| \leq \epsilon_g$, so the enclosure $\hat{g}_i \pm (\kappa_2 r + \epsilon_g)$ contains $\partial_i G(x)$, and the argument of proposition 5.12 applies. On the patch, $\nabla_{\bar{S}\bar{S}}^2 G \geq \nabla_{\bar{S}\bar{S}}^2 G(\bar{x}_S, c^P) - \kappa_3 r^P I \geq \hat{H} - (\epsilon_H + \kappa_3 r^P) I \succ g_0 I$. For success, the witnesses satisfy $g_0 \|y - a\|^2 \leq 4E_j$, so $\|y - a\|_2 \leq h_j \sqrt{nL/(2g_0)}$ and every hull lies within $h_j(1 + \sqrt{nL/(2g_0)}) \leq Ah_j$ of a . As in proposition 5.13, integer hulls are the correct singletons, and an active coordinate with $\partial_i G(a) > \tau$ has enclosure lower end at least $\partial_i G(a) - 2\kappa_2 r - 2\epsilon_g \geq \tau - \tau/2 - \tau/4 > 0$. The free Hessian at a is at least $2g_0 I$, so $\hat{H} - (\kappa_3 r^P + \epsilon_H + g_0) I \geq (g_0 - 2\kappa_3 r^P - 2\epsilon_H) I \geq \frac{g_0}{4} I$. $\square$

Lemma D.2 (Active retained gradients on implicit graphs). *In case (I), let $D_I = \max\{2, d, e\}$ and $K_I = \max\{1, n_c 3^{n_c} N_Z D_I^{n_c+2m}\}$. Conditionally on every η and for every $\tau > 0$, the probability that $g_* > 0$ and the unique minimizer a of $G = G_\eta + \gamma^\top x$ has a continuous coordinate at a bound with $|\partial_i G(a)| \leq \tau$ is at most $K_I(\tau/\sigma + 1/M)$.*

Proof. Fix an integer assignment z , a face of the continuous retained box given by a partition $I_C = L \cup U \cup F$, and an index $i \in L \cup U$; condition on η and on γ_l , $l \neq i$. Let $\tilde{x}(u)$ be the retained point with integer part z , bounds on $L \cup U$, and $u \in \mathbb{R}^F$ on the free set; put $k = |F|$ and $\mathcal{L} = F_0(\tilde{x}(u), y) + \gamma^\top \tilde{x}(u) + \eta^\top y + \sum_j \lambda_j q_j(\tilde{x}(u)_{S_j}, y_j)$. The system

$$q_j(\tilde{x}(u)_{S_j}, y_j) = 0, \quad \partial_{y_j} \mathcal{L} = 0, \quad \partial_{u_l} \mathcal{L} = 0 \quad (j \leq m, l \in F)$$

has $k+2m$ equations in the unknowns (u, y, λ) , each of degree at most D_I , and does not involve γ_i . If $k+2m = 0$, its unique empty tuple is counted with weight one. Otherwise by lemma C.1 it has at most D_I^{k+2m} nonsingular zeros. For a zero $\varrho = (u, y, \lambda)$ put $b_\varrho = \partial_{x_i} F_0(\tilde{x}(u), y) + \sum_j \lambda_j \partial_{x_i} q_j(\tilde{x}(u)_{S_j}, y_j)$, a complex number fixed by the conditioning. The set of real γ_i with $|\gamma_i + b_\varrho| \leq \tau$ lies in an interval of length 2τ , so it has conditional probability at most $\tau/\sigma + 1/M$. At a zero that comes from a graph point, $\partial_{y_j} q_j \geq \mu_j > 0$, the multipliers are $\lambda_j = -(\partial_{y_j} F_0 + \eta_j)/\partial_{y_j} q_j$, and the chain rule with $\partial_i \psi_j = -\partial_{x_i} q_j/\partial_{y_j} q_j$ shows that the reduced derivative there is $\partial_i G = \gamma_i + b_\varrho$.

Suppose $g_* > 0$ and let a be the minimizer, with its integer part, bounds and free set as above. Then $(u, y, \lambda) = (a_F, \psi(a), \lambda(a))$ is a zero, because the reduced gradient on F vanishes. Write $C = [q_u \ q_y]$ for the constraint Jacobian, with q_y diagonal and invertible, $Z = [I; -q_y^{-1} q_u]$, and $H = \nabla_{(u,y)}^2 \mathcal{L}$. Since $\mathcal{L}(u, \psi(u), \lambda)$ equals the reduced objective for fixed λ and $\partial_y \mathcal{L} = 0$ at the zero, the chain rule gives $Z^\top H Z = \nabla_{FF}^2 G(a) \succeq 2g_* I$, by two-sided growth along the free directions. If the Jacobian of the system, $\begin{bmatrix} H & C^\top \\ C & 0 \end{bmatrix}$, annihilates (v, μ) , then $Cv = 0$ gives $v = Zw$, and multiplying the first block row by $Z^\top$ gives $Z^\top H Z w = 0$, so $w = 0$, $v = 0$, and $q_y^\top \mu = 0$ gives $\mu = 0$. The zero is therefore nonsingular, and the event is contained in the union of the events counted above, over at most $N_Z 3^{n_c n_c}$ choices. $\square$

Proof of theorem 6.4, case (I). Schedule. Let B be the factor of theorem 3.20 for the base instance with domain formula consisting of the bounds and integer constraints of X , the bounds $y_j \in V_j$, and the equations $q_j = 0$, and objective F_0 tilted on all coordinates. Let C_{tail} be the constant of lemma 3.17 for this formula, with n noisy variables x , m further variables y , and degree $\max\{d, e\}$; and K_I as in lemma D.2. Put $\rho = 1/(4B)$, $g_0 = \rho\sigma/(2W)$, $\tau = \rho\sigma/(2K_I)$, $A = 2 + nL/g_0$, $\epsilon_g = \tau/8$, $\epsilon_H = g_0/8$, let J be the least level satisfying the condition of lemma D.1, and let M be the least power of two with $M \geq \max\{2, 2^J, 4nC_{\text{tail}}/\rho, 2K_I/\rho\}$. All these have polynomial encoding length, and κ_2, κ_3 are uniform in η (lemma 6.6).

Tails. Conditionally on η , the set $\mathfrak{X}$ is compact and described by the formula above, $F = F_0 + \eta^\top y$ has rational coefficients, and the reduced objective $\min_y \{F(x, y) : (x, y) \in \mathfrak{X}\}$ is $G_\eta(x)$, because the fiber is the single point $\psi(x)$. Hence lemma 3.17(a),(b) give $\mathbb{P}\{g_* < g_0 \mid \eta\} \leq g_0 W/\sigma + 2nC_{\text{tail}}/M \leq \rho$, and lemma D.2 gives probability at most ρ for small active gradients. By lemma D.1, closure fails at level J with probability at most $2\rho = 1/(2B)$, conditionally on every η .

Correctness and work. Closure outputs are correct by lemmas 6.7 and D.1 and the strong convexity of the reduced patch. Otherwise theorem 3.20 is applied to $F_{\gamma, \eta}$ on $\mathfrak{X}$. The work of the levels is bounded as in the proof of theorem 5.2, with the count above and the polynomial cost per certified row; the fallback contributes $\text{poly}(I)$ in expectation by lemma 3.18(a).

Evaluation. First handle a point patch. Its retained coordinates are rational and fixed, but its dependent coordinates can still be irrational. Use lemma 6.6 to enclose the sampled objective to width 2^{-q} and to refine each dependent root to width at most $2^{-q-1}/\max\{1, m\}$. Their exact roots

give the feasible lift; their rational midpoints give an ambient approximation within $2^{-q}/2$ of it. This costs $\text{poly}(I + q)$, with $q + O(\log(n + m + 1))$ coordinate precision and the oracle's polynomial encoded allowance for objective evaluation. No positive-dimensional convex optimization is invoked.

For a positive-dimensional ordinary patch apply [lemma C.2](#) to the reduced objective on the patch box, with $Z = I$, coordinatewise clipping as repair, $\nu = \|\cdot\|_2$, chain-rule bounds V and G , and the approximate oracle of [lemma 6.6](#); its curvature hypothesis holds with $g = g_0$. With $\eta_{\text{GLS}} = \min\{2^{-q}, \frac{g_0}{2}(2^{-q}/(2K_\psi))^2\}$ it returns a rational $\tilde{x}$ with $\|\tilde{x} - \hat{x}\|_2 \leq 2^{-q}/(2K_\psi)$ and a value interval of width at most 2^{-q} . By (20) the exactly feasible lift $(\tilde{x}, \psi(\tilde{x}))$, given by $\tilde{x}$ and root brackets, lies within $2^{-q}/2$ of the optimizer, and refining the brackets to total width $2^{-q}/2$ gives a rational point within 2^{-q} . On fallback draws, let $G' \geq \max\{1, \sup_{\hat{x}} \|\nabla G\|_1\}$ be a rational chain-rule bound, refine each retained continuous coordinate to error at most $2^{-q}/(4n \max\{1, K_\psi, G'\})$, clip it to its original interval, and identify the integer labels exactly as in [proposition 5.18\(b\)](#). The retained distance is at most $2^{-q}/(4K_\psi)$ and the objective gap at most $2^{-q}/4$. Attach the exact chart roots, refine their brackets to total width $2^{-q}/2$, and refine the optimal value to width $2^{-q}/2$. These give both the feasible lift with a certified objective gap and the ambient rational approximation. All extra precision is $q + \text{poly}(I)$ and costs $B \text{poly}(I + q)$ under [theorem 3.20](#). $\square$

D.2 Affine-state recurrences

Proof of theorem 6.11. Feasible set. For every free point the recurrence defines one trajectory, and by (22) and induction on k , $s_k \in I_k$ for all k , because $u_k \in \text{hull } U_k$. Conversely every feasible trajectory arises in this way. The identity (23) follows by substituting $s_N = a_{N-1}s_{N-1} + \chi_{N-1}(u_{N-1})$ into $(c_N + \eta_N)s_N$, which turns the coefficient of s_{N-1} into $c_{N-1} + \eta_{N-1} + a_{N-1}\lambda_N = \lambda_{N-1}$, and repeating down to $\lambda_1 s_1 = \lambda_1(a_0 s_0 + \chi_0(u_0))$. Hence, with constants from substituted fixed coordinates carried separately, the sampled objective of a trajectory equals $Q_\eta(z) + \xi^\top z$ at its free point.

Substitute fixed free coordinates before computing widths or a schedule. If none remains, use $M = 2$, draw the ambient coefficients once, and return the unique rational trajectory and its exact sampled value. The remaining argument concerns a nonempty free box with positive coordinate widths.

Uniform data. By induction from $|\lambda_N| \leq C_s + \sigma$ and $|\lambda_k| \leq C_s + \sigma + a|\lambda_{k+1}|$, every $|\lambda_k| \leq \Lambda$. The added terms are unary, so the coordinate curvature, the Hessian row sums and the third-derivative row sums of Q_η exceed those of q by at most ΛG_2 , ΛG_2 and ΛG_3 , which gives (24) and uniform κ_2, κ_3 . The adjoints are computed by N affine steps, so their bit lengths are sums of the lengths of the a_k, c_k and η_k , polynomial in $I + b$. Producing the coefficients of Q_η also costs $\text{poly}_d(I + b)$; neither height nor production cost is asserted to be bounded by the literal $I + b$. The fallback of [lemma 6.1\(b\)](#) is [theorem 3.20](#) applied to the base instance with variables (s, u) , domain formula $\bigwedge_k [s_{k+1} = a_k s_k + \chi_k(u_k)]$ together with the variable bounds and integer constraints, and objective F_0 tilted on all coordinates; its minimizers are the trajectories of the minimizers of $Q_\eta + \xi^\top z$. [Lemma 6.1](#) with the free-variable decomposition gives (i) and (ii).

Evaluation. For a rational free point, the states are rational: each step evaluates a fixed-degree polynomial at its own control and adds an affine function of the previous state, so bit lengths add over the horizon instead of being raised to powers, and the trajectory has length polynomial in I and the length of the free point. It is exactly feasible, and integer labels are exact. For two free points, $|\Delta s_{k+1}| \leq a|\Delta s_k| + G_1|\Delta u_k|$, so the vector $(|\Delta s_k|)_{k \geq 1}$ is dominated by a lower-triangular Toeplitz matrix with entries a^l applied to $(|\Delta s_0|, G_1|\Delta u_0|, \dots)$. Its row and column sums are at most $1/(1 - a)$, so its spectral norm is at most $1/(1 - a)$, and the trajectory map has Lipschitz constant at most $K_{\text{traj}} = 1 + (1 + G_1)/(1 - a)$. The free evaluation of [proposition 5.18](#), with $\log_2 K_{\text{traj}}$ extra bits, gives (iii); objective values and gaps transfer exactly by (23). $\square$

D.3 Products of simplices

At levels with $h_j \geq 1$ the cell of a block is its original simplex and its corners and grid are the original simplex vertices: $0, e_i$ for an inequality block and e_i for an equality block. At levels with $h = h_j = 1/m < 1$ the cells are the nonempty sets $C = \Delta_b \cap \prod_{i \in b} [hk_i, h(k_i + 1)]$, with $k_i \in \{0, \dots, m-1\}$, or the same with Δ_b^- ; their corners are their points in $(h\mathbb{Z})^b$. These bounded indices exclude exterior cubes that would otherwise add duplicate boundary cells. Integer coordinates use the cells of [section 5.2](#). A bag cell is a product of block cells and integer cells.

Lemma D.3 (Feasible block rounding). *Every block cell is the convex hull of its corners, which are original simplex vertices when $h_j \geq 1$ and cube corners when $h_j < 1$. For $x \in \Omega_j$ and a cell $C_0 \in \mathcal{W}_j(\beta_0)$ containing x_{β_0} , there is a random $Y \in A_j$ with Y_{β_0} a corner of C_0 , $Y_i = x_i$ whenever x_i is a grid value, and $\mathbb{E}F_\gamma(Y) \leq F_\gamma(x) + nLh^2/8$.*

Proof. For $h_j < 1$, with $z_i = x_i/h - k_i$, a cell becomes $\{z \in [0, 1]^b : \sum_i z_i \leq \rho\}$, or the same with equality, where $\rho = m - \sum_i k_i$ is an integer. At a vertex, b linearly independent constraints are tight. If the sum constraint is not among them, every coordinate is 0 or 1; if it is, at least $b-1$ coordinates are 0 or 1, and the remaining one equals ρ minus an integer and lies in $[0, 1]$. So all vertices are cube corners, and the bounded polytope is their convex hull. A whole simplex has the vertices 0 and e_i , or e_i .

For each block b , choose a cell $C_b \ni x_b$ (the one of C_0 if $b \subseteq \beta_0$), write x_b as a convex combination of the corners of C_b , and let Y_b equal each corner with its coefficient, independently over blocks; round integer coordinates as in [lemma 5.6](#). Then $\mathbb{E}Y_b = x_b$, and each coordinate of Y_b takes the two values $hk_i, h(k_i + 1)$, so its variance is at most $h^2/4$ (at most $1/4 \leq h^2/4$ while $h \geq 1$). If $x_i = hk_i$, then $Y_i \geq hk_i$ with mean hk_i forces $Y_i = x_i$; the same holds at the upper cube face. Hence Y_b lies in every level- j cell of b that contains x_b : coordinates of x_b strictly inside a grid interval have the same interval in every such cell, coordinates on a grid value stay fixed, and Y_b is feasible. As a grid point of that cell it is one of its corners. Every bag cell containing x_β therefore contains Y_β as a corner, so $Y \in A_j$. Rounding the blocks one after another, the segment from x_b to Y_b lies in C_b , so [\(25\)](#) and Taylor's theorem give $\mathbb{E}[F_\gamma \mid \text{earlier blocks}] \leq F_\gamma + \frac{L}{2} \mathbb{E} \|Y_b - x_b\|^2$, and summing over blocks and integer coordinates gives the allowance. $\square$

With this lemma the proof of [proposition 5.7](#) applies verbatim.

Lemma D.4 (Count by faces). *Let $j \leq J$ and $M \geq 2^J$. For a bag β , the set $N_j(\beta)$ of feasible bag grid tuples v with $V_\beta(v) + \gamma_\beta^\top v \leq f^* + 2E_j$ satisfies*

$$\mathbb{E} |N_j(\beta)| \leq 4^{|\beta|} \left[4 + \frac{(2 + n/2)L \max\{1, w_{\max}\}}{2\sigma} \right]^{|\beta|},$$

and $|\mathcal{W}'_j(\beta)| \leq 2^{|\beta|} |N_j(\beta)|$ on every draw.

Proof. If $h_j \geq 1$, each block has at most $|b| + 1 \leq 2^{|b|}$ grid vertices. Use their trivial probability bound one and the integer-coordinate count of [proposition 5.9](#); this is already bounded by the displayed product. Thus the following face comparisons concern $h_j < 1$ only.

Because bags contain whole blocks, the domain outside β is a fixed product, and V_β is defined as in [\(12\)](#). For a feasible bag tuple v and a direction d inside one block $b \subseteq \beta$ with $v \pm hd$ feasible, the minimizer argument of [lemma 5.8](#) and [\(25\)](#) give $V_\beta(v + hd) + V_\beta(v - hd) - 2V_\beta(v) \leq Lh^2 \|d\|^2$.

Fix v and a block $b \subseteq \beta$, let S be the positive support of v_b , and let r_b be the dimension of the smallest face of the simplex containing v_b . If the budget is slack, $r_b = |S|$, and the directions

$\mathbf{d} = e_i$, $i \in \mathbf{S}$, satisfy $v \pm h\mathbf{d}$ feasible, because positive coordinates and the budget slack are positive multiples of h . If the budget is tight (always for equality blocks), $r_b = |\mathbf{S}| - 1$; with the least index $\iota \in \mathbf{S}$ as anchor, the directions $e_i - e_\iota$, $i \in \mathbf{S} \setminus \{\iota\}$, are feasible in both signs. Condition on the noise outside β , on the anchor coefficients of tight blocks, and on the coefficients of the zero coordinates. If $v \in N_j(\beta)$, comparison with $v \pm h\mathbf{d}$ as in [lemma 5.8](#) confines $\gamma^\top \mathbf{d}$, which is γ_i or γ_i minus a conditioned anchor, to an interval determined by the conditioned values and v , of length at most $Lh \|\mathbf{d}\|^2 + 4E_j/h \leq (2 + n/2)Lh$. Each direction involves a different remaining coefficient, so the conditional probability is at most $[(2 + n/2)Lh/(2\sigma) + 1/M]^{r_b}$ for the block, times the integer factors of [proposition 5.9](#).

A face of dimension r has $\binom{1/h-1}{r} \leq h^{-r}$ grid points in its relative interior: positive multiples of h in the slack case with total at most $1 - h$, and compositions of 1 in the tight case. The simplex has at most $2^{|b|+1}$ faces, and $1/(Mh) = 2^j/(Ms) \leq 1$. Hence the expected contribution of the block is at most $2^{|b|+1}[1 + (2 + n/2)L/(2\sigma)]^{|b|} \leq 4^{|b|}[4 + (2 + n/2)L/(2\sigma)]^{|b|}$; while $h \geq 1$ the block grid consists of at most $|b| + 1$ vertices. The probabilities of different blocks and integer coordinates multiply after the conditioning, and summing over tuples factorizes, which proves the bound. Every retained cell has a corner in $N_j(\beta)$ by [proposition 5.7\(d\)](#), and a grid tuple is a corner of at most $2^{|b|}$ cells, since it lies in at most $2^{|b|}$ cubes per block. $\square$

Lemma D.5 (Simplex closure). *Let Q be the intersected hull box of level j , with midpoint c and largest half-width r , and let κ_2, κ_3 hold on the enclosing coordinate box. The following recorded equalities hold at every global minimizer:*

- (a) *for an inequality block, $x_i = 0$ if $\partial_i F_\gamma(c) - \kappa_2 r > 0$;*
- (b) *for an inequality block, $\sum_{i \in b} x_i = 1$ if $\partial_l F_\gamma(c) + \kappa_2 r < 0$ for some $l \in b$;*
- (c) *for any block, $x_i = 0$ if $\partial_i F_\gamma(c) - \partial_l F_\gamma(c) - 2\kappa_2 r > 0$ for some $l \in b$;*
- (d) *x_i equals the point of Q_i if Q_i is a singleton.*

If every integer coordinate is recorded, let P be the set of feasible points that satisfy the recorded equalities and lie in Q . After the substitutions described in the proof, let Z be the tangent basis with columns e_i for free coordinates of blocks with a slack budget and $e_i - e_\iota$ for the free coordinates other than an anchor ι of blocks with an equality or recorded tight budget, let $c^P \in P$ be the center constructed in the proof and r^P the largest width of Q . If

$$Z^\top \nabla^2 F_\gamma(c^P) Z - (\kappa_3 r^P + g_0) Z^\top Z \succ 0, \quad (46)$$

then $v^\top \nabla^2 F_\gamma(x) v \geq g_0 \|v\|^2$ for $x \in P$ and v in the column space of Z , and the unique minimizer of F_γ on P is the unique global minimizer. If the patch has tangent dimension zero, return its rational point before constructing a center or testing a matrix. Conversely, if $g_ \geq g_0$, every multiplier of an active inequality at the minimizer a exceeds τ , $A = 2 + nL/g_0$, and $h_j \leq \min\{1/(4A), \tau/(8\kappa_2 A), g_0/(8\kappa_3 A)\}$, then every active inequality at a and every integer coordinate is recorded, and closure returns a point or (46) holds. Active inequalities are zero-coordinate constraints and tight budgets of inequality blocks; original equality budgets have no multiplier-margin requirement.*

Proof. Soundness. Every minimizer x^* lies in Q , and $|\partial_i F_\gamma(x) - \partial_i F_\gamma(c)| \leq \kappa_2 r$ for $x \in Q$. In (a), $\partial_i F_\gamma(x^*) > 0$, and if $x_i^* > 0$ then decreasing x_i is feasible in an inequality block and decreases the objective. In (b), $\partial_l F_\gamma(x^*) < 0$, and if the budget were slack, increasing x_l would be feasible. In (c), $\partial_i F_\gamma(x^*) > \partial_l F_\gamma(x^*)$, and if $x_i^* > 0$, moving mass from x_i to x_l keeps every simplex constraint and

decreases the objective. These directions are feasible in the original simplex; the hull is used only to bound derivatives. Case (d) holds because Q contains every minimizer.

The patch. In each block let F be the unrecorded coordinates with bounds $l_i < u_i$ from Q and $[0, 1]$, and β_b the budget left after substituting fixed coordinates, with $\sum_F x_i \leq \beta_b$, or $= \beta_b$ for an equality block or recorded tight budget. If $F = \emptyset$, drop this block before any division: containment of a global minimizer certifies that its fixed values satisfy its budget. If an equality budget has $|F| = 1$, fix its remaining coordinate to β_b and drop the block. If $\sum_F l_i = \beta_b$, the block is forced to l ; if the budget is an equality and $\sum_F u_i = \beta_b$, it is forced to u ; substitute such blocks. If no tangent direction remains, P is a rational point and closure returns it. Otherwise, every remaining block has a positive sum of coordinate widths. For an equality budget put $\lambda = (\beta_b - \sum l_i) / \sum (u_i - l_i) \in (0, 1)$, and for an inequality budget $\lambda = \min\{\frac{1}{2}, (\beta_b - \sum l_i) / (2 \sum (u_i - l_i))\}$, and let $c_F^P = l + \lambda(u - l)$. Every bound and every inequality budget is strict at c^P , so the affine hull of P is cut out exactly by the recorded and equality budgets, and Z spans its direction space. If no free coordinate remains, P is a single point, which is the unique minimizer. For $x \in P$, $\|x - c^P\|_\infty \leq r^P$, so $Z^T \nabla^2 F_\gamma(x) Z \succeq Z^T \nabla^2 F_\gamma(c^P) Z - \kappa_3 r^P Z^T Z \succ g_0 Z^T Z$, which is the claimed bound with $v = Zt$. The patch contains every minimizer, so its unique minimizer is the unique global one.

Success. The witnesses lie within $\frac{h_j}{2} \sqrt{nL/g_0}$ of a , so every hull lies within Ah_j of a : $r \leq Ah_j$ and $r^P \leq 2Ah_j$. Integer hulls are the correct singletons because $Ah_j \leq 1/4$. KKT conditions at a read $\partial_i F_\gamma(a) + \lambda_b - \lambda_i = 0$, with $\lambda_i \geq 0$ for zero-coordinate inequalities and $\lambda_b \geq 0$ for inequality budgets, while the multiplier λ_b of an original equality budget is unrestricted. They are uniquely determined because the active normals, together with original equality normals, are linearly independent. In a slack inequality block, a zero coordinate has $\partial_i F_\gamma(a) = \lambda_i > \tau$, and test (a) fires since $\partial_i F_\gamma(c) - \kappa_2 r \geq \lambda_i - 2\kappa_2 Ah_j > 0$. In a block with a tight inequality budget, a positive coordinate l has $\partial_l F_\gamma(a) = -\lambda_b < -\tau$, so test (b) fires. In every block with a tight or equality budget, a zero coordinate i has $\partial_i F_\gamma(a) - \partial_l F_\gamma(a) = \lambda_i > \tau$, so test (c) fires since its error is at most $4\kappa_2 Ah_j \leq \tau/2$. No lower bound on the positive coordinate is needed. All original active inequalities have now been recorded, and further singleton fixing is sound by containment. Hence P lies in the smallest face of a , the column space of Z lies in the direction space of that face, and two-sided growth gives $Z^T \nabla^2 F_\gamma(a) Z \succeq 2g_0 Z^T Z$. Since $\|c^P - a\|_\infty \leq 2Ah_j$, the matrix of (46) is at least $(g_0 - 4\kappa_3 Ah_j) Z^T Z \succeq \frac{g_0}{2} Z^T Z$. $\square$

Lemma D.6 (Multiplier margins on simplices). *Let $D = \max\{1, d - 1\}$ and*

$$K_\Delta = \max\{1, 2n_c N_Z 3^{n_c} (2D)^{n_c}\}.$$

For $\tau > 0$, the probability that $g_ > 0$ and some active inequality at the unique minimizer has multiplier at most τ is at most $K_\Delta(\tau/\sigma + 1/M)$. Only zero-coordinate inequalities and tight budgets of inequality blocks enter this event; equality-budget multipliers are excluded.*

Proof. Fix integer labels z , a face of every block (support and tightness, at most $3^{|b|}$ choices), and one active inequality $\varkappa$ of the resulting product face Φ (at most $2n_c$ choices). Let $t \in \mathbb{R}^k$ be the tangent coordinates: the support coordinates of slack blocks and the support coordinates other than the anchor ι (least support index) of tight blocks. On Φ the linear noise equals a constant plus $\theta^T t$, where θ consists of the γ_i of slack supports and the differences $\gamma_i - \gamma_\iota$ of tight supports. The stationarity system of Φ , $\nabla_t[F_0(x(t))] + \theta = 0$, has at most D^k nonsingular zeros by lemma C.1 and depends on the noise only through θ . For $k = 0$ there is one empty stationary tuple, counted with weight $D^0 = 1$, without applying the positive-dimensional zero lemma. At a point of Φ the active-inequality multipliers are $\lambda_b = -\partial_i F_0 - \gamma_\iota$ for a tight inequality budget,

$\lambda_i = \partial_i F_0 - \partial_l F_0 + \gamma_i - \gamma_l$ for a zero coordinate of a tight block, and $\lambda_i = \partial_i F_0 + \gamma_i$ for a zero coordinate of a slack block. Thus $\lambda_{\mathcal{K}}$ is affine with slope ± 1 in its *normal coefficient* $\nu_{\mathcal{K}}$, namely γ_l or γ_i , once θ and the other coefficients are fixed.

The map from γ to (θ, ν) , where ν lists the original coefficients not differenced, is injective. Its transformed coordinates are not independent under the grid law, so we count tuples. A difference of two grid values takes at most $2M - 1$ values. For each of the at most $(2M - 1)^k M^{n-k-1}$ values of the transformed coordinates other than $\nu_{\mathcal{K}}$, the zeros are fixed, and for each zero the condition $|\lambda_{\mathcal{K}}| \leq \tau$ allows at most $\tau(M - 1)/\sigma + 1$ grid values of $\nu_{\mathcal{K}}$. Dividing by M^n , the probability that the event occurs at some nonsingular zero is at most $2^k D^k (\tau/\sigma + 1/M)$.

If $g_* > 0$, let a be the minimizer and Φ its smallest face. Its tangent coordinates are a zero of the system, nonsingular because two-sided growth along tangent directions gives a positive definite tangent Hessian. The event is therefore contained in the union over $N_Z 3^{n_c} 2n_c$ choices of the events above, with $k \leq n_c$. $\square$

Proof of theorem 6.15. Schedule. Let B be the factor of theorem 3.20 for the product of simplices and integer intervals, and C_{tail} the constant of lemma 3.17 for its linear description and label disjunctions. Put

$$\rho = \frac{1}{4B}, \quad g_0 = \frac{\rho\sigma}{2W}, \quad \tau = \frac{\rho\sigma}{2K_{\Delta}}, \quad A = 2 + \frac{nL}{g_0},$$

let J be the least level with $h_J \leq \min\{1/(4A), \tau/(8\kappa_2 A), g_0/(8\kappa_3 A)\}$, and let M be the least power of two with $M \geq \max\{2, 2^J, 4nC_{\text{tail}}/\rho, 2K_{\Delta}/\rho\}$. Every coordinate width is 1 for simplex coordinates and w_i for integers, so lemma 3.17(b) gives $\mathbb{P}\{g_* < g_0\} \leq \rho$, and lemma D.6 gives probability at most ρ for small multipliers. By lemma D.5 closure succeeds by level J outside these events, and is sound on every draw; otherwise the fallback runs on the same draw.

Work. The cells have at most $2^{|b|}$ corners, children and incident cells per block; lemma D.4 and the argument of theorem 5.2 give the expected work, and the fallback contributes $\text{poly}_d(I)$ by lemma 3.18(a).

Evaluation. If the tangent dimension is zero, return the rational point and its exactly evaluated objective; no convex-body oracle is used. Otherwise apply lemma C.2 to the patch P in the free coordinates, with $c = c^P$, the basis Z , $\zeta = 2k$ (since $\|Z\|_2 \leq \|Z\|_F \leq \sqrt{2k}$), $R = \max\{1, k\}$, and $r = \min\{\frac{1}{2}, \min_s \sigma_s / (k \|\alpha_s\|_1)\}$, where $\alpha_s^T x \leq \alpha_{0,s}$ runs over the bounds and inequality budgets of P and $\sigma_s > 0$ is its slack at c^P : every coordinate of Zt is a coordinate of t or minus a sum of them, so $|\alpha_s^T Zt| \leq k \|\alpha_s\|_1 \|t\|_2$. The repair map is the Euclidean projection onto P , computed blockwise: on a block with bounds $l \leq x \leq u$ and budget $\underline{\beta} \leq \sum x \leq \beta$ (with $\underline{\beta} = \beta$ for equality budgets and $\underline{\beta} = -\infty$ otherwise), the projection of w is $x_i = \min\{u_i, \max\{l_i, w_i - \theta\}\}$, with $\theta = 0$ if this satisfies the budget and otherwise the solution of $\sum_i x_i(\theta)$ equal to the violated endpoint. The sum is continuous, nonincreasing and piecewise linear in θ with breakpoints $w_i - u_i$ and $w_i - l_i$; sorting them and solving one linear equation gives a rational θ in polynomial time. Projection onto a convex set is nonexpansive, so $\nu = \|\cdot\|_2$. The curvature hypothesis holds with $g = g_0$ by lemma D.5. Use objective tolerance $\min\{2^{-q}, (g_0/2)2^{-2q}\}$, so the lemma controls physical Euclidean distance and the feasible objective gap. On fallback draws, let $G' \geq \max\{1, \sup \|\nabla F_{\gamma}\|_2\}$ bound the gradient on the enclosing box. Refine each continuous coordinate to error at most $2^{-q-1}/(nG')$, project each block onto its original simplex in the same way, and keep integer labels exact. Nonexpansiveness gives physical distance at most $2^{-q-1}/G'$ and objective gap at most 2^{-q-1} . A value enclosure of width 2^{-q-1} then completes the certified interval. These refinements cost $B \text{poly}_d(I + q)$ and have expectation $\text{poly}_d(I + q)$. $\square$

D.4 Order constraints

Write $x = (x_C, x_Z)$ for continuous and binary coordinates. The slice for a binary assignment $\mathbf{b}$ is $P_{\mathbf{b}} = \{x_C : (x_C, \mathbf{b}) \in X\}$, an order polytope on I_C intersected with the bounds $x_i = 0$ for edges (i, l) with $\mathbf{b}_l = 0$ and $x_i = 1$ for edges (l, i) with $\mathbf{b}_l = 1$.

Lemma D.7 (Zero-one vertices). *Every vertex of $P_{\mathbf{b}}$, and of every face of it, is a zero-one vector.*

Proof. Let x be a point of $P_{\mathbf{b}}$ with a value $t \in (0, 1)$, and let G be the set of coordinates with value t . Moving all coordinates of G by $\pm\epsilon$ keeps every edge inside G tight, keeps every edge between G and other coordinates strict for small ϵ , and keeps the bounds; so x is the midpoint of two feasible points and not a vertex. A face is a polytope whose vertices are vertices of $P_{\mathbf{b}}$. $\square$

Rounding for theorem 6.17. For fixed $U \in [0, 1]$ the scalar map $t \mapsto h(\lfloor t/h \rfloor + \mathbf{1}\{U < \text{frac}(t/h)\})$ is nondecreasing, fixes the grid values, including 0 and 1, and maps each grid interval onto its endpoints. Applied to all continuous coordinates of $x \in \Omega_j$, it therefore preserves every order constraint and every binary value, keeps every coordinate strictly inside a grid interval within that interval and every grid coordinate fixed, and so keeps every bag tuple a corner of every candidate cell containing it. For U uniform, every coordinate is mean preserving with variance at most $h^2/4$. The segment from x to Y lies in $[0, 1]^n$, so the full Hessian bound gives $\mathbb{E}F_{\gamma}(Y) \leq F_{\gamma}(x) + \frac{\bar{L}}{2}\mathbb{E}\|Y - x\|^2 \leq F_{\gamma}(x) + n_c \bar{L}h^2/8$. Every edge lies in a bag and rows violating an assigned edge are removed, so allowed grid points are feasible. The proof of proposition 5.7 then applies with $E_j = n_c \bar{L}h_j^2/8$. $\square$

Fix a bag β , write $\beta_C = \beta \cap I_C$ with $c_{\beta} = |\beta_C|$ and $\beta_Z = \beta \cap I_Z$, and let $O = [n] \setminus \beta$. For a binary row $\mathbf{b} \in \{0, 1\}^{\beta_Z}$ and $v \in [0, 1]^{\beta_C}$, put

$$V_{\mathbf{b}}(v) = \gamma_{\beta_Z}^{\top} \mathbf{b} + \min\{F_0(v, \mathbf{b}, z) + \gamma_O^{\top} z : (v, \mathbf{b}, z) \in X\},$$

with $\min \emptyset = +\infty$; it depends on γ_O and γ_{β_Z} only. A *standard order simplex* is $S_{\pi} = \{v : 0 \leq v_{\pi(1)} \leq \dots \leq v_{\pi(c_{\beta})} \leq 1\}$ for a permutation π of β_C . Its vertices form a chain of zero-one vectors, and a face with vertices $\mathbf{a}_0 < \dots < \mathbf{a}_k$ in this chain has *edge directions* $\mathbf{d}_l = \mathbf{a}_l - \mathbf{a}_{l-1}$, $l = 1, \dots, k$, with disjoint nonempty supports.

Lemma D.8 (Endpoint-preserving transport). *Let v lie in the relative interior of a face Φ of S_{π} , with $V_{\mathbf{b}}(v) < \infty$. There is an affine map $w \mapsto x(w)$ on the affine hull of Φ such that, for every $w \in \Phi$, $x(w) \in X$ has continuous bag part w and binary bag part $\mathbf{b}$; $x(v)$ attains $V_{\mathbf{b}}(v)$; and along every edge direction $\mathbf{d}_l$ of Φ each coordinate of x changes at a rate in $[0, 1]$, binary coordinates at rate 0. Consequently, whenever $v \pm \theta \mathbf{d}_l \in \Phi$,*

$$V_{\mathbf{b}}(v + \theta \mathbf{d}_l) + V_{\mathbf{b}}(v - \theta \mathbf{d}_l) - 2V_{\mathbf{b}}(v) \leq n_c \bar{L} \theta^2.$$

Proof. On the relative interior of Φ , coordinates of β_C are partitioned into groups of equal value: a group at 0, a group at 1 (either may be empty), and k interior groups with values $0 < r_1 < \dots < r_k < 1$; this partition is the same at every relative-interior point, and on the closed face the groups stay equal and ordered, $0 \leq s_1(w) \leq \dots \leq s_k(w) \leq 1$, where $s_l(w)$ is the common value of group l at w , a linear function of w . Moving along $\mathbf{d}_l$ changes exactly one group value at unit rate: writing $v = \sum_l \lambda_l \mathbf{a}_l$ with positive weights, the value of the l th interior group is $\sum_{i \geq l} \lambda_i$, and $\mathbf{d}_l$ moves weight from λ_{l-1} to λ_l .

Let $x^* = (v, \mathbf{b}, z^*)$ attain $V_{\mathbf{b}}(v)$; it exists by compactness. Put $r_0 = 0$, $r_{k+1} = 1$, $s_0 = 0$, $s_{k+1} = 1$, and let T_w be the piecewise-affine map of $[0, 1]$ with $T_w(r_l) = s_l(w)$, affine between consecutive knots. Define $x(w)$ by applying T_w to every coordinate of x^* . For $w \in \Phi$, T_w is nondecreasing with

$T_w(0) = 0$ and $T_w(1) = 1$, so $x(w)$ satisfies every order constraint, keeps every binary value, lies in $[0, 1]^n$, and maps every bag coordinate of group l to $s_l(w) = w_i$; thus $x(w) \in X$ with bag parts $(w, \mathbf{b})$, and $x(v) = x^*$. A coordinate of x^* in $[r_l, r_{l+1}]$ equals $(1 - \vartheta)r_l + \vartheta r_{l+1}$ for a fixed $\vartheta \in [0, 1]$ and is mapped to $(1 - \vartheta)s_l(w) + \vartheta s_{l+1}(w)$, which is affine in w ; along $\mathbf{d}_l$ its rate is $1 - \vartheta$, ϑ , or 0. Binary coordinates sit at the fixed knots 0 and 1.

The function $\varphi(w) = \gamma_{\beta_Z}^\top \mathbf{b} + F_0(x(w)) + \gamma_O^\top x(w)_O$ satisfies $\varphi(v) = V_{\mathbf{b}}(v)$ and $\varphi \geq V_{\mathbf{b}}$ on Φ . The linear noise term is affine in w , and the derivative $\mathbf{x}'$ of x along $\mathbf{d}_l$ has at most n_c nonzero entries, all in $[0, 1]$, so $\frac{d^2}{d\theta^2} \varphi(v + \theta \mathbf{d}_l) = \mathbf{x}'^\top \nabla^2 F_0 \mathbf{x}' \leq \bar{L} \|\mathbf{x}'\|^2 \leq n_c \bar{L}$. Comparing $V_{\mathbf{b}}$ with its upper support φ at $v \pm \theta \mathbf{d}_l$ gives the inequality. $\square$

The transported points need not exhaust the fiber over w : at endpoint faces new binary labels can become feasible, and the projection of the mixed domain onto the continuous bag coordinates need not be convex. Neither matters, because only the upper support at $v \pm \theta \mathbf{d}_l$ is used.

Proposition D.9 (Order count). *Let $h = h_j = 2^{-j}$ with $Mh \geq 1$, and let $N_j(\beta)$ be the set of feasible bag grid tuples $(v, \mathbf{b})$ with $V_{\mathbf{b}}(v) + \gamma_{\beta_C}^\top v \leq f^* + 2E_j$. Then*

$$\mathbb{E} |N_j(\beta)| \leq 2^{|\beta_Z|} c_\beta! (c_\beta + 1) \left[2 + \frac{3n_c \bar{L}}{4\sigma} \right]^{c_\beta},$$

and $|\mathcal{W}'_j(\beta)| \leq 2^{|\beta|} |N_j(\beta)|$ on every draw.

Proof. Fix $\mathbf{b}$ and a feasible v , choose π with $v \in S_\pi$, and let Φ be the face of S_π whose relative interior contains v , of dimension k . Its barycentric weights are positive multiples of h , so $v \pm h \mathbf{d}_l \in \Phi$ for every edge direction. Condition on all coefficients except those of β_C . If $(v, \mathbf{b}) \in N_j(\beta)$, the argument of lemma 5.8, with the transported points of lemma D.8 as feasible competitors and $\eta = 2E_j = n_c \bar{L} h^2 / 4$, confines $\gamma^\top \mathbf{d}_l$ to an interval of length at most $(n_c \bar{L} h^2 + 2\eta)/h = \frac{3}{2} n_c \bar{L} h$ that is determined by the conditioned values and the tuple. Choose one coordinate in the support of each $\mathbf{d}_l$ and condition also on the other continuous bag coefficients; the supports are disjoint, so the k remaining coefficients are independent and each lies in an interval of that length. The conditional probability is at most $[h(\frac{3n_c \bar{L}}{4\sigma} + \frac{1}{Mh})]^k$. A k -face has $\binom{1/h-1}{k} \leq h^{-k}/k!$ relative-interior grid points, S_π has $\binom{c_\beta+1}{k+1}$ faces of dimension k , there are at most $c_\beta!$ permutations, and at most $2^{|\beta_Z|}$ binary rows; tuples lying in several simplices are counted more than once, which only enlarges the bound. With $\mathbf{A} = \frac{3n_c \bar{L}}{4\sigma} + 1 \geq \frac{3n_c \bar{L}}{4\sigma} + \frac{1}{Mh}$,

$$\mathbb{E} |N_j(\beta)| \leq 2^{|\beta_Z|} c_\beta! \sum_{k=0}^{c_\beta} \binom{c_\beta+1}{k+1} \frac{\mathbf{A}^k}{k!} \leq 2^{|\beta_Z|} c_\beta! (c_\beta + 1) (1 + \mathbf{A})^{c_\beta},$$

using $\binom{c+1}{k+1}/k! = (c+1)\binom{c}{k}/(k+1)! \leq (c+1)\binom{c}{k}$. The second claim follows from proposition 5.7(d), since a retained cell has an allowed witness y with $V_{\mathbf{b}}(v) + \gamma_{\beta_C}^\top v \leq F_\gamma(y)$. $\square$

Lemma D.10 (Linear-programming exposure). *Suppose every global minimizer has binary part $\mathbf{b}$, let Q_C be a box containing their continuous parts, c its midpoint, r its largest half-width, $g = \nabla_C F_\gamma(c, \mathbf{b})$ and $\delta = \kappa_2 r$, where κ_2 bounds the continuous Hessian row sums on $[0, 1]^n$. For a proposed equality $\mathbf{e}$ (an edge $x_i = x_l$ with opposite face $P_{\mathbf{e}}^- = P_{\mathbf{b}} \cap \{x_i = 0, x_l = 1\}$, or a bound $x_i = 0$ or $x_i = 1$ with opposite face $P_{\mathbf{b}} \cap \{x_i = 1\}$ or $P_{\mathbf{b}} \cap \{x_i = 0\}$) with nonempty opposite face, put $\Delta_{\mathbf{e}}(v) = \min_{P_{\mathbf{e}}^-} v^\top x - \min_{P_{\mathbf{b}}} v^\top x$. If $\Delta_{\mathbf{e}}(g) > n_c \delta$, the equality holds at every global minimizer. For all v, w , $|\Delta_{\mathbf{e}}(v) - \Delta_{\mathbf{e}}(w)| \leq n_c \|v - w\|_\infty$. If the opposite face is empty, the equality holds on all of $P_{\mathbf{b}}$.*

Proof. Let $x^* = (a, \mathbf{b})$ be a global minimizer and $\mathbf{h} = \nabla_C F_\gamma(a, \mathbf{b})$. Since a minimizes $F_\gamma(\cdot, \mathbf{b})$ on the convex set $P_{\mathbf{b}}$, $\mathbf{h}^\top(x - a) \geq 0$ there, so a lies in the face $\Psi = \arg \min_{P_{\mathbf{b}}} \mathbf{h}^\top x$, whose vertices are zero-one by [lemma D.7](#). If an edge equality fails at a , then $a_i < a_l$, and not all vertices of Ψ have equal i th and l th coordinates; a vertex $\mathbf{v}$ with $v_i \neq v_l$ has $v_i = 0$ and $v_l = 1$, so $\mathbf{v} \in P_{\mathbf{e}}^-$. For a bound, a vertex of Ψ with the opposite value exists likewise. Let $\mathbf{w}$ be a zero-one minimizer of $g^\top x$ on $P_{\mathbf{b}}$. Since $\mathbf{h}^\top \mathbf{v} \leq \mathbf{h}^\top \mathbf{w}$, $\|\mathbf{v} - \mathbf{w}\|_1 \leq n_c$, and $\|g - \mathbf{h}\|_\infty \leq \kappa_2 \|c - a\|_\infty \leq \delta$,

$$\Delta_{\mathbf{e}}(g) \leq g^\top(\mathbf{v} - \mathbf{w}) = \mathbf{h}^\top(\mathbf{v} - \mathbf{w}) + (g - \mathbf{h})^\top(\mathbf{v} - \mathbf{w}) \leq n_c \delta,$$

a contradiction. For the Lipschitz bound, let $\mathbf{x}_1$ be a zero-one minimizer of $w^\top x$ on $P_{\mathbf{e}}^-$ and $\mathbf{x}_2$ one of $v^\top x$ on $P_{\mathbf{b}}$; then $\Delta_{\mathbf{e}}(v) - \Delta_{\mathbf{e}}(w) \leq (v - w)^\top(\mathbf{x}_1 - \mathbf{x}_2) \leq n_c \|v - w\|_\infty$, and symmetrically. If $P_{\mathbf{e}}^-$ is empty, no vertex of $P_{\mathbf{b}}$ violates the equality, so it holds on the convex hull of the vertices. $\square$

The closure test. After retention at level $j \geq 1$: form the hulls; fail unless every binary hull is a singleton, which fixes $\mathbf{b}$; apply [lemma D.10](#) to every continuous edge and every continuous bound, using rational linear programs; merge coordinates joined by certified edge equalities into blocks, fix blocks with certified bounds or bounds forced by $\mathbf{b}$, intersect the hull intervals of the members, contract directed cycles, and propagate bounds, replacing each lower bound by the largest lower bound of its predecessors and each upper bound by the smallest upper bound of its successors; fix blocks whose interval is a single point and repeat until nothing changes; remove tautological rows. This yields $x_C = Dy + \bar{x}_C$ with disjoint zero-one columns, a box $[\mathbf{L}, \mathbf{U}]$ in y with $\mathbf{L} < \mathbf{U}$, and an acyclic set of remaining order inequalities $y_u \leq y_{u'}$. If no block remains, return the point. Otherwise, with c_y the midpoint and r_y the largest half-width of the box, test

$$D^\top \nabla_{CC}^2 F_\gamma(Dc_y + \bar{x}_C, \mathbf{b})D - (\kappa_3 r_y + g_0)D^\top D \succ 0, \quad (47)$$

where κ_3 bounds the continuous third-derivative row sums on $[0, 1]^n$.

Lemma D.11 (Order closure). *The patch $P_y = \{y \in [\mathbf{L}, \mathbf{U}] : y_u \leq y_{u'} \text{ for the remaining inequalities}\}$ contains the block values of every global minimizer and is full-dimensional. If (47) holds, then $\nabla_y^2 F_\gamma(Dy + \bar{x}_C, \mathbf{b}) \succeq g_0 D^\top D \succeq g_0 I$ on P_y , and the unique minimizer on P_y is the unique global minimizer. Conversely, if $g_* \geq g_0$, every active proposal at the minimizer with a nonempty opposite face has exposure gap larger than τ , $A = 2 + n_c \bar{L}/g_0$, and $h_j \leq \min\{1/(4A), \tau/(4n_c \kappa_2 A), g_0/(4\kappa_3 A)\}$, then the test of level j succeeds.*

Proof. Containment. Binary singleton hulls fix $\mathbf{b}$ at every minimizer, the certified equalities hold by [lemma D.10](#), and merging, cycle contraction, bound propagation and singleton fixing are equivalent reformulations of $P_{\mathbf{b}} \cap Q_C$ with those equalities.

Full dimension. After propagation the bounds are monotone along the remaining inequalities: $\mathbf{L}_u \leq \mathbf{L}_{u'}$ and $\mathbf{U}_u \leq \mathbf{U}_{u'}$ for $u \rightarrow u'$. For an inequality $u \rightarrow u'$, the point that equals $\mathbf{U}$ on the blocks reachable from u' and $\mathbf{L}$ elsewhere is feasible, and u is not reachable from u' because the remaining graph is acyclic: an inequality from an unreachable block w to a reachable w' has $\mathbf{L}_w \leq \mathbf{L}_{w'} \leq \mathbf{U}_{w'}$, and none goes from a reachable to an unreachable block. At this point $y_u = \mathbf{L}_u$ and $y_{u'} = \mathbf{U}_{u'}$. If $\mathbf{L}_u = \mathbf{U}_{u'}$, then $\mathbf{L}_u \leq \mathbf{L}_{u'} \leq \mathbf{U}_{u'}$ and $\mathbf{L}_u \leq \mathbf{U}_u \leq \mathbf{U}_{u'}$ force singleton intervals, which were removed. So no inequality and, by the points $\mathbf{L}$ and $\mathbf{U}$, no bound is tight on all of P_y , and P_y has nonempty interior.

Certificate. For y, y' in the box, $\|D(y - y')\|_\infty = \|y - y'\|_\infty$ and $D^\top(\nabla^2 F_\gamma(x) - \nabla^2 F_\gamma(x'))D \succeq -\kappa_3 \|x - x'\|_\infty D^\top D$, so (47) gives the claimed bound, and $D^\top D \succeq I$ because every block has at least one coordinate.

Success. Witnesses lie within $\frac{h_j}{2}\sqrt{n_c\bar{L}/g_0}$ of the minimizer $x^* = (a, \mathbf{b}^*)$, so hulls lie within Ah_j of it, and binary hulls are the singletons $\mathbf{b}^*$ because $Ah_j \leq 1/4$ and binary cells are singletons at levels $j \geq 1$. By lemma D.10, $\Delta_e(g) \geq \Delta_e(\mathbf{h}) - n_c\kappa_2r > \tau - \tau/4 > n_c\delta$ for every active proposal with a finite gap, so all of them are certified. Proposals with an empty opposite face hold on all of $P_{\mathbf{b}}$. Hence every point of P_y satisfies every constraint that is tight at a , the patch lies in the smallest face of a in $P_{\mathbf{b}}$, and the columns of D times directions of P_y are directions of that face. Two-sided growth along them gives $D^\top \nabla^2 F_\gamma(x^*)D \succeq 2g_0 D^\top D$. With $\|Dc_y + \bar{x}_C - a\|_\infty \leq r_y \leq Ah_j$, the matrix in (47) is at least $(g_0 - 2\kappa_3 Ah_j)D^\top D \succeq \frac{g_0}{2} D^\top D$. $\square$

This weighted test measures curvature in the metric of the original coordinates. An unweighted version that subtracts $(n_c\kappa_3r_y + g_0)I$ is also sound, but the two must not be mixed by dropping the factor $D^\top D$ from one and the factor n_c from the other.

Lemma D.12 (Exposure gaps). *Let $D = \max\{1, d-1\}$ and $K_{\leq} = \max\{1, 2^{n_z}(m+2n_c)2^{m+2n_c}D^{n_c}\}$. For $\tau > 0$, the probability that $g_* > 0$ and some proposal that is active at the minimizer and has a nonempty opposite face has exposure gap $\Delta_e(\nabla_C F_\gamma(x^*)) \leq \tau$ is at most $K_{\leq}(\tau/\sigma + 2/M)$.*

Proof. Fix a binary assignment $\mathbf{b}$, a face Φ of $P_{\mathbf{b}}$ (determined by a subset of its at most $m+2n_c$ inequalities), and a proposal $\mathbf{e}$ that holds on Φ . Write the affine hull of Φ as $x = x_\Phi + B\zeta$ with B of full column rank. Its stationarity system $B^\top(\nabla_C F_0(x_\Phi + B\zeta, \mathbf{b}) + \gamma_C) = 0$ has at most D^{n_c} nonsingular zeros by lemma C.1. If Φ has dimension zero there is one empty stationary tuple, counted with weight one.

For an edge $\mathbf{e} = (i, l)$, condition on all coefficients except γ_i, γ_l , and write $\gamma_i = t$, $\gamma_l = s - t$. Every direction of Φ has equal i th and l th entries, so $B^\top(e_i - e_l) = 0$ and the system depends on s but not on t . Fix s and a zero, with point x_ρ in the relative interior of Φ . Whenever $(x_\rho, \mathbf{b})$ is a global minimizer with smallest face Φ , the slice gradient is $\mathbf{h}(t) = \mathbf{h}(0) + t(e_i - e_l)$, every vertex $\mathbf{w}$ of Φ minimizes $\mathbf{h}(t)^\top x$ on $P_{\mathbf{b}}$ (the minimizing face contains x_ρ and hence Φ), and every point of $P_{\mathbf{e}}^-$ has $x_i = 0$, $x_l = 1$, while $\mathbf{w}_i = \mathbf{w}_l$. So $\Delta_e(\mathbf{h}(t)) = \beta_\rho - t$ with β_ρ independent of t , and the event $\Delta_e \leq \tau$ confines t to an interval of length τ , since gaps are nonnegative. For each of the at most $2M-1$ values of s , the admissible values of t are spaced by $2\sigma/(M-1)$, so at most $\tau(M-1)/(2\sigma) + 1$ of them lie in the interval of each zero of that fiber. Summing over the fibers and dividing by M^2 , the probability for one face and edge is at most $D^{n_c}(2M-1)[\tau(M-1)/(2\sigma) + 1]/M^2 \leq D^{n_c}(\tau/\sigma + 2/M)$. A tuple-count on the fibers of s is used here because the conditional law of t given s is not uniformly spread. For a bound $x_i = 0$, vary γ_i alone: $B^\top e_i = 0$, and $\Delta_e = \beta_\rho + \gamma_i$; the upper bound is symmetric; the probability is at most $\tau/(2\sigma) + 1/M$.

If $g_* > 0$, the minimizer $(a, \mathbf{b})$ has a smallest face Φ in $P_{\mathbf{b}}$, its zero is nonsingular because two-sided growth gives $B^\top \nabla_{CC}^2 F_\gamma B \succeq 2g_* B^\top B \succ 0$, and the active proposals are those holding on Φ . The union over at most $2^{n_z} \cdot 2^{m+2n_c} \cdot (m+2n_c)$ choices gives the bound. Edges between a continuous and a binary coordinate are bounds of the slice and are covered; binary coefficients are fixed throughout. $\square$

Proof of theorem 6.17. Schedule. Let B be the factor of theorem 3.20 for the order polytope with binary constraints, C_{tail} the constant of lemma 3.17 for its description, $\rho = 1/(4B)$, $g_0 = \rho\sigma/(2n)$, $\tau = \rho\sigma/(2K_{\leq})$, $A = 2 + n_c\bar{L}/g_0$, $J \geq 1$ the least level satisfying the condition of lemma D.11, and M the least power of two with $M \geq \max\{2, 2^J, 4nC_{\text{tail}}/\rho, 4K_{\leq}/\rho\}$. All widths are at most one, so lemma 3.17(b) gives $\mathbb{P}\{g_* < g_0\} \leq \rho$, and lemma D.12 gives probability at most $K_{\leq}\tau/\sigma + 2K_{\leq}/M \leq \rho$ for small gaps. Outside these events closure succeeds by level J ; it is sound on every draw; otherwise the fallback runs on the same draw.

Work. Binary cells are $\{0, 1\}$ at level 0 and singletons afterwards, so cells have at most $2^{|\beta|}$ children and corners and a grid tuple lies in at most $2^{|\beta|}$ cells. [Proposition D.9](#) with $Mh_j \geq 1$ for $j \leq J$, and the argument of [theorem 5.2](#), give (26); the linear programs and matrix tests of a level cost $\text{poly}_d(I)$, and the fallback contributes $\text{poly}_d(I)$ in expectation.

Evaluation. If no block remains, evaluate the returned rational point and its sampled value exactly. Otherwise apply [lemma C.2](#) in the block coordinates y , with $Z = I$ and $\zeta = 1$. A rational center and inner radius come from the linear program $\max\{\vartheta : \alpha_s^\top y + \vartheta \|\alpha_s\|_1 \leq \alpha_{0,s} \text{ for all rows, } \vartheta \leq \frac{1}{2}\}$ over the rows of P_y : its optimal vertex (y_c, ϑ^*) is rational of polynomial length, and $\vartheta^* > 0$ by full dimensionality; the ball of radius ϑ^* about y_c lies in P_y since $|\alpha_s^\top u| \leq \|\alpha_s\|_1 \|u\|_2$. Take $R = \max\{1, \dim y\}$. The repair map clips to $[L, U]$ and then replaces every coordinate by the largest clipped value over its predecessors, itself included. The result is feasible, because propagated upper bounds are monotone along the inequalities. If $\bar{y} \in P_y$ and $\|z - \bar{y}\|_\infty = \epsilon$, clipping keeps every coordinate within ϵ of $\bar{y}$, the repaired value is at least the clipped value and hence at least $\bar{y}_u - \epsilon$, and it equals a clipped predecessor value at most $\bar{y}_w + \epsilon \leq \bar{y}_u + \epsilon$; so the repair is nonexpansive in $\|\cdot\|_\infty$, and we take $\nu = \|\cdot\|_\infty$ with a monomial bound on $\|D^\top \nabla_C F_\gamma\|_1 \leq \|\nabla_C F_\gamma\|_1$. The curvature hypothesis holds with $g = g_0$ by [lemma D.11](#). Since $\|D\|_2 \leq \sqrt{n_c}$, the objective tolerance $\min\{2^{-q}, (g_0/(2n_c))2^{-2q}\}$ gives distance 2^{-q} in the original coordinates. On fallback draws, identify the binary labels by refining to error below $\frac{1}{2}$, propagate their bounds through the order graph, clip the continuous approximations to the propagated intervals, and take predecessor maxima; this keeps the labels and every order constraint and does not increase the ∞ -norm error. Choose each continuous coordinate's error at most $2^{-q-1}/(nG')$, where $G' \geq \max\{1, \sup_{[0,1]^n} \|\nabla_C F_\gamma\|_1\}$. Then the physical Euclidean distance is at most 2^{-q-1} and the feasible objective gap at most 2^{-q-1} . Refining the optimal value to width 2^{-q-1} supplies the certified value interval, in $B \text{poly}_d(I + q)$ bit work. Without propagation, a slightly positive approximation of a continuous predecessor could move a binary zero. $\square$

E Proofs for conditional recourse

This appendix proves the results of [section 7](#). [Appendix E.1](#) lists the external tools. [Appendices E.2](#) to [E.4](#) prove the common machinery, [appendix E.5](#) proves the ambient-noise theorems, [appendix E.6](#) proves the core-only theorem, and [appendix E.7](#) verifies the examples.

E.1 Conventions and tools

Norms are Euclidean unless marked otherwise. For a symmetric matrix the operator norm is at most the largest absolute row sum. Hence, with K_2, K_3 as in (34), every principal submatrix and every off-diagonal block of $\nabla^2 F_0(x)$ has operator norm at most K_2 , and

$$\left\| \nabla^2 F_0(x) - \nabla^2 F_0(y) \right\| \leq K_3 \|x - y\|_\infty, \quad \left| D^3 F_0(x)[p, p', p''] \right| \leq K_3 \|p\| \|p'\| \|p''\| \quad (48)$$

for $x, y \in [0, 1]^n$ and vectors p, p', p'' . For the first bound, the (i, j) entry of the difference is at most $\sum_l \sup |\partial_{ijl} F_0| |x_l - y_l|$, so every row sum is at most $K_3 \|x - y\|_\infty$. For the second, the symmetric matrix $(\sum_l \partial_{ijl} F_0 p_l'')_{ij}$ has row sums at most $K_3 \|p''\|_\infty \leq K_3 \|p''\|$. The same row-sum argument gives $|\partial_i F_0(x) - \partial_i F_0(y)| \leq K_2 \|x - y\|_\infty$. Monomial coefficient sums give rational K_2, K_3, G of polynomial bit length for fixed d .

We use the following results. Items (T2)–(T5) are proved in [section 3](#) and its appendix; we restate the forms used.

- (T1) *Corrected corners and local counts* (lemma 3.7 and theorem 3.4). If f has upper coordinate curvature at most L on a box and C is a cell with side lengths η_i , then $\min_{\text{corners}} F_c \leq F_c(x) + \frac{1}{8} \sum_i L \eta_i^2$ for every $x \in C$ and every c . For the uniform grid with m_i parts of width h_i per coordinate and independent coefficients with interval concentrations π_i , the expected number of nodes v with $F_c(v) \leq F_c(v \pm h_i e_i) + \delta$ for all interior coordinates is at most $\prod_i [2 + (m_i - 1) \min\{1, \pi_i(L h_i + 2\delta/h_i)\}]$, also conditionally on any random element independent of c on which f may depend.
- (T2) *Growth tail* (theorem 3.16). If $Z \subset \mathbb{R}^p$ is compact with coordinate widths w_i , $f : Z \rightarrow \mathbb{R}$ is lower semicontinuous, and c is uniform on $[-\sigma, \sigma]^p$, then the point-growth modulus of $f + c^\top y$ satisfies $\Pr\{g_* < \varepsilon\} \leq \varepsilon \sum_i w_i / \sigma$.
- (T3) *Finite-law tails* (lemma 3.17). Let $\mathcal{D} \subset \mathbb{R}^{p+m}$ be compact and described by at most s polynomial sign conditions of degree at most d (bounded integer coordinates by disjunctions), let F have degree at most d , and perturb only the first p coordinates. For the reduced objective $y \mapsto \min\{F(y, y') : (y, y') \in \mathcal{D}\}$ on the projection of $\mathcal{D}$, with coefficients independent with law $U_{\sigma, M}$, $\Pr\{g_* < \varepsilon\} \leq S\varepsilon/\sigma + 2pC_{\text{tail}}/M$, where S is the sum of the projected widths and C_{tail} is base-computable with $\log C_{\text{tail}} \leq \text{poly}_d(I)$ when $\log(s+1) \leq \text{poly}(I)$. If $m = 0$ and $\mathcal{D} = [0, 1]^n$, the probability that $g_* > 0$ and the minimizer has a coordinate at a bound with $|\partial_i F_\gamma| \leq \tau$ is at most $N_{\text{act}}(\tau/\sigma + 1/M)$, $N_{\text{act}} = n3^n \max\{1, d-1\}^n$.
- (T4) *Exact fallback* (theorem 3.20). For a fixed-degree polynomial on an explicit compact mixed semialgebraic domain with base length I , there are a base-computable $B = 2^{\text{poly}_d(I)}$ and a fixed exponent such that, for all rational linear coefficients of bit length at most b , a deterministic algorithm returns, within $B \text{poly}_d(I+b)$ bit operations, an algebraic output (definition 2.5(d)) of the lexicographically least global minimizer and of the optimal value: one isolated real root and rational polynomial maps for every continuous coordinate and for the value, with integer coordinates listed. Its length is at most $B \text{poly}_d(I+b)$. Enclosing each coordinate to width $2^{-q - \lceil \log_2(n+1) \rceil}$ and the value to width 2^{-q} gives the Euclidean point and value evaluation contract in $B \text{poly}_d(I+b+q)$ work.
- (T5) *Rare fallback* (lemma 3.18(a)). If the fallback costs at most $B P(L_{\max})$ on every draw and runs with probability at most $1/B$, it contributes at most $P(L_{\max})$ to the expected work. The parameters are chosen in the order B , thresholds, stopping depth J , grid size M .
- (T6) *Fixed-block quantifier elimination* [37, Theorem 1.1]. A formula with ω quantifier blocks of positive sizes $n_1, \dots, n_\omega$, $\ell \geq 1$ free variables, $s \geq 1$ atoms, and degree at most $d_0 \geq 1$, with arbitrary real coefficients, is equivalent to a quantifier-free formula in disjunctive normal form whose numbers of disjuncts and of atoms per disjunct, and whose atom degrees, are at most $\max\{2, sd_0\}^{2^{c\omega\ell}} \prod_b n_b$ for an absolute constant c . If the coefficients are integers of bit length at most τ , the output coefficients have bit length at most $(\tau+1) \max\{2, sd_0\}^{2^{c\omega\ell}} \prod_b n_b$. Empty blocks are omitted; singleton domains are evaluated directly before invoking this statement.
- (T7) *Tubes around algebraic sets* [9, Theorem 1.1]. If $Z \subseteq \mathbb{R}^q$ is a real algebraic set defined by polynomials of degree at most D with $\dim Z \leq m$, and X is uniform in a Euclidean ball of radius T , then for every $\epsilon > 0$

$$\Pr\{\text{dist}(X, Z) \leq \epsilon\} \leq 4 \left(\frac{4qD\epsilon}{T} \right)^{q-m} \left(1 + \frac{(4D+1)\epsilon}{T} \right)^m.$$

- (T8) *Convex values and patch evaluation* (lemma C.2, based on weak convex optimization [24, Corollary 4.2.7]). For a convex polynomial f of fixed degree on a rational box P with nonempty interior relative to its free coordinates, and a rational $\eta \in (0, 1]$, one computes a rational $y \in P$ and rationals $a \leq \min_P f \leq f(y) \leq U \leq a + \eta$ in time polynomial in the length of the data and $\log(1/\eta)$; here the oracle of that lemma is exact rational evaluation of f and ∇f , and the repair map is coordinatewise clipping to P , which is nonexpansive. If $\nabla^2 f \succeq gI$ on P with rational $g > 0$, the unique minimizer y^* satisfies $\|y - y^*\| \leq \sqrt{2\eta/g}$.
- (T9) *Nonsingular zeros* (lemma C.1). A system of q polynomial equations of degree at most D in q complex variables has at most D^q zeros with nonsingular Jacobian.
- (T10) *Quadratic faces* (lemma C.3). For a rational quadratic on a mixed box with n_c continuous coordinates and N_Z integer labels, solving the stationary system of every face whose free Hessian block is positive definite yields a rational global minimizer, in $3^{n_c} N_Z \text{poly}(I + b)$ bit operations.

E.2 Conditional values and the core search

Proof of lemma 7.2. (a) The term $\gamma_C^\top v$ does not depend on z , so $V(v) - \gamma_C^\top v = \min_{z \in Y} [F_0(v, z) + \gamma_R^\top z]$.

(b) Fix $z \in Y$. Along the segment $t \mapsto (v + te_i, z)$, which stays in $[0, 1]^k \times \text{conv } Y$, the polynomial $\phi_z(t) = F_\gamma(v + te_i, z)$ has $\phi_z'' = \partial_{ii} F_0 \leq L$, so $\phi_z(t) - Lt^2/2$ is concave. The function $V(v + te_i) - Lt^2/2 = \min_{z \in Y} [\phi_z(t) - Lt^2/2]$ is a pointwise minimum of concave functions over an index set that does not depend on t , hence concave.

(c) Let $z \in Y'$ attain $V_{Y'}(w)$; it exists because Y' is compact. The segment from (w, z) to (v, z) lies in $[0, 1]^k \times \text{conv } Y$, so the mean value theorem and Hölder's inequality give $V_{Y'}(v) \leq F_\gamma(v, z) \leq F_\gamma(w, z) + G \|v - w\|_\infty = V_{Y'}(w) + G \|v - w\|_\infty$. Exchange v and w . $\square$

By lemma 7.2(a),(b), the function $\bar{V} = V - \gamma_C^\top v$ has upper coordinate curvature at most L on $[0, 1]^k$ and does not depend on γ_C . Applied to $f = \bar{V}$ and $c = \gamma_C$, the first part of (T1) gives (30): every point v of a level- j cell has a corner c with $V(c) \leq V(v) + E_j$.

Proof of proposition 7.5. Write $\ell(v), u(v)$ for the oracle's answers at a queried corner, with $\ell = u = V$ in exact mode. With η_j from (31), every answer at level j satisfies $\ell(v) \leq V(v) \leq u(v) \leq \ell(v) + \eta_j$, and $\varepsilon_j = E_j + \eta_j$. Level 0 generates the cell $[0, 1]^k$, queries its corners, and applies (32); level $j \geq 1$ generates the children of the cells retained at level $j - 1$. For $k = 0$ the only cell is the empty core point, $E_j = 0$, and the arguments below apply verbatim with the single corner.

(a) Let (a, z^*) be a global minimizer; then $V(a) = f^*$. Suppose a cell generated at level j contains a ; at level 0 this is $[0, 1]^k$. By (30) some corner c has $V(c) \leq f^* + E_j$, so $\min_{\text{corners}} \ell - E_j \leq V(c) - E_j \leq f^*$. Every upper value is the value of a feasible point, so $U_j \geq f^*$, and the cell is retained. A child of it contains a and is generated at level $j + 1$. Induction proves (a).

(b) With c as in (a), $U_j \leq u(c) \leq V(c) + \eta_j \leq f^* + \varepsilon_j$. The cell that generated the corner c_j satisfies $\min_{\text{corners}} \ell - E_j \leq \ell(c_j) \leq u(c_j) = U_j$, so it is retained.

(c) A retained cell has a corner v with $\ell(v) - E_j \leq U_j$, so $V(v) \leq \ell(v) + \eta_j \leq U_j + E_j + \eta_j \leq f^* + 2\varepsilon_j$.

Count. For $k = 0$ the count is one. Let $k \geq 1$, so that $\varepsilon_j = E_j$ in exact mode and $\varepsilon_j = 2E_j$ in certified mode. Fix $j \leq J$ and $h = h_j$, and let N_j be the number of points v of the deterministic grid $(h\mathbb{Z})^k \cap [0, 1]^k$ with $V(v) \leq f^* + 2\varepsilon_j$. Such a point satisfies $V(v) \leq V(v \pm he_i) + 2\varepsilon_j$ for every coordinate with $0 < v_i < 1$, because its grid neighbors are feasible. Apply the second part of (T1) to $f = \bar{V}$, $c = \gamma_C$, $m_i = 2^j$, $h_i = h$, and $\delta = 2\varepsilon_j$, conditionally on γ_R , which is independent of γ_C .

The relevant interval length is $Lh + 4\varepsilon_j/h$, which equals λh with $\lambda = L(1 + k/2)$ in exact mode and $\lambda = L(1 + k)$ in certified mode, and (2) gives $\pi_i(\lambda h) \leq \lambda h/(2\sigma) + 1/M$. Since $h^{-1} = 2^j \leq 2^J \leq M$,

$$\mathbb{E}[N_j \mid \gamma_R] \leq \prod_{i=1}^k \left[2 + (h^{-1} - 1) \left(\frac{\lambda h}{2\sigma} + \frac{1}{M} \right) \right] \leq \left[3 + \frac{\lambda}{2\sigma} \right]^k,$$

which is Q_{ex} or Q_{ap} .

Oracle calls. By (c), each retained cell at level j has a corner counted by N_j , and a grid point is a corner of at most 2^k cells of level j ; so at most $2^k N_j$ cells are retained. Level $j \geq 1$ generates at most $2^k \cdot 2^k N_{j-1}$ cells and queries at most $8^k N_{j-1}$ corners; level 0 queries 2^k corners. Taking expectations gives at most $2^k + 8^k JQ$ calls through level J . $\square$

The count compares the adaptive search with a deterministic grid fixed before sampling: retained cells are charged to grid points, and the event that a grid point is near-optimal is stated through the true function V . The oracle's answers may depend on all noise coefficients; they enter only through (a)–(c), which hold for every admissible answer.

E.3 Containment and closure

Proof of lemma 7.6. Cover. If $z \in Y \setminus P$, some coordinate violates its patch interval. If $z_i < \max\{\ell_i, z_{c,i} - r\}$, then $z_i < z_{c,i} - r$ because $z_i \geq \ell_i$; hence $\ell_i \leq z_i < z_{c,i} - r$, the lower slab Y_i^- is formed, and it contains z . The upper case is symmetric.

Containment. Let $v \in \mathcal{H}$ and let z minimize $F_\gamma(v, \cdot)$ on Y . If $z \notin P$, then z lies in some slab S . Since $c, v \in \mathcal{H}$, $\|v - c\|_\infty \leq w_\infty(\mathcal{H})$. By lemma 7.2(c) applied to S and to Y ,

$$\begin{aligned} V(v) = F_\gamma(v, z) &\geq V_S(v) \geq V_S(c) - Gw_\infty(\mathcal{H}) \geq \lambda_{\text{out}} - Gw_\infty(\mathcal{H}) \\ &> U + Gw_\infty(\mathcal{H}) \\ &\geq V(c) + Gw_\infty(\mathcal{H}) \geq V(v), \end{aligned}$$

a contradiction. If (a, z^*) is a global minimizer, then z^* minimizes $F_\gamma(a, \cdot)$ on Y ; when $a \in \mathcal{H}$ it follows that $z^* \in P$. $\square$

Proof of lemma 7.7. (a) Let x^* be a global minimizer; it lies in Q . If $\partial_i F_\gamma > 0$ on Q , the original lower bound ℓ_i lies in the i th interval Q_i , and $x_i^* > \ell_i$, then the segment from x^* to $x^* - (x_i^* - \ell_i)e_i$ lies in $Q \subseteq X$ and F_γ strictly decreases along it, which is impossible. The midpoint test is sufficient because $|\partial_i F_\gamma(x) - \partial_i F_\gamma(m)| \leq K_2 r_Q$ on Q (appendix E.1). For a quadratic, $\partial_i F_\gamma$ is affine and its extreme values on Q are attained at vertices chosen coordinatewise. Each fixing is justified with respect to the same box Q , so all of them hold simultaneously for every global minimizer.

(b) Let $\mathcal{U}$ be the set of unfixed coordinates. For $x \in Q'$, (48) gives $\nabla_{\mathcal{U}}^2 F_0(x) \succeq \nabla_{\mathcal{U}}^2 F_0(m') - K_3 r_{Q'} I \succeq g_0 I$. Thus F_γ is g_0 -strongly convex on the box Q' and has a unique minimizer there. By (a), $Q' \subseteq X$ contains every global minimizer, so $\min_{Q'} F_\gamma = f^*$, and the unique minimizer on Q' is the unique global minimizer.

(c) As in (b), $\min_{Q'} F_\gamma = f^*$, and the restriction is a convex quadratic, so every minimizer on Q' attains f^* . $\square$

E.4 Certified convex recourse

Proof of lemma 7.12. Fix a query (v, Y', γ, η) and let $f = F_\gamma(v, \cdot)$ on Y' , after substituting the coordinates whose interval is a single point; if none remains, return the value. Otherwise f is a

convex polynomial on the box $B = Y'$ with explicit rational coefficients of polynomial length, and monomial bounds give a rational $K_f \geq \max\{1, \max_B \|\nabla^2 f\| \max(1, \text{diam } B)^2\}$ of polynomial length. Let $\delta = \min\{\eta/2, \eta^2/(8K_f), 1\}$.

A feasible near-minimizer. By (T8), with accuracy δ , one computes a rational $y \in B$ with $f(y) \leq \min_B f + \delta$. That lemma uses weak convex optimization over a capped epigraph and repairs the near-feasibility of its output by clipping; it uses convexity of f only.

The tangent certificate. Put $u = f(y)$ and $\ell = f(y) + \min_{z \in B} \nabla f(y)^\top (z - y)$, an explicit choice of one endpoint per coordinate. By convexity $f(z) \geq f(y) + \nabla f(y)^\top (z - y) \geq \ell$ on B , so $\ell \leq \min_B f$. Let $\bar{z}$ attain the linear minimum and $\Delta = u - \ell = -\nabla f(y)^\top (\bar{z} - y) \geq 0$. For $t \in (0, 1]$ the point $y + t(\bar{z} - y)$ lies in B , and Taylor's theorem gives $\min_B f \leq f(y) - t\Delta + K_f t^2/2$. Since $f(y) \leq \min_B f + \delta$, $\Delta \leq \delta/t + K_f t/2$. Choose $t = \min\{1, \eta/(2K_f)\}$. If $\eta \leq 2K_f$, the two terms are each at most $\eta/4$; otherwise $t = 1$ and $\Delta \leq \eta/2 + \eta/4$. In both cases $u - \ell \leq \eta$. All quantities have bit length polynomial in the query length and the length of η , so the bit work is polynomial. A verifier recomputes $f(y)$, $\nabla f(y)$, and the endpoint minimum, and needs neither the optimization trace nor a modulus of strong convexity. $\square$

E.5 Proofs of the ambient-noise theorems

We prove theorems 7.9 and 7.13 together. In the quadratic case $F_0 = \frac{1}{2}x^\top Ax + a^\top x + c_0$, the oracle runs in exact mode, $d = 2$, and K_3 is not used.

Base choices. Compute K_2, K_3 from (34) and $G = \max\{1, \sum_{i \in C} (\sup_{[0,1]^n} |\partial_i F_0| + \sigma)\}$, which satisfies the hypothesis of lemma 7.2(c) for every noise vector in $[-\sigma, \sigma]^n$. Let $B = \max\{2, 3^n\}$ in the quadratic case ((T10) with $N_Z = 1$) and the factor of (T4) in the polynomial case, enlarged so that it also bounds q -bit evaluation. Let N_{act} and C_{tail} be as in (T3) for $[0, 1]^n$, and choose

$$\begin{aligned} \rho &= \frac{1}{4B}, & g_0 &= \frac{\rho\sigma}{2n}, & \tau &= \frac{\rho\sigma}{2N_{\text{act}}}, \\ r &= \min\left\{\frac{1}{8}, \frac{\tau}{16K_2}, \frac{g_0}{4K_3}\right\}, & A_0 &= 2 + \frac{kL + 8}{g_0}, \\ J &= \min\left\{j \geq 0 : 2^{-j} \leq \min\left\{\frac{r}{4A_0}, \frac{g_0 r^2}{16GA_0}\right\}\right\}, \\ M &\geq \max\left\{2, 2^J, \frac{4nC_{\text{tail}}}{\rho}, \frac{2N_{\text{act}}}{\rho}\right\} \quad (\text{least power of two}). \end{aligned} \tag{49}$$

In the quadratic case omit the term $g_0/(4K_3)$. All these numbers are computed from the base instance, have bit length polynomial in I , and $J, \log_2 M \leq \text{poly}_d(I)$.

The algorithm. For $j = 0, 1, \dots, J$:

- (1) Run level j of the core search: exact mode in the quadratic case, certified mode with accuracy η_j of (31) otherwise. If $k = 0$ this is one oracle call with accuracy 4^{-j} ; U_j is its upper value, and $\mathcal{H}_j$ is the empty core point.
- (2) Form the patch P of radius r about z_j and its slabs. For each slab obtain its exact value (quadratic case) or a certified lower bound with accuracy $\eta_{\text{out}} = g_0 r^2/16$; let λ_{out} be the least of these. If (33) fails with $U = U_j$ and $\mathcal{H} = \mathcal{H}_j$, go to the next level.

- (3) Apply the sign tests of lemma 7.7(a) on $Q = \mathcal{H}_j \times P$ and obtain Q' . If no coordinate is unfixed, return the fixed point and its value. In the quadratic case test whether the Hessian block of the unfixed coordinates is positive semidefinite; if so, solve the convex quadratic program on Q' exactly [31] and return its solution. In the polynomial case test lemma 7.7(b) with g_0 ; if it passes, return (S, z_S, Q', g_0) . Otherwise go to the next level.

If no level returns, run the fallback on the same draw: (T10) in the quadratic case and (T4) otherwise.

Correctness on every draw. By proposition 7.5(a),(b), $\mathcal{H}_j$ contains the core of every global minimizer and the corner c_j , and $U_j \geq V(c_j)$. Each certified lower bound is at most the true slab value. Hence lemma 7.6 shows that every global minimizer lies in $\mathcal{H}_j \times P$ when step (2) passes, and lemma 7.7 shows that the outputs of step (3) are correct. The fallback is exact. No step assumes growth, uniqueness, or any probabilistic event.

Lemma E.1 (Stopping). *Suppose F_γ has a unique global minimizer $x^* = (a, z^*)$ with point growth at least g_0 , and every coordinate of x^* at a bound has $|\partial_i F_\gamma(x^*)| > \tau$. Then the algorithm returns at a level $j \leq J$.*

Proof. Assume no earlier level returned; we show that level J returns. Put $h = 2^{-J}$ and $E = E_J$.

Hull and incumbent. Every retained cell has a corner v with $V(v) \leq f^* + 4E$ (proposition 7.5(c); $2E$ in exact mode), and $V(v) - f^* \geq g_0 \|v - a\|^2$. Hence $\|v - a\| \leq h\sqrt{kL/(2g_0)}$, and every point of the cell is within h of v in each coordinate. Thus $\mathcal{H}_J \subseteq a + [-A_0h, A_0h]^k$, because $1 + \sqrt{y} \leq 2 + y$. By proposition 7.5(b) the incumbent value satisfies $\varepsilon_{\text{inc}} = U_J - f^* \leq \varepsilon_J$, which is at most $2E$ for $k \geq 1$ and equals $\eta_J = 4^{-J} = h^2$ for $k = 0$; the empty-core case thus uses the positive allowance of (31), not an exact incumbent. Growth gives $\|(c_J, z_J) - x^*\|^2 \leq \varepsilon_{\text{inc}}/g_0$. Since $A_0 \geq \max\{2, kL/g_0, 8/g_0\}$, this yields $\|(c_J, z_J) - x^*\| \leq A_0h \leq r/4$ and $\varepsilon_{\text{inc}} \leq g_0r^2/16$; for $k \geq 1$ the last inequality is $kLh^2/4 \leq kLr^2/(64A_0^2) \leq g_0r^2/16$, and for $k = 0$ it is $h^2 \leq r^2/(16A_0^2) \leq g_0r^2/256$.

Containment. A point z of a slab differs from z_J by at least r in one coordinate, hence from z^* by at least $3r/4$. Growth gives $F_\gamma(c_J, z) - f^* \geq 9g_0r^2/16$, so $\lambda_{\text{out}} \geq f^* + 9g_0r^2/16 - \eta_{\text{out}} = f^* + g_0r^2/2$ (or $f^* + 9g_0r^2/16$ in exact mode). Therefore $\lambda_{\text{out}} - U_J \geq 7g_0r^2/16$, while $2Gw_\infty(\mathcal{H}_J) \leq 4GA_0h \leq g_0r^2/4$. Step (2) passes.

Sign tests. Every point of $Q = \mathcal{H}_J \times P$ is within $5r/4$ of x^* in the maximum norm, and $r_Q \leq r$. Let x_i^* be at a bound, say the lower one, with $\partial_i F_\gamma(x^*) > \tau$; that bound lies in Q_i because $x^* \in Q$. The midpoint test value is at least $\tau - \frac{5}{4}K_2r - K_2r \geq \tau - \frac{9}{64}\tau > 0$, so the coordinate is fixed; in the quadratic case the exact test passes because $\partial_i F_\gamma \geq \tau - \frac{5}{4}K_2r > 0$ on Q . A coordinate strictly inside its bounds at x^* has zero derivative at the point $x^* \in Q$ and cannot pass a strict sign test. Hence Q' fixes exactly the coordinates that are at a bound at x^* , and the unfixed set $\mathcal{U}$ consists of the coordinates strictly inside.

Hessian test. For p supported on $\mathcal{U}$ and small t , $F_\gamma(x^* + tp) - f^* \geq g_0t^2 \|p\|^2$ and $\nabla_{\mathcal{U}} F_\gamma(x^*) = 0$, so $\nabla_{\mathcal{U}}^2 F_0(x^*) \succeq 2g_0I$. In the quadratic case the semidefinite test passes. Otherwise $x^* \in Q'$ gives $\|m' - x^*\|_\infty \leq r_{Q'} \leq r$, and (48) gives $\nabla_{\mathcal{U}}^2 F_0(m') - (K_3r_{Q'} + g_0)I \succeq (g_0 - 2K_3r)I \succeq \frac{g_0}{2}I$ because $r \leq g_0/(4K_3)$. $\square$

Proof of theorems 7.9 and 7.13. Correctness was shown above. Let $\mathcal{B}$ be the event that the hypoth-

esis of lemma E.1 fails. By (T2) and (T3) applied to F_0 on $[0, 1]^n$ with ambient noise,

$$\begin{aligned} \Pr\{g_* < g_0\} &\leq \frac{ng_0}{\sigma} + \frac{2nC_{\text{tail}}}{M} \leq \frac{\rho}{2} + \frac{\rho}{2}, \\ \Pr\{g_* \geq g_0 \text{ and an active gradient has magnitude at most } \tau\} \\ &\leq N_{\text{act}}\left(\frac{\tau}{\sigma} + \frac{1}{M}\right) \leq \rho, \end{aligned}$$

so $\Pr(\mathcal{B}) \leq 2\rho = 1/(2B)$, and the fallback runs only on $\mathcal{B}$.

Work. Since $M \geq 2^J$, proposition 7.5 bounds the expected number of core oracle calls by $2^k + 8^k JQ$. Every query has length $\text{poly}_d(I)$: core corners are dyadic with at most J bits per coordinate, sampled coefficients have $O(I + \log M)$ bits, the accuracies E_j and η_{out} have polynomial length, and the oracle's outputs, hence the patch endpoints $z_{j,i} \pm r$, have polynomial length. Each oracle call therefore costs $\text{poly}_d(I)$, with an exponent independent of k . Each level adds at most $2n_R$ slab calls and $\text{poly}_d(I)$ work for the sign and Hessian tests and the final convex solve. The fallback costs $B \text{poly}_d(I + \log M)$ and runs with probability at most $1/(2B)$ ((T5)). Summing, the expected work is $[2^k + 8^k JQ + (J + 1)n_R] \text{poly}_d(I) \leq 8^k Q \text{poly}_d(I)$, with $Q = Q_{\text{ex}}$ or Q_{ap} . The global proof record consists of the oracle answers with their checkable certificates (exact values, or tangent bounds (35)), the retained lists, the slab bounds, and the test data; its expected size obeys the same bound.

Outputs. In the quadratic case the returned point is rational of polynomial length on every draw, by Kozlov et al. [31] on the ordinary branch and (T10) on the fallback branch. In the polynomial case the descriptor (S, z_S, Q', g_0) consists of dyadic hull endpoints, patch endpoints of polynomial length, and the base number g_0 ; (T8), run with $\eta = \min\{2^{-q}, (g_0/2)2^{-2q}\}$, evaluates it in $\text{poly}_d(I + q)$. The fallback branch returns the algebraic output of (T4), of length $B \text{poly}_d(I + \log M)$. For its q -bit evaluation, enclose every coordinate with width $2^{-q - \lceil \log_2 n \rceil}$ and the value with width 2^{-q} ; the midpoints of the coordinate enclosures, clipped to $[0, 1]^n$, are feasible and within Euclidean distance 2^{-q} of the represented minimizer. This costs $B \text{poly}_d(I + \log M + q)$, and the branch occurs with probability at most $1/(2B)$, which gives the expected evaluation bound. $\square$

Proof of corollary 7.10. Given a query $(v, \prod_{i \in R} [\ell'_i, u'_i], \gamma)$, substitute v and every residual coordinate whose interval is a point. For each remaining coordinate write $z_i = \ell'_i + (u'_i - \ell'_i)y_i$ with $y_i \in [0, 1]$. The result is $y^\top \tilde{A}y + \tilde{c}^\top y + \text{const}$ with $\tilde{A}_{ij} = \frac{1}{2}A_{ij}(u'_i - \ell'_i)(u'_j - \ell'_j)$, so its interaction graph is a subgraph of the residual graph, a forest. The algorithm of Del Pia and Khajavirad [19, Theorem 1 and Lemma 19] returns a rational optimal y of polynomial bit length using $O(n^2)$ arithmetic operations on numbers of polynomial length, hence in polynomial bit time with an absolute exponent. Mapping back and evaluating gives the oracle's answer. Diagonal entries of A_{RR} of either sign are allowed. $\square$

Proof of proposition 7.14. The Hessian blocks are $\partial_{tt}F = \lambda$, $\nabla_z \partial_t F = T \nabla P(z)$, and $\nabla_{zz}^2 F = T t \nabla^2 P(z)$ with $\nabla^2 P = 12 \text{diag}(z_i^2) + 12(\sum_i z_i)^2 \mathbf{1}\mathbf{1}^\top \succeq 0$. So the residual problem is convex for every $t \in [0, 1]$, also after linear perturbations and box restrictions.

Suppose $F + \frac{1}{2}(t, z)^\top K(t, z)$ is convex on the box. Its Hessian $\nabla^2 F + K$ is positive semidefinite in the interior and, by continuity, on the closed box, in particular at $t = 0$, where $\nabla^2 F(0, z) = \begin{pmatrix} \lambda & T \nabla P(z)^\top \\ T \nabla P(z) & 0 \end{pmatrix}$. Let $u \in \ker K_{zz}$ and $w = (0, u)$. Then $w^\top K w = 0$, so $K w = 0$, and $w^\top (\nabla^2 F(0, z) + K) w = 0$, so $(\nabla^2 F(0, z) + K) w = 0$ and $\nabla^2 F(0, z) w = 0$. Its first component gives $T \nabla P(z)^\top u = 0$ for all $z \in [0, 1]^m$. Since $\nabla P(e_i) = 4(e_i + \mathbf{1})$ and the matrix $I + \mathbf{1}\mathbf{1}^\top$ is invertible, $u = 0$. Thus the positive semidefinite matrix K_{zz} is nonsingular, and $\text{rank } K \geq m$.

For the remark after the proposition: on points with $t > 0$ and all $z_i > 0$, $\nabla^2 P \succ 0$, and Euler's identities for the quartic form P give $\nabla^2 P(z)z = 3 \nabla P(z)$ and $\nabla P(z)^\top z = 4P(z)$, hence

$\nabla P^\top (\nabla^2 P)^{-1} \nabla P = \frac{4}{3}P$. Adding $Ct \log t$ makes the Schur complement of the residual block equal to $\lambda + C/t - \frac{4T}{3t}P(z) \geq \lambda$ when $C \geq \frac{4}{3}T \max P$. Continuity extends convexity to the closed box, where $t \log t$ is extended by 0 at $t = 0$. $\square$

E.6 Core-only noise

Throughout, the hypotheses of [theorem 7.15](#) hold, $Y = [0, 1]^{n_R}$, $H = K_2/\mu$, and $H_V = K_2(1 + H)$. The residual selector is $s(v) = \arg \min_{z \in Y} F_0(v, z)$, and $V_0(v) = F_0(v, s(v))$. Core-only noise changes neither s nor the residual active sets, and $V_\gamma = V_0 + \gamma^\top v$.

Lemma E.2 (Selector, value, and growth lift).

- (a) The selector is unique and $\|s(v) - s(w)\| \leq H \|v - w\|$.
- (b) V_0 is differentiable on $[0, 1]^k$ (one-sided at the boundary), with $\nabla V_0(v) = \nabla_v F_0(v, s(v))$, and ∇V_0 is H_V -Lipschitz.
- (c) If $V_\gamma(v) - V_\gamma(a) \geq g_V \|v - a\|^2$ on $[0, 1]^k$ with $g_V > 0$, then $x^* = (a, s(a))$ is the unique global minimizer of F_γ and $F_\gamma(x) - F_\gamma(x^*) \geq g_F \|x - x^*\|^2$ on X , where $g_F = \min\{g_V/(1 + 2H^2), \mu/4\}$.

Proof. (a) Uniqueness follows from strong convexity. Write ∇_z for the residual gradient. The optimality conditions $\nabla_z F_0(v, s(v))^\top (y - s(v)) \geq 0$ and $\nabla_z F_0(w, s(w))^\top (y - s(w)) \geq 0$ for $y \in Y$, with $y = s(w)$ and $y = s(v)$ respectively, add to $[\nabla_z F_0(w, s(w)) - \nabla_z F_0(v, s(v))]^\top (s(w) - s(v)) \leq 0$. Split the bracket at $(w, s(v))$. Strong monotonicity bounds the first part below by $\mu \|s(w) - s(v)\|^2$, and the cross block of the Hessian bounds the second part by $K_2 \|w - v\| \|s(w) - s(v)\|$.

(b) Taylor's theorem in v at fixed residual points gives

$$\begin{aligned} V_0(w) &\leq F_0(w, s(v)) \\ &\leq V_0(v) + \nabla_v F_0(v, s(v))^\top (w - v) + \frac{K_2}{2} \|w - v\|^2, \\ V_0(w) &= F_0(w, s(w)) \\ &\geq V_0(v) + \nabla_v F_0(v, s(w))^\top (w - v) - \frac{K_2}{2} \|w - v\|^2. \end{aligned}$$

By (a), $\|\nabla_v F_0(v, s(w)) - \nabla_v F_0(v, s(v))\| \leq K_2 H \|w - v\|$. Hence

$$|V_0(w) - V_0(v) - \nabla_v F_0(v, s(v))^\top (w - v)| \leq K_2 \left(\frac{1}{2} + H\right) \|w - v\|^2,$$

which proves differentiability. The Lipschitz bound follows by splitting at $(w, s(v))$ and using (a).

(c) For $x = (v, z) \in X$, strong convexity in z and the optimality of $s(v)$ on the box give $F_\gamma(v, z) - V_\gamma(v) \geq \frac{\mu}{2} \|z - s(v)\|^2$, so $F_\gamma(x) - f^* \geq g_V \|v - a\|^2 + \frac{\mu}{2} \|z - s(v)\|^2$. By (a), $\|v - a\|^2 + \|z - s(a)\|^2 \leq (1 + 2H^2) \|v - a\|^2 + 2 \|z - s(v)\|^2$, and the choice of g_F completes the proof. $\square$

Lemma E.3 (Core-only tails). *Let g_V be the point growth modulus of V_γ on $[0, 1]^k$ (zero if the optimal core is not unique), and let the core coefficients be independent with law $U_{\sigma, M}$. There is a base-computable C_{tail} with $\log C_{\text{tail}} \leq \text{poly}_d(I)$ such that*

- (a) $\Pr\{g_V < t\} \leq kt/\sigma + 2kC_{\text{tail}}/M$ for every $t > 0$;
- (b) $\Pr\{g_V > 0 \text{ and some core coordinate at a bound at } x^* \text{ has } |\partial_i F_\gamma(x^*)| \leq \tau\} \leq N_C(\tau/\sigma + 1/M)$, where $N_C = \max\{1, k3^n \max(1, d-1)^n\}$.

Proof. (a) The function V_0 is the reduced objective of F_0 on $\mathcal{D} = [0, 1]^{k+n_R}$ with the core as perturbed coordinates, and g_V is its point-growth modulus under the core noise. (T3) with $p = k$, $m = n_R$, $s = 2n$, and $S = k$ gives (a).

(b) A face Φ of $[0, 1]^n$ fixes some coordinates at bounds and leaves a set $\mathcal{F}_\Phi$ free. Let $i \in C$ be fixed on Φ and condition on all core coefficients except γ_i . The stationary system $\partial_j F_0(x) + \gamma_j[j \in C] = 0$, $j \in \mathcal{F}_\Phi$, on Φ does not involve γ_i , and its equations have degree at most $\max(1, d-1)$. By (T9) it has at most $\max(1, d-1)^n$ solutions with nonsingular Jacobian; the empty system has the single vertex. At such a solution x , $\partial_i F_\gamma(x) = \partial_i F_0(x) + \gamma_i$ lies in $[-\tau, \tau]$ with conditional probability at most $\tau/\sigma + 1/M$ by (2). A union bound over at most 3^n faces, k core coordinates, and the solutions gives the bound for the event that some such solution has a small core gradient. If $g_V > 0$, lemma E.2(c) gives full growth $g_F > 0$ at x^* ; on the face Φ^* whose relative interior contains x^* , two-sided directions give $\nabla_{\mathcal{F}\mathcal{F}}^2 F_0(x^*) \succeq 2g_F I$, so x^* is a nonsingular solution of the system of Φ^* , and the event of (b) is contained in the union. $\square$

Proof of proposition 7.16. Identify K with a q -dimensional box by deleting its fixed coordinates.

(a) The selector s is continuous by lemma E.2(a) and semialgebraic: by residual convexity, $y = s(v)$ is equivalent to the box Karush–Kuhn–Tucker formula

$$\mathcal{G}(v, y) : v \in K \wedge \bigwedge_{i \in R} [(y_i = \ell_i \wedge \partial_{z_i} F_0 \geq 0) \vee (\ell_i \leq y_i \leq u_i \wedge \partial_{z_i} F_0 = 0) \vee (y_i = u_i \wedge \partial_{z_i} F_0 \leq 0)],$$

with derivatives evaluated at (v, y) . Hence each A_i^ℓ, A_i^u is a closed semialgebraic subset of K , and $\text{bd}_K A = A \setminus \text{int}_K A$ is closed and semialgebraic. It has empty interior in $\mathbb{R}^q$: an open set inside it would lie in $\text{int}_K A$. A semialgebraic subset of $\mathbb{R}^q$ with empty interior has dimension at most $q-1$ [10], and so does ∂K ; hence $\dim S_K \leq q-1$, and S_K is compact. The map $\nabla_K V_0$ is continuous by lemma E.2(b) and semialgebraic, and semialgebraic maps do not increase dimension, so $\mathcal{E}_K$ is compact with $\dim \mathcal{E}_K \leq q-1$.

Let $a \in \text{relint } K$ and $0 < \eta' < \text{dist}(a, S_K)$, and let $\mathcal{B}$ be the relative open ball of radius η' about a . A segment from a to a point of $\mathcal{B}$ outside K would cross $\partial K \subseteq S_K$, so $\mathcal{B} \subseteq \text{relint } K$. If $\mathcal{B}$ met both A_i^ℓ and its complement, connectedness of $\mathcal{B}$ would give a point of $\mathcal{B}$ in $\text{bd}_K A_i^\ell \subseteq S_K$. So the status of each residual coordinate is constant on $\mathcal{B}$.

(b) For $v \in K$, $v \in \text{bd}_K A_i^\ell$ if and only if

$$\exists y \forall \varepsilon \exists (w, z) : \mathcal{G}(v, y) \wedge y_i = \ell_i \wedge [\varepsilon \leq 0 \vee \{\mathcal{G}(w, z) \wedge \|w - v\|^2 < \varepsilon^2 \wedge z_i > \ell_i\}],$$

because A_i^ℓ is closed, so its relative boundary consists of its points every relative neighborhood of which meets $K \setminus A_i^\ell$. The image $-\nabla_K V_0(\text{bd}_K A_i^\ell)$ is the set of $\beta \in \mathbb{R}^q$ with

$$\exists (v, y) \forall \varepsilon \exists (w, z) : [\text{the matrix above}] \wedge \beta + \nabla_v F_0(v, y)_K = 0.$$

Upper boundaries use $y_i = u_i$ and $z_i < u_i$, and the image of ∂K is $\exists (v, y) : \mathcal{G}(v, y) \wedge v \in \partial K \wedge \beta + \nabla_v F_0(v, y)_K = 0$. Each of these at most $2n_R + 1$ formulas has at most three blocks of at most $N + 1$ variables, $N = q + n_R \leq n$, q free variables, at most $s_0 = 50(N + 1)$ atoms, and degree at most $d_0 = \max\{2, d\}$. By (T6) each is equivalent to a quantifier-free formula with at most $Q = (s_0 d_0)^{a(N+1)^3}$ disjuncts, Q atoms per disjunct, and atom degrees at most Q , for an absolute constant a ; this bound does not depend on coefficient sizes. Let P_j be the product of the distinct nonconstant atom polynomials of the j th formula ($P_j = 1$ if there are none). If an image point β_0 had $P_j(\beta_0) \neq 0$, every atom would have constant sign near β_0 , the formula would hold on a neighborhood, and the image would have nonempty interior, contradicting (a). So each image lies in $Z(P_j)$, and $\deg P_j \leq Q^3$. The product of the P_j is a nonzero polynomial of degree at

most $D_* = (2n_R + 1)Q^3$ whose zero set contains $\mathcal{E}_K$, and $\log D_* \leq \text{poly}_d(I)$. The algorithm never computes these polynomials.

(c) Let P_K be the polynomial of (b). For a ball of radius T and $\epsilon \leq T$, (T7) with $m = q - 1$ gives the probability bound $16qD_*(4D_* + 2)^{q-1}\epsilon/T$ for $Z(P_K)$. The cube $[-R_0, R_0]^q$ lies in the ball of radius $T = \sqrt{q}R_0$, whose volume is at most $q^{q/2}(2R_0)^q$. Hence for $\epsilon \leq R_0$ the normalized volume of $\{x \in [-R_0, R_0]^q : \text{dist}(x, Z(P_K)) \leq \epsilon\}$ is at most $16q^{(q+1)/2}D_*(4D_* + 2)^{q-1}\epsilon/R_0 \leq C(q, D_*)\epsilon/R_0$; since $C \geq 1$, the bound $\min\{1, C\epsilon/R_0\}$ holds for all $\epsilon \geq 0$. Now let $h = \sigma/(M - 1)$ and $R_0 = \sigma + h$. The cubes $g + [-h, h]^q$ centered at the M^q grid points have disjoint interiors and tile $[-R_0, R_0]^q$, and each has volume $(2R_0)^q/M^q$. If $\text{dist}(g, \mathcal{E}_K) \leq \delta$, every point of its cube is within $\delta + \sqrt{q}h$ of $Z(P_K)$. Therefore

$$\Pr\{\text{dist}(\gamma_K, \mathcal{E}_K) \leq \delta\} \leq \min\left\{1, C(q, D_*) \frac{\delta + \sqrt{q}h}{R_0}\right\}, \quad \frac{\delta + \sqrt{q}h}{R_0} = \frac{M-1}{M} \frac{\delta}{\sigma} + \frac{\sqrt{q}}{M}.$$

The bound includes the atoms that lie on $\mathcal{E}_K$. □

Proof of proposition 7.17. Let $\mathcal{B}$ be the relative open ball of radius η about a , let $\mathcal{A}$ be the residual coordinates at a bound on $\mathcal{B}$, with bound values $\bar{a}$, and let $\mathcal{F}$ be the residual coordinates strictly inside on $\mathcal{B}$. Write $\varsigma_i = 1$ for a lower and $\varsigma_i = -1$ for an upper bound, and v for the free core coordinates of K .

The branch. For $v \in \mathcal{B}$, $s(v) = (s_{\mathcal{F}}(v), \bar{a})$ and $\nabla_{\mathcal{F}} F_0(v, s_{\mathcal{F}}(v), \bar{a}) = 0$. Since $\nabla_{\mathcal{F}\mathcal{F}}^2 F_0 \succeq \mu I$, the implicit function theorem gives near each point of $\mathcal{B}$ a smooth solution of this equation, which coincides with $s_{\mathcal{F}}$ because the strongly monotone map $\nabla_{\mathcal{F}} F_0(v, \cdot, \bar{a})$ has at most one zero in the box. Hence $s_{\mathcal{F}}$, V_0 , and the multipliers $\lambda_i(v) = \varsigma_i \partial_{z_i} F_0(v, s(v))$, $i \in \mathcal{A}$, are smooth on $\mathcal{B}$, and $\lambda_i \geq 0$ there by the optimality conditions of the residual problem.

Derivative bounds. Let $W(v) = (v, s_{\mathcal{F}}(v), \bar{a})$. Differentiating $\nabla_{\mathcal{F}} F_0(W(v)) = 0$ once and twice gives

$$Ds_{\mathcal{F}} = -(\nabla_{\mathcal{F}\mathcal{F}}^2 F_0)^{-1} \nabla_{\mathcal{F}v}^2 F_0, \\ \nabla_{\mathcal{F}\mathcal{F}}^2 F_0 D^2 s_{\mathcal{F}}[p, p'] = -D^3 F_0[\cdot_{\mathcal{F}}, DWp, DWp'].$$

So $\|Ds_{\mathcal{F}}\| \leq H$, $\|DW\| \leq 1 + H$, and $\|D^2 s_{\mathcal{F}}\| \leq K_3(1 + H)^2/\mu$ by (48). Since

$$D^2 \lambda_i[p, p'] = \varsigma_i (D^3 F_0[e_i, DWp, DWp'] + \nabla_{i\mathcal{F}}^2 F_0 D^2 s_{\mathcal{F}}[p, p']),$$

$$\begin{aligned} \|\nabla^2 \lambda_i\| &\leq K_3(1 + H)^2 + K_2 K_3(1 + H)^2/\mu \\ &= K_3(1 + H)^3 \leq K_{\lambda} \quad \text{on } \mathcal{B}. \end{aligned}$$

Small multipliers have small gradients. Let $i \in \mathcal{J}$, so $0 \leq \lambda_i(a) \leq \theta \leq K_{\lambda}\eta^2/2$. If $\lambda_i(a) = 0$, then a minimizes the nonnegative function λ_i on the open ball, so $\nabla \lambda_i(a) = 0$. Otherwise let $\beta_i = \|\nabla \lambda_i(a)\|$, assume $\beta_i > 0$, and move from a in the direction $-\nabla \lambda_i(a)/\beta_i$. For $0 < t < \eta$, Taylor's theorem and nonnegativity give $0 \leq \lambda_i(a) - t\beta_i + K_{\lambda}t^2/2$. Letting t increase to $\min\{\eta, \sqrt{2\lambda_i(a)/K_{\lambda}}\} = \sqrt{2\lambda_i(a)/K_{\lambda}}$ gives $\beta_i^2 \leq 2K_{\lambda}\lambda_i(a)$. Hence the matrix B with columns $\varsigma_i \nabla \lambda_i(a)$, $i \in \mathcal{J}$, satisfies $\|B\|^2 \leq \|B\|_F^2 \leq 2n_R K_{\lambda} \theta \leq \mu g/2$.

Elimination. Near $(a, \bar{a}_{\mathcal{J}})$ define $\Pi(v, y) = F_0(v, \phi(v, y), y, \bar{a}_{\mathcal{A} \setminus \mathcal{J}})$, where ϕ solves $\nabla_{\mathcal{F}} F_0 = 0$ by the implicit function theorem. Its Hessian at $(a, \bar{a}_{\mathcal{J}})$ is the Schur complement of the $\mathcal{F}\mathcal{F}$ block in the Hessian of F_0 at x^* restricted to $(\mathcal{F}, v, \mathcal{J})$. Its blocks are as follows. Since $V_0(v) = \Pi(v, \bar{a}_{\mathcal{J}})$ on $\mathcal{B}$, the vv block is $\nabla_K^2 V_0(a) = \nabla_K^2 V_{\gamma}(a) \succeq 2gI$. By the envelope identity, $\partial_{y_i} \Pi(v, \bar{a}_{\mathcal{J}}) = \partial_{z_i} F_0(v, s(v)) =$

$\varsigma_i \lambda_i(v)$, so the block of mixed (v, y) derivatives is B . The yy block is a Schur complement of the residual Hessian and is $\succeq \mu I$, since for every u it equals a minimum over w of the residual quadratic form at (w, u) . For vectors (p, p') , Young's inequality gives

$$p^\top \nabla_K^2 V_0 p + 2p^\top B p' + p'^\top \Pi_{yy} p' \geq 2g \|p\|^2 - g \|p\|^2 - \frac{\|B p'\|^2}{g} + \mu \|p'\|^2 \geq g \|p\|^2 + \frac{\mu}{2} \|p'\|^2.$$

So $\nabla^2 \Pi \succeq \delta I$ with $\delta = \min\{g, \mu/2\}$.

Restoring the eliminated coordinates. Write the Hessian of F_0 at x^* on $(\mathcal{F}, q)$, $q = (v, y)$, as $\begin{pmatrix} D & E \\ E^\top & G \end{pmatrix}$ with $D \succeq \mu I$ and Schur complement $\nabla^2 \Pi \succeq \delta I$. For (w, q) put $\hat{w} = w + D^{-1} E q$; then the quadratic form equals $\hat{w}^\top D \hat{w} + q^\top \nabla^2 \Pi q \geq \mu \|\hat{w}\|^2 + \delta \|q\|^2$. Since $\|D^{-1} E\| \leq K_2/\mu = H$, $\|w\|^2 + \|q\|^2 \leq 2 \|\hat{w}\|^2 + (1 + 2H^2) \|q\|^2$, and therefore the form is at least $\nu(\|w\|^2 + \|q\|^2)$. Principal submatrices of a matrix $\succeq \nu I$ are $\succeq \nu I$. $\square$

Proof of theorem 7.15. Base choices. Let B be the factor of (T4) for F_0 on $[0, 1]^n$ (it applies to every rational linear term, in particular to core-only noise), enlarged to bound q -bit evaluation. Let C_{tail} and N_C be as in lemma E.3, let D_* be as in proposition 7.16, and put $C_S = 3^k \max_{1 \leq q \leq k} C(q, D_*)$. Choose

$$\begin{aligned} \rho &= \frac{1}{6B}, \quad g = \frac{\rho\sigma}{2k}, \quad \tau = \frac{\rho\sigma}{2N_C}, \quad \eta = \min\left\{\frac{1}{4}, \frac{\rho\sigma}{4C_S H_V}\right\}, \quad K_\lambda = \max\{1, K_3(1+H)^3\}, \\ \theta &= \min\left\{\frac{K_\lambda \eta^2}{2}, \frac{\mu g}{4n_R K_\lambda}\right\}, \quad \nu = \min\left\{\frac{\mu}{2}, \frac{\min\{g, \mu/2\}}{1+2H^2}\right\}, \quad g_* = \min\left\{\frac{g}{1+2H^2}, \frac{\mu}{4}, \frac{\nu}{4}\right\}, \\ r &= \min\left\{\frac{1}{8}, \frac{\tau}{16K_2}, \frac{\theta}{16K_2}, \frac{g_*}{4K_3}\right\}, \quad A_0 = 2 + \frac{kL+8}{g_*}, \end{aligned} \quad (50)$$

let J be the least $j \geq 0$ with $2^{-j} \leq \min\{r/(4A_0), g_* r^2/(16GA_0)\}$, and let M be the least power of two with $M \geq \max\{2, 2^J, 4kC_{\text{tail}}/\rho, 2N_C/\rho, 2kC_S/\rho\}$. All these are base-computable with polynomial bit length, and $J, \log_2 M \leq \text{poly}_d(I)$.

Algorithm and correctness. Run the algorithm of appendix E.5 in certified mode with g_* in place of g_0 and the parameters above: the core search with accuracy E_j , slab bounds with accuracy $g_* r^2/16$, sign tests on all original bounds (core and residual), and the Hessian test with modulus g_* . Every residual coordinate remains eligible for fixing only at an original bound. The fallback of (T4) runs on the same draw if no level returns. Correctness on every draw follows exactly as in appendix E.5.

The good event. Let $\mathcal{B}_1 = \{g_V < g\}$, let $\mathcal{B}_2$ be the event of lemma E.3(b), and let $\mathcal{B}_3$ be the event that $g_V > 0$ and the optimal core a lies in the relative interior of a face K with $q \geq 1$ free coordinates and $\text{dist}(a, S_K) \leq 2\eta$. On $\mathcal{B}_3$, a minimizes V_γ along K and V_γ is differentiable, so $\gamma_K = -\nabla_K V_0(a)$; for a nearest point $\hat{v} \in S_K$, $\text{dist}(\gamma_K, \mathcal{E}_K) \leq \|\nabla_K V_0(a) - \nabla_K V_0(\hat{v})\| \leq 2H_V \eta$. By lemma E.3, proposition 7.16(c), a union over the at most 3^k faces of the core box, and the choices above,

$$\Pr(\mathcal{B}_1) \leq \frac{kg}{\sigma} + \frac{2kC_{\text{tail}}}{M} \leq \rho, \quad \Pr(\mathcal{B}_2) \leq N_C \left(\frac{\tau}{\sigma} + \frac{1}{M} \right) \leq \rho, \quad \Pr(\mathcal{B}_3) \leq C_S \left(\frac{2H_V \eta}{\sigma} + \frac{k}{M} \right) \leq \rho.$$

So the bad event has probability at most $3\rho = 1/(2B)$.

Stopping on the good event. Outside $\mathcal{B}_1 \cup \mathcal{B}_2 \cup \mathcal{B}_3$ the optimal core a is unique with projected growth at least g , and by lemma E.2(c) $x^* = (a, s(a))$ has full point growth at least $g_F \geq g_*$. The hull, incumbent, and containment estimates of lemma E.1 hold with g_* in place of g_0 (the hull estimate uses the projected growth $g \geq g_*$), so at level J step (2) passes and every point of

Q is within $5r/4$ of x^* . The sign tests fix every core coordinate at a bound, since its derivative exceeds τ in magnitude, and every active residual coordinate with multiplier $\lambda_i > \theta$, since the midpoint test value is at least $\lambda_i - \frac{9}{4}K_2r > \theta - \frac{9}{64}\theta$. Coordinates strictly inside their bounds at x^* are not fixed. If $q \geq 1$, outside $\mathcal{B}_3$ the residual status is constant on the relative ball of radius η about a , $\nabla_K V_\gamma(a) = 0$, and growth gives $\nabla_K^2 V_\gamma(a) \succeq 2gI$; [proposition 7.17](#) then gives a Hessian at least νI on the unfixed coordinates. If $q = 0$, all core coordinates are fixed and the unfixed coordinates are residual, with Hessian at least $\mu I \succeq \nu I$. In both cases $\nu \geq 4g_*$, and (48) gives $\nabla^2 F_0(m') - (K_3 r_{Q'} + g_*)I \succeq (4g_* - 2K_3 r - g_*)I \succeq \frac{5}{2}g_*I$. The algorithm returns by level J .

Work and outputs. The expected-count bound of [proposition 7.5](#) needs only the independent core coefficients and $M \geq 2^J$. The rest of the accounting, the proof record, and the evaluation bounds are as in the proof of [theorem 7.13](#). $\square$

E.7 Examples

Verification of [example 7.3](#). For each v , the least feasible z is $|v - \frac{1}{3}| \leq \frac{2}{3}$, so $V(v) = |v - \frac{1}{3}|$. Since $2^j \not\equiv 0 \pmod{3}$, the number $2^j/3$ has fractional part $\frac{1}{3}$ or $\frac{2}{3}$, so $\frac{1}{3}$ lies in the interior of a level- j cell whose endpoints are at distances $h_j/3$ and $2h_j/3$ from it. Every grid value is at least $h_j/3$, so $U_j - E_j \geq h_j/3 - kLh_j^2/8 > 0 = f^*$ when $h_j < 8/(3kL)$. $\square$

Verification of [example 7.4](#) and its variant. Each leaf term $z^2 - \frac{v}{2}z$ is minimized over $[0, 1]$ at $z = v/4$ with value $-v^2/16$, which gives V . Over $\{0, \frac{1}{2}, 1\}$ its minimum is 0 for $v \in \{0, \frac{1}{2}, 1\}$ (attained at $z = 0$, and also at $z = \frac{1}{2}$ when $v = 1$), so $V_h(v) = v^2 + \frac{15}{16}v$ on the grid, which gives the three values, and $V_h(v) - V(v) = m \min_z (z - v/4)^2$. The four bag corners $(v, z_1) \in \{\frac{1}{2}, 1\} \times \{0, \frac{1}{2}\}$ have min-marginals $\frac{23}{32}, \frac{23}{32} + \frac{1}{8}, \frac{31}{16}, \frac{31}{16}$, and the grid minimum is $V_h(0) = 0$. Since $\frac{23}{32} - \frac{1}{8} > 0$, the bag-local rule discards the cell, while $\frac{23}{32} - \frac{33}{16} < 0$.

For the variant let $d \geq 8$, $m = d^2$, $h = \frac{1}{4}$, and $F_0(v, z) = (v - \frac{1}{2})^2 + \sum_{i=1}^m (z_i - \frac{1}{8} - (v - \frac{1}{2})/d)^2$ on $[0, 1]^{m+1}$. Its Hessian has eigenvalue 2 with multiplicity $m - 1$ and eigenvalues $3 \pm \sqrt{5}$ on the span of e_v and the normalized sum of the leaf directions, so it is uniformly positive definite, $L = 4$ is valid, and the star graph has coloring number two. With $t = v - \frac{1}{2}$, every ideal leaf value $\frac{1}{8} + t/d$ lies in $[\frac{1}{16}, \frac{3}{16}]$, its nearest point of $\{0, \frac{1}{4}, \dots, 1\}$ is 0 or $\frac{1}{4}$ at distance $\frac{1}{8} - |t|/d$, and

$$V(v) = t^2, \quad V_h(v) = \frac{d^2}{64} - \frac{d|t|}{4} + 2t^2.$$

The grid minimum is $U = \frac{d^2}{64} - \frac{d}{8} + \frac{1}{2}$ at $t = \pm \frac{1}{2}$. The cell $v \in [\frac{1}{2}, \frac{3}{4}]$, $z_1 \in [0, \frac{1}{4}]$ contains the optimizer $(\frac{1}{2}, \frac{1}{8}, \dots, \frac{1}{8})$, and its least corner min-marginal is $q_C = \frac{d^2}{64} - \frac{d}{16} + \frac{1}{8}$, at $(v, z_1) = (\frac{3}{4}, \frac{1}{4})$. Then $q_C - U = \frac{d}{16} - \frac{3}{8} \geq \frac{1}{8} > \frac{1}{16} = |B|Lh^2/8$, so the bag-local rule discards the cell. Finally $(V_h - V)(\frac{3}{4}) - (V_h - V)(\frac{1}{2}) = -(d - 1)h^2$. $\square$

Verification of [example 7.19](#). Write $\varrho = z_1 - vz_2$. The residual Hessian is $2(1, -v)^\top(1, -v) \succeq 0$ and $\partial_{vv}F_0 = 2 + 2z_2^2 \leq 4$. Taking $z = (0, 0)$ shows $V_\gamma(v) = (v - \frac{1}{2})^2 + \gamma v = (v - a_\gamma)^2 + f^*$, which gives the optimal core, its growth constant, and the optimal set $\{\varrho = 0, v = a_\gamma\}$. For $p = (0, v, 1)$ one computes $p^\top \nabla^2 F_0 p = 0$ and $\nabla^2 F_0 p = (-2\varrho, 0, 0)$. A positive semidefinite matrix with $p^\top H p = 0$ has $H p = 0$, so the Hessian is not positive semidefinite where $\varrho \neq 0$; every full-dimensional box contains such points.

For the strictly convex variant put $\alpha = z_1 - \frac{1}{2}$, $\beta = z_2 - \frac{1}{2}$, and $\varrho = \alpha - v\beta$. The residual problem $\min(\alpha - v\beta)^2 + \beta^4$ is strictly convex: on a segment either β varies, and β^4 is strictly convex, or β is constant, and the square is strictly convex in α . Its unique minimizer is $\alpha = \beta = 0$, so V_γ and a_γ are unchanged and the optimizer $(a_\gamma, \frac{1}{2}, \frac{1}{2})$ is unique and interior. At points with $\beta = 0$ and $\alpha \neq 0$,

the vector $p = (0, v, 1)$ in the coordinates (v, α, β) satisfies $p^\top H p = 0$ and $H p = (-2\alpha, 0, 0)$, and the direction $p + \alpha e_v$ has $(p + \alpha e_v)^\top H (p + \alpha e_v) = -2\alpha^2 < 0$. Such points lie arbitrarily close to the optimizer. $\square$

Verification of examples 7.20 and 7.21. For $F_0 = \phi(v) + y^2$, $\partial_y F_0 = 2y \geq 0$ with equality only at $y = 0$, so the selector is 0 and the multiplier $\partial_y F_0(v, 0)$ vanishes for all v . For the coupled variant, $\partial_y F_0 = 2(y + (v - \frac{1}{2})^2) \geq 0$ on $y \geq 0$, the selector is 0, the multiplier is $2(v - \frac{1}{2})^2$, and $V_0(v) = (v - \frac{1}{2})^2 + (v - \frac{1}{2})^4$.

In example 7.21, $\partial_y F_\gamma = v + y \geq 0$ on the box, with equality only at $(0, 0)$, so the selector is 0 and $V_\gamma(v) = \frac{1}{4}v^2 + \gamma v$. The Hessian has determinant $\frac{1}{2} - 1 < 0$. The multiplier is $\lambda(v) = v$. For $\gamma > 0$, V_γ is increasing on $[0, 1]$, so the optimum is $(0, 0)$ with $\partial_v F_\gamma(0, 0) = \gamma$. $\square$

F Proofs for integer recourse

This appendix proves the results of section 8. Appendix F.1 proves the exact box solver, appendix F.2 the native integer theorems and the oracle statements, appendix F.3 the flow results, appendix F.4 the totally unimodular extension, and appendices F.5 and F.6 the strong-noise and low-rank results. The tools (T1)–(T10) of appendix E.1 are used freely. We also use the following univariate facts [10]: for integer polynomials of degree at most δ and coefficient bit length at most τ , greatest common divisors, exact quotients, squarefree parts, modular inverses, resultants, isolation of all real roots by disjoint rational intervals, the sign of another such polynomial at an isolated root, and refinement of an isolating interval to width 2^{-q} take $\text{poly}(\delta, \tau, q)$ bit operations with an absolute exponent. Algebraic degrees are polynomial in δ , and intermediate bit lengths are polynomial in δ, τ, q . Fast complex-disk isolation and refinement for integer polynomials are also available [33, arXiv v2, Theorem 5]. Every nonzero root t of an integer polynomial $\sum_j a_j t^j$ with $a_0 \neq 0$ satisfies $|t| \geq |a_0| / (|a_0| + \max_{j \geq 1} |a_j|)$, by Cauchy's bound applied to the reversed polynomial.

F.1 The exact box solver

Let f and $B = \prod_{i=1}^k [\ell_i, u_i]$ be as in theorem 8.1. Substitute the coordinates with $\ell_i = u_i$; then every interval is nondegenerate. A *face* Φ is a partition of $[k]$ into sets L, U, F of coordinates fixed at the lower bound, fixed at the upper bound, and free. Write $r = |F|$ and $f_\Phi(y)$, $y \in \mathbb{R}^r$, for f with the fixed coordinates substituted; its degree is at most d . Let D be the least even integer greater than d , let $a = D - 1 \geq d$, and let z be an indeterminate. For a face with $r \geq 1$ put

$$g_i(y) = y_i^a + \frac{z}{D} \partial_i f_\Phi(y) \quad (i \in F). \quad (51)$$

For a real $z = 1/\varepsilon > 0$ these are the stationary equations of $f_\Phi(y) + \varepsilon \sum_{i \in F} y_i^D$ divided by εD , and $f + \varepsilon \sum_{i=1}^k x_i^D$ restricted to the face is this function plus a constant. Let $\mathcal{M} = \{y^e : 0 \leq e_i < a\}$ and $N = |\mathcal{M}| = a^r$.

The algorithm. For every face: if $r = 0$, the vertex is a candidate. If $r \geq 1$:

- (1) Compute the normal forms of all monomials of total degree at most $R = r(a - 1) + 1$ with respect to (51), in order of increasing degree: a monomial in $\mathcal{M}$ is its own normal form, and otherwise $\text{NF}(y_i^a y^{e'}) = -\frac{z}{D} \sum_{\varrho} c_{\varrho} \text{NF}(y^{\varrho} y^{e'})$, where $\partial_i f_\Phi = \sum_{\varrho} c_{\varrho} y^{\varrho}$. Read off the matrices $X_i(z) \in \mathbb{Q}[z]^{N \times N}$ of multiplication by y_i in the basis $\mathcal{M}$.

- (2) For $t = 0, 1, \dots, Q_r - 1$, $Q_r = 1 + (r - 1)(N + \binom{N}{2})$, let $\lambda = (1, t, \dots, t^{r-1})$ and $\mathbf{X}_\lambda = \sum_i \lambda_i \mathbf{X}_i$. Compute $\chi(z, T) = \det(TI - \mathbf{X}_\lambda)$ and, for $i \in \mathbb{F}$, $\chi_i(z, T) = -\partial_u \det(TI - \mathbf{X}_\lambda - u\mathbf{X}_i)|_{u=0}$, by interpolation in (z, T, u) .
- (3) Let $K_\lambda = \deg_z \chi$ and $p = [z^{K_\lambda}] \chi \in \mathbb{Q}[T]$. Skip the form if p is constant, if some $\deg_z \chi_i > K_\lambda$, if $G = \gcd(p, p')$ does not divide $b_i = [z^{K_\lambda}] \chi_i$ for some i , or if $\gcd(p'/G, p/G) \neq 1$. Otherwise let $P = p/G$, $A = p'/G$, and $B_i = b_i/G$. Multiply P by a positive common denominator of its coefficients so that $P \in \mathbb{Z}[T]$, and put $r_i = B_i A^{-1} \bmod P$. This scaling changes neither the roots nor the quotient ideal.
- (4) For every real root θ of P , keep the point with free coordinates $r_i(\theta)$ and the fixed coordinates of the face if it lies in B ; the membership test consists of signs of $r_i - \ell_i$ and $u_i - r_i$ at θ .

Among all kept points and vertices, select the first one of least value in a fixed order, compared exactly by [lemma F.7](#). Return its integer polynomial P , the isolating interval of its root θ , the maps r_i of the free coordinates and constant maps for the fixed ones, the value map r_f of [lemma F.7](#), and, as the auxiliary representation, the value polynomial of that lemma with an isolating interval of the value. A selected vertex is returned with $P(T) = T$, $\theta = 0$, and constant maps.

Lemma F.1 (Quotient algebra). *Let $\mathbb{F}$ be a field containing $\mathbb{Q}$ and $\zeta \in \mathbb{F}$, or let $\mathbb{F}$ contain $\mathbb{Q}(z)$ and $\zeta = z$. Under every degree-compatible monomial order, the polynomials g_i with $z = \zeta$ form a Gröbner basis of the ideal $\mathcal{I}_\zeta \subseteq \mathbb{F}[y]$ they generate, $\mathcal{M}$ is a basis of $\mathbb{F}[y]/\mathcal{I}_\zeta$, and the matrices of multiplication by y_i in this basis are $\mathbf{X}_i(\zeta)$. They commute.*

Proof. The term y_i^a has coefficient one and degree a , and every other term of g_i has degree at most $d - 1 < a$. So the leading monomials are y_i^a whatever ζ is, and they are pairwise coprime. By Buchberger's first criterion all S-polynomials reduce to zero, so the g_i form a Gröbner basis, and the monomials not divisible by any y_i^a , namely $\mathcal{M}$, form a basis of the quotient [16]. Reduction by the g_i divides only by the leading coefficient one, so the normal form of a monomial is the specialization at ζ of the normal form computed over $\mathbb{Q}[z]$. Each recursion step of the algorithm lowers the total degree by at least $a - (d - 1) \geq 1$, so the normal forms of all monomials of degree at most R , which include $y_i \mathbf{m}$ for $\mathbf{m} \in \mathcal{M}$, are computed. Multiplication operators of a commutative algebra commute. $\square$

Let $\mathcal{P}$ be the field of Puiseux series $\bigcup_{m \geq 1} \overline{\mathbb{Q}}((\varepsilon^{1/m}))$; it is algebraically closed [10]. Embed $\mathbb{Q}(z)$ in $\mathcal{P}$ by $z \mapsto \varepsilon^{-1}$. For $s \in \mathcal{P}$ let $\text{val}(s)$ be its least exponent, and for a vector or a polynomial in T the least valuation of its entries or coefficients.

Lemma F.2 (Joint eigenvalues). *There are points $\xi^{(1)}, \dots, \xi^{(N)} \in \mathcal{P}^r$, not necessarily distinct, such that every $\xi^{(t)}$ is a common zero of the g_i (with $z = \varepsilon^{-1}$) and $\det(TI - \sum_i \lambda_i \mathbf{X}_i(\varepsilon^{-1})) = \prod_{t=1}^N (T - \lambda^\top \xi^{(t)})$ for every $\lambda \in \mathcal{P}^r$.*

Proof. Commuting matrices over an algebraically closed field are simultaneously upper triangularizable [27]. Let Π triangularize all $\mathbf{X}_i(\varepsilon^{-1})$, and let $\xi_i^{(t)}$ be the t th diagonal entry of $\Pi^{-1} \mathbf{X}_i \Pi$. Then $\Pi^{-1} \mathbf{X}_\lambda \Pi$ is triangular with diagonal entries $\lambda^\top \xi^{(t)}$, which gives the determinant. For a polynomial h , let $\mathbf{X}_h$ be the matrix of multiplication by h ; the map sending h to the t th diagonal entry of $\Pi^{-1} \mathbf{X}_h \Pi$ is a ring homomorphism, because diagonal entries of sums and products of upper triangular matrices are sums and products of diagonal entries. It sends y_i to $\xi_i^{(t)}$ and every element of the ideal to 0. So $g_i(\xi^{(t)}) = 0$. $\square$

Call t *bounded* if $\text{val}(\xi^{(t)}) \geq 0$, and let $v^{(t)} \in \overline{\mathbb{Q}}^r$ be the vector of its ε^0 -coefficients, its *limit*. Otherwise let $s_t = -\text{val}(\xi^{(t)}) > 0$ and let $w^{(t)} \neq 0$ be the vector of ε^{-s_t} -coefficients, its *leading pole*. Let $\mathcal{L}$ be the set of distinct limits, $\mu_v \geq 1$ the number of bounded t with limit v , $\kappa = \sum_{t \text{ unbounded}} s_t$, and $C(\lambda) = \prod_{t \text{ unbounded}} (-\lambda^\top w^{(t)})$, a nonzero polynomial. Write $\chi(\lambda, z, T) = \det(TI - \sum_i \lambda_i X_i(z)) = \sum_m c_m(\lambda, T) z^m$ with $c_m \in \mathbb{Q}[\lambda, T]$.

Lemma F.3 (The leading coefficient). *The number κ is an integer, $c_m = 0$ for $m > \kappa$, and $c_\kappa(\lambda, T) = C(\lambda) \prod_{v \in \mathcal{L}} (T - \lambda^\top v)^{\mu_v}$ as polynomials in (λ, T) .*

Proof. Let $\lambda \in \overline{\mathbb{Q}}^r$ satisfy $\lambda^\top w^{(t)} \neq 0$ for every unbounded t . For unbounded t , $\lambda^\top \xi^{(t)}$ has valuation $-s_t$ and leading coefficient $\lambda^\top w^{(t)}$, so $T - \lambda^\top \xi^{(t)} = (-\lambda^\top \xi^{(t)})(1 - T/\lambda^\top \xi^{(t)})$, where the second factor is 1 plus a polynomial of positive valuation. For bounded t , $T - \lambda^\top \xi^{(t)}$ is $T - \lambda^\top v^{(t)}$ plus a polynomial of positive valuation. By lemma F.2,

$$\varepsilon^\kappa \chi(\lambda, \varepsilon^{-1}, T) = C(\lambda) \prod_{v \in \mathcal{L}} (T - \lambda^\top v)^{\mu_v} + \Omega(T), \quad \text{val}(\Omega) > 0.$$

The left side equals $\sum_m c_m(\lambda, T) \varepsilon^{\kappa-m}$. Comparing the coefficients of each power of T shows that $c_m(\lambda, \cdot) = 0$ when $\kappa - m < 0$, and that the ε^0 -part of the left side equals the nonzero first term on the right; hence some m equals κ , and $c_\kappa(\lambda, T) = C(\lambda) \prod_v (T - \lambda^\top v)^{\mu_v}$. These identities hold on the complement of finitely many hyperplanes in $\overline{\mathbb{Q}}^r$, so they hold as polynomial identities. $\square$

Call a form $\lambda \in \mathbb{Q}^r$ *good* for the face if $\lambda^\top w^{(t)} \neq 0$ for every unbounded t and $\lambda^\top v \neq \lambda^\top v'$ for distinct $v, v' \in \mathcal{L}$.

Lemma F.4 (Recovery from one root). *If λ is good and $\mathcal{L} \neq \emptyset$, the form passes step (3), and for every $v \in \mathcal{L}$ the number $\theta = \lambda^\top v$ is a root of P with $r_i(\theta) = v_i$ for all i . If θ is real, v is real.*

Proof. By lemma F.3, $K_\lambda = \kappa$ because $C(\lambda) \neq 0$, and $p = C(\lambda) \prod_v (T - \alpha_v)^{\mu_v}$ with distinct $\alpha_v = \lambda^\top v$. Since $\chi(\lambda + u e_i, z, T) = \det(TI - X_\lambda - u X_i)$, we have $\chi_i = -\partial_{\lambda_i} \chi$; because $c_m \equiv 0$ for $m > \kappa$, also $\deg_z \chi_i \leq \kappa$ and $b_i = -\partial_{\lambda_i} c_\kappa$. Differentiating the identity of lemma F.3,

$$b_i = -\frac{\partial C}{\partial \lambda_i} \prod_u (T - \alpha_u)^{\mu_u} + C(\lambda) \sum_v \mu_v v_i (T - \alpha_v)^{\mu_v-1} \prod_{u \neq v} (T - \alpha_u)^{\mu_u},$$

while $p' = C(\lambda) \sum_v \mu_v (T - \alpha_v)^{\mu_v-1} \prod_{u \neq v} (T - \alpha_u)^{\mu_u}$ and $G = \gcd(p, p')$ is a constant times $\prod_v (T - \alpha_v)^{\mu_v-1}$. So G divides b_i , P is a constant times $\prod_v (T - \alpha_v)$, and with a common constant $\kappa' \neq 0$,

$$A(\alpha_v) = \kappa' \mu_v \prod_{u \neq v} (\alpha_v - \alpha_u) \neq 0, \quad B_i(\alpha_v) = \kappa' \mu_v v_i \prod_{u \neq v} (\alpha_v - \alpha_u),$$

because the first term of b_i keeps a factor $T - \alpha_v$ after division by G , and $\mu_v \neq 0$ in characteristic zero. Thus $\gcd(A, P) = 1$ and $r_i(\alpha_v) = B_i(\alpha_v)/A(\alpha_v) = v_i$. The polynomials A and B_i have rational coefficients, so a real θ gives a real v . $\square$

Lemma F.5 (Limits of real critical points). *Let $\varepsilon_m \downarrow 0$, and let $y^{(m)} \in \mathbb{R}^r$ be common zeros of the g_i with $z = 1/\varepsilon_m$ that converge to y^* . Then $y^* \in \mathcal{L}$.*

Proof. Let $\lambda \in \mathbb{Q}^r$. The row vector $e = (\mathbf{m}(y^{(m)}))_{\mathbf{m} \in \mathcal{M}}$ is nonzero, because $1 \in \mathcal{M}$, and satisfies $e X_i(1/\varepsilon_m) = y_i^{(m)} e$: evaluation at a common zero is well defined on the quotient. So $\lambda^\top y^{(m)}$ is an eigenvalue of $X_\lambda(1/\varepsilon_m)$, and by lemma F.3

$$0 = \varepsilon_m^\kappa \chi(\lambda, 1/\varepsilon_m, \lambda^\top y^{(m)}) = \sum_{m' \leq \kappa} c_{m'}(\lambda, \lambda^\top y^{(m)}) \varepsilon_m^{\kappa-m'}.$$

The arguments are bounded, so letting $m \rightarrow \infty$ gives $c_\kappa(\lambda, \lambda^\top y^*) = 0$, that is, $C(\lambda) \prod_{v \in \mathcal{L}} (\lambda^\top (y^* - v))^{\mu_v} = 0$ for all $\lambda \in \mathbb{Q}^r$. The polynomial on the left vanishes on the Zariski-dense set $\mathbb{Q}^r$, so it is zero; $C \neq 0$ and the polynomial ring is a domain, so some linear factor $\lambda^\top (y^* - v)$ is identically zero. Hence $y^* = v$. If $\mathcal{L}$ were empty the same argument would give $C \equiv 0$. $\square$

Lemma F.6 (Good forms). *Among $\lambda(t) = (1, t, \dots, t^{r-1})$, $t = 0, \dots, Q_r - 1$, at least one form is good.*

Proof. A form fails for t if $\lambda(t)^\top w = 0$ for one of at most N leading poles w , or $\lambda(t)^\top (v - v') = 0$ for one of at most $\binom{N}{2}$ pairs of distinct limits. Each condition says that a nonzero polynomial in t of degree at most $r - 1$, with coefficient vector w or $v - v'$, vanishes; it has at most $r - 1$ roots. So at most $(r - 1)(N + \binom{N}{2}) < Q_r$ integers fail. $\square$

Proof of correctness in theorem 8.1. For $\varepsilon > 0$ let $x(\varepsilon)$ minimize $f + \varepsilon \sum_i x_i^D$ on B . Along a sequence $\varepsilon_m \downarrow 0$ choose a convergent subsequence with limit $\hat{x}$; since the perturbation converges to zero uniformly on B , $\hat{x}$ minimizes f . Pass to a further subsequence on which every $x(\varepsilon_m)$ lies in the relative interior of one face Φ . If Φ is a vertex, $\hat{x}$ is that vertex, a candidate. Otherwise the free part of $x(\varepsilon_m)$ is an interior minimizer of the deformed face function, hence a real common zero of (51) with $z = 1/\varepsilon_m$. By lemma F.5 its limit lies in $\mathcal{L}_\Phi$; by lemmas F.4 and F.6 some enumerated form recovers it exactly from a real root of P , and it lies in B . So $\hat{x}$ is kept. A bad form that passes step (3) still yields well-defined maps, because A is invertible modulo P , but its reconstructed points need not be critical limits; for instance, its leading coefficient can retain the bounded projection of a branch that escapes to infinity. Step (4) keeps such a point only if it lies in B . Every kept point is therefore feasible and has value at least $\min_B f$, so a kept point of least value is a global minimizer. $\square$

Lemma F.7 (Values and comparisons). *For a kept point with root θ of P and coordinates $r_i(\theta)$ (fixed coordinates substituted), let $r_f = f(r(T)) \bmod P$ be the remainder of $f(r_1(T), \dots, r_k(T))$ on division by P . Then $\deg r_f < \deg P$ and $r_f(\theta) = f(r(\theta))$, the value of the point. Write $r_f = h/q_0$ with $h \in \mathbb{Z}[T]$ and $q_0 \in \mathbb{Z} \setminus \{0\}$. The resultant $\text{Res}_T(P(T), q_0 W - h(T)) \in \mathbb{Z}[W]$ is nonzero, has degree at most $\deg P$, and has the value as a root, which is identified by refining the interval of θ . Two such values are compared exactly, including equality, by isolating the real roots of the squarefree part of the product of their two value polynomials and locating both values among them.*

Proof. P divides $f(r(T)) - r_f(T)$, and $P(\theta) = 0$, so $r_f(\theta) = f(r(\theta))$. Up to a nonzero constant factor, the resultant is $\prod_{\vartheta} (q_0 W - h(\vartheta))$ over the roots ϑ of P , a nonzero polynomial of degree $\deg P$ with the root $h(\theta)/q_0$. Refining the interval of θ encloses $h(\theta)/q_0$ by interval evaluation; by the root separation bound of the squarefree part, polynomially many bits suffice to single out its root. Equal values are the same root of the squarefree product, so ties are decided exactly. $\square$

The value map r_f belongs to the output, so the optimizer and the value are polynomials in the same root, as definition 2.5(d) requires. The value polynomial and its isolating interval are an auxiliary representation for comparisons; composing them with the coordinate representation is never needed.

Lemma F.8 (Cost). *The algorithm uses $c_d^k \text{poly}_d(H)$ bit operations for a constant c_d that depends only on d . Its output, and the representations of the value, have length $c_d^k \text{poly}_d(H)$, and q -bit refinements cost $c_d^k \text{poly}_d(H + q)$.*

Proof. Fix a face with $r \geq 1$ free coordinates. There are $\binom{r+R}{r} \leq 2^{r+R} \leq 2^{Dr}$ monomials of degree at most R and $N = a^r \leq 2^{Dr}$. Clear the denominators of $\partial_i f_\Phi / D$ by one integer of $O_d(H)$ bits. Each recursion step multiplies by z and by integer coefficients of $O_d(H)$ bits; a normal form is a sum over reduction paths of at most R steps, of which there are at most $(s+1)^R$, where $s = \text{poly}_d(r)$ bounds the number of monomials of a derivative. So every normal form has z -degree at most R and coefficients of $O_d(R(H + \log(s+1)))$ bits; memoization computes all of them without enumerating paths, in $2^{O_d(r)} \text{poly}_d(H)$ operations. The forms have $O(r^2 \log(N+1))$ -bit entries. Each determinant has order N , degree at most N in T and u and at most RN in z , and is interpolated from $(RN+1)(N+1)^2$ rational determinants of polynomial length; so χ and the χ_i have $\text{poly}(N, H, r)$ -bit coefficients. The $Q_r = O(rN^2)$ forms, the univariate steps (3)–(4) on polynomials of degree at most N , the at most N real roots per form, and the value and comparison steps of [lemma F.7](#) each cost $\text{poly}(N, H, r)$; the value map is obtained from at most H monomials of degree at most d , each a product of at most d maps reduced modulo P . Summing over at most 3^k faces gives $c_d^k \text{poly}_d(H)$ with c_d depending on D only.

Refinement. Refine the interval of θ and evaluate the r_i and r_f by interval arithmetic with derivative bounds on the interval; polynomially many additional bits give every coordinate within $2^{-q - \lceil \log_2(k+1) \rceil}$, hence a point within Euclidean distance 2^{-q} , and the value within 2^{-q} . Clipping a rational approximation to B does not increase its distance to the exact point, which lies in B . With a rational bound $G_f \geq \max\{1, \sup_B \|\nabla f\|_1\}$ from monomial sums, coordinates within $2^{-q}/(2G_f)$ give a feasible rational point whose value exceeds the optimum by at most 2^{-q-1} ; enclosing the value with width 2^{-q-1} gives a lower bound, and so a certified gap 2^{-q} . The value gap alone would not bound the distance to x^* ; the two refinements are separate. $\square$

Lemma F.9 (An explicit constant). *Implement the algorithm with schoolbook integer arithmetic, fraction-free Bareiss determinants, tensor interpolation on consecutive integer nodes, subresultant greatest common divisors and Bézout coefficients, and squarefree Sturm sequences for isolation, sign determination, and refinement. Then its bit work for dimension k , input length H , and q refinement bits is at most $C_d 2^{10000Dk} (H+q+1)^{10000}$ for a constant C_d depending only on d . In particular $c_d = 2^{10000D}$ is a valid constant in [theorem 8.1](#) and, with a larger C_d , in [corollary 8.2](#).*

Proof. Put $U = 2^{Dk}(H+q+d+2)$. The ledger is deliberately loose. On a face with $r \leq k$ free coordinates the number of memoized monomials and N are at most 2^{Dk} ; normal forms have z -degree $O_d(k)$ and coefficient bit length $O_d((H+k+1)^2)$; there are $O(rN^2)$ forms with $O_d(k^2)$ -bit coefficients; determinants have order N and degrees $N, N, O_d(kN)$, so their interpolation grids have $O_d(U^4)$ nodes, sampled values have bit length $O_d(U^5)$, and interpolated coefficients $O_d(U^8)$. Subresultant and Sylvester bounds keep greatest common divisors, inverses, and the r_i within degree and bit length $O_d(U^{12})$; composition with f and the value resultant stay within $O_d(U^{18})$; Cauchy and squarefree separation bounds, derivative bounds for evaluation, and the q extra bits require precision $O_d(U^{25})$. Every primitive call therefore has input length at most $C_d U^{50}$, and the number of primitive calls, over at most $3^k \leq 2^{Dk}$ faces, forms, coordinates, nodes, roots, and comparisons, is at most $C_d U^{10}$.

For a primitive call let L be the combined binary length of its dense arithmetic data, including rational endpoints and interpolation nodes, and let q_{req} be the largest requested refinement precision. Put $\ell = 2 + L + q_{\text{req}}$. Thus precision enters numerically, rather than only through the length of its binary encoding. The preceding bounds give $\ell \leq C_d U^{50}$. We next justify a universal primitive-work bound $A(\ell+2)^{40}$, which is smaller than the exponent-100 bound used below. The fixed factor A is absorbed into the global prefactor C_d .

Schoolbook multiplication and long division on B -bit integers cost $O(B^2)$. Euclidean gcd uses at most $O(B)$ divisions, so rational arithmetic with reduction costs at most $O(B^3)$. After clearing

denominators, input polynomial degrees are at most ℓ and heights at most $K\ell^2$, for a fixed absolute constant K . Hadamard's inequality bounds Sylvester minors by $K\ell^3$ bits. The subresultant identities show that each nonzero primitive Euclidean remainder is proportional to a subresultant; skipping defective degrees gives the same bound. Therefore primitive gcd and signed Sturm data have heights at most $K\ell^4$. Ordinary rational long division, followed by division by positive contents, computes these sequences in at most $K\ell^3$ coefficient operations, with intermediate lengths at most $K\ell^6$. This gives at most $K\ell^{21}$ bit work, including exact quotients and squarefree parts. The classical subresultant identities and Sturm correctness are used here [10]; the numerical work estimate follows from this operation count.

Bézout coefficients and modular inverses can be obtained from a consistent Sylvester linear system with at most $2\ell + O(1)$ unknowns. Exact pivoting and fraction-free determinants bound the intermediate lengths by $K\ell^8$. At most $K\ell^2$ determinants, each requiring $O(\ell^3)$ exact operations, give a bound below $K\ell^{30}$. Bareiss determinants, resultants, and consecutive-node Lagrange interpolation satisfy the same loose bound: intermediate entries are minors, and products of node differences have polynomial height. Tensor interpolation has at most three variables and is a fixed number of successive univariate interpolations.

For isolation, let p be a primitive squarefree polynomial of degree $n \leq \ell$ and height $\tau' \leq K\ell^4$. Degrees zero and one are handled directly. For $n \geq 2$, every root has absolute value at most $R = 2^{\tau'+2}$. The discriminant is a nonzero integer. If ζ, η are two closest roots, its product formula gives

$$1 \leq |a_n|^{2n-2} |\zeta - \eta|^2 (2R)^{n(n-1)-2},$$

where a_n is the leading coefficient. Hence a sufficiently large fixed integer K gives a separation bound 2^{-S} with $S = K(n+1)^2(\tau'+1) \leq K\ell^6$. Bisect $[-R, R]$, subdividing only intervals that contain a real root by exact Sturm counts. At any depth at most n open intervals contain roots; roots at endpoints or midpoints are recorded once as point intervals. One-sided variations, obtained from the first nonzero derivative at a zero endpoint, handle these cases. After $K(\tau' + S + q_{\text{req}} + 1)$ levels, distinct roots are separated and the isolators have the requested width. Thus at most $K\ell^7$ variation evaluations suffice, at endpoints of at most $K\ell^6$ bits. Homogenized Horner evaluation and endpoint derivative tests use at most $K\ell^3$ integer operations per evaluation, with operands of at most $K\ell^7$ bits. Their total cost is below $K\ell^{30}$, including rational normalization. This also bounds refinement of a given isolator.

For a sign test at a stored root, gcd membership first detects a zero value. Otherwise apply the same separation and isolation argument to the squarefree part of the product of the two input polynomials. The stored root is then separated from all other roots, and the sign is evaluated at a rational point in its interval. The product has degree at most 2ℓ and height at most $K\ell^2$ before squarefree extraction, so the same ledger applies with a larger fixed K . Comparing two represented values uses their two value polynomials in exactly this way, including equality. All the listed primitives therefore fit $A(\ell+2)^{40}$ and, in particular, $A(\ell+2)^{100}$. The program constants are fixed independently of dimension, degree and precision. The total work is at most

$$C'_d U^{10} (C_d U^{50} + 2)^{100} \leq C''_d U^{5010} \leq C'''_d 2^{10000Dk} (H + q + 1)^{10000}.$$

□

The constant is an explicit proved bound used to define noise laws and fallback budgets before sampling. A smaller constant with a proved bound of the same form may replace it.

Proof of corollary 8.2. Enumerate the integer labels, substitute each, and apply [theorem 8.1](#); substitution keeps the degree and gives coefficients of length $\text{poly}_d(H)$. Keep the current best value

and compare each new value with it by [lemma F.7](#). Each comparison involves two univariate polynomials of degree at most $(D-1)^k$, so the total is $c_d^k \prod_i N_i \text{poly}_d(H)$. The explicit budget above includes these comparisons; only its global prefactor is enlarged. Only the winner's representation is retained; isolating its roots afresh in their own polynomials removes any extra precision used in comparisons. $\square$

F.2 Native integer recourse and integer oracles

Throughout, $G \geq \max\{1, \sup \sum_{i \in C} |\partial_{v_i} F_\gamma|\}$ over $[0, 1]^k \times \prod_i [\ell_i, u_i]$ and all noise in $[-\sigma, \sigma]^n$, computed by monomial bounds; it satisfies the hypothesis of [lemma 7.2\(c\)](#).

Lemma F.10 (Competing labels). *Let $\mathcal{H}$ be a core box, $c \in \mathcal{H}$, and $z \in Y$ with $F_\gamma(c, z) = V(c)$. The sets $Y_i^- = \{z' \in Y : z'_i \leq z_i - 1\}$ and $Y_i^+ = \{z' \in Y : z'_i \geq z_i + 1\}$ cover $Y \setminus \{z\}$. If (38) holds with $V_{\text{other}}(c)$ the least exact value over the nonempty ones ($+\infty$ if none), then z is the unique minimizer of $F_\gamma(v, \cdot)$ on Y for every $v \in \mathcal{H}$. If moreover $\mathcal{H}$ contains the core of every global minimizer, every global minimizer lies in $\mathcal{H} \times \{z\}$ and $\min_{v \in [0, 1]^k} F_\gamma(v, z) = f^*$.*

Proof. If $Y = \{z\}$, the conclusions are immediate. Otherwise, a label $z' \neq z$ differs in some coordinate, and by integrality $z'_i \leq z_i - 1$ or $z'_i \geq z_i + 1$. For $v \in \mathcal{H}$, [lemma 7.2\(c\)](#), applied to $Y \setminus \{z\}$ and to $\{z\}$, gives

$$\begin{aligned} V_{Y \setminus \{z\}}(v) &\geq V_{\text{other}}(c) - Gw_\infty(\mathcal{H}) \\ &> V(c) + Gw_\infty(\mathcal{H}) \\ &= F_\gamma(c, z) + Gw_\infty(\mathcal{H}) \geq F_\gamma(v, z). \end{aligned}$$

A global minimizer (a, z^*) with $a \in \mathcal{H}$ therefore has $z^* = z$. The slice $[0, 1]^k \times \{z\}$ lies in X and contains (a, z) , so its minimum is f^* . $\square$

Lemma F.11 (Stopping for labels). *Suppose F_γ has a unique global minimizer $x^* = (a, a_R)$ with point growth at least g_0 on X , let $A_0 = 2 + kL/g_0$, and let $h = 2^{-j}$ satisfy*

$$h \leq \min\left\{\frac{1}{4A_0}, \frac{g_0}{16GA_0}\right\}.$$

Then the label test at level j passes, with $z_j = a_R$.

Proof. By [proposition 7.5](#) in exact mode, every retained cell has a corner v with $V(v) \leq f^* + 2E_j$, and growth gives $\|v - a\| \leq h\sqrt{kL/(4g_0)}$; so $\mathcal{H}_j \subseteq a + [-A_0h, A_0h]^k$ and $w_\infty(\mathcal{H}_j) \leq 2A_0h$. The incumbent has $F_\gamma(c_j, z_j) \leq f^* + E_j$, so $\|(c_j, z_j) - x^*\| \leq \sqrt{E_j/g_0} \leq A_0h \leq \frac{1}{4}$, which forces $z_j = a_R$ because distinct labels are at distance at least one. Every other label z' has $F_\gamma(c_j, z') - f^* \geq g_0 \|z' - a_R\|^2 \geq g_0$, so $V_{\text{other}}(c_j) - V(c_j) \geq g_0 - E_j \geq \frac{7}{8}g_0$, since $E_j = kLh^2/8 \leq g_0/256$. Finally $2Gw_\infty(\mathcal{H}_j) \leq 4GA_0h \leq g_0/4$. $\square$

Proof of theorem 8.4. Base choices. Let $B = \max\{2, N_Z C_d c_d^k\}$, where C_d, c_d are those of [lemma F.9](#) enlarged by [corollary 8.2](#), so that enumerating all labels, checking feasibility, solving each core slice, and comparing values costs at most $B(I + \log M + 1)^e$ for a fixed exponent e . Let $S = k + \sum_i (u_i - \ell_i)$, $n = k + n_R$, and let C_{tail} be the constant of (T3) for $\mathcal{D} = X$, described by the box, the rows of $Az \leq b$, and one disjunction per integer coordinate, with all n coordinates perturbed; $\log C_{\text{tail}} \leq \text{poly}_d(I)$. Choose $\rho = 1/(4B)$, $g_0 = \rho\sigma/(2S)$, $A_0 = 2 + kL/g_0$, the least $J \geq 0$ with $2^{-J} \leq \min\{1/(4A_0), g_0/(16GA_0)\}$, and the least power of two $M \geq \max\{2, 2^J, 4nC_{\text{tail}}/\rho\}$. All have polynomial bit length, and $J, \log_2 M \leq \text{poly}_d(I)$.

Algorithm and correctness. At levels $j = 0, \dots, J$ run the core search in exact mode with the integer oracle, and test (38) at (c_j, z_j) with at most $2n_R$ further oracle calls on tightened bounds. When it passes, apply [theorem 8.1](#) to $F_\gamma(\cdot, z_j)$ on $[0, 1]^k$ and return. Otherwise, after level J , enumerate the labels on the same draw as in [corollary 8.2](#). By [proposition 7.5\(a\),\(b\)](#) and [lemma F.10](#), every return is correct.

Probability. By [lemma F.11](#), the fallback runs only if $g_* < g_0$, which by (T3) has probability at most $Sg_0/\sigma + 2nC_{\text{tail}}/M \leq \rho = 1/(4B)$.

Work and output. By [proposition 7.5](#) the expected number of core calls is at most $2^k + 8^k JQ_{\text{ex}}$; each call, including the label calls, has polynomial query length and costs $\text{poly}_d(I)$. The core slice has coefficients of length $\text{poly}_d(I + \log M) = \text{poly}_d(I)$, so the completion costs $c_d^k \text{poly}_d(I)$ and returns a representation of that length. The fallback contributes $\text{poly}_d(I)$ in expectation by (T5), and returns one label with its re-isolated core representation, of the same length bound. The proof record contains the oracle answers, retained lists, label values, and the solver's computation; its expected size obeys the work bound. For quadratic F_0 replace [theorem 8.1](#) by (T10) on the core box $(3^k \text{poly}(I), \text{rational output})$ and take $B = \max\{2, N_Z 3^k\}$. $\square$

Proof of corollary 8.5. Use the base choices above with two changes. Let

$$N' = \max\{1, k3^k N_Z \max(1, d-1)^k\}, \quad \tau = \frac{\rho\sigma}{2N'},$$

and require additionally

$$2^{-J} \leq \min\left\{\frac{\tau}{8K_2 A_0}, \frac{g_0}{4K_3 A_0}\right\}, \quad M \geq \frac{2N'}{\rho},$$

where K_2, K_3 are as in (34) over the real box. After (38) passes at level j , apply the sign tests of [lemma 7.7\(a\)](#) to the core coordinates of $f = F_\gamma(\cdot, z_j)$ on $Q = \mathcal{H}_j$, which contains the core of every global minimizer by [lemma F.10](#), and then the Hessian test of [lemma 7.7\(b\)](#) with modulus g_0 on the remaining core box Q' . If both pass, return z_j and the patch (Q', g_0) ; its minimizer is the global optimizer. Otherwise continue; the fallback is that of [theorem 8.4](#).

The extra event. Fix a label w in the bounding box, a face of the core cube, and a core coordinate i fixed on it, and condition on all coefficients but γ_i . The free core stationary system of $F_\gamma(\cdot, w)$ on the face does not involve γ_i and has at most $\max(1, d-1)^k$ nonsingular solutions ((T9)); at each, the core derivative lies in $[-\tau, \tau]$ with probability at most $\tau/\sigma + 1/M$. On $g_* > 0$ the optimal core is a nonsingular solution for $w = a_R$ on its face, by two-sided growth. Hence the probability that $g_* \geq g_0$ and some active core derivative has magnitude at most τ is at most $N'(\tau/\sigma + 1/M) \leq \rho$.

Stopping. Outside both events, at level J the label test passes ([lemma F.11](#)), $\mathcal{H}_J \subseteq a + [-A_0 h, A_0 h]^k$, and the midpoint test of a core coordinate at a bound has value at least $\tau - 2K_2 A_0 h \geq \frac{3}{4}\tau$; free core coordinates have zero derivative at a and are not fixed. Growth with the label fixed gives $\nabla_{\mathcal{U}}^2 f(a) \succeq 2g_0 I$ on the free core coordinates, and the Hessian test value is at least $g_0 - 2K_3 A_0 h \geq g_0/2$. The failure probability is at most $2\rho = 1/(2B)$, there is no c_d^k completion term, and a q -bit evaluation uses (T8). $\square$

Proof of proposition 8.6. Write $Az = b$ as $Az \geq b$, $-Az \geq -b$; with the bound rows, the constraint matrix remains totally unimodular, and tightening bounds keeps it so. At a rational core point v and rational linear coefficients, each $t \mapsto f_i(v, t) + \gamma_i t$ is a convex polynomial on $[\ell_i, u_i]$; extending it beyond each endpoint by the tangent line there gives a convex function on $\mathbb{R}$ that agrees on the interval and has rational values of polynomial length at rational points. The proximity-scaling algorithm of Hochbaum and Shanthikumar [25, Algorithm 4.2 and Theorem 4.3] solves separable

convex integer minimization over totally unimodular systems exactly, using a number of linear programs logarithmic in the numerical bounds, each with an optimal extreme point computed exactly in polynomial bit time [24], and function evaluations at prescribed points. All data have polynomial length, so the oracle runs in polynomial bit time, and the exact value is the evaluation at the returned integer point. Infeasibility is detected by the linear program of the first stage. $\square$

Proof of lemma 8.7. Sufficiency. Let $z' \in Y$ and $\Delta = z' - z$. It is a circulation; route $\Delta_a > 0$ units on the forward residual copy of a and $-\Delta_a$ units on the backward copy when $\Delta_a < 0$; both copies are available because z' satisfies the bounds. By discrete convexity, $f_a(z'_a) - f_a(z_a)$ is at least the residual arc cost times the number of units. Summing, $F(z') - F(z)$ is at least the cost of this nonnegative residual circulation, which equals its reduced cost because potential terms cancel around a circulation, and so is nonnegative.

Necessity. A simple residual cycle using at most one copy of each original arc admits one unit of feasible augmentation. Its objective change is exactly the sum of its unit marginal costs, which is nonnegative by optimality. This includes two-cycles on distinct arcs and one-arc loops. The only simple cycle using both residual copies of an original arc is their two-cycle, whose cost is nonnegative by convexity. Thus no simple residual cycle has negative cost. Hence shortest-path distances π from an added source joined to every node by zero-cost arcs are finite and satisfy $c_e + \pi_p - \pi_{p'} \geq 0$ on every residual arc from p to p' . A shortest path uses at most s arcs. When unit marginal costs are rational of bit length at most H_m , its length, and hence each potential, has bit length polynomial in s and H_m . $\square$

F.3 Flows under core-only noise

Proof of corollary 8.8. The feasible residual set is fixed, so lemma 7.2(b) and lemma 3.7 apply to the conditional value V . Every core point belongs to a grid cell of side $h = 1/m$, and a corner of that cell has conditional value at most $V(v) + kLh^2/8$. Thus $\min_{G_m} \ell - kLh^2/8 \leq \min V = f^*$. If v_0 minimizes the lower answers on G_m , then the least upper answer satisfies $F_\gamma(\hat{v}, \hat{z}) \leq u(v_0) \leq \ell(v_0) + \eta$; subtracting the displayed lower bound proves the interval and gap claims. Feasibility comes from the returned completion. The choice of m makes $kLh^2/8$ at most ε . For integer flows the exact oracle of proposition 8.6 has $\eta = 0$; approximate calls have the separately specified allowance η . No residual variable is rounded. $\square$

We treat networks and totally unimodular systems together. Let $Y = \{z \in \mathbb{Z}^{n_R} : Az = b, \ell \leq z \leq u\}$ with A totally unimodular; a network is the case of a node-arc incidence matrix. The objective is $F_\gamma(v, z) = \phi(v) + \sum_i f_i(v, z_i) + \gamma^\top v$ with $f_i(v, \cdot)$ convex on $[\ell_i, u_i]$. A *multiplier map* is a polynomial vector $\psi(w) \in \mathbb{R}^{n_R}$ such that $\psi(w)^\top z$ is constant on Y for every w ; for a network $\psi_a(w) = \pi_{p'}(w) - \pi_p(w)$ for an arc $a = (p, p')$, and for a totally unimodular system $\psi(w) = A^\top \lambda(w)$. The *adjusted costs* are $g_i(w, t) = f_i(w, t) - \psi_i(w)t$, and

$$F_\gamma(w, z) - F_\gamma(w, z') = \sum_i [g_i(w, z_i) - g_i(w, z'_i)] \quad (z, z' \in Y). \quad (52)$$

Definition F.12 (Charts). For a network, a *chart* is determined by a feasible flow z , a spanning arborescence $\mathcal{T}$ rooted at an added source in the residual network of z augmented by zero-cost source arcs, an arc a , and an integer t with $t, t+1 \in [\ell_a, u_a]$. Its potentials $\pi^\mathcal{T}(v)$ are the sums of the residual unit marginals of z along the tree paths, and its *chart polynomial* is $m(v) = f_a(v, t+1) - f_a(v, t) + \pi_p^\mathcal{T}(v) - \pi_{p'}^\mathcal{T}(v)$. The family $\mathcal{C}$ of all charts has at most $N_Z(2n_R + s)^s R_1$ members, $R_1 = \sum_a (u_a - \ell_a)$.

A unit difference $f_a(v, t+1) - f_a(v, t)$ has core degree at most $d-1$, so chart polynomials and potentials have core degree at most $d-1$, adjusted costs have total degree at most d , and all chart coefficients have bit length at most a base bound $H_0 = \text{poly}_d(I)$: labels have base length, and tree paths have at most s arcs. A tree chosen by the algorithm is a member of $\mathcal{C}$, because the family contains every structurally valid arborescence of every label. The residual reduced costs of a flow under the potentials of such a tree are, up to sign, chart polynomials.

Proof of theorem 8.9. Algorithm. Run the core search in exact mode with the convex flow oracle. At level j , at the incumbent corner c_j with flow z_j , compute shortest-path distances in the residual network at c_j , choose a spanning arborescence of the subgraph of tight arcs (zero reduced cost) by graph search, and form its potentials $\pi(v)$. For every arc of the residual network of z_j decide exactly whether its reduced cost is nonnegative on $\mathcal{H}_j$, by minimizing it with theorem 8.1 and comparing the algebraic minimum with zero; the source arcs serve only to build the tree and are not tested. If all pass, lemma 8.7 shows that z_j is optimal at every point of $\mathcal{H}_j$, which contains every optimal core; so the slice $[0, 1]^k \times \{z_j\}$ contains a global minimizer, and theorem 8.1 solves it. After level J use label enumeration as in theorem 8.4. Every return is correct, with or without the interiority premise.

Exceptional set. Let $S \subseteq [0, 1]^k$ be the union of the zero sets of the chart polynomials that are not identically zero, and $\mathcal{E} = \bigcup_w [-\nabla F_w](S)$ over feasible labels w , where $F_w(v) = \phi(v) + \sum_a f_a(v, w_a)$. Both are compact of dimension at most $k-1$. For one chart polynomial m and one label w , the set $\{\beta : \exists x \in [0, 1]^k, m(x) = 0, \beta + \nabla F_w(x) = 0\}$ is defined by a formula with one block of k variables, k free variables, $O(k)$ atoms, and degree at most $\max\{d, 2\}$. By (T6) it is a Boolean combination of at most $\Xi_d(k)$ atoms of degree at most $\Xi_d(k)$, with $\Xi_d(k) = (k+1)^{O_d(k^2)}$; as in the proof of proposition 7.16(b), the product of its nonconstant atoms is a nonzero polynomial of degree at most $\Xi_d(k)^3$ vanishing on it. So $\mathcal{E}$ lies in the zero set of a nonzero polynomial of degree at most $D_{\text{int}} = \max\{1, N_Z |\mathcal{C}| \Xi_d(k)^3\}$, and $\log D_{\text{int}} \leq \text{poly}_d(I)$. Nothing of this is computed by the algorithm.

Base choices. Let B be the label-enumeration budget of theorem 8.4, $H_2 \geq 1$ a monomial bound on $\|\nabla^2 F_w\|$ over the core box for all labels in the bounding box, $C_{\text{tube}} = C(k, D_{\text{int}})$ with C from proposition 7.16, and C_{growth} the constant of (T3) for the reduced objective V_0 (core perturbed, flows as the eliminated block). Choose $\rho = 1/(4B)$, $g_0 = \rho\sigma/(2k)$, $A_0 = 2+kL/g_0$, $\delta = \rho\sigma/(4C_{\text{tube}}H_2)$, the least J with $2^{-J} \leq \delta/(2kA_0)$, and the least power of two $M \geq \max\{2, 2^J, 4kC_{\text{growth}}/\rho, 4kC_{\text{tube}}/\rho\}$.

Probability. Let a be an optimal core and w an optimal flow at a . Then $F_w + \gamma^\top v \geq V_\gamma$ with equality at a , so a minimizes this smooth function on the box; by the interiority premise $\gamma = -\nabla F_w(a)$, and $\text{dist}(\gamma, \mathcal{E}) \leq H_2 \text{dist}(a, S)$. By (T3) and the grid tube bound in the proof of proposition 7.16(c), the probability that the projected growth is below g_0 or that $\text{dist}(\gamma, \mathcal{E}) \leq H_2\delta$ is at most $\rho + C_{\text{tube}}(H_2\delta/\sigma + \sqrt{k}/M) \leq \frac{3}{2}\rho < 1/(2B)$.

Stopping. Outside this event the optimal core a is unique, as in lemma F.11 every point of $\mathcal{H}_J$ is within Euclidean distance $kA_0h_J \leq \delta/2$ of a , and $\text{dist}(a, S) > \delta$; so $\mathcal{H}_J \cap S = \emptyset$. Each reduced cost of the computed tree is, up to sign, a chart polynomial. If it is identically zero it passes. Otherwise it has no zero on the connected box $\mathcal{H}_J$, so it has constant sign there; it is nonnegative and, since $c_J \notin S$, nonzero at c_J ; hence it is positive on $\mathcal{H}_J$. The test passes by level J .

Work. Each level makes at most $2n_R$ exact box minimizations, of cost $c_d^k \text{poly}_d(I)$ in total over the J levels after absorbing J and n_R , plus shortest paths and the core search. With one completion and the expected fallback, the expected work is $[8^k Q_{\text{ex}} + c_d^k] \text{poly}_d(I)$. Quadratic objectives use (T10) for the box minimizations and the completion. $\square$

Proof of lemma 8.10. We prove it for totally unimodular A ; networks are a special case. Let $z \in Y$,

choose $\bar{z} \in Y_0$ nearest to z in the ℓ_1 norm, and put $x = z - \bar{z} \in \ker A \cap \mathbb{Z}^{n_R}$. *Conformal circuits.* If $x \neq 0$, choose $y \in \ker A$ conformal to x (same sign wherever $y_i \neq 0$) with minimal nonempty support S . If the vectors of $\ker A$ supported in S formed a space of dimension at least two, adding a small multiple of a nonparallel one and following until a coordinate vanishes would give a conformal kernel vector with smaller support. So that space is a line. Its primitive generator is given by signed maximal minors of the columns in S , which lie in $\{0, \pm 1\}$ by total unimodularity, so a conformal circuit $c \in \{0, \pm 1\}^{n_R}$ with support S exists. Subtracting the largest integer multiple $\min_{i \in S} |x_i|$ of c keeps x integral and conformal and removes a coordinate. Repeating gives $x = \sum_j \theta_j c^{(j)}$ with positive integers θ_j and conformal circuits $c^{(j)}$.

Charging. Each $\bar{z} + c^{(j)}$ satisfies $Az = b$ and lies between $\bar{z}$ and z coordinatewise, hence within the bounds. If it also had all coordinates in the intervals I_i , it would belong to Y_0 and be closer to z . So some coordinate i_j has $\bar{z}_{i_j} \in I_{i_j}$ but $\bar{z}_{i_j} + c_{i_j}^{(j)} \notin I_{i_j}$: $\bar{z}_{i_j}$ is the endpoint of I_{i_j} toward z_{i_j} , and $|x_{i_j}| = \text{dist}(z_{i_j}, I_{i_j})$. By conformality the multiplicities charged to one coordinate i sum to at most $|x_i|$. Each circuit has at most n_R nonzero entries, so $\|x\|_1 \leq n_R \sum_j \theta_j \leq n_R \sum_i \text{dist}(z_i, I_i)$. $\square$

Lemma F.13 (Optimal-face certificate, general form). *Let $\mathcal{A}$, $\mathcal{D}$, $\mathcal{D}_{\mathcal{A}}$, c , ς_i , T_1 , T_∞ , $R_{\mathcal{A}}$, H , K be as in [theorem 8.11](#), with K bounding $|\partial_{v_i} \partial_{z_i} f_i|$. Let $z^0 \in Y$ be optimal at c and let ψ be a polynomial multiplier map such that every $g_i(c, \cdot)$ is minimized over the integers of $[\ell_i, u_i]$ at z_i^0 . Let I_i be the integer minimizers of $g_i(c, \cdot)$, $Y_0 = \{z \in Y : z_i \in I_i\}$, and assume (a)–(c) of [theorem 8.11](#) for these adjusted costs. Then the conclusions of [theorem 8.11](#) hold.*

Proof. By [\(52\)](#), Y_0 is the optimal set at c . For $w \in \mathcal{D}_{\mathcal{A}}$, each $g_i(w, \cdot)$ is convex, constant on I_i by (a), and has first outside differences at least μ_{out} by (b), so $g_i(w, t) - g_i(w, z_i^0) \geq \mu_{\text{out}} \text{dist}(t, I_i)$ for integers t in the native interval. With [\(52\)](#),

$$F_\gamma(w, z) - V(w) \geq \mu_{\text{out}} v_I(z), \quad v_I(z) = \sum_i \text{dist}(z_i, I_i), \quad (53)$$

and every flow of Y_0 is optimal at w . Now let $v \in \mathcal{D}$ with face projection w , write $v = w + \sum_i \varsigma_i d_i e_i$ with $d \geq 0$ supported on the normal coordinates, and let $z \in Y$. Choose $\bar{z} \in Y_0$ by [lemma 8.10](#). By the derivative bounds, $\varsigma_i \partial_{v_i} F_\gamma(w, z) \geq \varsigma_i \partial_{v_i} F_\gamma(c, \bar{z}) - H R_{\mathcal{A}} - K \|z - \bar{z}\|_1 \geq \beta - H R_{\mathcal{A}} - K n_R v_I(z)$, and Taylor's theorem in the core gives

$$F_\gamma(v, z) - V(w) \geq (\mu_{\text{out}} - K n_R \|d\|_1) v_I(z) + (\beta - H R_{\mathcal{A}}) \|d\|_1 - \frac{H}{2} \|d\|_2^2 \geq \left(\beta - H R_{\mathcal{A}} - \frac{H}{2} T_\infty \right) \|d\|_1,$$

using $\|d\|_1 \leq T_1$, $\|d\|_2^2 \leq T_\infty \|d\|_1$, and (b). The coefficient is positive by (c), so off the face $F_\gamma(v, z) > V(w)$, the value of a feasible point on the face. If $\mathcal{D}$ contains every optimal core, every optimal core lies in $\mathcal{D}_{\mathcal{A}}$, and $z^0 \in Y_0$ is optimal there, so the slice of z^0 contains a global minimizer. $\square$

Proof of [theorem 8.11](#). For a network take $\psi_a(w) = \pi_{p'}(w) - \pi_p(w)$ from the tree potentials. At c they certify optimality ([lemma 8.7](#)), which says exactly that each $g_a(c, \cdot)$ is minimized at z_a^0 , and $\psi(w)^\top z$ is constant on Y because supplies are fixed. The intervals I_a are found by binary search on the nondecreasing differences $g_a(c, t+1) - g_a(c, t)$. For (c): on an interval with at least three integers, convexity and equal values at three of its points make $g_a(c, \cdot)$ constant on its convex hull, so $f_a(c, \cdot)$ is affine there and $\partial_{tt} f_a(c, t) = 0$; since $\partial_{tt} f_a(c + h \varsigma_i e_i, t) \geq 0$ for small $h \geq 0$, the inward derivative $\varsigma_i \partial_{v_i} f_a(c, \cdot)$ is convex on that interval. With linear interpolation on two-point intervals and substitution on singletons, and tangent extensions, β_i is the optimum of a separable convex flow problem on the tightened bounds ([proposition 8.6](#)). The polynomial identities in (a) need

$d + 1$ integer representatives per interval because the adjusted cost has total degree at most d . [Lemma F.13](#) gives the conclusions. $\square$

Lemma F.14 (Value margins away from zeros). *Let $p \neq 0$ be a rational polynomial of degree at most d in $q \leq k$ variables with coefficients of bit length at most H_0 , let $\delta > 0$ be rational, let Z_p be its zero set in $[0, 1]^q$, and let $\mathcal{K} = \{x \in [0, 1]^q : \text{dist}(x, Z_p) \geq \delta\}$ (the whole cube if $Z_p = \emptyset$). If $\mathcal{K} \neq \emptyset$, then $\min_{\mathcal{K}} |p| \geq 2^{-(B_0+1)}$ with $B_0 \leq \Xi'_d(k) \text{poly}_d(H_0 + \langle \delta \rangle + 1)$ and $\Xi'_d(k) = (k + 1)^{O_d(k^2)}$.*

Proof. For $q = 0$ the value is a nonzero rational with denominator at most 2^{H_0} . Let $q \geq 1$. By compactness $m = \min_{\mathcal{K}} |p| > 0$. The formula

$$\exists x \forall y : x \in [0, 1]^q \wedge [\neg(y \in [0, 1]^q \wedge p(y) = 0) \vee \|x - y\|^2 \geq \delta^2] \wedge 0 < p(x)^2 < t$$

has truth set (m^2, ∞) : a minimizer witnesses every $t > m^2$, and a witness lies in $\mathcal{K}$, so no $t \leq m^2$ is witnessed. It has two blocks of q variables, one free variable, $O(q)$ atoms, and degree at most $2 \max\{d, 1\}$. Clearing denominators gives integer coefficients of bit length at most $\binom{q+d}{d} \text{poly}(H_0 + \langle \delta \rangle)$. By [\(T6\)](#), an equivalent quantifier-free formula has integer coefficients of bit length at most B_0 of the stated form. The boundary point m^2 is a root of a nonconstant atom polynomial, since otherwise all atom signs would be constant near m^2 . Remove a power of t from it; the remaining integer polynomial has nonzero constant term and coefficients of absolute value at most 2^{B_0} , so Cauchy's bound gives $m^2 \geq 1/(1 + 2^{B_0}) \geq 2^{-(B_0+1)}$, and $m \geq 2^{-(B_0+1)/2}$. $\square$

Lemma F.15 (Normal margins). *For a face $\mathcal{A}$ of $[0, 1]^k$, let $a_{\mathcal{A}}$ be the lexicographically least minimizer of V_{γ} on $\mathcal{A}$, Y^* the set of flows optimal at $a_{\mathcal{A}}$, and, for a normal coordinate i of $\mathcal{A}$, $\beta_{\mathcal{A},i}^* = \min_{z \in Y^*} \varsigma_i \partial_{v_i} F_{\gamma}(a_{\mathcal{A}}, z)$. Under core-only noise with law $U_{\sigma, M}$, $\Pr\{\exists \mathcal{A}, i : \beta_{\mathcal{A},i}^* \in [0, \tau]\} \leq k3^k(\tau/\sigma + 1/M)$. If the optimal core a is unique and lies in the relative interior of $\mathcal{A}$, then $a_{\mathcal{A}} = a$ and $\beta_{\mathcal{A},i}^* \geq 0$ for every normal i .*

Proof. On $\mathcal{A}$, V_{γ} equals V_0 plus the free noise terms plus a constant given by the normal noises, so $a_{\mathcal{A}}$ depends only on the free coefficients; Y^* depends only on $a_{\mathcal{A}}$, because core noise is constant in z . Hence $\beta_{\mathcal{A},i}^* = \bar{b} + \varsigma_i \gamma_i$ with $\bar{b}$ independent of γ_i given the other coefficients, and [\(2\)](#) bounds the conditional probability by $\tau/(2\sigma) + 1/M$. A union over at most $k3^k$ pairs gives the bound. If a is the unique optimal core and lies on $\mathcal{A}$, every minimizer on $\mathcal{A}$ is a global one, so $a_{\mathcal{A}} = a$. For $z \in Y^*$, $F_{\gamma}(\cdot, z) \geq V_{\gamma} \geq V_{\gamma}(a) = F_{\gamma}(a, z)$, so a minimizes the smooth function $F_{\gamma}(\cdot, z)$ on the box, and its inward derivatives are nonnegative. $\square$

Proof of [theorem 8.12](#). Algorithm. At each level $j \leq J$ and for each face $\mathcal{A}$ whose fixed bounds lie in the corresponding intervals of $\mathcal{H}_j$, let $\mathcal{D} = \mathcal{H}_j$, c the midpoint of $\mathcal{D}_{\mathcal{A}}$, compute an optimal flow, tree potentials, and intervals at c , and test (a)–(c) of [theorem 8.11](#); on success solve the whole core slice by [theorem 8.1](#). After level J enumerate labels. Correctness follows from [theorem 8.11](#) and [proposition 7.5\(a\)](#).

Analysis sets. For a face $\mathcal{A}$ with $q \geq 1$ free coordinates, let $S_{\mathcal{A}}$ be the union of the zero sets in the face of the chart polynomials whose restriction to the face is not identically zero, and $\mathcal{E}_{\mathcal{A}} = \bigcup_w [-\nabla_{\mathcal{A}} F_w](S_{\mathcal{A}})$. As in the interior case, $\mathcal{E}_{\mathcal{A}}$ lies in the zero set of a nonzero polynomial of degree at most $D = \max\{1, N_Z |\mathcal{C}| \Xi_d(k)^3\}$. Let $H \geq 1$ bound core Hessians and K the mixed derivatives, $C_{\text{tot}} = 3^k C(k, D)$, $N_{\text{nor}} = k3^k$, and let C_{growth} and B be as before. Choose $\rho = 1/(8B)$, $g_0 = \rho\sigma/(2k)$, $A_0 = 2 + kL/g_0$, $\delta = \min\{\frac{1}{4}, \rho\sigma/(4C_{\text{tot}}H)\}$, $\tau = \rho\sigma/(4N_{\text{nor}})$, and $\mu_0 = 2^{-(B_0+1)}$ from [lemma F.14](#) with $\delta/2$, uniformly for all chart polynomials restricted to all faces. Let J be least with

$$2^{-J} \leq \min\left\{\frac{\delta}{4kA_0}, \frac{\tau}{16kHA_0}, \frac{\mu_0}{4k \max\{1, n_R K\}A_0}\right\},$$

and M the least power of two with $M \geq \max\{2, 2^J, 4kC_{\text{growth}}/\rho, 4kC_{\text{tot}}/\rho, 4N_{\text{nor}}/\rho\}$. Then $J, \log_2 M \leq f_d(k) \text{poly}_d(I)$, because $\log(1/\mu_0)$ is of that form; no sampled quantity enters these choices.

Probability. The growth event costs at most ρ . At an optimal core a in the relative interior of a face $\mathcal{A}$ with $q \geq 1$, every optimal flow w makes $F_w|_{\mathcal{A}} + \gamma^\top v$ minimal at a in the free directions, so $\gamma_{\mathcal{A}} = -\nabla_{\mathcal{A}} F_w(a)$ and $\text{dist}(\gamma_{\mathcal{A}}, \mathcal{E}_{\mathcal{A}}) \leq H \text{dist}(a, S_{\mathcal{A}})$; the tube event that this is at most $H\delta$ for some face costs at most $C_{\text{tot}}(H\delta/\sigma + k/M) \leq \rho/2$. [Lemma F.15](#) costs at most $N_{\text{nor}}(\tau/\sigma + 1/M) \leq \rho/2$. The total is at most $2\rho = 1/(4B)$.

Stopping. Outside these events let a be the unique optimal core and $\mathcal{A}$ its face. At level J , $\mathcal{H}_J \subseteq a + [-A_0h, A_0h]^k$, so the face is tried; its normal widths are at most A_0h , $T_1 \leq kA_0h$, $T_\infty \leq A_0h$, $R_{\mathcal{A}} \leq 2kA_0h$, and $\mathcal{D}_{\mathcal{A}}$, including c , lies within Euclidean distance $kA_0h \leq \delta/4$ of a . If $q \geq 1$, then $\text{dist}(a, S_{\mathcal{A}}) > \delta$, so $\mathcal{D}_{\mathcal{A}}$ is at distance more than $\delta/2$ from $S_{\mathcal{A}}$. A within-interval unit difference of the computed adjusted costs is a chart polynomial vanishing at $c \notin S_{\mathcal{A}}$, hence identically zero on the face: (a) passes. A first outside difference is positive at c because the interval is the full set of minimizers, so it has no zero on the connected set $\mathcal{D}_{\mathcal{A}}$ and, by [lemma F.14](#), is at least $\mu_0 > \mu_0/4 \geq n_RKT_1$ there: (b) passes. For a vertex face the same holds with constant charts. Consequently Y_0 is the optimal set at every point of $\mathcal{D}_{\mathcal{A}}$, in particular at a , so $\beta_i \geq \beta_{\mathcal{A},i}^* - HkA_0h > \tau - HkA_0h$ by [lemma F.15](#), and $\beta - HR_{\mathcal{A}} - \frac{H}{2}T_\infty > \tau - 4kHA_0h \geq \frac{3}{4}\tau$: (c) passes.

Work and output. At most 3^k face attempts per level, each with polynomially many flow calls, interval searches, identities, and c_d^k -cost box minimizations on data of length $f_d(k) \text{poly}_d(I)$, give with the core search and the expected fallback the bound $f_d(k)Q_{\text{ex}} \text{poly}_d(I)$. The completion and the fallback return one flow and one common-root core representation of length $f_d(k) \text{poly}_d(I)$. $\square$

Lemma F.16 (Affine margins). *Let $p(x) = p_0 + \pi^\top x$ be a nonzero affine polynomial on $[0, 1]^q$ whose coefficients have numerators and denominators of bit length at most H_0 , and let Z_p be its zero set in the cube. If $Z_p \neq \emptyset$, then $|p(x)| \geq 2^{-H_0} \text{dist}(x, Z_p)$; otherwise $|p| \geq 2^{-(q+1)H_0}$ on the cube.*

Proof. Suppose first that $Z_p \neq \emptyset$. Then $\pi \neq 0$, since otherwise p would be a nonzero constant without zeros; let $\pi_{\min} \geq 2^{-H_0}$ be the least nonzero $|\pi_i|$. If $p(x) = 0$ there is nothing to prove. If $p(x) > 0$, define v by $v_i = 0$ for $\pi_i > 0$, $v_i = 1$ for $\pi_i < 0$, and $v_i = x_i$ for $\pi_i = 0$. This point of the cube minimizes p there, so $p(v) \leq 0$ because $Z_p \neq \emptyset$. Move from x to v one coordinate with $\pi_i \neq 0$ at a time. The path stays in the cube, and along each leg p decreases at rate $|\pi_i| \geq \pi_{\min}$ per unit length. Hence p reaches zero after an ℓ_1 path of length at most $p(x)/\pi_{\min}$, and this length bounds the Euclidean distance from x to Z_p . If $p(x) < 0$, apply the same argument to $-p$, which has the same coefficients up to sign and the same zero set. If $Z_p = \emptyset$, p has one sign on the cube and its least absolute value is attained at a vertex, where it is a nonzero rational whose denominator divides a product of at most $q + 1$ denominators. $\square$

Proof of corollary 8.13. All unit marginals $\psi_a(t+1) - \psi_a(t) + (W^\top v)_a$, all potentials, and all chart polynomials are affine in the core with base-bounded coefficients. Replace μ_0 in the proof of [theorem 8.12](#) by $2^{-(k+1)H_0}\delta/2$, valid by [lemma F.16](#) at distance $\delta/2$; it has polynomial length, so $J, \log_2 M \leq \text{poly}_d(I)$. The identities in (a) are coefficient comparisons, the minima in (b) are endpoint choices, and the inward derivative $s_i(\partial_i \phi(c) + \gamma_i + (Wz)_i)$ is linear in z , so β_i is a linear-cost flow problem on the tightened bounds. The tube degree bound still uses [\(T6\)](#) for the possibly nonlinear ϕ , but only its logarithm enters. The per-level work is polynomial and is absorbed by $8^k Q_{\text{ex}}$; the completion minimizes ϕ plus a linear function, at cost $c_d^k \text{poly}_d(I)$; quadratic ϕ gives rational output by [\(T10\)](#). $\square$

F.4 Totally unimodular systems

Substitute fixed coordinates, check consistency, and delete dependent rows; the remaining A is totally unimodular with full row rank m , and every native interval has positive length.

Lemma F.17 (A compact dual certificate). *Let z^0 be an integer optimum at a rational core point c . There is $\lambda \in \mathbb{R}^m$ with*

$$f_i(c, z_i^0) - f_i(c, z_i^0 - 1) \leq (A^\top \lambda)_i \quad (z_i^0 > \ell_i), \quad (A^\top \lambda)_i \leq f_i(c, z_i^0 + 1) - f_i(c, z_i^0) \quad (z_i^0 < u_i). \quad (54)$$

Every solution certifies optimality of z^0 . With a rational bound $\bar{V} \geq 1$ on the absolute values of all unit marginals over the core cube, a solution exists in $[-R, R]^m$, $R = m\bar{V} + 1$, and a vertex of (54) intersected with this box is computed in polynomial time; its active basis $\mathbf{C}$, a nonsingular submatrix of $A^\top$ with unit rows appended, has an inverse with entries in $\{0, \pm 1\}$.

Proof. Let $\hat{f}_i$ be the piecewise-linear interpolation of $f_i(c, \cdot)$ at the integers of $[\ell_i, u_i]$; it is convex. On each integer unit cell the function $\sum_i \hat{f}_i$ is affine and the cell intersected with $\{Az = b\}$ is a polytope with a totally unimodular matrix and integral right-hand side, so its vertices are integral. Hence the minimum of $\sum_i \hat{f}_i$ over the real polytope equals the integer minimum, and z^0 is a real optimum. Linear programming duality for the epigraph formulation gives λ such that z^0 minimizes $\sum_i [\hat{f}_i(z_i) - (A^\top \lambda)_i z_i]$ over the box; for convex piecewise-linear functions with integer breakpoints this is (54). Conversely, (54) and discrete convexity make each $f_i(c, t) - (A^\top \lambda)_i t$ minimal at z_i^0 , and $\lambda^\top Az = \lambda^\top b$ on Y .

Every column has an inequality, and a line in the solution set would be orthogonal to all columns, which full row rank excludes; so the nonempty polyhedron has a vertex, determined by m independent active rows forming a nonsingular signed submatrix of $A^\top$, whose inverse has entries in $\{0, \pm 1\}$ by Cramer's rule. Its coordinates are at most $m\bar{V}$ in absolute value, so the box contains it. A vertex of the bounded polyhedron is found by exact linear programming, and its active rows are extracted from the original inequalities and box rows; appending unit rows to $A^\top$ preserves total unimodularity. $\square$

Proof of theorem 8.14 and corollary 8.15. At a core point the algorithm uses the totally unimodular oracle of proposition 8.6, computes λ and its basis $\mathbf{C}$ by lemma F.17, and keeps the basis symbolically: $\lambda(v) = \mathbf{C}^{-1}h(v)$, where $h(v)$ consists of unit marginals and the constants $\pm R$. This is a multiplier map of core degree at most $d - 1$ with base-bounded coefficients, and the adjusted costs at c are minimized at z^0 . The chart family consists of the polynomials $f_i(v, t + 1) - f_i(v, t) - (A^\top \lambda(v))_i$ over all labels, ordered bases of at most $2n_R + 2m$ candidate rows, coordinates, and adjacent pairs; it has at most $N_Z(2n_R + 2m)^m R_1$ members, so all logarithms remain polynomial. Only the original adjusted marginals are tested on the hull; the artificial box rows serve to select a bounded vertex at c . Lemmas 8.10 and F.13 hold for A , and the proofs of theorem 8.12 and corollary 8.13 apply verbatim with these charts. For corollary 8.15, the slack system $Cz + s = b$, $0 \leq s_i \leq b_i - \min_{\ell \leq z \leq u} C_i z$, has the totally unimodular matrix $[C \ I]$ and projects bijectively onto Y ; slacks have zero cost and coupling, so the objective, its core derivatives, and L are unchanged, and the added bounds have polynomial length. $\square$

F.5 Strong fields

Proof of theorem 8.16. Enclosures. For a monomial derivative, the exact range of each univariate power on its interval is computed from the endpoints, adding 0 for an even positive power on an interval containing 0; the product of these intervals, scaled by the coefficient, and the sum over

monomials enclose $\partial_i F_0$ on the hull. All endpoints have polynomial length, and each q_i is computed by counting the grid indices of the closed interval $[-\bar{\partial}_i, -\underline{\partial}_i]$.

Pinning. If $\gamma_i > -\underline{\partial}_i$, then $\partial_i F_\gamma > 0$ on the whole continuous hull; for any feasible x with $x_i > \ell_i$, decreasing x_i to ℓ_i , continuously or between integer labels, strictly decreases F_γ , and the product domain keeps the point feasible. So every global minimizer has $x_i = \ell_i$; the case $\gamma_i < -\bar{\partial}_i$ is symmetric. The bad event is $\gamma_i \in [-\bar{\partial}_i, -\underline{\partial}_i]$, which leaves coordinate i unpinned by this test. These events depend on one coefficient each and are independent with probabilities q_i .

Separation. After substituting the pinned values, each remaining monomial has pairwise adjacent bad variables, which lie in one component of the induced subgraph. So $F_\gamma = c_\gamma + \sum_C f_{C,\gamma}$ on a product of component domains, and choosing a minimizer of each summand minimizes the sum. Each component is solved by [corollary 8.2](#), or for quadratics by [\(T10\)](#), in $A(C) \text{poly}_d(I + \log_2 M)$ with $A(C) = \prod_{i \in C} a_i$; the separated data have length $\text{poly}_d(I + \log_2 M)$.

Expectation. A connected set of size t containing a vertex v has a rooted ordered spanning tree with root v ; there are at most 4^{t-1} shapes and at most Δ_+^{t-1} choices of neighbors, so at most $(4\Delta_+)^{t-1}$ such sets. A connected set S is a bad component only if all its vertices are bad, with probability at most $\prod_{i \in S} q_i$. Hence

$$\mathbb{E} \sum_C A(C) \leq \sum_{S \text{ connected}} \prod_{i \in S} a_i q_i \leq \sum_v a_v q_v \sum_{t \geq 1} (4\Delta_+ \beta)^{t-1} = \frac{\sum_v a_v q_v}{1 - 4\Delta_+ \beta},$$

which gives [\(40\)](#). Under [\(41\)](#), [\(2\)](#) gives $q_i \leq R_i/(2\sigma) + 1/M$, so $a_i q_i \leq \frac{1}{16\Delta_+} + \frac{1}{16\Delta_+}$.

Refinement. Put $q' = q + \lceil \log_2(n+1) \rceil$. For each component separately, use [theorem 8.1](#) to obtain point error at most $2^{-q'}$ relative to its stored optimizer, value width at most $2^{-q'}$, and a feasible rational point with certified gap at most $2^{-q'}$. Clip continuous approximations to their boxes and keep integer labels exact. Combining the point approximations with pinned coordinates gives Euclidean error at most $\sqrt{n} 2^{-q'} \leq 2^{-q}$; adding value intervals and certified gaps gives width and gap at most $n 2^{-q'} \leq 2^{-q}$. The gap witnesses and the point approximations may be refined separately; no distance conclusion is inferred from an objective gap. This adds only $O(\log(n+1))$ bits. If all coordinates are pinned, the rational point and value are returned directly. The proof record (enclosures, pinning comparisons, partition, component computations) is checked within the same bound; independence and the noise condition enter only the expectation. $\square$

F.6 The low-rank lattice theorem

Proof of theorem 8.17. Expansion and scalar recourse. Work with the expanded coefficients of all listed pieces. Expanding a shifted or factored piece of degree at most the fixed d_* takes polynomial bit work and preserves polynomial coefficient length. Let Q_{den} be the denominator product specified in [section 8.6](#); all expanded coefficients have denominator dividing it. For a rational $a \in \mathbb{R}^k$ put $W(a) = \frac{\alpha}{2} \|a\|^2 + \min_{x \in X} [G(x) - \alpha a^\top T x]$. The inner problem separates into $\psi_j(t) = g_j(t) - \alpha (T^\top a)_j t$ on $\{L_j, \dots, U_j\}$. Convexity gives $2\psi_j(t+1) \leq \psi_j(t) + \psi_j(t+2)$, so its forward differences are nondecreasing even when consecutive arguments lie in different pieces. For each difference, locate the piece at each argument by exact comparisons with the rational knots and evaluate both polynomials. At a knot either adjacent piece gives the same value by continuity.

Binary search for the first t with $\psi_j(t+1) - \psi_j(t) \geq 0$ returns a minimizer; if none exists choose U_j . A singleton interval is evaluated directly. The neighboring signs at the returned point, when the neighbors exist, certify optimality by summing the nondecreasing differences to any other integer. Ties, constant pieces, affine pieces, and changes of derivative at knots are allowed. There are $O(1 + \log(U_j - L_j + 1))$ comparisons per coordinate, without enumerating its labels. This

returns $W(a)$ and an attaining integer witness $x(a)$. If $k = 0$, these scalar searches solve the original separable objective directly; henceforth assume $k \geq 1$.

Square completion. Since $\frac{\alpha}{2}\|a\|^2 - \alpha a^\top T x = \frac{\alpha}{2}\|a - T x\|^2 - \frac{\alpha}{2}\|T x\|^2$, $W(a) = \min_x [F(x) + \frac{\alpha}{2}\|a - T x\|^2]$, and $W - \frac{\alpha}{2}\|a\|^2$ is a minimum of affine functions, so W has upper coordinate curvature α . With $Z_\xi(a) = W(a) + \xi^\top a$ and $C_\xi = \|\xi\|^2/(2\alpha)$,

$$Z_\xi(a) = \min_{x \in X} \left[F_\xi(x) + \frac{\alpha}{2} \left\| a - T x + \frac{\xi}{\alpha} \right\|^2 \right] - C_\xi.$$

For an optimal x^* , $a = T x^* - \xi/\alpha$ lies in the box $\mathcal{A} = \prod_i [\ell_i - \sigma_i/\alpha, u_i + \sigma_i/\alpha]$, so $\min_{\mathcal{A}} Z_\xi = F_\xi^* - C_\xi$, and every witness satisfies $F_\xi(x(a)) \leq Z_\xi(a) + C_\xi$.

Search and count. Run the corrected-corner search of [theorem 3.9](#) on $\mathcal{A}$ with $f = W$, $c = \xi$, curvature α in every coordinate, and the isotropic balanced meshes of [definition 3.5](#) ($h_j = s2^{-j}$), through the predetermined level $J = \log_2 M$, with exact evaluations of Z_ξ . Since $m_{iJ} \leq 2^J = M$, [corollary 3.6](#) with the coordinatewise laws $U_{\sigma_i, M}$ gives expected level counts at most H_{rat} , and the expected number of evaluations is at most $2^k + 8^k H_{\text{rat}} J$. The comparison in [corollary 3.6](#) applies only to coordinates with interior grid nodes; coordinates not yet subdivided contribute their two endpoints. Zero, dependent, or constant rows of T cause no exception: each auxiliary width is positive because $\sigma_i > 0$.

Exact termination. By [theorem 3.9\(iii\)](#), the incumbent value satisfies $U_J \leq \min_{\mathcal{A}} Z_\xi + E_J$, and its witness has $0 \leq F_\xi(x_{\text{inc}}) - F_\xi^* \leq U_J + C_\xi - (\min_{\mathcal{A}} Z_\xi + C_\xi) \leq E_J$, and

$$E_J = \frac{\alpha}{8} \sum_i h_{iJ}^2 \leq \frac{k\alpha s^2}{8M^2} \leq \frac{1}{8D_0M} < \frac{1}{D_0(M-1)}.$$

Here $D_0 = 2Q_{\text{den}}^3$, and the choice $M \geq \max\{2, k\alpha s^2 D_0\}$ gives the second inequality. Every base denominator divides Q_{den} . For integer x , $g_j(x_j) \in \frac{1}{Q_{\text{den}}}\mathbb{Z}$, $\frac{\alpha}{2}\|T x\|^2 \in \frac{1}{2Q_{\text{den}}^3}\mathbb{Z}$, and, writing $\xi_i = \sigma_i(2K_i - (M-1))/(M-1)$ with the sampled index K_i , $\xi^\top T x \in \frac{1}{Q_{\text{den}}^2(M-1)}\mathbb{Z}$. So every value $F_\xi(x)$ lies in $\frac{1}{D_0(M-1)}\mathbb{Z}$. The optimum lies in this lattice because X is finite. The nonnegative incumbent gap is strictly smaller than its spacing, so it is zero and x_{inc} is optimal; its value is computed exactly. One common grid size for all rows is what makes the original values share the single denominator factor $M-1$; the auxiliary values Z_ξ and C_ξ can involve $(M-1)^2$, and no lattice property of them is used. An integer unary monomial is an integer, whatever its degree, and a rational knot only selects a piece; neither contributes another sampling denominator.

Bit lengths and work. The sum of the expanded denominator lengths bounds $\log Q_{\text{den}}$, so $\log D_0 = O(I)$. Row extrema are obtained by rational endpoint sums. The widths, s , and $k\alpha s^2 D_0$ therefore have polynomial length. Comparing rationals computes the least adequate power of two in polynomial work; it does not enumerate its support. Thus $J = \log_2 M \leq \text{poly}_{d_*}(I)$, sampling uses exactly kJ fair bits, and each ξ_i has polynomial length.

Every feasible integer has input-bounded bit length. Raising a B -bit integer to a power at most d_* uses $O(d_* B)$ bits; rational Horner evaluation on a listed piece has polynomial numerator and denominator length. Piece lookup compares rational knots with binary integers by integer products, and both lookup and the number of scalar comparisons are polynomial in I . This also covers negative labels and knots within a unit integer step. Continuity and convexity can be checked with rational endpoint evaluations and univariate sign determination on each piece; the input length counts the complete list.

At level $j \leq J$, each queried auxiliary coordinate is its base endpoint plus an integer index times w_i/m_{ij} , with polynomial length in $I + j$. Consequently $\alpha(T^\top a)_j$, all scalar differences, $W(a)$, $Z_\xi(a)$,

E_j , and each original witness value have uniformly polynomial length. Each query combines only polynomially many rationals; denominators do not accumulate from unrelated calls because an incumbent is selected by comparison and each new query is constructed from the base data and its mesh indices. Cell addresses have $O(kJ)$ bits. Every query and every cell therefore costs $\text{poly}_{d_*}(I)$ with an exponent independent of k , even on a draw visiting many cells. Multiplying this cost by $2^k + 8^k H_{\text{rat}} J$ proves the expected bound. The final integer vector has input-bounded length, and the same fixed-degree evaluation bound gives a polynomial-length rational optimal value on every draw. $\square$

G Proofs for the boundary results

This appendix proves [theorems 9.4](#) and [9.10](#) and [proposition 9.11](#).

G.1 Proof of [theorem 9.4](#)

One block. Let $n = m^2$, which is even. Choose $u \in \mathbb{Q}^n$ with $n/2$ entries $1/m$ and $n/2$ entries $-1/m$, such that $u_{n-1} = u_n = 1/m$, and let $B_+, B_- \subseteq \{1, \dots, n-2\}$ be the index sets of the remaining positive and negative entries, so $|B_+| = n/2 - 2$ and $|B_-| = n/2$. Then $\|u\| = 1$. Put $q = e_{n-1} - e_n$, which is orthogonal to u , and $c = \frac{3}{4}$, $\varepsilon = m^{-3}$,

$$A_1 = \alpha[c(uq^\top + qu^\top) + qq^\top],$$

and let $P_1 \subset \mathbb{R}^n$ be the polytope defined by

$$x_i \geq 0 \ (i \leq n-2), \quad \sum_{i \leq n-2} x_i = 4m, \quad x_{n-1} + x_n = 0, \quad |x_{n-1} - x_n| \leq \varepsilon.$$

The instance of the theorem consists of k copies: $A = \text{diag}(A_1, \dots, A_1)$, $T = \text{diag}(u^\top, \dots, u^\top)$, and $P = P_1^k$. Its data have encoding length polynomial in N .

Part (a). In the orthonormal basis $(u, q/\sqrt{2})$ of their common range, A_1 and $A_1 + \alpha uu^\top$ act as

$$\alpha \begin{pmatrix} 0 & c\sqrt{2} \\ c\sqrt{2} & 2 \end{pmatrix} \quad \text{and} \quad \alpha \begin{pmatrix} 1 & c\sqrt{2} \\ c\sqrt{2} & 2 \end{pmatrix},$$

and both vanish on the orthogonal complement. The second matrix has positive trace and determinant $\alpha^2(2 - 2c^2) = \frac{7}{8}\alpha^2 > 0$, so $A + \alpha T^\top T \succeq 0$. The first has eigenvalues $\alpha(1 \pm \sqrt{1 + 2c^2}) = \alpha(1 \pm \sqrt{17/8})$, exactly one of them negative, and $\nu = \alpha(\sqrt{17/8} - 1) > \alpha/4$ because $\sqrt{17/8} > \frac{5}{4}$. The rows of T have disjoint supports and unit norm, so $TT^\top = I_k$. For $x \in P_1$ write $y = u^\top x$ and $z = q^\top x$. Then $x_{n-1} = z/2$, $x_n = -z/2$, and $y = m^{-1}(\sum_{B_+} x_i - \sum_{B_-} x_i)$; since the nonnegative x_i , $i \leq n-2$, have sum $4m$, the pair (y, z) ranges exactly over $[-4, 4] \times [-\varepsilon, \varepsilon]$. Moreover $\frac{1}{2}x^\top A_1 x = \alpha(cyz + z^2/2)$.

The auxiliary function of one block. Let $\gamma \in \mathbb{R}^n$, $\xi = u^\top \gamma$, $\zeta = \gamma - u\xi$, and

$$S = \frac{m}{2} \left(\min_{B_+} \gamma_i - \min_{B_-} \gamma_i \right), \quad t = \frac{\gamma_{n-1} - \gamma_n}{2}.$$

We claim that, for a number K depending on γ only,

$$V(a) = K + \varphi(a), \quad \varphi(a) = \min_{|y| \leq 4, |z| \leq \varepsilon} \left[G(y, z) + \frac{\alpha}{2} \left(a + \frac{\xi}{\alpha} - y \right)^2 \right], \quad (55)$$

with $G(y, z) = \alpha(cyz + z^2/2) + Sy + tz$, where V is the function (45) for one block. Fix (y, z) and minimize $\zeta^\top x$ over the points of P_1 with these projections. The last two coordinates are determined and contribute $(\zeta_{n-1} - \zeta_n)z/2 = tz$, because $u_{n-1} = u_n$. The masses $\sum_{B_+} x_i = m(4+y)/2$ and $\sum_{B_-} x_i = m(4-y)/2$ are determined, and a linear function on a scaled simplex is minimized at a vertex, so the minimum is $\frac{m(4+y)}{2} \min_{B_+} \zeta_i + \frac{m(4-y)}{2} \min_{B_-} \zeta_i$. Since $\zeta_i = \gamma_i - \xi/m$ on B_+ and $\zeta_i = \gamma_i + \xi/m$ on B_- , this equals $K_0 + (S - \xi)y$ with K_0 independent of (y, z) . Hence

$$V(a) = K_0 + \xi a + \min_{y,z} \left[\alpha \left(cyz + \frac{z^2}{2} \right) + (S - \xi)y + tz + \frac{\alpha}{2}(a - y)^2 \right],$$

and $\frac{\alpha}{2}(a - y)^2 + \xi(a - y) = \frac{\alpha}{2}(a - y + \xi/\alpha)^2 - \xi^2/(2\alpha)$ gives (55) with $K = K_0 - \xi^2/(2\alpha)$.

The objective inside (55) is a quadratic in (y, z) with Hessian $\alpha \begin{pmatrix} 1 & c \\ c & 1 \end{pmatrix} \succ 0$. It has a unique minimizer $(y^*(a), z^*(a))$ on the rectangle, and by Danskin's theorem φ is differentiable with $\varphi'(a) = \alpha(a + \xi/\alpha - y^*(a))$.

A good event. Let $\mathcal{E}_1$ be the event that $\min_{B_+} \gamma_i \leq -1 + 4/n$, $\min_{B_-} \gamma_i \leq -1 + 4/n$, and $|\xi| \leq 2$. We show $\mathbb{P}(\mathcal{E}_1) > \frac{1}{4}$ under both laws. Let $\pi = \mathbb{P}\{\gamma_i \leq -1 + 4/n\}$. For the uniform law $\pi = 2/n$. For $U_{1,M}$ the atoms $-1 + 2j/(M-1)$ in this range are those with $j \leq 2(M-1)/n$, at least $2(M-1)/n$ of them, so $\pi \geq (2/n)(1 - 1/M) \geq 19/(10n)$ because $M \geq 32$. In both cases $\pi \geq 19/(10n)$. Using $1 - \pi \leq e^{-\pi}$, $|B_+| \geq 15n/32$ (as $n \geq 64$), and $|B_-| = n/2$,

$$\mathbb{P}\left\{\min_{B_+} \gamma_i \leq -1 + \frac{4}{n}\right\} \geq 1 - e^{-57/64} > \frac{7}{12}, \quad \mathbb{P}\left\{\min_{B_-} \gamma_i \leq -1 + \frac{4}{n}\right\} \geq 1 - e^{-19/20} > \frac{3}{5},$$

because $e^{57/64} > \frac{12}{5}$ and $e^{19/20} > \frac{5}{2}$. These two events depend on disjoint coordinates and are independent. Next, $\text{Var } \xi = \sum_i u_i^2 \text{Var } \gamma_i = \text{Var } \gamma_1$, which is $\frac{1}{3}$ for the uniform law and $(M+1)/(3(M-1)) \leq \frac{3}{8}$ for the grid law, so $\mathbb{P}\{|\xi| > 2\} \leq \frac{3}{32}$ by Chebyshev's inequality. Therefore $\mathbb{P}(\mathcal{E}_1) \geq \frac{7}{12} \cdot \frac{3}{5} - \frac{3}{32} = \frac{41}{160} > \frac{1}{4}$. No independence between ξ and the two minima is used.

Flatness on the good event. On $\mathcal{E}_1$, both minima lie in $[-1, -1 + 4/n]$, so $|S| \leq \frac{m}{2} \cdot \frac{4}{n} = \frac{2}{m}$; also $|t| \leq 1$ and $|\xi| \leq 2$.

Small derivative. Let $|a| \leq \frac{3}{2}$. For every $|z| \leq \varepsilon$, the unconstrained minimizer in y of the objective in (55) is $a + \xi/\alpha - (S + \alpha cz)/\alpha$, whose absolute value is at most $\frac{3}{2} + 2 + \frac{2}{m} + \varepsilon < 4$ because $\alpha \geq 1$. So $y^*(a)$ is this point, and $\varphi'(a) = S + \alpha cz^*(a)$. Since $\alpha \leq m$,

$$|V'(a)| = |\varphi'(a)| \leq \frac{2}{m} + \frac{3}{4} m \cdot m^{-3} < \frac{3}{m} \quad (|a| \leq \frac{3}{2}). \quad (56)$$

Small gap. Let $|a| \leq 1$. The point $(y, z) = (a + \xi/\alpha, 0)$ is feasible, since $|y| \leq 3$, so $\varphi(a) \leq G(a + \xi/\alpha, 0) \leq 3|S| \leq 6/m$. Every feasible (y, z) has $G(y, z) \geq -4\alpha c\varepsilon - 4|S| - \varepsilon \geq -3m^{-2} - 8m^{-1} - m^{-3}$, and $\min \varphi \geq \min G$. Hence

$$V(a) - \min V = \varphi(a) - \min \varphi < \frac{15}{m} \quad (|a| \leq 1). \quad (57)$$

Products and counts. For the k -block instance, A , T , and P split into blocks, so the minimization in (45) splits and $V(a) = \sum_{l=1}^k V^{(l)}(a_l)$, where $V^{(l)}$ is the one-block function of the l th block of noise coordinates. The block events $\mathcal{E}_1^{(l)}$ are independent, so their intersection $\mathcal{E}$ has probability greater than 4^{-k} . The allowance is $2E_h = \alpha kh^2/4$.

For (b)(i), let $8/\sqrt{\alpha m} \leq h \leq 16/\sqrt{\alpha m}$, so $2E_h \geq 16k/m$. On $\mathcal{E}$, every $v \in \mathbb{G}_h \cap [-1, 1]^k$ has $V(v) - \min V = \sum_l (V^{(l)}(v_l) - \min V^{(l)}) < 15k/m < 2E_h$ by (57). A grid of spacing h has at least

$2/h - 1 \geq \sqrt{\alpha m}/8 - 1 \geq \sqrt{\alpha m}/16$ nodes in $[-1, 1]$, because $\alpha m \geq 256$. So the expected count is at least $4^{-k}(\sqrt{\alpha m}/16)^k = (\sqrt{\alpha m}/64)^k$.

For (b)(ii), let $16/(\alpha m) \leq h \leq 32/(\alpha m)$. Then $h \leq \frac{1}{8}$, so $v_l \pm h \in [-\frac{3}{2}, \frac{3}{2}]$ for $v_l \in [-1, 1]$. On $\mathcal{E}$, (56) and the mean value theorem give, for every l and both signs,

$$V(v) - V(v \pm h e_l) = V^{(l)}(v_l) - V^{(l)}(v_l \pm h) \leq \frac{3h}{m} \leq \frac{\alpha h^2}{4} \leq 2E_h,$$

because $h \geq 12/(\alpha m)$. The grid has at least $2/h - 1 \geq \alpha m/16 - 1 \geq \alpha m/32$ nodes in $[-1, 1]$, so the expected count is at least $4^{-k}(\alpha m/32)^k = (\alpha m/128)^k$.

For (c), $\alpha = 1$ and $m = \sqrt{N/k}$ turn the two bounds into $64^{-k}(N/k)^{k/4}$ and $128^{-k}(N/k)^{k/2}$. The matrix A has $O(N)$ nonzero entries of $O(\log N)$ bits and P has $O(N)$ constraints, so even a dense encoding has $I \leq C_0 N^2 \log N$ for an absolute constant C_0 . The quantities recorded in R do not change with m . If either expected count were at most $f(k, R)I^c$, then fixing $k > 8c$ and letting m grow would contradict $64^{-k}(N/k)^{k/4} \leq f(k, R)(C_0 N^2 \log N)^c$.

Finally, a search grid of the low-rank route qualifies: it divides an auxiliary box that contains $[-\frac{3}{2}, \frac{3}{2}]$ in every coordinate by powers of two, and its meshes $w2^{-j}$ meet every interval $[t, 2t]$ with t at most the box width w , in particular both ranges of h in (b). The ambient diameter of P_1 is at least $4\sqrt{2}m$, since P_1 contains $4me_i$ for every $i \leq n - 2$; the theorem therefore does not contradict bounds that grow with the ambient diameter. $\square$

G.2 Proof of theorem 9.10

All three parts, including the core-only-noise and deterministic consequences, also appear in the unpublished companion manuscript [2]; they are reproduced here so that the proof is self-contained. Both use the same two tools.

Lemma G.1 (A sign test). *Let Φ be a polynomial on a box Z with $\nabla^2 \Phi \succeq \kappa I$ on Z for some $\kappa \in (0, 1]$, let z^0 be its unique minimizer on Z , and let ζ be an affine function with $\|\nabla \zeta\| \leq 1$. Put $\varepsilon = \kappa/4$ and, on $Z \times [0, 1]$,*

$$H(z, \theta) = \Phi(z) + \theta^2 + \varepsilon \theta \zeta(z).$$

Then $\nabla^2 H \succeq \frac{3}{4}\kappa I$, H has a unique minimizer (z^H, θ^H) , and $\theta^H > 0$ if and only if $\zeta(z^0) < 0$.

Proof. The Hessian of $\Phi(z) + \theta^2$ is at least κI , and the bilinear term has Hessian of operator norm $\varepsilon \|\nabla \zeta\| \leq \kappa/4$. This gives the modulus and uniqueness on the compact convex box. On the face $\theta = 0$, H equals Φ , whose unique minimizer is z^0 . At $(z^0, 0)$ the z -gradient of H is $\nabla \Phi(z^0)$, which satisfies the first-order conditions of Φ on Z , and $\partial_\theta H = \varepsilon \zeta(z^0)$. If $\zeta(z^0) \geq 0$, the point $(z^0, 0)$ satisfies the first-order conditions of the convex function H on its box, so it is the minimizer and $\theta^H = 0$. If $\zeta(z^0) < 0$, increasing θ decreases H , so $(z^0, 0)$ is not the minimizer; no other point of the face $\theta = 0$ is either, since a minimizer on that face minimizes Φ . Hence $\theta^H > 0$. $\square$

Lemma G.2 (A convex amplifier). *Let $H_\pm$ be polynomials on boxes $Z_\pm \times [0, 1]$, with last coordinates $\theta_\pm$, whose Hessians are at least $\mu_0 I$ for a supplied $\mu_0 > 0$, and whose unique minimizers have exactly one of θ_+^H, θ_-^H positive. Put $\eta = \mu_0/12$ and, with $y \in [0, 1]$,*

$$\Phi = H_+(z_+, \theta_+) + H_-(z_-, \theta_-) + \eta[\theta_+^2(y - 1)^2 + \theta_-^2 y^2].$$

Then Φ is convex on its box and has a unique minimizer, whose y -coordinate is 1 if $\theta_+^H > 0$ and 0 if $\theta_-^H > 0$.

Proof. For a constant c and a direction (π, ς) in (θ, y) , direct differentiation gives

$$(\pi, \varsigma)^\top \nabla^2 [\theta^2 (y - c)^2] (\pi, \varsigma) = 2(\theta \varsigma + 2(y - c)\pi)^2 - 6(y - c)^2 \pi^2.$$

With $c = 1$ for θ_+ , $c = 0$ for θ_- , and $|y - c| \leq 1$, the amplifier contributes at least $-6\eta(\pi_+^2 + \pi_-^2)$, where $\pi_\pm$ are the components of a direction along $\theta_\pm$. The base Hessian contributes at least μ_0 times the squared norm of the base part of the direction. So the Hessian quadratic form is at least $\frac{1}{2}\mu_0$ times that squared norm, and Φ is convex. No division by $\theta_\pm$ occurs, so this includes points where both amplitudes vanish.

The amplifier is nonnegative, so $\Phi \geq \min H_+ + \min H_-$. If $\theta_+^H > 0 = \theta_-^H$, the two base minimizers with $y = 1$ make the amplifier vanish, so the bound is attained; every minimizer must attain both base minima, which fixes all base coordinates, and must make the amplifier vanish, which forces $y = 1$. The case $\theta_-^H > 0$ is symmetric and forces $y = 0$. $\square$

Part (a): Square Root Sum. An instance consists of positive integers $a_1, \dots, a_n$ and an integer B . If $B < 0$, the answer is no. Compute powers of two R_i with $R_i^2 \leq a_i < 4R_i^2$, and put $c_i = a_i/R_i^2 \in [1, 4)$, $R = \max_i R_i$, $w_i = R_i/R \in (0, 1]$, and let K be the least power of two with $K \geq \max\{2, n\}$. Take a complete binary tree with K leaf slots. Each real leaf $i \leq n$ has a variable $u_i \in [1, 2]$ and signal $w_i u_i$; padded leaves have signal 0. Each internal node ν has a variable $v_\nu \in [0, 2]$, its own signal v_ν , and residual $\rho_\nu = v_\nu - \frac{1}{2}(\text{left signal} + \text{right signal})$. Let s be the root variable, Y the box of all these variables, and

$$F_{\text{tree}}(u, v) = \sum_{i=1}^n \left(\frac{u_i^3}{3} - c_i u_i \right) + \sum_{\nu} \rho_\nu^2.$$

Lemma G.3 (Averaging tree). $\nabla^2 F_{\text{tree}} \succeq \frac{1}{8}I$ on Y . The unique minimizer has $u_i = \sqrt{c_i}$, all residuals zero, and root value $s^0 = \sum_i \sqrt{a_i}/(KR) \in [0, 2]$. The bags consisting of an internal variable and its variable children, joined along the tree, form a tree decomposition of bag size at most three that covers every term.

Proof. The leaf term has derivative $u_i^2 - c_i$ and is minimized on $[1, 2]$ at $\sqrt{c_i} \in [1, 2]$. Setting each internal variable to the average of its children's signals makes every residual zero and keeps it in $[0, 2]$; this point minimizes every summand, and its root is the average of the K leaf signals, $K^{-1} \sum_i w_i \sqrt{c_i} = \sum_i \sqrt{a_i}/(KR)$. Write the residuals and the leaf variables of a direction d as $(I - P)d$, where a leaf row of P is zero and the row of an internal node holds the coefficients $\frac{1}{2}w_i$ or $\frac{1}{2}$ of its variable children. Every variable is a child of at most one node, so distinct rows of P have disjoint supports, $PP^\top$ is diagonal with entries at most $\frac{1}{2}$, and $\|P\| \leq 1/\sqrt{2}$. Since $2u_i \geq 2$,

$$d^\top \nabla^2 F_{\text{tree}} d \geq 2 \|(I - P)d\|^2 \geq 2(1 - 1/\sqrt{2})^2 \|d\|^2 \geq \frac{1}{8} \|d\|^2.$$

This also proves uniqueness. A leaf variable occurs only in its parent's bag, which covers its unary term and its parent's residual; adjacent bags share one internal variable. $\square$

Lemma G.4 (Integral radical sums). If positive integers $a_1, \dots, a_n$ satisfy $\sum_i \sqrt{a_i} \in \mathbb{Z}$, then every a_i is a perfect square.

Proof. Let $\mathcal{F} = \mathbb{Q}(\sqrt{a_1}, \dots, \sqrt{a_n})$ and $\text{Tr} = \text{Tr}_{\mathcal{F}/\mathbb{Q}}$. If a_i is not a square, $\mathbb{Q}(\sqrt{a_i})$ has degree two and the trace of $\sqrt{a_i}$ from it to $\mathbb{Q}$ is zero, so transitivity gives $\text{Tr}(\sqrt{a_i}) = 0$. A rational z has $\text{Tr}(z) = [\mathcal{F} : \mathbb{Q}]z$. Applying the trace to $\sum_i \sqrt{a_i} = B' \in \mathbb{Z}$ shows that the sum over the square radicands alone equals B' , so the sum of the positive numbers $\sqrt{a_i}$ over the nonsquares is zero, and there are no nonsquares. $\square$

The field $\mathcal{F}$ is used only in the proof; the reduction tests squareness by integer square roots. If every a_i is a square, decide the instance by integer arithmetic. If $B \geq 2KR$, the answer is yes, because $s^0 \leq 2$. Otherwise $b = B/(KR) \in [0, 2)$, and $s^0 \neq b$ by [lemma G.4](#). Take two copies of F_{tree} with root variables $s_{\pm}$, apply [lemma G.1](#) with $\kappa = \frac{1}{8}$ to the first copy with $\zeta_+ = b - s_+$ and to the second with $\zeta_- = s_- - b$, and apply [lemma G.2](#) with $\mu_0 = \frac{3}{32}$. Then $\theta_+^H > 0$ exactly when $s^0 > b$ and $\theta_-^H > 0$ exactly when $s^0 < b$, so the resulting quartic Φ is convex on its box and its unique minimizer has y -coordinate 1 if $\sum_i \sqrt{a_i} > B$ and 0 if $\sum_i \sqrt{a_i} < B$.

For the structure, take the two decompositions of [lemma G.3](#), add the bags $\{s_{\pm}, \theta_{\pm}\}$ at their roots and the bags $\{\theta_{\pm}, y\}$ attached to them, and join the two new bags; this is a tree decomposition of bag size three that covers every term, so the interaction graph has treewidth at most two. The diagonal Hessian entries are at most $\frac{9}{2}$ for leaf variables, $\frac{5}{2}$ for internal variables, $2 + 2\eta$ for $\theta_{\pm}$, and 4η for y , hence at most 5. All coordinate intervals have width at most two. The affine maps onto $[0, 1]$ preserve degree and scopes and multiply diagonal Hessian entries by at most four, giving the bound 20. Every term has degree at most four, at most three variables, and coefficients bounded by an absolute constant after the substitution, and every variable lies in at most four terms, so every nonconstant monomial of the expanded polynomial has a coefficient bounded by an absolute constant. The constant term need not be bounded, and deleting it does not change the minimizer. The construction takes polynomial time.

Part (b): PosSLP. A straight-line program is a sequence of integers $A_1, \dots, A_s$ in which each A_j is 0, 1, or $A_l \circ A_{l'}$ with $l, l' < j$ and $\circ \in \{+, -, \times\}$; its output is $A = A_s$. The integers can have exponentially many bits. The construction never computes them; it writes only bounded local formulas.

Lemma G.5 (Bounded pair encoding). *Define rational pairs (u_j, v_j) by $(0, \frac{1}{4})$ for the constant 0, $(\frac{1}{4}, \frac{1}{4})$ for the constant 1, and, for $A_j = A_l \pm A_{l'}$ and $A_j = A_l A_{l'}$ respectively,*

$$u_j = \frac{u_l v_{l'} \pm u_{l'} v_l}{4}, \quad v_j = \frac{v_l v_{l'}}{4}, \quad u_j = \frac{u_l u_{l'}}{4}, \quad v_j = \frac{v_l v_{l'}}{4}.$$

Then $v_j > 0$, $u_j = A_j v_j$, and $|u_j|, |v_j| \leq \frac{1}{4}$ for every j . The number $t = (2u_s - v_s)/4 = v_s(2A - 1)/4$ is nonzero, it is positive if and only if $A > 0$, and $|t| \leq \frac{3}{16}$.

Proof. By induction, v_j is a product of positive numbers, and the ratio u_j/v_j equals $u_l/v_l \pm u_{l'}/v_{l'}$ or $(u_l/v_l)(u_{l'}/v_{l'})$, which is A_j . If all earlier coordinates lie in $[-\frac{1}{4}, \frac{1}{4}]$, the new numerators have absolute value at most $2 \cdot \frac{1}{16} \cdot \frac{1}{4} = \frac{1}{32}$ or $\frac{1}{64}$. Since A is an integer, $2A - 1 \neq 0$, and $v_s > 0$. $\square$

List the coordinates $u_1, v_1, \dots, u_s, v_s, t$ as $x_1, \dots, x_N$, $N = 2s + 1$. Each is given by a rule $x_i = p_i(x_1, \dots, x_{i-1})$, where p_i is a constant, one of the bilinear formulas of [lemma G.5](#), or the affine formula for t . On the box $\mathcal{B} = [-\frac{1}{4}, \frac{1}{4}]^N$ these rules satisfy

$$|p_i| \leq \frac{1}{4}, \quad \|\nabla p_i\|_1 \leq 1, \quad \|\nabla^2 p_i\|_2 \leq 1. \quad (58)$$

Indeed, each bilinear rule has at most four gradient entries of absolute value at most $\frac{1}{16}$, or fewer and larger entries when $l = l'$, with $\|\nabla p_i\|_1 \leq \frac{1}{4}$; its Hessian has at most two nonzero entries in each row, of absolute value at most $\frac{1}{4}$, or a single entry at most $\frac{1}{2}$ when $l = l'$; and the rule for t is affine with $\|\nabla p_N\|_1 = \frac{3}{4}$. The exact assignment x^* , defined by $x_i^* = p_i(x_{<i}^*)$, lies in $\mathcal{B}$ by [lemma G.5](#), and $x_N^* = t^*$ has the sign of $A - \frac{1}{2}$.

Lemma G.6 (Weighted gate objective). *Let $G(x) = \sum_{i=1}^N 64^{N-i} (x_i - p_i(x))^2$. Then $\nabla^2 G \succeq I$ on $\mathcal{B}$, and x^* is the unique minimizer of G on $\mathcal{B}$.*

Proof. Write $a_i = 64^{N-i}$, $\rho_i = x_i - p_i(x)$, $D = \text{diag}(\sqrt{a_i})$, and let L be the strictly lower-triangular matrix with $L_{ij} = \partial_j p_i$. For a direction h ,

$$h^\top \nabla^2 G h = 2 \|D(I - L)h\|^2 - 2 \sum_i a_i \rho_i h^\top \nabla^2 p_i h.$$

Put $\hat{E} = DLD^{-1}$, so $\hat{E}_{ij} = 8^{j-i} L_{ij}$ for $j < i$. By (58) every row of $\hat{E}$ has absolute sum at most $\frac{1}{8}$, and every column has absolute sum at most $\sum_{l \geq 1} 8^{-l} = \frac{1}{7}$. Hence $\|\hat{E}\|_2 \leq (\frac{1}{8} \cdot \frac{1}{7})^{1/2} < \frac{1}{4}$ and

$$2 \|D(I - L)h\|^2 = 2 \|(I - \hat{E})Dh\|^2 \geq \frac{9}{8} \|Dh\|^2.$$

On $\mathcal{B}$, $|\rho_i| \leq \frac{1}{2}$, and p_i depends only on $x_{<i}$, so $|h^\top \nabla^2 p_i h| \leq \|h_{<i}\|^2$. Therefore the second term is at least

$$- \sum_i a_i \|h_{<i}\|^2 = - \sum_j h_j^2 \sum_{i>j} a_i \geq -\frac{1}{63} \sum_j a_j h_j^2.$$

Altogether $h^\top \nabla^2 G h \geq (\frac{9}{8} - \frac{1}{63}) \|Dh\|^2 \geq \|h\|^2$, since every $a_j \geq 1$. Because the rules are triangular, $G \geq 0$ vanishes exactly at x^* , which is therefore the unique minimizer. $\square$

Take two copies of G with outputs $t_\pm$, apply lemma G.1 with $\kappa = 1$, $\zeta_+ = -t_+$, and $\zeta_- = t_-$, and apply lemma G.2 with $\mu_0 = \frac{3}{4}$. Since $t^* \neq 0$, $\theta_+^H > 0$ exactly when $A > 0$ and $\theta_-^H > 0$ exactly when $A \leq 0$. The resulting quartic Φ is convex on its rational box, and its unique minimizer has y -coordinate 1 if $A > 0$ and 0 if $A \leq 0$. Every residual $x_i - p_i(x)$ has at most three monomials and every weight has $O(N)$ bits, so Φ is an explicit rational quartic with $O(N)$ monomials, computed in time polynomial in the size of the program.

Part (c). For F_γ , the core and residual terms are separate. The core term $(v - \frac{1}{2})^2 + \gamma v$ equals $(v - v_\gamma)^2$ plus a constant, so the core optimizer is $v_\gamma = (1 - \gamma)/2 \in [\frac{1}{4}, \frac{3}{4}]$, the projected growth is one, and $\partial_{vv} F_\gamma = 2$. The residual objective Φ does not depend on γ , so the unique optimizer is $(v_\gamma, \hat{x})$ on every draw. Given an instance of either problem, construct Φ , compute the law, draw γ , run the algorithm, and compute a point within distance $1/4$ of $(v_\gamma, \hat{x})$. Its y -coordinate is within $1/4$ of $\hat{x}_y$, so comparing it with $\frac{1}{2}$ decides the instance. The answer is correct on every draw, and the expected running time is polynomial. For the deterministic statement apply the algorithm to Φ directly. Finally, Square Root Sum is decidable in polynomial time with a PosSLP oracle [5], which gives the consequences for Square Root Sum also through part (b). $\square$

G.3 Proof of proposition 9.11

The atoms of $U_{1,M}$ are $-1 + 2j/(M-1)$, $0 \leq j < M$. Such an atom is at least $\frac{1}{2}$ exactly when $j \geq 3(M-1)/4$. For a power of two $M \geq 4$, $3M/4$ is an integer and $3M/4 - 1 < 3(M-1)/4 \leq 3M/4$, so the condition is $j \geq 3M/4$, which holds for $M/4$ indices. Each coordinate therefore has probability $\frac{1}{4}$, and by independence $\Pr(\mathcal{E}) = \frac{1}{16}$. The second derivatives of F_γ in the core are $\partial_{v_i v_i} = 1$, so $L = 1$ is valid.

Part (a). Minimizing $v_1 z_1 + v_2 z_2$ over the two labels gives $\min\{v_1, v_2\}$, so V_γ has the stated form. On $\mathcal{E}$ each $\gamma_i > 0$, and $v_i \geq v_i^2$ on $[0, 1]$, so $\gamma^\top v \geq \min_i \gamma_i \|v\|^2$; with $\min\{v_1, v_2\} \geq 0$ this gives $V_\gamma(v) \geq (\frac{1}{2} + \min_i \gamma_i) \|v\|^2 \geq \|v\|^2$. Since $V_\gamma(0) = 0$, the origin is the unique optimal core, and both labels have cost zero there.

Part (b). At $(a, 0)$ with $a > 0$ the label $(1, 0)$ costs a and $(0, 1)$ costs 0; at $(0, b)$ with $b > 0$ the opposite holds. The noise term is common to both labels and cancels from these comparisons. So

each label is strictly worse than the other at some point of the box, and no valid certificate can assert that one label is optimal throughout.

Part (c). The cell $[0, h]^2$ has the corner 0 with value $V_\gamma(0) = \min V_\gamma$, and the incumbent U_j is a feasible value, hence at least $\min V_\gamma$. So the lower bound of the cell, at most $V_\gamma(0) - h^2/4$, does not exceed U_j , and the cell is retained whenever its parent is. At level 0 the whole core box is retained, and induction gives the claim at every level.

The chain. With $m - 1$ further stages of two zero-cost parallel arcs, each feasible flow chooses one arc per stage, so there are 2^m flows. The conditional value is unchanged, so parts (a)–(c) hold, and on $\mathcal{E}$ every flow is optimal at the origin. In every box $[0, a] \times [0, b]$ the optimal flows at $(a, 0)$ and at $(0, b)$ use different first-stage arcs, so no single flow is certified on the retained hull at any level. A strategy that enumerates all flows when this certificate fails does at least 2^m work on $\mathcal{E}$, and its expected work is at least $2^m/16$. The network has $2m$ unit-capacity arcs and $m + 1$ nodes, so its encoding has length $O(m \log m)$. $\square$

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